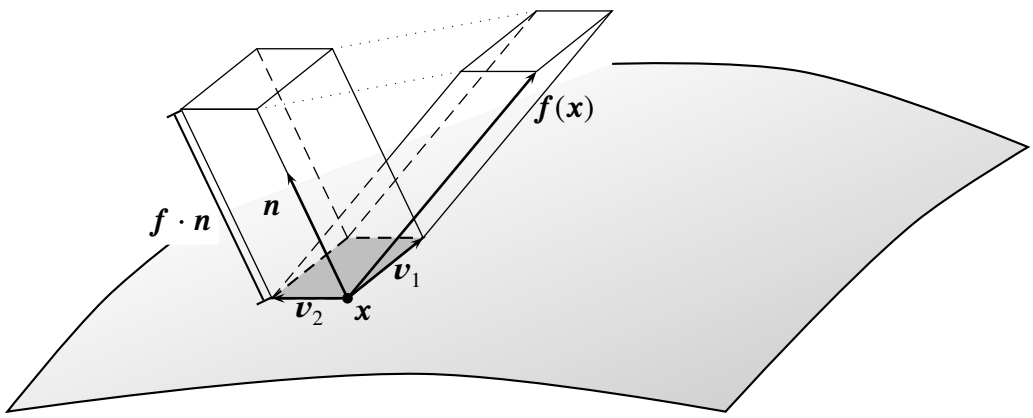
THE UNIVERSITY OF SYDNEY

Lecture Notes for MATH 2921

Vector Calculus (Advanced)



Daniel Daners

School of Mathematics and Statistics

Semester 1, 2023

Lecture Notes for
Vector Calculus

Daniel Daners

School of Mathematics and Statistics
The University of Sydney
NSW 2006, Australia

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Foreword

These notes have been designed to serve as a textbook to MATH2901: Vector Calculus and Complex Variables (Advanced). The main purpose is to present the theory and outline the ideas behind them. The emphasis is on providing an intuitive understanding and showing the relationship between seemingly different concepts, and not on complete and rigorous proofs. As a consequence the notes contain relatively few worked examples. More worked examples are given in lectures and tutorials.

We mainly work in two or three dimensions, but I have made sure that all results generalise without much difficulties to higher dimensions. In particular this is the case for the definition of line and surface integrals, which immediately generalise to integrals over "k-dimensional sub-manifolds" of $\mathbb{R}^N$.

Most of the material in Chapter 1 should be known from first year calculus and linear algebra, except that we work in higher dimension. A thorough understanding is necessary to be able to master the new concepts such as surface integrals or the flux of a vector field across a surface. We derive some new results on the area of parallelograms and parallelepipeds useful for the transformation formula and the definition of surface integrals.

A ☞ at a remark or an example indicates that it either contains an important special case, or it points out connections which are relevant for the understanding of later material. Important definitions are displayed in shadowed boxes

$$\boxed{\text{definition}}$$

and important or particularly useful formulae in oval boxes

$$\text{important formula.}$$

The lectures do not necessarily follow the order of the notes, and some sections may not be covered at all. I hope the notes serve as a useful complement to lectures.

Daniel Daners
Sydney, February 2002

Chapter 1

Functions of Several Variables

The aim of vector calculus is to study functions of several variables, or functions whose values are vectors. Such functions appear naturally in applications. As examples think of the temperature of a body varying with position, or the position of a particle travelling through the air. The first is a function of position described by three spatial components. The second is a function of time, but its value is a vector. We say it is vector valued. It also makes sense to study functions of several variables which are vector valued. Just think of the velocity of a fluid, which depends on position and time, and whose value can be thought of as a vector with three components. Hence we have to study a function of four variables, whose values are vectors with three components. There are many more models involving one or several quantities depending on a number of other quantities. Vector calculus provides a powerful mathematical tool to treat such problems.

1.1 Vectors

Most, but not all of the material presented in this section is a review of some facts about vectors you know from first year. We only extend some definitions to "higher dimensions." A more in depth presentation will be given in a course on linear algebra or matrix applications. As its name says, *vector calculus* is calculus with vectors, and thus a good working knowledge of vectors is essential.

1.1.1 Euclidean Space

For every positive integer N we introduce the set

$$\mathbb{R}^N := \{(x_1, x_2, \dots, x_N) \mid x_i \in \mathbb{R}, i = 1, \dots, N\}$$

of arrays of real numbers of length N. For $N = 1$ we set $\mathbb{R}^1 := \mathbb{R}$. If $N = 2$ we can interpret (x_1, x_2) as the coordinates of a point or the components of a vector in the plane as shown in Figure 1.1.1. Likewise for $\mathbb{R}^3$ as shown in Figure 1.1.1 we can interpret (x_1, x_2, x_3) as the coordinates of a point or the components of a vector in space.

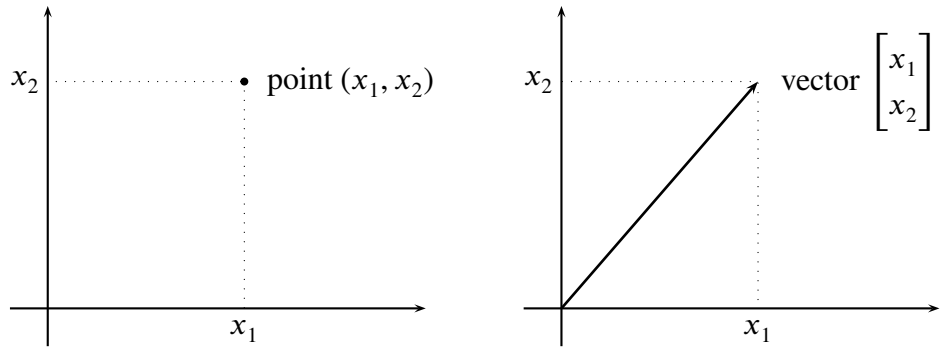
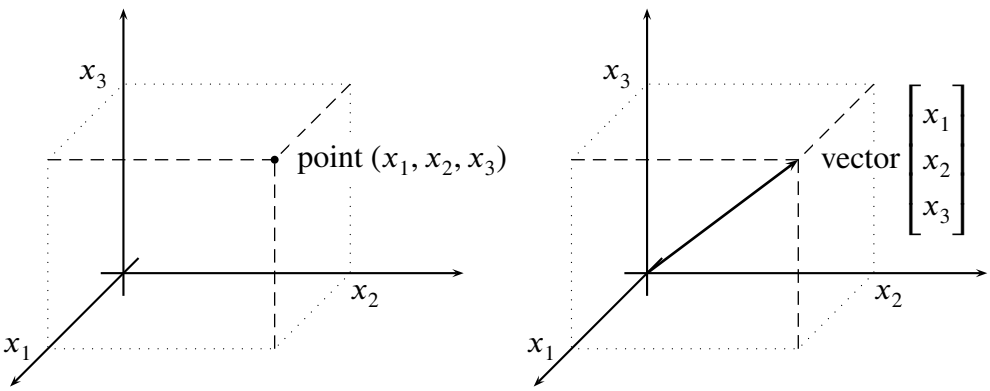


Figure 1.1.1: Coordinates of a point or components of a vector in the plane



In analogy to $N = 2$ or $N = 3$ we call the array

$$\boldsymbol{x} := (x_1, x_2, \dots, x_N)$$

an N-*vector* or simply a *vector*, and denote it by $\boldsymbol{x}$. The entries in such an array we call the *components* of a vector, or the *coordinates* of the point $\boldsymbol{x} \in \mathbb{R}^N$. We often write such a vector as a "column vector"

$$\boldsymbol{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_N \end{bmatrix}$$

as in case of vectors in the plane or in space. To save space we often write $(x_1, \dots, x_N)$ for the above column vectors. Note that, even for $N = 2$ or $N = 3$, the "coordinates" of a

"vector" do not need to represent the coordinates of a point in the plane or in space. They can represent other quantities, indicating the dependence of one quantity upon others. Hence the number of "components" is not restricted to the physical "dimensions." As an example consider the price of a car. It certainly depends on the price of the materials to manufacture it, the cost of labour, and the cost of shipping it to the dealer. Nobody would claim that these are the only cost factors, by limiting the length of an array to three! Try to think of other factors influencing the price of the car, give each a name, and you end up considering arrays of five, ten, or one hundred entries, which we then call components or coordinates.

In analogy to $\mathbb{R}^2$ and $\mathbb{R}^3$ we call N the *dimension* of $\mathbb{R}^N$, and call $\mathbb{R}^N$ the *N-dimensional* (*Euclidean*) *space*, or an *N-dimensional vector space*. Hence we use the word "space" in two different meanings. The first is as "continuous expanse in which things exist and move," to quote the description given in the "Oxford Dictionary" [6]. The second meaning is the mathematical definition of the set of all arrays of real numbers of a certain length. Hence there is nothing mysterious about four or ten dimensional "spaces." In linear algebra a precise mathematical definition of the "dimension" of a "vector space" will be given. Intuitively, it represents the number of (independent) "degrees of freedom" of a system, which can be more than three! Even in physics "higher dimensional" problems appear. As an example look at the equations describing the electric and magnetic fields. The Maxwell Equations show that they are not independent of each other, and a full description is only possible when looking at the two simultaneously. Hence it is convenient to look at one array of six components, the first three representing the electric, and the second three representing the magnetic field. The vector comprising the two has six components, and is usually called the "electro-magnetic field." The field in general depends on at least four variables, namely the three spatial components and time. It may depend on more variables, so for instance the position and velocity of moving charged particles. Hence the "field" depends on $3 + 1 + 3 + 3 = 10$ variables, and thus is a function defined on a subset of $\mathbb{R}^{10}$ with values in $\mathbb{R}^6$.

1.1.2 Addition and Multiplication by Scalars

Given two vectors $\boldsymbol{x} = (x_1, \dots, x_N)$ and $\boldsymbol{y} = (y_1, \dots, y_N)$ in $\mathbb{R}^N$ we define their sum by

$$\boldsymbol{x} + \boldsymbol{y} := (x_1 + y_1, \dots, x_N + y_N).$$

The operation is called the *addition* of vectors. Moreover, we set

$$-\boldsymbol{x} := (-x_1, \dots, -x_N)$$

and define $\boldsymbol{y} - \boldsymbol{x} := \boldsymbol{y} + (-\boldsymbol{x})$, and call it *subtraction* of vectors. The vector $-\boldsymbol{x}$ is sometimes called the *vector opposite to* $\boldsymbol{x}$. If $\boldsymbol{x}$ and $\boldsymbol{y}$ are vectors in the plane or in space the sum and difference can be interpreted geometrically as shown in Figure 1.1.2.

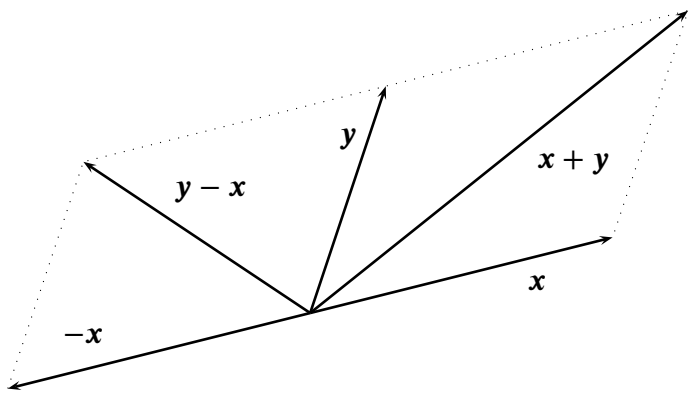


Figure 1.1.2: Sum and difference of two vectors.

We finally define the *zero vector* in $\mathbb{R}^N$ by

$$\mathbf{0} := (\underbrace{0, \dots, 0}_{N \text{ zeros}}).$$

Elements of $\mathbb{R}$ are often called *scalars* (as opposed to *vectors*). If α is a scalar and $\boldsymbol{x} = (x_1, \dots, x_N)$ a vector we define the *multiplication by scalars* by

$$\alpha\boldsymbol{x} := (\alpha x_1, \dots, \alpha x_N).$$

Geometrically this means that we "stretch" the vector $\boldsymbol{x}$ by the factor α. Note that the algebraic rules for addition and multiplication of N-vectors are *exactly the same* as those for vectors in the plane or in space. Given vectors $\boldsymbol{x}$, $\boldsymbol{y}$ and $\boldsymbol{z}$ in $\mathbb{R}^N$ and scalars $\alpha, \beta \in \mathbb{R}$

these rules are:

$$
\begin{aligned}
\boldsymbol{x}+\boldsymbol{y} &= \boldsymbol{y}+\boldsymbol{x} && \text{(commutative law)}\\
(\boldsymbol{x}+\boldsymbol{y})+\boldsymbol{z} &= \boldsymbol{y}+(\boldsymbol{x}+\boldsymbol{z}) && \text{(associative law)}\\
\boldsymbol{x}+\boldsymbol{0} &= \boldsymbol{x} && \text{(existence of a neutral element)}\\
\boldsymbol{x}+(-\boldsymbol{x}) &= \boldsymbol{0} && \text{(existence of an opposite vector)}\\
\alpha(\boldsymbol{x}+\boldsymbol{y}) &= \alpha\boldsymbol{x}+\alpha\boldsymbol{y} && \text{(distributive law)}\\
(\alpha+\beta)\boldsymbol{x} &= \alpha\boldsymbol{x}+\beta\boldsymbol{x} && \text{(distributive law)}\\
(\alpha\beta)\boldsymbol{x} &= \alpha(\beta\boldsymbol{x}) && \text{(associative law)}\\
1\boldsymbol{x} &= \boldsymbol{x} && \text{(existence of a neutral element)}
\end{aligned}
\tag{1.1.1}
$$

As you will learn in linear algebra, these are the axioms of what is usually called a "vector space" in mathematics.

1.1.3 Scalar Product and Norm

For two vectors $\boldsymbol{x} = (x_1, \dots, x_N)$ and $\boldsymbol{y} = (y_1, \dots, y_N)$ in $\mathbb{R}^N$ we define the *scalar product* by

$$\boxed{\boldsymbol{x}\cdot\boldsymbol{y} := \sum_{i=1}^{N} x_i y_i.}$$

It is also sometimes called *inner product* or *dot product*. It is called *scalar* product, because its result is a scalar, and not a vector. If $\boldsymbol{x}$, $\boldsymbol{y}$, $\boldsymbol{z}$ are vectors, and α a scalar it is easily verified from the above definition that

$$
\begin{aligned}
\boldsymbol{x}\cdot\boldsymbol{y} &= \boldsymbol{y}\cdot\boldsymbol{x} && \text{(commutative law)}\\
(\boldsymbol{x}+\boldsymbol{y})\cdot\boldsymbol{z} &= \boldsymbol{x}\cdot\boldsymbol{z}+\boldsymbol{y}\cdot\boldsymbol{z} && \text{(distributive law)}\\
\alpha(\boldsymbol{x}\cdot\boldsymbol{y}) &= (\alpha\boldsymbol{x})\cdot\boldsymbol{y} = \boldsymbol{x}\cdot(\alpha\boldsymbol{y}).
\end{aligned}
\tag{1.1.2}
$$

We next define the *norm* of a vector $\boldsymbol{x} = (x_1, \dots, x_N)$ by

$$\boxed{\|\boldsymbol{x}\| := \sqrt{\boldsymbol{x}\cdot\boldsymbol{x}} = \sqrt{\sum_{i=1}^{N} x_i^2}.}$$

Alternatively we say $\|\boldsymbol{x}\|$ is the *length* or the *magnitude* of the vector $\boldsymbol{x}$. As a sum of squares $\boldsymbol{x}\cdot\boldsymbol{x}$ is always nonnegative, and zero if and only if $\boldsymbol{x}$ is the zero vector. If $\|\boldsymbol{x}\| = 1$ we call $\boldsymbol{x}$ a *unit vector*. For vectors in the plane and in space it follows from Pythagoras' theorem that $\|\boldsymbol{x}\|$ is the *length* of $\boldsymbol{x}$. Even for $N \geq 3$ we still talk about the "length" of the vector $\boldsymbol{x}$, meaning its norm.

Remark 1.1.1 If we assume that $\boldsymbol{x}$ and $\boldsymbol{y}$ are 'column vectors,' that is, $1 \times N$-matrices, then

$$\boldsymbol{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_N \end{bmatrix} \qquad \boldsymbol{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_N \end{bmatrix}$$

then by using matrix multiplications

$$\boldsymbol{x} \cdot \boldsymbol{y} = \boldsymbol{x}^T \boldsymbol{y} \qquad \text{and} \qquad \|\boldsymbol{x}\| = \sqrt{\boldsymbol{x}^T \boldsymbol{x}}.$$

Here, as usual, $\boldsymbol{x}^T$ means the transposed matrix to $\boldsymbol{x}$.

There is another formula for the scalar product, giving it a geometric interpretation. It is often used to define the dot product for vectors in $\mathbb{R}^2$ and $\mathbb{R}^3$.

Proposition 1.1.2 *If $\theta \in [0, \pi]$ is the angle between the vectors $\boldsymbol{x}$ and $\boldsymbol{y}$ then*

$$\boxed{\boldsymbol{x} \cdot \boldsymbol{y} = \|\boldsymbol{x}\|\|\boldsymbol{y}\| \cos\theta.} \tag{1.1.3}$$

If $\boldsymbol{x} \neq \boldsymbol{0}$ and $\boldsymbol{y} \neq \boldsymbol{0}$ then

$$\boxed{\cos\theta = \frac{\boldsymbol{x} \cdot \boldsymbol{y}}{\|\boldsymbol{x}\|\|\boldsymbol{y}\|}.} \tag{1.1.4}$$

Proof. The formula follows from the law of cosines. Indeed, applying the law of cosines to the triangle with side lengths $\|\boldsymbol{x}\|$, $\|\boldsymbol{y}\|$ and $\|\boldsymbol{y} - \boldsymbol{x}\|$ as shown in Figure 1.1.3 we have

$$\|\boldsymbol{y} - \boldsymbol{x}\|^2 = \|\boldsymbol{x}\|^2 + \|\boldsymbol{y}\|^2 - 2\|\boldsymbol{x}\|\|\boldsymbol{y}\| \cos\theta.$$

Using the rules (1.1.2) for the scalar product, and the definition of the norm we see that

$$\|\boldsymbol{y} - \boldsymbol{x}\|^2 = \|\boldsymbol{x}\|^2 + \|\boldsymbol{y}\|^2 - 2\boldsymbol{x} \cdot \boldsymbol{y}.$$

Substituting this into the previous formula yields (1.1.3). ■

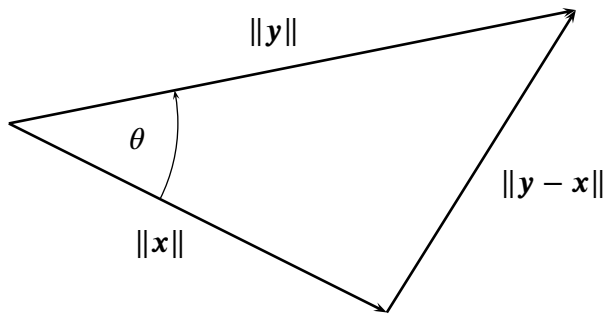


Figure 1.1.3: Triangle formed by $\boldsymbol{x}$, $\boldsymbol{y}$ and $\boldsymbol{y}-\boldsymbol{x}$.

☞ **Remark 1.1.3** One important consequence of (1.1.3), which we will use extensively later, is, that the scalar product allows us to compute the projection of one vector into the direction of another vector. More precisely,

$$\boldsymbol{p} = \|\boldsymbol{y}\| \cos\theta \frac{\boldsymbol{x}}{\|\boldsymbol{x}\|} = \frac{\boldsymbol{x}\cdot\boldsymbol{y}}{\|\boldsymbol{x}\|^2}\,\boldsymbol{x} \tag{1.1.5}$$

is the projection of $\boldsymbol{y}$ into the direction of $\boldsymbol{x}$ as shown in Figure 1.1.4). We call $\|\boldsymbol{p}\|$ the *component of* $\boldsymbol{y}$ *in the direction of* $\boldsymbol{x}$.

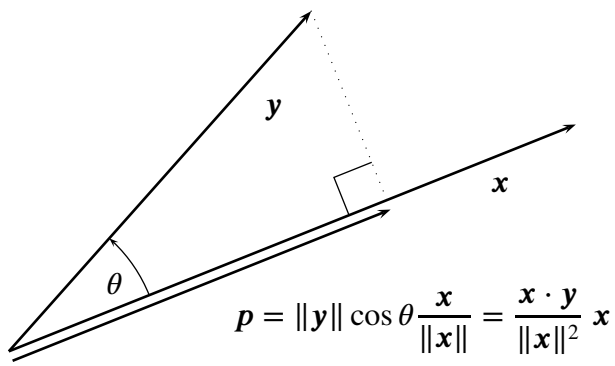


Figure 1.1.4: Projection of $\boldsymbol{y}$ in the direction of $\boldsymbol{x}$.

It is tempting to *define* the angle between two vectors in $\mathbb{R}^N$ ($N > 3$) by

$$\cos\theta = \frac{\boldsymbol{x}\cdot\boldsymbol{y}}{\|\boldsymbol{x}\|\|\boldsymbol{y}\|}.$$

Note however that $\cos\theta$ needs to be between -1 and 1. If we can show that $|\boldsymbol{x}\cdot\boldsymbol{y}| \leq \|\boldsymbol{x}\|\|\boldsymbol{y}\|$ in general then we can indeed define angles between N-vectors. The above inequality turns out to be true always and is often called the "Cauchy-Schwarz" inequality.

Theorem 1.1.4 (Cauchy-Schwarz Inequality) *For all* $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^N$ *we have*

$$\boxed{|\boldsymbol{x}\cdot\boldsymbol{y}| \leq \|\boldsymbol{x}\|\|\boldsymbol{y}\|.} \tag{1.1.6}$$

Moreover, equality holds if and only if $\boldsymbol{x}$ *and* $\boldsymbol{y}$ *are linearly dependent.*

Proof. If $\boldsymbol{x} = \boldsymbol{0}$ or $\boldsymbol{y} = \boldsymbol{0}$ the inequality is obvious and $\boldsymbol{x}$ and $\boldsymbol{y}$ are linearly dependent. Hence assume that $\boldsymbol{x} \neq \boldsymbol{0}$ and $\boldsymbol{y} \neq \boldsymbol{0}$. We can then define

$$\boldsymbol{n} = \boldsymbol{y} - \frac{\boldsymbol{x} \cdot \boldsymbol{y}}{\|\boldsymbol{x}\|^2}\,\boldsymbol{x}.$$

A geometric interpretation of $\boldsymbol{n}$ is shown in Figure 1.1.5. Using the algebraic rules

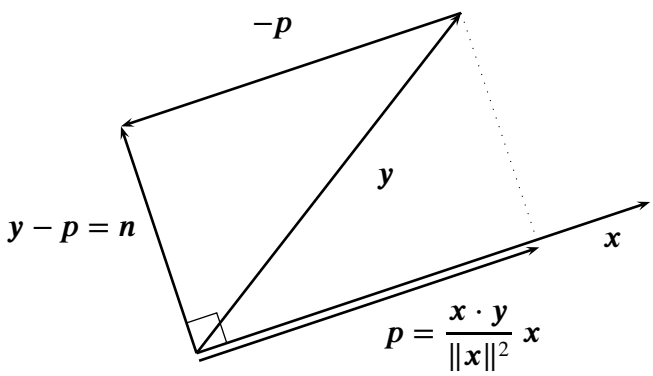


Figure 1.1.5: Geometric interpretation of $\boldsymbol{n}$.

(1.1.2) of the scalar product and the definition of the norm we get

$$\begin{aligned} 0 \leq \|\boldsymbol{n}\|^2 &= \boldsymbol{y} \cdot \boldsymbol{y} - 2\frac{\boldsymbol{x} \cdot \boldsymbol{y}}{\|\boldsymbol{x}\|^2}\boldsymbol{x} \cdot \boldsymbol{y} + \frac{(\boldsymbol{x} \cdot \boldsymbol{y})^2}{\|\boldsymbol{x}\|^4}\boldsymbol{x} \cdot \boldsymbol{x} \\ &= \|\boldsymbol{y}\|^2 - 2\frac{(\boldsymbol{x} \cdot \boldsymbol{y})^2}{\|\boldsymbol{x}\|^2} + \frac{(\boldsymbol{x} \cdot \boldsymbol{y})^2}{\|\boldsymbol{x}\|^4}\|\boldsymbol{x}\|^2 = \|\boldsymbol{y}\|^2 - \frac{(\boldsymbol{x} \cdot \boldsymbol{y})^2}{\|\boldsymbol{x}\|^2}. \end{aligned}$$

Therefore $(\boldsymbol{x} \cdot \boldsymbol{y})^2 \leq \|\boldsymbol{x}\|^2\|\boldsymbol{y}\|^2$, and by taking square roots we find (1.1.6). Clearly $\|\boldsymbol{n}\| = 0$ if and only if

$$\boldsymbol{y} = \frac{\boldsymbol{x} \cdot \boldsymbol{y}}{\|\boldsymbol{x}\|^2}\,\boldsymbol{x},$$

that is, $\boldsymbol{y} = \alpha\boldsymbol{x}$ for some $\alpha \in \mathbb{R}$, that is, $\boldsymbol{x}$ and $\boldsymbol{y}$ are linearly independent. This completes the proof of the theorem. ■

It is common practice in mathematics to make a fact in some particular situation a definition in a more general situation. Here we proved that in the plane or in space the cosine of the angle between two vectors is given by (1.1.4). For higher dimension no angles are defined, so we take (1.1.4) as a *definition* of the angle.

Definition 1.1.5 (Angles, Orthogonal Vectors) If $\boldsymbol{x}$ and $\boldsymbol{y}$ are two (non-zero) vectors in $\mathbb{R}^N$ (N arbitrary) we define the angle, θ, between $\boldsymbol{x}$ and $\boldsymbol{y}$ by

$$\cos\theta := \frac{\boldsymbol{x} \cdot \boldsymbol{y}}{\|\boldsymbol{x}\|\|\boldsymbol{y}\|}.$$

(By the Cauchy-Schwarz inequality the above is always in the interval $[-1, 1]$, so the definition makes sense.) We say $\boldsymbol{x}$ and $\boldsymbol{y}$ are *orthogonal* or *perpendicular* if $\boldsymbol{x} \cdot \boldsymbol{y} = 0$.

We next summarise the main properties of the norm.

Proposition 1.1.6 *Let $\boldsymbol{x}$, $\boldsymbol{y} \in \mathbb{R}^N$ and $\alpha \in \mathbb{R}$. Then*

- $\|x\| \geq 0$, *and* $\|x\| = 0$ *if and only if* $\boldsymbol{x} = \boldsymbol{0}$.
- $\|\alpha \boldsymbol{x}\| = |\alpha| \|\boldsymbol{x}\|$.
- $\|\boldsymbol{x} + \boldsymbol{y}\| \leq \|\boldsymbol{x}\| + \|\boldsymbol{y}\|$ *(Triangle Inequality).*

Proof. The first two properties follow easily from the definition of the norm. To prove the last one we use (1.1.2), the definition of the norm, and inequality (1.1.6) to see that

$$\begin{aligned}\|\boldsymbol{x} + \boldsymbol{y}\|^2 = (\boldsymbol{x} + \boldsymbol{y}) \cdot (\boldsymbol{x} + \boldsymbol{y}) &= \|\boldsymbol{x}\|^2 + \|\boldsymbol{y}\|^2 + 2\boldsymbol{x} \cdot \boldsymbol{y} \\ &\leq \|\boldsymbol{x}\|^2 + \|\boldsymbol{y}\|^2 + 2\|\boldsymbol{x}\|\|\boldsymbol{y}\| = \big(\|\boldsymbol{x}\| + \|\boldsymbol{y}\|\big)^2.\end{aligned}$$

Taking the square root, the inequality follows. ■

The last of the above properties reflects the fact that the total length of two edges in a triangle is bigger than the length of the third edge. For this reason it is called the *triangle inequality*.

1.1.4 The Area of a Parallelogram

The purpose of this section is to derive a formula for the area of a parallelogram spanned by two vectors in the plane, in space or more generally in $\mathbb{R}^N$. The result is essential for the understanding of the transformation formula for double integrals and the definition of surface integrals! The area of a parallelogram is given by the product of its base and its height. If we look at the parallelogram spanned by two vectors $\boldsymbol{x}$ and $\boldsymbol{y}$ then the base is $\|\boldsymbol{x}\|$, and the height is $\|\boldsymbol{n}\|$, where $\boldsymbol{n}$ is the projection of the vector $\boldsymbol{y}$ into the direction orthogonal to $\boldsymbol{x}$ as shown in Figure 1.1.6. To compute $\|\boldsymbol{n}\|$ note that

$$\begin{aligned}\|\boldsymbol{n}\|^2 = \|\boldsymbol{y} - \boldsymbol{p}\|^2 &= \boldsymbol{y} \cdot \boldsymbol{y} - 2\frac{\boldsymbol{x} \cdot \boldsymbol{y}}{\|\boldsymbol{x}\|^2}\boldsymbol{x} \cdot \boldsymbol{y} + \frac{(\boldsymbol{x} \cdot \boldsymbol{y})^2}{\|\boldsymbol{x}\|^4}\boldsymbol{x} \cdot \boldsymbol{x} \\ &= \|\boldsymbol{y}\|^2 - 2\frac{(\boldsymbol{x} \cdot \boldsymbol{y})^2}{\|\boldsymbol{x}\|^2} + \frac{(\boldsymbol{x} \cdot \boldsymbol{y})^2}{\|\boldsymbol{x}\|^4}\|\boldsymbol{x}\|^2 = \|\boldsymbol{y}\|^2 - \frac{(\boldsymbol{x} \cdot \boldsymbol{y})^2}{\|\boldsymbol{x}\|^2}.\end{aligned}$$

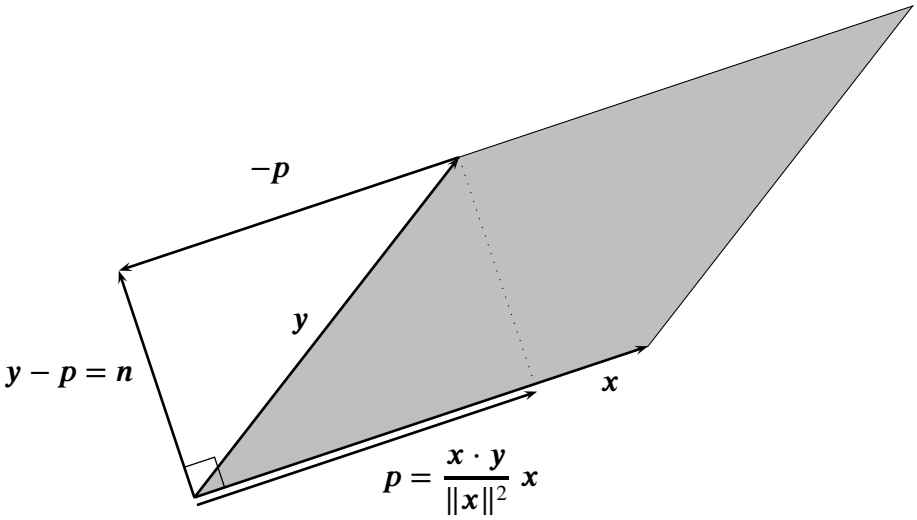


Figure 1.1.6: The Parallelogram spanned by $\boldsymbol{x}$ and $\boldsymbol{y}$.

(This is the same computation as done in the proof of the Cauchy-Schwarz inequality). Hence the area, A, of the parallelogram under consideration is

$$A = \|\boldsymbol{x}\|\|\boldsymbol{n}\| = \|\boldsymbol{x}\|\sqrt{\|\boldsymbol{y}\|^2 - \frac{(\boldsymbol{x}\cdot\boldsymbol{y})^2}{\|\boldsymbol{x}\|^2}} = \sqrt{\|\boldsymbol{x}\|^2\|\boldsymbol{y}\|^2 - (\boldsymbol{x}\cdot\boldsymbol{y})^2}. \tag{1.1.7}$$

By definition of the determinant of a 2×2-matrix and the norm of a vector we see that

$$A^2 = \det\begin{bmatrix} \|\boldsymbol{x}\|^2 & \boldsymbol{x}\cdot\boldsymbol{y} \\ \boldsymbol{x}\cdot\boldsymbol{y} & \|\boldsymbol{y}\|^2 \end{bmatrix} = \det\begin{bmatrix} \boldsymbol{x}\cdot\boldsymbol{x} & \boldsymbol{x}\cdot\boldsymbol{y} \\ \boldsymbol{x}\cdot\boldsymbol{y} & \boldsymbol{y}\cdot\boldsymbol{y} \end{bmatrix}.$$

Now let $\boldsymbol{J} := \begin{bmatrix} \boldsymbol{x} & \boldsymbol{y} \end{bmatrix}$ denote the matrix with columns $\boldsymbol{x}$ and $\boldsymbol{y}$, and let $\boldsymbol{J}^T$ be its transposed matrix. Taking into account Remark 1.1.1 on page 6 we find that

$$\begin{bmatrix} \boldsymbol{x}^T\boldsymbol{x} & \boldsymbol{x}^T\boldsymbol{y} \\ \boldsymbol{x}^T\boldsymbol{y} & \boldsymbol{y}^T\boldsymbol{y} \end{bmatrix} = \boldsymbol{J}^T\boldsymbol{J}.$$

Hence we have proved the following proposition.

Proposition 1.1.7 *Suppose that $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^N$. Then the area of the parallelogram spanned by $\boldsymbol{x}$ and $\boldsymbol{y}$ is*

$$\boxed{A = \sqrt{\det(J^T J)},}$$

where $J := \begin{bmatrix} \boldsymbol{x} & \boldsymbol{y} \end{bmatrix}$ is the $N \times 2$-matrix with columns $\boldsymbol{x}$ and $\boldsymbol{y}$. If $N = 2$, that is, for vectors in the plane we have

$$\boxed{A = |\det(J)|.}$$

The last assertion follows as $\det(J^T J) = \det J^T \det J = (\det J)^2$. Note that it is essential that you compute $J^T J$ and not $J J^T$

Example 1.1.8 Compute the area of the parallelogram spanned by $(1, 2, 4)$ and $(1, 1, 3)$.
Solution: We form the matrix J with the given vectors as its columns and its transposed matrix:

$$J = \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 4 & 3 \end{bmatrix} \qquad J^T = \begin{bmatrix} 1 & 2 & 4 \\ 1 & 1 & 3 \end{bmatrix}.$$

Then we form their product:

$$J^T J = \begin{bmatrix} 1 & 2 & 4 \\ 1 & 1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 4 & 3 \end{bmatrix} = \begin{bmatrix} 21 & 15 \\ 15 & 11 \end{bmatrix}.$$

Finally we take the determinant of the last matrix:

$$\det \begin{bmatrix} 21 & 15 \\ 15 & 11 \end{bmatrix} = 231 - 225 = 6.$$

According to the above proposition the area of the parallelogram is $\sqrt{\det(J^T J)} = \sqrt{6}$ square units.

Example 1.1.9 Compute the area of the parallelogram spanned by the vectors $(1, 3)$ and $(2, 4)$.
Solution: According to the above proposition the area is given by

$$\left| \det \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \right| = |4 - 6| = 2 \text{ square units.}$$

1.1.5 The Vector Product

We now introduce another way of multiplying two vectors, the *vector product*, also sometimes called the *cross product*. It is called *vector* product because the resulting object is a vector. In contrast to the scalar product it is *only defined in* $\mathbb{R}^3$. If two vectors $\boldsymbol{x} = (x_1, x_2, x_3)$ and $\boldsymbol{y} = (y_1, y_2, y_3)$ in $\mathbb{R}^3$ are given then we define

$$\boxed{\boldsymbol{x} \times \boldsymbol{y} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \times \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} x_2 y_3 - x_3 y_2 \\ x_3 y_1 - x_1 y_3 \\ x_1 y_2 - x_2 y_1 \end{bmatrix}.} \tag{1.1.8}$$

Note that the second and third components are obtained from the first by a cyclic permutation of the indices $1 \to 2 \to 3 \to 1$. There is an easy way to remember the definition of the vector product. If we set $\boldsymbol{e}_1 = (1, 0, 0)$, $\boldsymbol{e}_2 = (0, 1, 0)$ and $\boldsymbol{e}_3 = (0, 0, 1)$ then, formally,

$$\boldsymbol{x} \times \boldsymbol{y} = \det \begin{bmatrix} \boldsymbol{e}_1 & \boldsymbol{e}_2 & \boldsymbol{e}_3 \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{bmatrix}$$

expanded along the first row. There is another way to express the components of $\boldsymbol{v} = (v_1, v_2, v_3) = \boldsymbol{x} \times \boldsymbol{y}$, namely

$$v_1 = \det \begin{bmatrix} x_2 & y_2 \\ x_3 & y_3 \end{bmatrix}, \qquad v_2 = -\det \begin{bmatrix} x_1 & y_1 \\ x_3 & y_3 \end{bmatrix}, \qquad v_3 = \det \begin{bmatrix} x_1 & y_1 \\ x_2 & y_2 \end{bmatrix}. \tag{1.1.9}$$

We next state the main algebraic rules for the vector product. If $\boldsymbol{x}$, $\boldsymbol{y}$ and $\boldsymbol{z}$ are vectors in $\mathbb{R}^3$ and α a scalar, then it is easily verified from the above definition that

$$\begin{aligned} \boldsymbol{x} \times \boldsymbol{y} &= -(\boldsymbol{y} \times \boldsymbol{x}) \\ \boldsymbol{x} \times (\boldsymbol{y} + \boldsymbol{z}) &= \boldsymbol{x} \times \boldsymbol{y} + \boldsymbol{x} \times \boldsymbol{z} \\ \alpha(\boldsymbol{x} \times \boldsymbol{y}) &= (\alpha \boldsymbol{x}) \times \boldsymbol{y} = \boldsymbol{x} \times (\alpha \boldsymbol{y}). \end{aligned} \tag{1.1.10}$$

Note that $\boldsymbol{x} \times \boldsymbol{y}$ is *not* the same as $\boldsymbol{y} \times \boldsymbol{x}$, the sign changes! As $\boldsymbol{x} \times \boldsymbol{y}$ is a vector we can form the scalar product with a third vector. The following lemma shows how to compute such a scalar product.

Lemma 1.1.10 *Suppose that* $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathbb{R}^3$. *Then*

$$\boxed{(\boldsymbol{x} \times \boldsymbol{y}) \cdot \boldsymbol{z} = \det \begin{bmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{bmatrix}} \tag{1.1.11}$$

Proof. By (1.1.9) and the definition of the scalar product we have

$$\begin{aligned}(\boldsymbol{x} \times \boldsymbol{y}) \cdot \boldsymbol{z} &= z_1 \det \begin{bmatrix} x_2 & y_2 \\ x_3 & y_3 \end{bmatrix} - z_2 \det \begin{bmatrix} x_1 & y_1 \\ x_3 & y_3 \end{bmatrix} + z_3 \det \begin{bmatrix} x_1 & y_1 \\ x_2 & y_2 \end{bmatrix} \\ &= \det \begin{bmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{bmatrix}\end{aligned}$$

if we expand the last determinant along the third column. ■

The vector product has, as the scalar product, a geometric meaning. We will use this property extensively later in the course. Given three vectors $\boldsymbol{x}$, $\boldsymbol{y}$ and $\boldsymbol{z}$ in $\mathbb{R}^3$, not all parallel to some plane, recall that the triplet $(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z})$ is *positively oriented* or *right handed* if

$$\det \begin{bmatrix} \boldsymbol{x} & \boldsymbol{y} & \boldsymbol{z} \end{bmatrix} = \det \begin{bmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{bmatrix} > 0.$$

Geometrically this means that if we take a screw driver, align it with $\boldsymbol{z}$ and turn it in such a way that $\boldsymbol{x}$ is moved towards $\boldsymbol{y}$ through the smaller angle, then the screw moves in the direction of $\boldsymbol{z}$. The situation is depicted in Figure 1.1.7.

Proposition 1.1.11 *Suppose that* $x, y \in \mathbb{R}^3$. *Then*

$$\boxed{\boldsymbol{x} \cdot (\boldsymbol{x} \times \boldsymbol{y}) = \boldsymbol{y} \cdot (\boldsymbol{x} \times \boldsymbol{y}) = 0,} \tag{1.1.12}$$

and

$$\boxed{\|\boldsymbol{x} \times \boldsymbol{y}\| = \|\boldsymbol{x}\|\|\boldsymbol{y}\| \sin\theta = \sqrt{\|\boldsymbol{x}\|^2\|\boldsymbol{y}\|^2 - (\boldsymbol{x} \cdot \boldsymbol{y})^2},} \tag{1.1.13}$$

where θ *is the angle between* $\boldsymbol{x}$ *and* $\boldsymbol{y}$. *Hence,* $\boldsymbol{x} \times \boldsymbol{y}$ *is perpendicular to* $\boldsymbol{x}$ *and* $\boldsymbol{y}$, *and* $\|\boldsymbol{x} \times \boldsymbol{y}\|$ *represents the area of the parallelogram spanned by* $\boldsymbol{x}$ *and* $\boldsymbol{y}$. *Finally, the triple* $(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{x} \times \boldsymbol{y})$ *is positively oriented (see Figure 1.1.8).*

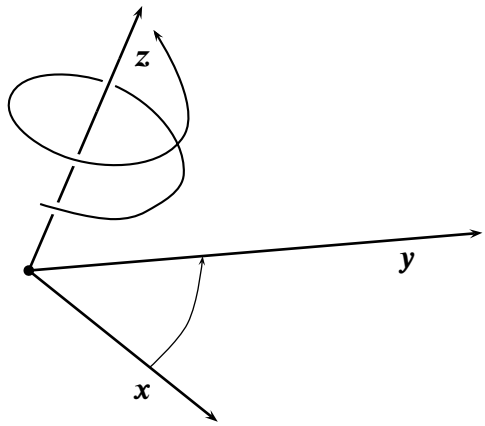


Figure 1.1.7: A positively oriented triplet $(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z})$

Proof. According to (1.1.7) on page 10 the area, A, of the parallelogram spanned by $\boldsymbol{x}$ and $\boldsymbol{y}$ is given by

$$A = \sqrt{\|\boldsymbol{x}\|^2\|\boldsymbol{y}\|^2 - (\boldsymbol{x}\cdot\boldsymbol{y})^2}.$$

We then use the definition of the scalar product, the norm and the vector product, and rearrange the terms in the square root:

$$\begin{aligned}
A^2 &= \|\boldsymbol{x}\|^2\|\boldsymbol{y}\|^2 - (\boldsymbol{x}\cdot\boldsymbol{y})^2 \\
&= (x_1^2 + x_2^2 + x_3^2)(y_1^2 + y_2^2 + y_3^2) - (x_1y_1 + x_2y_2 + x_3y_3)^2 \\
&= x_1^2y_1^2 + x_1^2y_2^2 + x_1^2y_3^2 + x_2^2y_1^2 + x_2^2y_2^2 + x_2^2y_3^2 + x_3^2y_1^2 + x_3^2y_2^2 + x_3^2y_3^2 \\
&\qquad - (x_1^2y_1^2 + x_2^2y_2^2 + x_3^2y_3^2 + 2x_1x_2y_1y_2 + 2x_1x_3y_1y_3 + 2x_2x_3y_2y_3) \\
&= x_1^2y_2^2 + x_1^2y_3^2 + x_2^2y_1^2 + x_2^2y_3^2 + x_3^2y_1^2 + x_3^2y_2^2 \\
&\qquad - 2x_1x_2y_1y_2 - 2x_1x_3y_1y_3 - 2x_2x_3y_2y_3 \\
&= (x_2y_3 - x_3y_2)^2 + (x_3y_1 - x_1y_3)^2 + (x_1y_2 - x_2y_1)^2 \\
&= \|\boldsymbol{x}\times\boldsymbol{y}\|^2.
\end{aligned}$$

This shows (1.1.13). To show that the triplet $(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{x}\times\boldsymbol{y})$ is positively oriented we apply (1.1.11) to the matrix with columns $\boldsymbol{x}$, $\boldsymbol{y}$ and $\boldsymbol{z} = \boldsymbol{x}\times\boldsymbol{y}$. Then using the definition of the norm

$$\det\begin{bmatrix} x_1 & y_1 & x_2y_3 - x_3y_2 \\ x_2 & y_2 & x_3y_1 - x_1y_3 \\ x_3 & y_3 & x_1y_2 - x_2y_1 \end{bmatrix} = (\boldsymbol{x}\times\boldsymbol{y})\cdot(\boldsymbol{x}\times\boldsymbol{y}) = \|\boldsymbol{x}\times\boldsymbol{y}\|^2 \geq 0,$$

and thus the triplet is positively oriented. ■

Remark 1.1.12 In particular, the above proposition shows that the vector product is independent of the particular coordinate system we choose. Often the vector product is

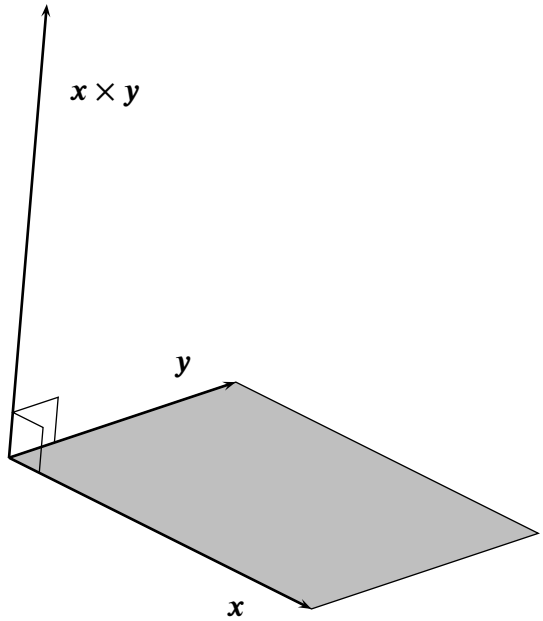


Figure 1.1.8: Geometric interpretation of the vector product.

defined by its geometric properties, and then the representation (1.1.8) is derived from them.

☞ **Remark 1.1.13** Let us compare the scalar and vector products. They are both products of two vectors. The result of the former is a scalar, and the result of the latter is a vector, giving the products their names. Another difference is that the vector product is *only defined in* $\mathbb{R}^3$, whereas the scalar product is defined in $\mathbb{R}^N$ for all N. Given vectors $\boldsymbol{x}$ and $\boldsymbol{y}$ it follows from (1.1.3) and (1.1.13) that

- $\boldsymbol{x}$ and $\boldsymbol{y}$ are orthogonal (perpendicular) if and only if $\boldsymbol{x} \cdot \boldsymbol{y} = 0$;
- $\boldsymbol{x}$ and $\boldsymbol{y}$ are parallel if and only if $\boldsymbol{x} \times \boldsymbol{y} = \boldsymbol{0}$.

We now want to derive a formula to compute the volume of a parallelepiped spanned by three vectors in $\mathbb{R}^3$.

☞ **Application 1.1.14** Consider the parallelepiped spanned by the vectors $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathbb{R}^3$ as shown in Figure 1.1.9. We want to compute its volume. The volume is the area of its base times its height. The height is $\|\boldsymbol{z}\||\cos\theta|$, where θ is the angle between $\boldsymbol{z}$ and the vector $\boldsymbol{x} \times \boldsymbol{y}$ perpendicular to the plane spanned by $\boldsymbol{x}$ and $\boldsymbol{y}$. By (1.1.4) on page page 6

$$\|\boldsymbol{z}\||\cos\theta| = \|\boldsymbol{z}\|\frac{|\boldsymbol{z} \cdot (\boldsymbol{x} \times \boldsymbol{y})|}{\|\boldsymbol{z}\|\|\boldsymbol{x} \times \boldsymbol{y}\|} = \frac{|\boldsymbol{z} \cdot (\boldsymbol{x} \times \boldsymbol{y})|}{\|\boldsymbol{x} \times \boldsymbol{y}\|}$$

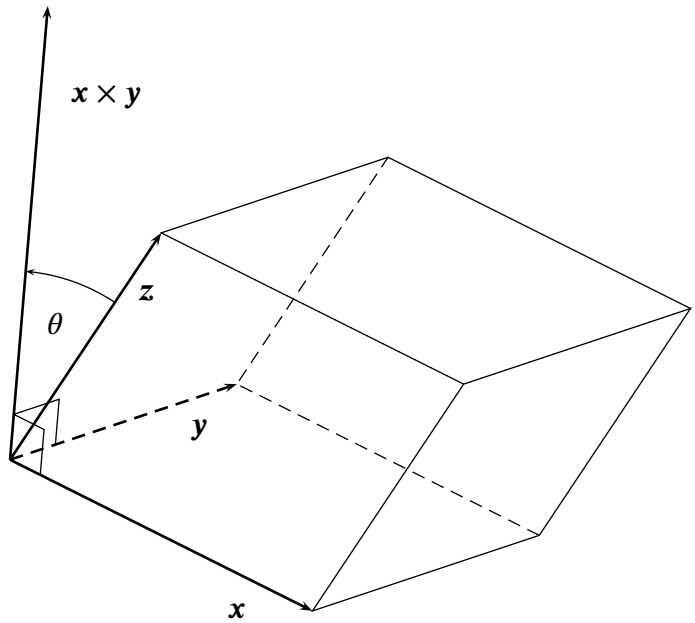


Figure 1.1.9: A parallelepiped spanned by three vectors

Using that the surface area of the parallelogram spanned by $\boldsymbol{x}$ and $\boldsymbol{y}$ is $\|\boldsymbol{x} \times \boldsymbol{y}\|$ (see Proposition 1.1.11) we find that the volume of the parallelepiped is

$$V = \frac{|\boldsymbol{z} \cdot (\boldsymbol{x} \times \boldsymbol{y})|}{\|\boldsymbol{x} \times \boldsymbol{y}\|} \|\boldsymbol{x} \times \boldsymbol{y}\| = |\boldsymbol{z} \cdot (\boldsymbol{x} \times \boldsymbol{y})|.$$

Using Lemma 1.1.10 on page 12 we see that

$$\boldsymbol{z} \cdot (\boldsymbol{x} \times \boldsymbol{y}) = \det \begin{bmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{bmatrix}.$$

Therefore the volume, V, of the parallelepiped spanned by the three vectors $\boldsymbol{x}$, $\boldsymbol{y}$ and $\boldsymbol{z}$ is given by

$$\boxed{V = \left| \det \begin{bmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{bmatrix} \right|.}$$

As for 2×2 matrices this provides a geometrical interpretation of the determinant of a 3×3 matrix. It is, up to a sign, the volume of the parallelepiped spanned by the columns of the matrix.

Remark 1.1.15 One can also define a "volume" or "measure" for subsets of $\mathbb{R}^N$ if $N > 3$. One starts with "rectangular boxes," and defines their volume to be the product

of its N sides as one does in the plane and in space. Then one can look at parallelepipeds. It turns out that the "volume" of an N-dimensional parallelepiped spanned by the N (linearly independent) vectors $\boldsymbol{v}_1, \dots, \boldsymbol{v}_N$ is given by the absolute value of $\det[\boldsymbol{v}_1 \dots \boldsymbol{v}_N]$ as in case of $N = 2$ or $N = 3$. More generally we can compute the "k-dimensional" volume of k vectors $\boldsymbol{v}_1, \dots, \boldsymbol{v}_k$ in $\mathbb{R}^N$, generalising the formula given in Proposition 1.1.7. We form the matrix $\boldsymbol{J}$ having columns $\boldsymbol{v}_1, \dots, \boldsymbol{v}_k$. It turns out that the volume required is $\sqrt{\det(\boldsymbol{J}^T\boldsymbol{J})}$. For an elementary proof of this fact see for instance Bretscher [3, Section 5.3].

1.2 Functions of Several Variables

Suppose that $D \subset \mathbb{R}^N$. We consider functions defined on D with values in $\mathbb{R}$. A function, f, as usual assigns to every $\boldsymbol{x} \in D$ a *unique* value denoted by $f(\boldsymbol{x})$. We call D the domain of f, and $\mathbb{R}$ is its range. If $N > 3$ there is no way to visualise the functions. In the following we discuss ways to visualise functions of two and three variables.

1.2.1 Graphs

Given a function from a set $D \subset \mathbb{R}^2$ with values in $\mathbb{R}$ we can plot its value on the x_3-axis as shown in Figure 1.2.1. The resulting object is a surface, called the graph of f. Formally, it is the set

$$\operatorname{graph}(f) := \left\{ \left(x_1, x_2, f(x_1, x_2)\right) \mid (x_1, x_2) \in D \right\}$$

in $\mathbb{R}^3$. As an example consider the graph of $f(x, y) := 5y(1 - y^2)(1/2 - x^2) - x$ in

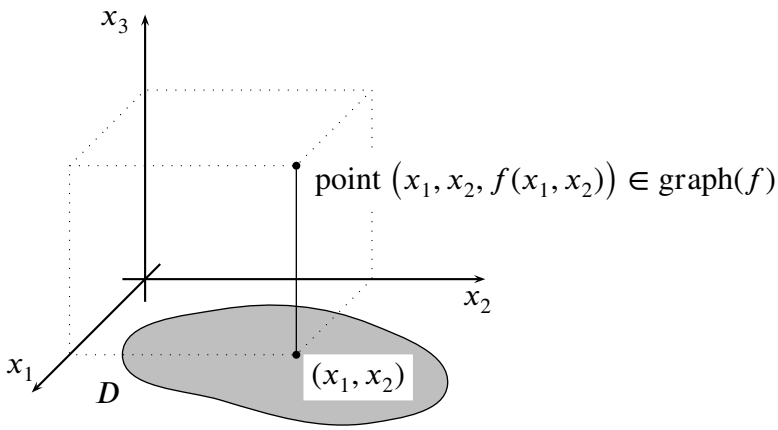


Figure 1.2.1: To obtain the graph of f plot $f(\boldsymbol{x})$ above $\boldsymbol{x} \in D$.

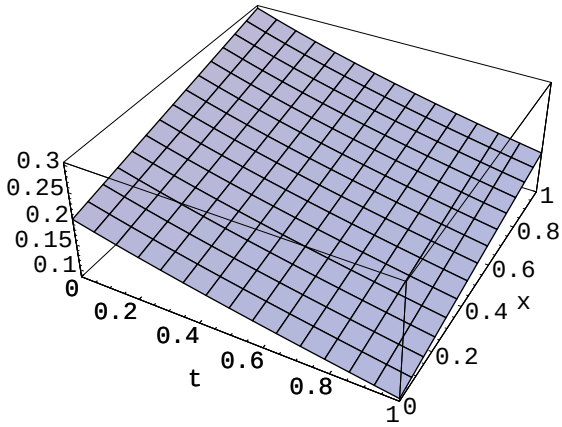


Figure 1.2.2: Graph of $f(x, y) := 5y(1 - y^2)(1/2 - x^2) - x$.

Figure 1.2.2. If $D \subset \mathbb{R}^N$ we still call the set

$$\text{graph}(f) := \left\{ \left(\boldsymbol{x}, f(\boldsymbol{x}) \right) \mid \boldsymbol{x} \in D \right\} \subset \mathbb{R}^{N+1}$$

the *graph of f over D*, even if we cannot visualise it!

1.2.2 Contour Maps

Looking at Figure 1.2.2 we see that the graph looks like the topography of a mountain range. On maps the topography is usually represented by lines of equal altitude. We can do this as well in our case. We specify a "level" c and look at the set of all $(x, y) \in D$ such that $f(x, y) = c$. Such a line is called a *contour line*. Drawing several such lines for equally spaced levels we get what we will call a *contour map*. The contour map of the function from Figure 1.2.2 is depicted in Figure 1.2.3. The numbers on the contour lines indicate their levels.

Example 1.2.1 Here are some examples on how to find contour lines.

1. Find the contour lines $f(x, y) = x^2 + y^2 = c$ for $c = 0, \pm 1$.

   Solution: The contour "line" for $c = 0$ just consists of one point, namely $(0, 0)$. As $x^2 + y^2 \geq 0$ for all (x, y) the contour line for $c = -1$ is empty. For $c = 1$ the contour line is a circle about the origin with radius one. For general $c > 0$ the contour line is a circle of radius $\sqrt{c}$ as shown in Figure 1.2.4.

2. Sketch the contour map of the function $f(x, y) = x^2 - y^2$.

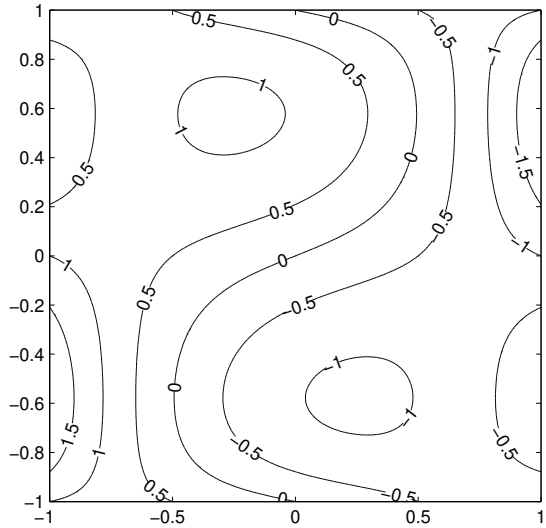


Figure 1.2.3: Contour map of $f(x, y) := 5y(1 - y^2)(1/2 - x^2) - x$.

Solution: We have to solve the equations $c = x^2 - y^2$. For $c = 0$ we have the lines $y = \pm x$. If $c > 0$ then we have $x = \pm\sqrt{y^2 + c}$, so we get a pair of hyperbolas. Similarly, if $c < 0$ we have $y = \pm\sqrt{x^2 - c}$ which is defined everywhere since $-c > 0$. Hence we have another pair of hyperbolas, but turned by 90° as shown in Figure 1.2.5

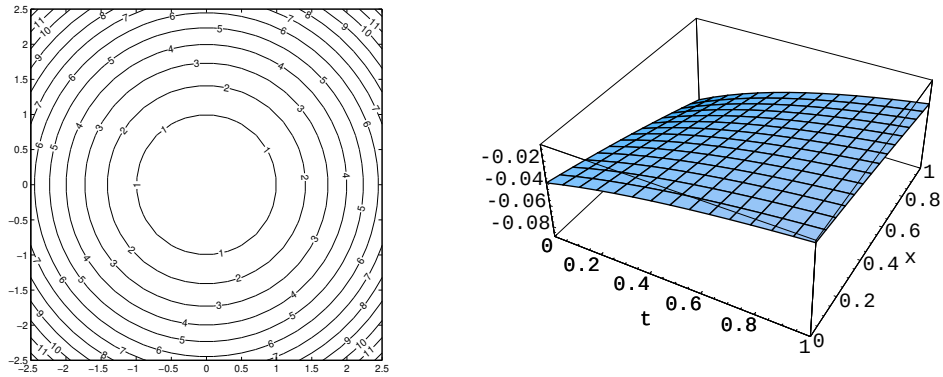


Figure 1.2.4: Contour map and graph of $f(x, y) = x^2 + y^2$.

1.2.3 Level Sets

We can generalise the idea of contour lines to higher dimensions, and consider what we call *level sets*. If f is a function defined on a set $D \subset \mathbb{R}^N$ with values in $\mathbb{R}$ we call the

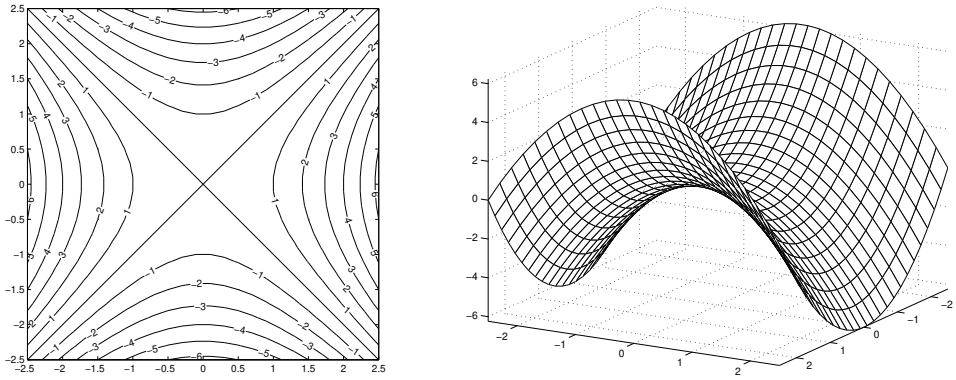


Figure 1.2.5: Contour map and graph of $f(x, y) := x^2 - y^2$.

set

$$f^{-1}[c] := \{\boldsymbol{x} \in D \mid f(\boldsymbol{x}) = c\}$$

a level set (to the level c). If $N = 2$ then $f^{-1}[c]$ are the contour lines from the previous subsection. For $N = 3$ we can still visualise the level sets. In general they are surfaces in $\mathbb{R}^3$, and often called *level surfaces*.

Example 1.2.2 As a first example look at the function $f(x, y, z) = x^2/4 + y^2/9 + z^2$ defined on $\mathbb{R}^3$. To find the level surface, say to the level $c = 1$, we can look at cross-sections along some coordinate plains. Let us take the plane $z = 0$. Then $f(x, y, 0) = x^2/4+y^2/9 = 1$, which is the equation of an ellipse with axis of length 2 and 3. Similarly we consider $y = 0$, which leads to the ellipse $f(x, 0, z) = x^2/4 + z^2 = 1$ with axis 2 and 1. Doing the same on the third coordinate plane, $x = 0$, we get the ellipse $f(x, y, z) = y^2/9 + z^2$ with axis 3 and 1. The resulting surface is the ellipsoid shown in Figure 1.2.6.

Example 1.2.3 We next consider some level sets of $f(x, y, z) = x^2 + y^2 - z^2$. Observe that the function is rotationally symmetric about the z-axis. Hence the level surfaces $f^{-1}[c]$ can be obtained by revolving the curve $f(x, 0, z) = c$ about the z-axis. We determine the surfaces for the levels $c = 0, \pm 1$. If $c = 0$ then $f(x, 0, z) = 0 = x^2 - z^2$ is given by $z = \pm x$. Hence $f^{-1}[0]$ is a double cone centred about the z-axis as shown in Figure 1.2.7. For $c = 1$ we have $f(x, 0, z) = 1 = x^2 - z^2$, implying that $x = \pm\sqrt{1 + z^2}$. Revolving about the about the z-axis we get the corresponding surface in Figure 1.2.7. Proceeding similarly for $c = -1$ we get $f(x, 0, z) = -1 = x^2 - z^2$, or $z = \pm\sqrt{1 + x^2}$. These are the same hyperbolas as before but turned by 90°. As a result the corresponding level surface of f shown in Figure 1.2.7 consists of two shells.

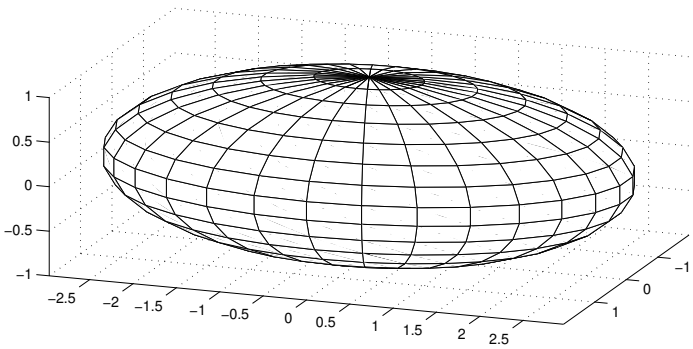
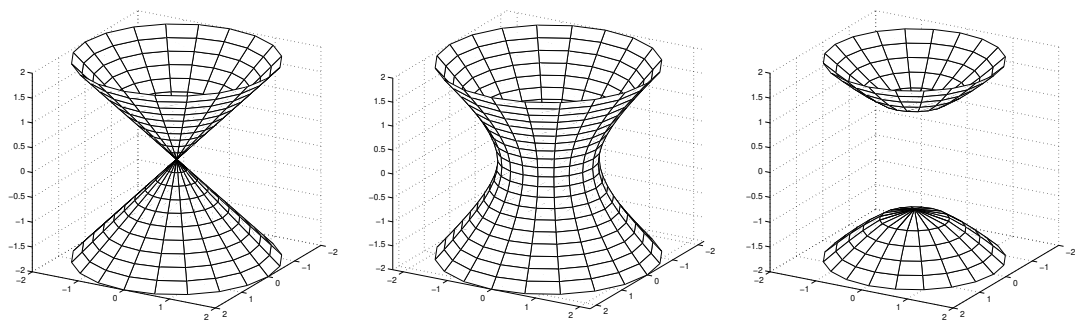


Figure 1.2.6: The level surface $f(x, y, z) = x^2/4 + y^2/9 + z^2 = 1$.



$c = 0$ $c = 1$ $c = -1$

Figure 1.2.7: Level surfaces $f(x, y, z) = x^2 + y^2 - z^2 = c$.

1.2.4 Vector Valued Functions

We now look at vector valued functions, that is, functions defined on a subset $D \subset \mathbb{R}^N$ taking values in $\mathbb{R}^k$. Such a function assigns every $\boldsymbol{x} \in D$ a unique vector $\boldsymbol{f}(\boldsymbol{x})$. As a result $\boldsymbol{f}(\boldsymbol{x})$ has k components, so $\boldsymbol{f}(\boldsymbol{x}) = (f_1(\boldsymbol{x}), \dots, f_k(\boldsymbol{x}))$. Every component is a function of N variables with values in $\mathbb{R}$ as considered before. We call f_i, $i = 1, \dots, k$, the *component functions* of $\boldsymbol{f}$. We can visualise functions of one or two variables with values in $\mathbb{R}^2$ or $\mathbb{R}^3$.

Suppose that $N = 1$ and $k = 2$ then $\boldsymbol{f}(t) = (f_1(t), f_2(t))$ is defined for instance for t in an interval $I \subset \mathbb{R}$. Similarly, if $N = 1$ and $k = 3$ then $\boldsymbol{f}(t) = (f_1(t), f_2(t), f_3(t))$ for $t \in I$. In both cases $\boldsymbol{f}(t)$ can be thought of as the position of a particle at time t moving in the plane or in space, respectively. We can represent $\boldsymbol{f}$ by drawing the path along which the particle travels as shown in Figure 1.2.8. We talk about a curve.

Now consider a function defined on a subset of $D \subset \mathbb{R}^2$ with values in $\mathbb{R}^3$. Again we

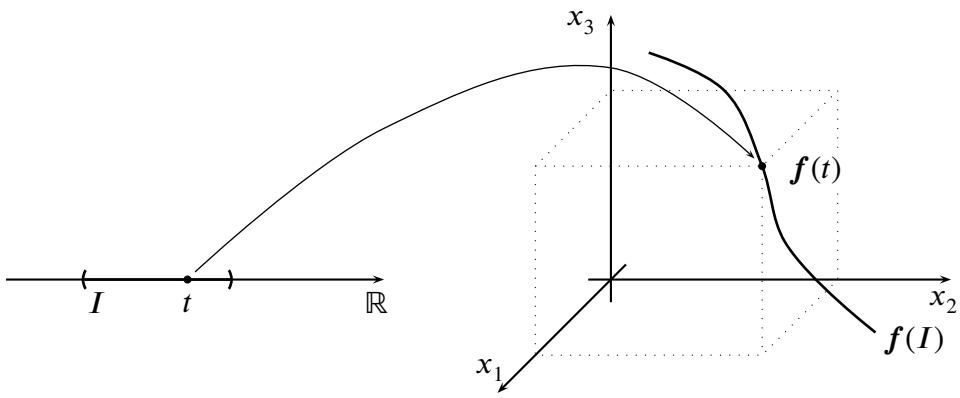


Figure 1.2.8: A curve in $\mathbb{R}^3$: The image of an interval in space.

can plot the vectors $\boldsymbol{f}(\boldsymbol{x})$ for $\boldsymbol{x} \in D$. The resulting set is a surface in $\mathbb{R}^3$ as shown in Figure 1.2.9. We will discuss curves and surfaces later in more detail.

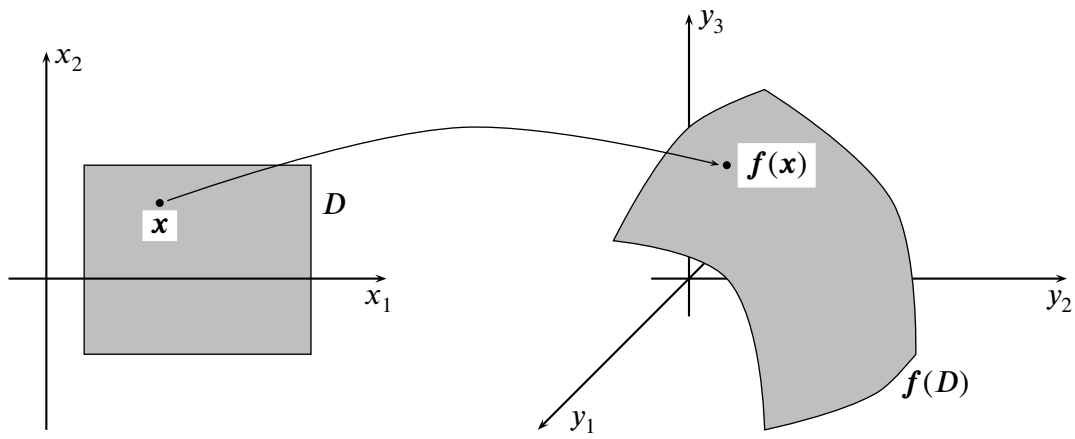


Figure 1.2.9: A surface in $\mathbb{R}^3$: The image of a plane domain in space.

A general vector valued function of N variables has the form

$$\boldsymbol{f}(\boldsymbol{x}) = \begin{bmatrix} f_1(x_1, \dots, x_N) \\ \vdots \\ f_k(x_1, \dots, x_N) \end{bmatrix},$$

and is defined on a subset of $\mathbb{R}^N$. Most properties of vector valued functions are determined by the properties of each component function.

1.3 Limits and Continuity

In this section we discuss limits of sequences of vectors, and continuity of functions of several variables. The ideas are essentially the same as for the corresponding notions

for sequences of real numbers and functions of one variable, respectively. Hence, before reading the present section it is advisable to review the relevant material (sequences, limits, continuity of functions) from first year calculus.

1.3.1 Sequences of Vectors

Given vectors $\boldsymbol{x}_n = (x_{1n}, \dots, x_{Nn})$ and $\boldsymbol{x} = (x_1, \dots, x_N)$ in $\mathbb{R}^N$ we say that $\boldsymbol{x}_n$ *converges to* $\boldsymbol{x}$ *as* n *goes to infinity* if *all* components x_{in} of $\boldsymbol{x}_n$ converge to the corresponding component x_i of $\boldsymbol{x}$. In symbols

$$\boxed{\boldsymbol{x}_n \xrightarrow{n\to\infty} \boldsymbol{x} \text{ if and only if } x_{in} \xrightarrow{n\to\infty} x_i \text{ for all } i = 1, \dots, N.} \tag{1.3.1}$$

We say that $\boldsymbol{x}$ is the *limit* of the sequence $(\boldsymbol{x}_n)$, and write

$$\lim_{n\to\infty} \boldsymbol{x}_n = \boldsymbol{x}$$

as in case of sequences of numbers if the limit exists. If a sequence of real numbers (t_n) converges to t then we know that $|t_n - t| \xrightarrow{n\to\infty} 0$ and vice versa. A similar fact is true for sequences of vectors as the following proposition shows.

Proposition 1.3.1 *Let* $(\boldsymbol{x}_n)_{n\in\mathbb{N}}$ *be a sequence of vectors in* $\mathbb{R}^N$. *Then*

$$\boxed{\lim_{n\to\infty} \boldsymbol{x}_n = \boldsymbol{x} \textit{ if and only if } \lim_{n\to\infty} \|\boldsymbol{x}_n - \boldsymbol{x}\| = 0.}$$

Proof. By definition of the norm we have

$$\|\boldsymbol{x}_n - \boldsymbol{x}\| = \Big(\sum_{i=1}^{N} |x_{in} - x_i|^2\Big)^{1/2}. \tag{1.3.2}$$

If $\boldsymbol{x}_n \xrightarrow{n\to\infty} \boldsymbol{x}$ then by (1.3.1) we know that $x_{in} \xrightarrow{n\to\infty} x_i$ for every $i = 1, \dots, N$, and thus every term on the right hand side of (1.3.2) goes to zero. This shows that $\|\boldsymbol{x}_n - \boldsymbol{x}\| \xrightarrow{n\to\infty} 0$ if $\boldsymbol{x}_n \xrightarrow{n\to\infty} \boldsymbol{x}$. Now assume that $\|\boldsymbol{x}_n - \boldsymbol{x}\| \xrightarrow{n\to\infty} 0$. Observe that every term in the sum on the right hand side of (1.3.2) is non-negative. Hence

$$0 \le |x_{in} - x_i| \le \Big(\sum_{i=1}^{N} |x_{in} - x_i|^2\Big)^{1/2} = \|\boldsymbol{x}_n - \boldsymbol{x}\|$$

for every $i = 1, \dots, N$. By assumption $\|\boldsymbol{x}_n - \boldsymbol{x}\| \xrightarrow{n\to\infty} 0$, and thus by the "squeezing lemma" $|x_{in} - x_i| \xrightarrow{n\to\infty} 0$ for all $i = 1, \dots, N$. This shows that thus $x_{in} \xrightarrow{n\to\infty} x_i$ for all $i = 1, \dots, N$ as required. ■

As for sequences in $\mathbb{R}$ we can give an ε-characterisation of convergence. It can also be used as a definition of convergence of a sequence. The proof is similar to the proof of the above proposition and left as an exercise.

Proposition 1.3.2 *A sequence $(\boldsymbol{x}_n)$ in $\mathbb{R}^N$ converges to $\boldsymbol{x}_0 \in \mathbb{R}^N$ if and only if for every $\varepsilon > 0$ there exists $n_0 \in \mathbb{N}$ such that $\|\boldsymbol{x}_n - \boldsymbol{x}_0\| < \varepsilon$ for all $n > n_0$.*

Limits of vectors have the same properties as limits of sequences of real numbers.

Proposition 1.3.3 *Suppose that $(\boldsymbol{x}_n)$, $(\boldsymbol{y}_n)$ are sequences in $\mathbb{R}^N$ with limits $\boldsymbol{x}$ and $\boldsymbol{y}$, respectively. Moreover, assume that (α_n) is a sequence in $\mathbb{R}$ with limit α. Then*

- $\lim\limits_{n\to\infty}(\boldsymbol{x}_n + \boldsymbol{y}_n) = \lim\limits_{n\to\infty} \boldsymbol{x}_n + \lim\limits_{n\to\infty} \boldsymbol{y}_n = \boldsymbol{x} + \boldsymbol{y}$
- $\lim\limits_{n\to\infty} \alpha_n \boldsymbol{x}_n = (\lim\limits_{n\to\infty} \alpha_n)(\lim\limits_{n\to\infty} \boldsymbol{x}_n) = \alpha \boldsymbol{x}$
- $\lim\limits_{n\to\infty}(\boldsymbol{x}_n \cdot \boldsymbol{y}_n) = (\lim\limits_{n\to\infty} \boldsymbol{x}_n) \cdot (\lim\limits_{n\to\infty} \boldsymbol{y}_n) = \boldsymbol{x} \cdot \boldsymbol{y}$
- *If $N = 3$ then* $\lim\limits_{n\to\infty}(\boldsymbol{x}_n \times \boldsymbol{y}_n) = (\lim\limits_{n\to\infty} \boldsymbol{x}_n) \times (\lim\limits_{n\to\infty} \boldsymbol{y}_n) = \boldsymbol{x} \times \boldsymbol{y}$

Proof. The proof follows directly from the properties of sequences in $\mathbb{R}$ and the definition of the multiplication by scalars, the scalar and the vector product. ■

Example 1.3.4 1. Find the limit of

$$\boldsymbol{x}_n = \left(\frac{n-1}{n+1}, 2^{-n}, \sqrt[n]{n}\right).$$

Solution: We must find the limit of every component. We have

$$\lim_{n\to\infty} \frac{n-1}{n+1} = 1, \qquad \lim_{n\to\infty} 2^{-n} = 0 \qquad \text{and} \qquad \lim_{n\to\infty} \sqrt[n]{n} = 1.$$

Hence $\lim_{n\to\infty} \boldsymbol{x}_n = (1, 0, 1)$.

2. Find the limit of $\boldsymbol{x}_n = ((-1)^n, 1, 1/n)$.

   Solution: We again have to check whether *every* component converges. This is not the case for the present sequence as $(-1)^n$ does not converge. Hence the sequence $\boldsymbol{x}_n$ does not converge.

3. Find the limit of

$$\boldsymbol{x}_n = \left(\frac{n}{1+n^2}, e^{-n}, \frac{\ln n}{n}, n^{-n}\right) \in \mathbb{R}^4.$$

Solution: We must find the limit of every component. We have

$$\lim_{n\to\infty} \frac{n}{1+n^2} = 0, \quad \lim_{n\to\infty} e^{-n} = 0, \quad \lim_{n\to\infty} \frac{\ln n}{n} = 0 \quad \text{and} \quad \lim_{n\to\infty} n^{-n} = 0.$$

Hence $\lim_{n\to\infty} \boldsymbol{x}_n = (0,0,0,0)$.

1.3.2 Open and Closed Sets

You are certainly familiar with the notion of open and closed intervals in $\mathbb{R}$. In this section we want to generalise these notions to subsets of $\mathbb{R}^N$. We call an interval $I \subset \mathbb{R}$ *open*, if its endpoints do not belong to I. Hence, if we pick an arbitrary point in $x_0 \in I$ and move a little bit to the left or to the right we stay in the interval. Mathematically we can express this by saying that for every $x_0 \in I$ there exists a small interval $(x_0-\varepsilon, x_0+\varepsilon)$ which is *entirely* contained in I. If I is an arbitrary interval we call such a point $x_0 \in I$ an *interior point* of I. The same ideas can be applied to subsets of $\mathbb{R}^N$. A point $\boldsymbol{x}_0$ in $U \subset \mathbb{R}^N$ is called an *interior point* if we can move a little bit in *every* direction without immediately leaving U. Mathematically we can express this by requiring that a small ball about $\boldsymbol{x_0}$ be *entirely* contained in the set U.

Definition 1.3.5 (Open N-ball) Given $\boldsymbol{x}_0 \in \mathbb{R}^N$ and $r > 0$ we set

$$B_r(\boldsymbol{x}_0) := \left\{\boldsymbol{x} \in \mathbb{R}^N : \|\boldsymbol{x} - \boldsymbol{x}_0\| < r\right\}.$$

We call the set $B_r(\boldsymbol{x}_0)$ the *open ball* in $\mathbb{R}^N$ (or the *open N-ball*) with radius r centred at $\boldsymbol{x}_0$.

If $N = 1$ then $B_r(x_0)$ is simply the interval $(x_0 - r, x_0 + r)$. If $N = 2$ then $B_r(\boldsymbol{x}_0)$ is an (open) disc centred at $\boldsymbol{x}_0$ with radius r, and for $N = 3$ it is really a ball. We could distinguish between these different cases and expressions, but it is more convenient to call $B_r(\boldsymbol{x}_0)$ simply a "ball," no matter what the 'dimension' of the space is. As motivated before we make the following definition.

Definition 1.3.6 (Interior point) We call $\boldsymbol{x}_0$ an *interior point* of $U \subset \mathbb{R}^N$ if there exists $\varepsilon > 0$ such that $B_\varepsilon(\boldsymbol{x}_0) \subset U$. The set of all interior points of U is called the *interior of* U. We denote the interior of a set by $\mathrm{Int}(U)$.

Certain sets have only interior points. They are called open sets:

Definition 1.3.7 (Open set) A set $U \subset \mathbb{R}^N$ is said to be an *open set* if all its points are interior points. In other words, U is open if and only if $U = \mathrm{Int}(U)$.

If I is an interval, its endpoints may or may not belong to it. In any case they play a special role. They are the points where we can 'enter' or 'exit' the interval with an arbitrarily small move. The set of all endpoints of an interval we also call the boundary of the interval. For subsets of $\mathbb{R}^N$ the boundary is defined the same way, namely as the set of points, where we can 'enter' or 'exit' the set with an arbitrarily small move. Here is a precise mathematical definition.

Definition 1.3.8 (Boundary of a set) Let $U \subset \mathbb{R}^N$. Points in $\mathbb{R}^N$ which are not interior points of U or its complement are called *boundary points* of U. The set of boundary points of U is denoted by ∂U, and is called the *boundary of* U.

In Figure 1.3.1 interior and boundary points of a set in $\mathbb{R}^2$ are shown, together with a ball centred at them as required in the definition. Note that, as we get close to the boundary of the set the balls about an interior point might be very small. Moreover, as

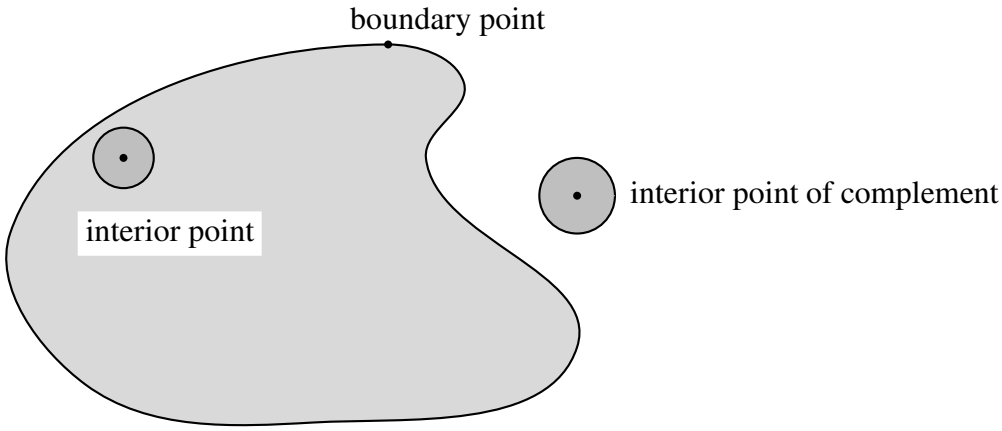


Figure 1.3.1: Interior and boundary points of a set and its complement.

with the endpoints of an interval, the boundary of a set may or may not belong to the set. Sometimes only part of the boundary of a set belongs to the set, as with a half open interval.

Definition 1.3.9 (Closed set) A set $U \subset \mathbb{R}^N$ is called a *closed set* or *closed* if $\partial U \subset U$. Equivalently, we can say U is closed if the complement of U is open.

Sometimes it is useful to look at the 'closure' of a set.

Definition 1.3.10 (Closure) If $U \subset \mathbb{R}^N$ then $\partial U \cup U$ is called the *closure of* U. We denote the closure of U by $\overline{U}$.

Remark 1.3.11 The closure of a set $U \subset \mathbb{R}^N$ can also be characterised as the smallest closed set containing U. Moreover, note that $U = \overline{U}$ if U is a closed set, and that $U = \mathrm{Int}(U)$ if U is open.

Example 1.3.12 We want to give now some examples illustrating the above definitions.

1. The intervals $(1, 4)$, $(2, \infty)$ or $(-\infty, 10)$ are open sets in $\mathbb{R}$, whereas $[1, 4]$, $[2, \infty)$ or $(-\infty, 10]$ are closed sets in $\mathbb{R}$. The last three sets are in fact the closures of the first three sets.

2. The set U:=$\{1/n \mid n \in \mathbb{N}\}$ is not open and not closed in $\mathbb{R}$. Zero does not belong to the set U, but for every $\varepsilon > 0$ the intersection $U \cap (-\varepsilon, \varepsilon)$ is non-empty. Hence zero is not an interior point of the complement. It is also not an interior point, and thus it belongs to the boundary of U. The set $U \cup \{0\}$ is a closed set, in fact it is the closure of U. The set does not have interior points as every interval about a point in U, no matter how small, contains a point not in U. Hence $\overline{U} = U \cup \{0\} = \partial U$. Visualise the situation by drawing a picture!

3. The set $U := \{(x, y) \in \mathbb{R}^2 \mid x^2 - 4y^2 > 1\}$ is an open set in $\mathbb{R}^2$. It can be characterised as the set of points, (x, y), satisfying $|x| > \sqrt{4y^2 + 1}$. The shaded part in Figure 1.3.2 is the set, and the dashed line is its boundary.

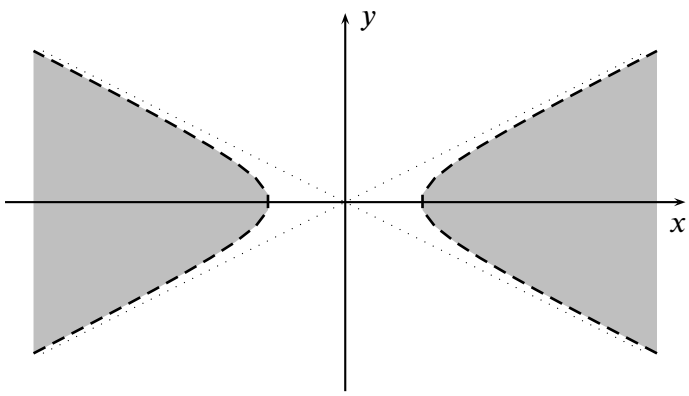


Figure 1.3.2: The set $U := \{(x, y) \in \mathbb{R}^2 \mid x^2 - 4y^2 > 1\}$.

4. The line segment $\{(x, 0) \in \mathbb{R}^2 \mid x \in (0, 5)\}$ is not open and not closed in $\mathbb{R}^2$. It has no interior. Its boundary is $\{(x, 0) \in \mathbb{R}^2 \mid x \in [0, 5]\}$. This is in contrast to the interval $(0, 5)$ in $\mathbb{R}$ which is open, and whose boundary is the set $\{0, 5\}$. Hence 'boundary,' 'interior' and 'closure' depend on the underlying space.

1.3.3 Continuous Functions

Using the notion of convergence from the previous subsection we now define what we mean by a continuous function of several variables. Note that the definition looks the same as in case of a function of one variable. Before we give a definition of continuity we introduce limits.

Definition 1.3.13 Suppose that $\boldsymbol{f}$ is a function defined on a set $D \subset \mathbb{R}^N$ with values in $\mathbb{R}^k$. If $\boldsymbol{x}_0$ is in the closure of D we write

$$\lim_{\boldsymbol{x}\to\boldsymbol{x}_0} \boldsymbol{f}(\boldsymbol{x}) = \boldsymbol{y}_0$$

if for every $\varepsilon > 0$ there exists $\delta > 0$ such that $\|\boldsymbol{f}(\boldsymbol{x}) - \boldsymbol{y}_0\| < \varepsilon$ for all $\boldsymbol{x} \in D$ for which $0 < \|\boldsymbol{x} - \boldsymbol{x}_0\| < \delta$. We also write

$$\boldsymbol{f}(\boldsymbol{x}) \to \boldsymbol{y}_0 \text{ as } \boldsymbol{x} \to \boldsymbol{x}_0 \qquad \text{or} \qquad \boldsymbol{f}(\boldsymbol{x}) \xrightarrow{\boldsymbol{x}\to\boldsymbol{x}_0} \boldsymbol{y}_0,$$

and say that $\boldsymbol{f}(\boldsymbol{x})$ converges to $\boldsymbol{y}_0$ as $\boldsymbol{x}$ tends to $\boldsymbol{x}_0$, or $\boldsymbol{y}_0$ is the limit of $\boldsymbol{f}(\boldsymbol{x})$ as $\boldsymbol{x}$ tends to $\boldsymbol{x}_0$.

As for functions of one variable there is a characterisation of limits involving sequences.

Proposition 1.3.14 *Suppose that f is a function defined on a set $D \subset \mathbb{R}^N$ with values in $\mathbb{R}^k$. Then $\boldsymbol{f}(\boldsymbol{x}) \to \boldsymbol{y}_0$ as $\boldsymbol{x} \to \boldsymbol{x}_0$ if and only if $\boldsymbol{f}(\boldsymbol{x}_n) \to \boldsymbol{y}_0$ for* every *sequence of points $\boldsymbol{x}_n \in D$ with $\boldsymbol{x}_n \to \boldsymbol{x}_0$.*

We omit the proof of the above proposition and state some properties of limits. Compare with functions of one variable!

Proposition 1.3.15 *Suppose that $\boldsymbol{f}, \boldsymbol{g}$ are functions defined on a domain $D \subset \mathbb{R}^N$ with values in $\mathbb{R}^k$, and that $u : D \to \mathbb{R}$ is a scalar valued function. Moreover, suppose that $\boldsymbol{f}(\boldsymbol{x}) \to \boldsymbol{a}$, $\boldsymbol{g}(\boldsymbol{x}) \to \boldsymbol{b}$ and $u(\boldsymbol{x}) \to \alpha$ as $\boldsymbol{x} \to \boldsymbol{x}_0$. Then*

- $\lim\limits_{\boldsymbol{x}\to\boldsymbol{x}_0} \big(\boldsymbol{f}(\boldsymbol{x}) + \boldsymbol{g}(\boldsymbol{x})\big) = \lim\limits_{\boldsymbol{x}\to\boldsymbol{x}_0} \boldsymbol{f}(\boldsymbol{x}) + \lim\limits_{\boldsymbol{x}\to\boldsymbol{x}_0} \boldsymbol{g}(\boldsymbol{x}) = \boldsymbol{a} + \boldsymbol{b}$;
- $\lim\limits_{\boldsymbol{x}\to\boldsymbol{x}_0} u(\boldsymbol{x})\boldsymbol{f}(\boldsymbol{x}) = \lim\limits_{\boldsymbol{x}\to\boldsymbol{x}_0} u(\boldsymbol{x}) \lim\limits_{\boldsymbol{x}\to\boldsymbol{x}_0} \boldsymbol{f}(\boldsymbol{x}) = \alpha\boldsymbol{a}$;
- $\lim\limits_{\boldsymbol{x}\to\boldsymbol{x}_0} \big(\boldsymbol{f}(\boldsymbol{x}) \cdot \boldsymbol{g}(\boldsymbol{x})\big) = (\lim\limits_{\boldsymbol{x}\to\boldsymbol{x}_0} (\boldsymbol{f}(\boldsymbol{x})) \cdot (\lim\limits_{\boldsymbol{x}\to\boldsymbol{x}_0} \boldsymbol{g}(\boldsymbol{x})) = \boldsymbol{a} \cdot \boldsymbol{b}$;
- *If $k = 3$ then*
  $\lim\limits_{\boldsymbol{x}\to\boldsymbol{x}_0} \big(\boldsymbol{f}(\boldsymbol{x}) \times \boldsymbol{g}(\boldsymbol{x})\big) = (\lim\limits_{\boldsymbol{x}\to\boldsymbol{x}_0} \boldsymbol{f}(\boldsymbol{x})) \times (\lim\limits_{\boldsymbol{x}\to\boldsymbol{x}_0} \boldsymbol{g}(\boldsymbol{x})) = \boldsymbol{a} \times \boldsymbol{b}$.

We now define what we mean by a continuous function. As with the definition of limits, compare it to what you know about continuity of functions of one variable.

Definition 1.3.16 Suppose that $\boldsymbol{f}$ is a function defined on a set $D \subset \mathbb{R}^N$ with values in $\mathbb{R}^k$. We say that f is continuous at a point $\boldsymbol{x}_0 \in D$ if

$$\lim_{\boldsymbol{x} \to \boldsymbol{x}_0} \boldsymbol{f}(\boldsymbol{x}) = \boldsymbol{f}(\boldsymbol{x}_0).$$

We say that f is continuous if f is continuous at every point in its domain D.

Depending on the situation the following characterisations of continuity are useful.

Proposition 1.3.17 *Suppose that $\boldsymbol{f}$ is a function defined on a set $D \subset \mathbb{R}^N$ with values in $\mathbb{R}^k$. Then f is continuous at $\boldsymbol{x}_0 \in D$ if and only if one of the following conditions are satisfied:*

- *For all $\varepsilon > 0$ there exists $\delta > 0$ such that $\|\boldsymbol{f}(\boldsymbol{x}) - \boldsymbol{f}(\boldsymbol{x}_0)\| < \varepsilon$ for all $\boldsymbol{x} \in D$ satisfying $\|\boldsymbol{x} - \boldsymbol{x}_0\| < \delta$. ($\varepsilon$-$\delta$-characterisation of limit)*
- *$\boldsymbol{f}(\boldsymbol{x}_n) \to \boldsymbol{f}(\boldsymbol{x}_0)$ as $n \to \infty$ for* every *sequence $\boldsymbol{x}_0$ in D for which $\boldsymbol{x}_n \to \boldsymbol{x}_0$.*
- *Every component function f_i of $\boldsymbol{f} = (f_1, \dots, f_k)$ is continuous at $\boldsymbol{x}_0$.*

The proof of the above facts follow from Proposition 1.3.14 and the definition of limits as discussed in Section 1.3.1. Next we give a few examples of continuous and non-continuous functions.

Example 1.3.18 Consider $f(x, y) := x^2 - y^2$ defined on $\mathbb{R}^2$. We claim that f is continuous. To see this pick a point $(x, y) \in \mathbb{R}^N$. If (x_n, y_n) is an *arbitrary* sequence we need to show that $f(x_n, y_n) = x_n^2 - y_n^2 \to x^2 - y^2 = f(x, y)$. This is quite clear since $(x_n, y_n) \to (x, y)$ means that $x_n \to x$ and $y_n \to y$. Hence by the continuity of the square function and addition we have $x_n^2 - y_n^2 \to x^2 - y^2$, showing that f is continuous.

Example 1.3.19 Show that f given by $f(x, y, z, t) := (x^2 + y^2 + z^2)e^{-t}$ is a continuous function on $\mathbb{R}^4$.
Solution: If $(x_n, y_n, z_n, t_n) \to (x, y, z, t)$, then in particular $x_n \to x$, $y_n \to y$, $z_n \to z$ and $t_n \to t$. As f consists of products and sums of the four variables as in the previous example, and the exponential function is continuous we have indeed $f(x_n, y_n, z_n, t_n) \to f(x, y, z, t)$. Hence f is continuous.

Example 1.3.20 Determine the domain of definition of $f(x, y) := x/y$, and determine whether f is continuous on its domain.
Solution: The natural domain of f is $D = \{(x, y) \in \mathbb{R}^2 \mid y \neq 0\}$. Geometrically this is the plane with the x-axis removed. If $y \neq 0$ and $(x_n, y_n) \to (x, y)$ then clearly $x_n/y_n \to x/y$, and thus f is continuous on its domain.

Example 1.3.21 Define the function f on $\mathbb{R}^2$ by setting

$$f(s,t) := \begin{cases} \dfrac{st^2}{s^2+t^2} & \text{if } (s,t) \neq (0,0), \\ 0 & \text{if } (s,t) = (0,0). \end{cases}$$

Determine whether f is continuous or not.
Solution: Along the same lines as in the previous examples we can show that f is continuous at every point $(s, t) \neq (0, 0)$. The difficulty is at $(0, 0)$. We need to show that

$$\left|\frac{s_n t_n^2}{s_n^2 + t_n^2} - 0\right| = \left|\frac{s_n t_n^2}{s_n^2 + t_n^2}\right| \xrightarrow{n\to\infty} 0 \tag{1.3.3}$$

whenever $(s_n, t_n) \xrightarrow{n\to\infty} (0, 0)$. If a, b are non-negative real numbers, then clearly $0 \leq (a - b)^2 = a^2 + b^2 - 2ab$. This implies that

$$ab \leq \frac{1}{2}\big(a^2 + b^2\big).$$

Applying the inequality to estimate $|s_n||t_n|$ we get

$$0 \leq \left|\frac{s_n t_n^2}{s_n^2 + t_n^2}\right| = |s_n||t_n|\frac{|t_n|}{s_n^2 + t_n^2} \leq \frac{1}{2}\big(s_n^2 + t_n^2\big)\frac{|t_n|}{s_n^2 + t_n^2} = \frac{1}{2}|t_n|.$$

If $(s_n, t_n) \xrightarrow{n\to\infty} (0, 0)$ then in particular $|t_n| \to 0$, and thus by the "squeezing lemma" (1.3.3) holds as $(s_n, t_n) \xrightarrow{n\to\infty} (0, 0)$. This shows that f is also continuous at $(0, 0)$, and therefore continuous everywhere. The graph of f is shown in Figure 1.3.3.

Example 1.3.22 Define the function f on $\mathbb{R}^2$ by setting

$$f(x,y) := \begin{cases} \dfrac{xy^2}{x^2+y^4} & \text{if } (x,y) \neq (0,0), \\ 0 & \text{if } (x,y) = (0,0). \end{cases} \tag{1.3.4}$$

Determine whether f is continuous or not.

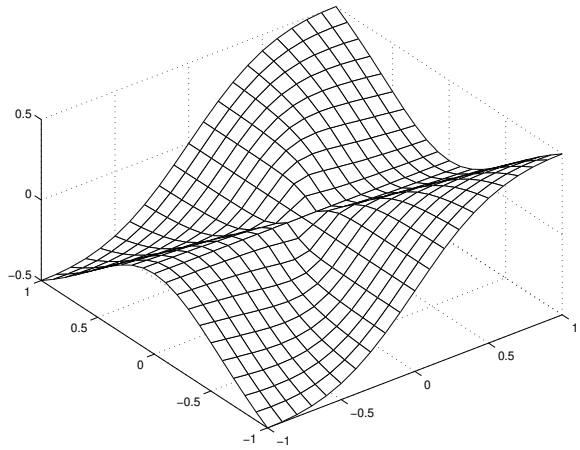


Figure 1.3.3: A function continuous at $(0, 0)$.

Solution: Along the same lines as in the previous examples we can show that f is continuous at every point $(x, y) \neq (0, 0)$. The difficulty is at $(0, 0)$. We need to check whether

$$\left|\frac{x_n y_n^2}{x_n^2 + y_n^4} - 0\right| = \left|\frac{y_n y_n^2}{x_n^2 + y_n^4}\right| \xrightarrow{n\to\infty} 0$$

for *every* sequence with $(x_n, y_n) \to (0, 0)$. If we find one particular sequence for which the limit is not zero then the function is not continuous. Let (y_n) be a sequence converging to zero and set $x_n := y_n^2$. Then $(x_n, y_n) = (y_n^2, y_n) \to (0, 0)$, and

$$f(x_n, y_n) = f(y_n^2, y_n) = \frac{y_n^2 y_n^2}{y_n^4 + y_n^4} = \frac{1}{2}$$

for all $n \in \mathbb{N}$. Hence $f(x_n, y_n) \to 1/2 \neq 0 = f(0, 0)$ for that particular sequence, and so f is not continuous at $(0, 0)$.

☞ **Remark 1.3.23** 1. One could try and make the function (1.3.4) into a continuous one by modifying it at $(0, 0)$. This does not work as for instance if $y_n \to 0$ then $f(0, y_n) = 0 \to 0$, but $f(y_n^2, y_n) = 0 \to 1/2$, so the limits are different for the sequence $(0, y_n)$ and (y_n^2, y_n).
2. When checking whether f is continuous at $(0, 0)$ we cannot take separate limits, that is first the limit as $x \to 0$ and then the limit as $y \to 0$. For the function (1.3.4) we get

$$\lim_{x\to 0} \frac{xy^2}{x^2 + y^4} = 0,$$

and thus

$$\lim_{y\to 0} \lim_{x\to 0} \frac{xy^2}{x^2 + y^4} = 0,$$

but as we know the limit is not zero for *all* sequences.
3. The example shows even more. To show continuity, we could try to see whether it is sufficient to take limits along radial straight lines. This means we just consider sequences of the form (tx, ty) as $t \to 0$ for every $(x, y) \neq (0, 0)$. If we consider such a sequence we get

$$\lim_{t\to 0} f(tx, ty) = \frac{t^3xy^2}{t^2x^2 + t^4y^4} = \frac{txy^2}{x^2 + t^2y^4} = 0.$$

no matter whether $x = 0$ or not. Hence we really need to check convergence for *every sequence*! From the graph given in Figure 1.3.4 one can clearly see this very well. Along the curve (y^2, y) the graph has a ridge (valley) of constant height (depth) going

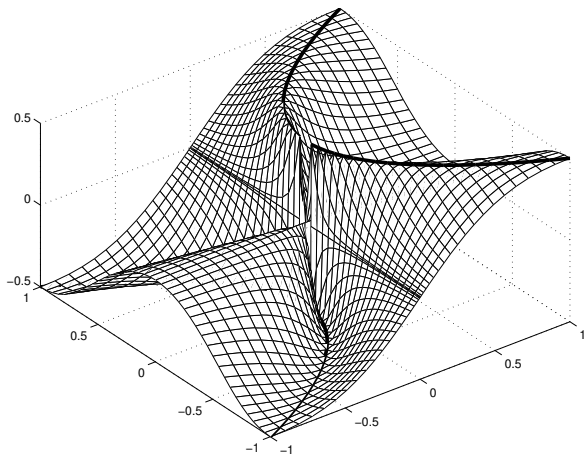


Figure 1.3.4: A function discontinuous at $(0, 0)$.

into $(0, 0)$, but never completely closing up. (Due to the insufficient resolution of the plot near $(0, 0)$ there is quite a gap between the ridges. In reality, however, the gap is only one point!) Compare the graph to the graph of the similar function from Example 1.3.21 in Figure 1.3.3

Example 1.3.24 Define $\boldsymbol{f}(x, t) = (x^2 + t, e^{-x}\cos t)$ for $x, t \in \mathbb{R}^2$. The function $\boldsymbol{f}$ is a vector valued function of two variables. To check whether it is continuous we need to determine whether every component function is continuous. The two component functions are given by $f_1(x, t) = x^2 + t$ and $f_2(x, t) = e^{-x}\cos t$. It is easily checked that they both are continuous, and thus $\boldsymbol{f}$ is continuous.

Example 1.3.25 Is $(1/x, x)$ continuous at $x = 0$?
Solution: $1/x$ is not defined at $x = 0$, so 0 is not in the domain of definition of the given vector valued function. Hence it does not make sense to talk about continuity of $(1/x, x)$

at 0.

1.4 Partial Derivatives

Given a function of several variables we can fix all of them except for one. We then have a function of one variable, and can apply all we know about such functions. In particular we can differentiate it if possible. The resulting derivatives are called the "partial derivatives" of the function. In this section we briefly want to review the basic definitions and facts on partial derivatives.

1.4.1 Basic Definitions

Consider a function f with domain $D \subset \mathbb{R}^N$ with values in $\mathbb{R}$. For every $i = 1, \dots, n$ we can then define a function of one variable, t, by setting

$$g_i(t) = f(a_1, \dots, a_{i-1}, a_i + t, a_{i+1}, \dots, a_n).$$

If $N = 2$ then the graph of this function is a cross-section of the graph of f in the i-th coordinate direction as seen in Figure 1.4.1. If $\boldsymbol{a}$ is an interior point of D then g_i is defined

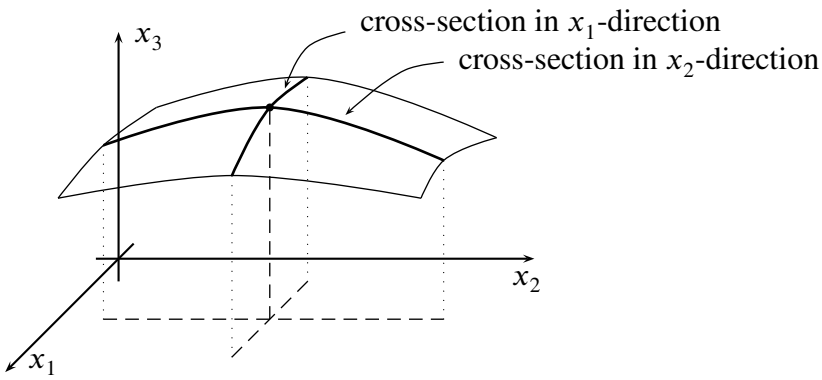


Figure 1.4.1: Cross-sections of the graph of a function through a given point

on a small interval $I = (-\varepsilon, \varepsilon)$, and zero is an interior point. From the calculus of one variables we know that $g_i'(0)$ is the slope of the graph of the function g_i at $t = 0$. Hence $g_i'(0)$ is the slope of the cross-section of the graph of f through $\boldsymbol{a}$ in the x_i-direction. This motivates the following definition.

Definition 1.4.1 (Partial derivative) Suppose that $\boldsymbol{a}$ is an interior point of $D \subset \mathbb{R}^N$, and that f is a function on D with values in $\mathbb{R}$. We define the *partial derivative of f with respect to x_i at $\boldsymbol{a}$* by

$$\frac{\partial}{\partial x_i} f(\boldsymbol{a}) := g_i'(0) = \lim_{t \to 0} \frac{1}{t}\Big(f(a_1, \dots, a_i + t, \dots, a_N) - f(a_1, \dots, a_i, \dots, a_N)\Big)$$

whenever the limit exists.

☞ **Remark 1.4.2** There are other possible notations for partial derivatives. We also write

$$\frac{\partial}{\partial x_i} f(\boldsymbol{a}) = \frac{\partial f}{\partial x_i}(\boldsymbol{a}) = f_{x_i}(\boldsymbol{a}) = \partial_{x_i} f(\boldsymbol{a}) = \partial_i f(\boldsymbol{a}).$$

Look at the symbol for the partial derivative carefully, and practise its writing a little bit.

$$\text{Write} \quad \frac{\partial}{\partial x_i} \quad \text{and NOT} \quad \frac{\delta}{\delta x_i} \quad \text{or} \quad \frac{d}{d x_i}$$

as it is often seen among beginning students. The last two symbols have a meaning on their own right, the first being the 'variation,' and the other being the 'total derivative' of a function. An example showing the difference between partial and total derivatives will be shown in Example 1.4.21.

Given a function of N variables we can form a vector having as its components the partial derivatives of that function.

Definition 1.4.3 (Gradient) Suppose f is a function defined on $D \subset \mathbb{R}^N$ having partial derivatives with respect to all N variables at the interior point $\boldsymbol{a}$. Then the vector

$$\boxed{\operatorname{grad} f(\boldsymbol{a}) = \nabla f(\boldsymbol{a}) = \Big(\frac{\partial f}{\partial x_1}(\boldsymbol{a}), \dots, \frac{\partial f}{\partial x_N}(\boldsymbol{a})\Big)}$$

is called the *gradient of f at $\boldsymbol{a}$*.

We conclude this section by some examples.

Example 1.4.4 Consider $f(x, y) := x^2 - y^2$ defined on $\mathbb{R}^2$. To compute its derivatives we treat one variable as a constant, and differentiate with respect to the other:

$$\frac{\partial}{\partial x} f(x, y) = 2x, \qquad \frac{\partial}{\partial y} f(x, y) = -2y.$$

The gradient of the function is

$$\nabla f(x, y) = (2x, -2y) = 2(x, -y).$$

Example 1.4.5 Compute the partial derivatives and the gradient of the function f given by $f(x, y, z, t) := (x^2 + y^2 + z^2)e^{-t}$.
Solution: As in the previous example we treat all variables as constants except for one. Hence the partial derivatives are

$$\frac{\partial f}{\partial x} = 2xe^{-t}, \quad \frac{\partial f}{\partial y} = 2ye^{-t}, \quad \frac{\partial f}{\partial z} = 2ze^{-t}, \quad \frac{\partial f}{\partial t} = -(x^2 + y^2 + z^2)e^{-t}.$$

The gradient is

$$\begin{aligned}\operatorname{grad} f(x, y, z, t) &= (2xe^{-t}, 2ye^{-t}, 2ze^{-t}, -(x^2 + y^2 + z^2)e^{-t}) \\ &= \big(2x, 2y, 2z, -(x^2 + y^2 + z^2)\big)e^{-t}.\end{aligned}$$

Example 1.4.6 Compute the partial derivatives of $f(x, y) := x/y$ on its domain of definition.
Solution: We saw in Example 1.3.20 that $\mathbb{R}^2$ minus the x-axis is the domain of the function under consideration. Hence, if $x \neq 0$ then

$$\frac{\partial}{\partial x} f(x, y) = \frac{1}{y}, \qquad \frac{\partial}{\partial y} f(x, y) = -\frac{x}{y^2}.$$

Example 1.4.7 Compute the gradient of $f(x, t) := x^t$ defined for $x > 0$ and $t \in \mathbb{R}$.
Solution: We first compute the partial derivatives:

$$\frac{\partial}{\partial x} f(x, t) = tx^{t-1}, \qquad \frac{\partial}{\partial t} f(x, t) = \frac{\partial}{\partial t} e^{t \ln x} = e^{t \ln x} \ln x = x^t \ln x.$$

Hence the gradient of f is

$$\operatorname{grad} f(x, t) = (tx^{t-1}, x^t \ln x) = x^t(tx^{-1}, \ln x).$$

Example 1.4.8 Define the function f on $\mathbb{R}^2$ by setting

$$f(s, t) := \begin{cases} \dfrac{st^2}{s^2 + t^2} & \text{if } (s, t) \neq (0, 0), \\ 0 & \text{if } (s, t) = (0, 0). \end{cases}$$

Determine whether f has partial derivatives at $(0, 0)$. Determine the gradient at $(0, 0)$ if it exists.
Solution: By definition of the partial derivative with respect to s we have

$$\frac{\partial f}{\partial s} f(0, 0) = \lim_{h \to 0} \frac{1}{h}\Big(\frac{(0 + h)0^2}{(0 + h)^2 + 0^2} - 0\Big) = \lim_{h \to 0} 0 = 0.$$

Similarly we have

$$\frac{\partial f}{\partial t} f(0,0) = \lim_{h\to 0} \frac{1}{h}\Big(\frac{0(0+h)^2}{0^2+(0+h)^2} - 0\Big) = \lim_{h\to 0} 0 = 0.$$

Hence both partial derivatives exist, and thus grad $f(0,0) = (0,0)$. Look at the cross-sections through the graph of f shown in Figure 1.3.3 along the coordinate axis to visualise the result.

Example 1.4.9 Consider the function f on $\mathbb{R}^2$ defined by

$$f(x,y) := \begin{cases} \dfrac{xy^2}{x^2+y^4} & \text{if } (x,y) \neq (0,0), \\ 0 & \text{if } (x,y) = (0,0). \end{cases}$$

Determine whether f has partial derivatives at $(0,0)$, and determine them if possible.
Solution: By the definition of partial derivatives we have

$$\frac{\partial f}{\partial x} f(0,0) = \lim_{t\to 0} \frac{1}{t}\Big(\frac{t0^2}{t^2+0^4} - 0\Big) = \lim_{t\to 0} 0 = 0$$

and

$$\frac{\partial f}{\partial y} f(0,0) = \lim_{t\to 0} \frac{1}{t}\Big(\frac{0t^2}{0^2+t^4} - 0\Big) = \lim_{t\to 0} 0 = 0.$$

Hence both partial derivatives exist and are zero.

Remark 1.4.10 For functions of one variable we know that differentiability at a point implies continuity of the function at that point. Note however, that the existence of all partial derivatives of a function of several variables *does not* imply the continuity of the function at the corresponding point: The function considered in the previous example has partial derivatives at $(0,0)$, but is discontinuous at $(0,0)$ as shown in Example 1.3.22.

1.4.2 Derivatives of Vector Valued Functions

Suppose that $\boldsymbol{g}$ is a vector valued function with domain $D \subset \mathbb{R}^N$ with values in $\mathbb{R}^k$. We denote its component functions by g_i, $i = 1, \dots, k$ and define

$$\boxed{\frac{\partial}{\partial x_j}\boldsymbol{g}(\boldsymbol{x}) = \frac{\partial}{\partial x_j}\begin{bmatrix} g_1(\boldsymbol{x}) \\ \vdots \\ g_k(\boldsymbol{x}) \end{bmatrix} := \begin{bmatrix} \frac{\partial}{\partial x_j} g_1(\boldsymbol{x}) \\ \vdots \\ \frac{\partial}{\partial x_j} g_k(\boldsymbol{x}) \end{bmatrix}.}$$

If $N = 1$, and the variable is t we set

$$\frac{d}{dt}\boldsymbol{g}(t) = \boldsymbol{g}'(t) = \frac{d}{dt}\begin{bmatrix} g_1(t) \\ \vdots \\ g_k(t) \end{bmatrix} := \begin{bmatrix} g_1'(t) \\ \vdots \\ g_k'(t) \end{bmatrix},$$

that is, we differentiate component-wise. We now want to derive an interpretation of $\boldsymbol{g}'(t)$. To do so it is useful to think of $\boldsymbol{g}(t)$ to be the position of moving particle at time t. Then the difference coefficient

$$\frac{1}{\Delta t}\Big(\boldsymbol{g}(t + \Delta t) - \boldsymbol{g}(t)\Big)$$

points in the direction of the line L shown in Figure 1.4.2. The magnitude of that vector is the approximate speed of the particle moving from $\boldsymbol{g}(t)$ to $\boldsymbol{g}(t + \Delta t)$. Passing to the

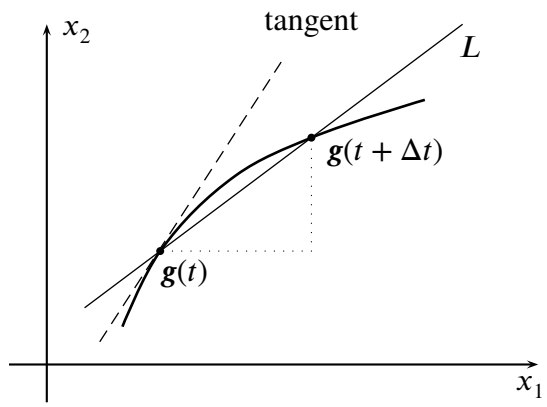


Figure 1.4.2: Secants approaching the tangent of a curve.

limit we see that

$$\frac{1}{\Delta t}\Big(\boldsymbol{g}(t + \Delta t) - \boldsymbol{g}(t)\Big) = \begin{bmatrix} \frac{g_1(t+\Delta t)-g_1(t)}{\Delta t} \\ \vdots \\ \frac{g_k(t+\Delta t)-g_k(t)}{\Delta t} \end{bmatrix} \xrightarrow{\Delta t \to 0} \begin{bmatrix} g_1'(t) \\ \vdots \\ g_k'(t) \end{bmatrix}.$$

Let us summarise what we found in the following proposition.

Proposition 1.4.11 *Under the above assumption*

$$\boxed{\textit{the vector } \boldsymbol{g}'(t) \textit{ is tangential to the path described by } \boldsymbol{g}(t).} \tag{1.4.1}$$

Moreover,

$$\boxed{\lim_{\Delta t \to 0} \frac{\|\boldsymbol{g}(t+\Delta t) - \boldsymbol{g}(t)\|}{\Delta t} = \|\boldsymbol{g}'(t)\|} \tag{1.4.2}$$

☞ **Remark 1.4.12** If we consider $\boldsymbol{g}(t)$ as the position of a particle at time t, then

$$\frac{\|\boldsymbol{g}(t+\Delta t) - \boldsymbol{g}(t)\|}{\Delta t}$$

is the approximate speed of that particle travelling from $\boldsymbol{g}(t)$ to $\boldsymbol{g}(t+\Delta t)$. Now (1.4.2) tells us that $\|\boldsymbol{g}'(t)\|$ is very close to that speed. The direction in which the particle travels is the tangential direction given by $\boldsymbol{g}'(t)$. In other words, $\boldsymbol{g}'(t)$ is the velocity of the particle at $\boldsymbol{g}(t)$. The distance travelled in a small time interval is therefore

$$\|\boldsymbol{g}(t+\Delta t) - \boldsymbol{g}(t)\| \approxeq \|\boldsymbol{g}'(t)\|\Delta t.$$

Note that the above approximate equality holds no matter what k is.

We can also use the above to find a representation of the tangent line.

Proposition 1.4.13 *Suppose that $I \subset \mathbb{R}$ is an interval, and $\boldsymbol{g} : I \to \mathbb{R}^N$ is a differentiable function. Then*

$$\boxed{\boldsymbol{r}(s) := \boldsymbol{g}(t) + s\boldsymbol{g}'(t), \qquad s \in \mathbb{R}}$$

is an equation for the tangent line to the curve described by $\boldsymbol{g}(t)$ ($t \in I$) at $\boldsymbol{g}(t)$.

For vectors we have three different kinds of products: the multiplication by scalars and the scalar product in $\mathbb{R}^N$ for general N, and the vector product in $\mathbb{R}^3$. We conclude this section by proving a 'product rule' for these products.

Theorem 1.4.14 (Generalised product rule) *Suppose that $\boldsymbol{f}$ and $\boldsymbol{g}$ are vector valued continuous functions defined on an open set $D \subset \mathbb{R}^N$ with values in $\mathbb{R}^k$, and that u is a continuous function on D with values in $\mathbb{R}$. Then*

- $\dfrac{\partial}{\partial x_i}\big(u(\boldsymbol{x})\boldsymbol{f}(\boldsymbol{x})\big) = \Big(\dfrac{\partial u}{\partial x_i}(\boldsymbol{x})\Big)\boldsymbol{f}(\boldsymbol{x}) + u(\boldsymbol{x})\dfrac{\partial \boldsymbol{f}}{\partial x_i}(\boldsymbol{x})$
- $\dfrac{\partial}{\partial x_i}\big(\boldsymbol{f}(\boldsymbol{x}) \cdot \boldsymbol{g}(\boldsymbol{x})\big) = \Big(\dfrac{\partial \boldsymbol{f}}{\partial x_i}(\boldsymbol{x})\Big) \cdot \boldsymbol{g}(\boldsymbol{x}) + \boldsymbol{f}(\boldsymbol{x}) \cdot \dfrac{\partial \boldsymbol{g}}{\partial x_i}(\boldsymbol{x})$
- *If* $k = 3$ *then*
  $\dfrac{\partial}{\partial x_i}\big(\boldsymbol{f}(\boldsymbol{x}) \times \boldsymbol{g}(\boldsymbol{x})\big) = \Big(\dfrac{\partial \boldsymbol{f}}{\partial x_i}(\boldsymbol{x})\Big) \times \boldsymbol{g}(\boldsymbol{x}) + \boldsymbol{f}(\boldsymbol{x}) \times \dfrac{\partial \boldsymbol{g}}{\partial x_i}(\boldsymbol{x})$

whenever all relevant derivatives exist.

Proof. The proof of all the above formulae are analogous to the classical product rule. As an example we give a proof of the first. By definition of the partial derivative we need to compute the limit of the difference quotient

$$D(t) := \frac{1}{t}\Big(u(\boldsymbol{x} + t\boldsymbol{e}_i)\boldsymbol{f}(\boldsymbol{x} + t\boldsymbol{e}_i) - u(\boldsymbol{x})\boldsymbol{f}(\boldsymbol{x})\Big)$$

as $t \to 0$, where $\boldsymbol{e}_i$ is the vector

$$\boldsymbol{e}_i := (0, \dots, 0, \underset{\substack{\uparrow \\ i\text{-th coordinate}}}{1}, 0, \dots, 0).$$

By adding and subtracting a term we can rewrite $D(t)$ by

$$D(t) = \frac{1}{t}\Big(u(\boldsymbol{x} + t\boldsymbol{e}_i) - u(\boldsymbol{x})\Big)\boldsymbol{f}(\boldsymbol{x} + t\boldsymbol{e}_i) + u(x)\frac{1}{t}\Big(\boldsymbol{f}(\boldsymbol{x} + t\boldsymbol{e}_i) - \boldsymbol{f}(\boldsymbol{x})\Big).$$

Using the continuity of the functions and the definition of the partial derivatives we get from the above that

$$\lim_{t\to 0} D(t) = \Big(\frac{\partial u}{\partial x_i}(\boldsymbol{x})\Big)\boldsymbol{f}(\boldsymbol{x}) + u(\boldsymbol{x})\frac{\partial \boldsymbol{f}}{\partial x_i}(\boldsymbol{x})$$

as required. ■

1.4.3 The Chain Rule

If f, g are differentiable functions of one variable and $g(x)$ is in the domain of f then the *chain rule* asserts that

$$[f \circ g(x)]' = f'(g(x))g'(x),$$

as you know from first year calculus. We now want to generalise the above formula if f is a function of several variables. One key requirement was that g has its values in the

domain of f. Hence, if f is defined on a subset $D \subset \mathbb{R}^N$, then g must to be a function with values in $\mathbb{R}^N$, that is, a vector valued function. In general $\boldsymbol{g}$ could be a function of several variables as well.

In what follows we assume that f is a function of $\boldsymbol{x} = (x_1, \dots, x_N)$, and that $\boldsymbol{g}$ a vector valued function with values in $\mathbb{R}^N$ of the variables $\boldsymbol{y} = (y_1, \dots, y_k)$. As usual, we denote the component functions of $\boldsymbol{g}$ by g_i, $i = 1, \dots, N$. We are now in a position to state one possible version of the chain rule in vector calculus.

Theorem 1.4.15 (Chain rule) *Suppose that f, $\boldsymbol{g}$ are as above. Moreover, assume that $\boldsymbol{y}$ is an interior point of the domain of $f \circ \boldsymbol{g}$, that $\boldsymbol{g}$ has a partial derivative with respect to y_j, and that* grad $f(\boldsymbol{g}(\boldsymbol{y}))$ *exists and is continuous at $\boldsymbol{y}$. Then*

$$\boxed{\frac{\partial}{\partial y_j} f(\boldsymbol{g}(\boldsymbol{y})) = \operatorname{grad} f(\boldsymbol{g}(\boldsymbol{y})) \cdot \frac{\partial}{\partial y_j} \boldsymbol{g}(\boldsymbol{y}) = \sum_{i=1}^{N} \frac{\partial f}{\partial x_i}(\boldsymbol{g}(\boldsymbol{y})) \frac{\partial}{\partial y_j} g_i(\boldsymbol{y}).}$$

If $\boldsymbol{g}$ is a function of one variable, t, then

$$\boxed{\frac{d}{dt} f(\boldsymbol{g}(t)) = \operatorname{grad} f(\boldsymbol{g}(t)) \cdot \boldsymbol{g}'(t) = \sum_{i=1}^{N} \frac{\partial f}{\partial x_i}(\boldsymbol{g}(t)) \frac{d}{dt} g_i(t).}$$

Proof. By definition of the partial derivative of a function we treat all variables as constant except for one. Hence the proof of the first formula can be reduced to the proof of the second one. The general proof is quite tedious, so we only illustrate the main ideas if $N = 2$. We need to compute the limit as $h \to 0$ of

$$D(h) := \frac{1}{h}\big(f(g_1(t+h), g_2(t+h)) - f(g_1(t), g_2(t))\big).$$

To do so we rewrite the expression as

$$\begin{aligned} D(h) = {} & \frac{f(g_1(t+h), g_2(t+h)) - f(g_1(t), g_2(t+h))}{g_1(t+h) - g_1(t)} \quad \frac{g_1(t+h) - g_1(t)}{h} \\ & + \frac{f(g_1(t), g_2(t+h)) - f(g_1(t), g_2(t))}{g_2(t+h) - g_2(t)} \quad \frac{g_2(t+h) - g_2(t)}{h}, \end{aligned}$$

adding and subtracting a term. As g_i is differentiable at t it follows that

$$\lim_{h \to 0} \frac{g_i(t+h) - g_i(t)}{h} = g_i'(t)$$

for $i = 1, 2$. As a differentiable function of one variable is continuous, we have that $g_i(t+h) - g_i(t) \to 0$ as $h \to 0$. Hence, by definition of partial derivatives,

$$\frac{f(g_1(t), g_2(t+h)) - f(g_1(t), g_2(t))}{g_2(t+h) - g_2(t)} \xrightarrow{h\to 0} \frac{\partial f}{\partial x_2}(\boldsymbol{g}(t)).$$

To deal with the remaining term we apply the mean value theorem to the function $x_1 \mapsto f(x_1, g_2(t+h))$. By assumption, that function is differentiable and its derivative is continuous at $g_1(t)$. Thus, by the mean value theorem there exist $c_h \in \mathbb{R}$ such that

$$\begin{aligned} &f(g_1(t+h), g_2(t+h)) - f(g_1(t), g_2(t+h)) \\ &\qquad\qquad = \big(g_1(t+h) - g_1(t)\big)\frac{\partial f}{\partial x_1} f(c_h, g_2(t+h)) \end{aligned}$$

with $\Big|c_h - g_1(t))\Big| \leq \Big|g_1(t+h) - g_1(t))\Big|$. As g_1 is continuous at t it follows that $c_h \to g_1(t)) \to 0$ as $h \to 0$. Finally, by continuity of $\partial f/\partial x_1$ with respect to x_1 we conclude that

$$\frac{f(g_1(t+h), g_2(t+h)) - f(g_1(t), g_2(t+h))}{g_1(t+h) - g_1(t)} \xrightarrow{h\to 0} \frac{\partial f}{\partial x_1}(\boldsymbol{g}(t))$$

as $h \to 0$. If we put everything together it follows that

$$\lim_{h\to 0} D(h) = \frac{\partial f}{\partial x_1}(\boldsymbol{g}(t))g_1'(t) + \frac{\partial f}{\partial x_2}(\boldsymbol{g}(t))g_2'(t) = \operatorname{grad} f(\boldsymbol{g}(t)) \cdot \boldsymbol{g}'(t)$$

as required. For general N there are more terms to add and subtract, but the basic ideas stay the same. ■

If $\boldsymbol{f}$ is a vector valued function then the chain rule applies to every component function. We want to use this to derive a formula for the derivative of a composition $\boldsymbol{f} \circ \boldsymbol{g}$, which is a vector valued function. Before we do so we introduce the 'Jacobian matrix' of a function.

Definition 1.4.16 (Jacobian matrix) Suppose that $\boldsymbol{f}$ is a function defined on a subset of $\mathbb{R}^N$ with values in $\mathbb{R}^k$. If all entries exist, the matrix

$$\boxed{J_{\boldsymbol{f}}(\boldsymbol{x}) := \begin{bmatrix} \frac{\partial}{\partial x_1} f_1(\boldsymbol{x}) & \dots & \frac{\partial}{\partial x_N} f_1(\boldsymbol{x}) \\ \vdots & \ddots & \vdots \\ \frac{\partial}{\partial x_1} f_k(\boldsymbol{x}) & \dots & \frac{\partial}{\partial x_N} f_k(\boldsymbol{x}) \end{bmatrix}}$$

is called the *Jacobian matrix of* $\boldsymbol{f}$ *at* $\boldsymbol{x}$.

Note that $J_f(\boldsymbol{x}) = \operatorname{grad} f(\boldsymbol{x})$ if f is scalar valued, that is, $k = 1$. If we apply Theorem 1.4.15 to every component function of $\boldsymbol{f}$ then we get the following formula for the Jacobian matrix of a composition of functions.

Corollary 1.4.17 *Suppose that every component function of $\boldsymbol{f}$ and $\boldsymbol{g}$ satisfies the assumptions of Theorem 1.4.15. Then*

$$\boxed{J_{f\circ g}(\boldsymbol{y}) = J_f(\boldsymbol{g}(\boldsymbol{y}))J_g(\boldsymbol{y})}$$

In other words, the Jacobian matrix of a composition of vector valued function is the matrix product of the individual Jacobian matrices.

Example 1.4.18 Let $f(x, y, z) = x^2 - z\cos y$ and $\boldsymbol{g}(t) := (t, t^2, 1/t)$. To compute $(f\circ\boldsymbol{g})'$ we first note that $\operatorname{grad} f(x, y, z) = (2x, z\sin y, \cos y)$ and $\boldsymbol{g}'(t) = (1, 2t, -1/t^2)$. Then by the chain rule

$$\begin{aligned}\frac{d}{dt} f\circ\boldsymbol{g}(t) &= \operatorname{grad} f(t, t^2, 1/t)\cdot \boldsymbol{g}'(t)\\ &= (2t, t^{-1}\sin t^2, \cos t^2)\cdot(1, 2t, -1/t^2) = 2t + 2\sin t^2 - \frac{\cos t^2}{t^2}\end{aligned}$$

Example 1.4.19 Let $r(\boldsymbol{x}) := \|x\| = \sqrt{x_1^2 + x_2^2}$ and $\boldsymbol{g}(t) := (2\cos t, 3\sin t)$ for $t \in (0, 2\pi)$. Compute $(r\circ\boldsymbol{g})'$.
Solution: We start by computing the gradient of r:

$$\begin{aligned}\frac{\partial r}{\partial x_1}(x_1, x_2) &= \frac{\partial}{\partial x_1}\sqrt{x_1^2 + x_2^2} = \frac{2x_1}{2\sqrt{x_1^2 + x_2^2}} = \frac{x_1}{\sqrt{x_1^2 + x_2^2}},\\ \frac{\partial r}{\partial x_2}(x_1, x_2) &= \frac{\partial}{\partial x_2}\sqrt{x_1^2 + x_2^2} = \frac{2x_2}{2\sqrt{x_1^2 + x_2^2}} = \frac{x_2}{\sqrt{x_1^2 + x_2^2}}.\end{aligned}$$

Hence

$$\operatorname{grad} r(\boldsymbol{x}) = \frac{1}{\sqrt{x_1^2 + x_2^2}}(x_1, x_2) = \frac{\boldsymbol{x}}{r(\boldsymbol{x})}.$$

Next note that $\boldsymbol{g}'(t) = (-2\sin t, 3\cos t)$. By the chain rule

$$\begin{aligned}\frac{d}{dt} r\circ\boldsymbol{g}(t) &= \operatorname{grad} r(\boldsymbol{g}(t))\cdot \boldsymbol{g}'(t)\\ &= \frac{(2\cos t, 3\sin t)}{\sqrt{4\sin^2 t + 9\cos^2 t}}\cdot(-2\sin t, 3\cos t)\\ &= \frac{-4\sin t\cos t + 9\sin t\cos t}{\sqrt{4\sin^2 t + 9\cos^2 t}} = \frac{5\sin t\cos t}{\sqrt{4 + 5\cos^2 t}}\end{aligned}$$

Example 1.4.20 Compute the Jacobian matrix of

$$\boldsymbol{g}(s,t) := \left(\frac{-t}{s^2+t^2}, \frac{s}{s^2+t^2}\right).$$

Solution: We need both partial derivatives for every component function:

$$\begin{aligned}
\frac{\partial}{\partial s}\frac{-t}{s^2+t^2} &= \frac{2st}{(s^2+t^2)^2};\\
\frac{\partial}{\partial t}\frac{-t}{s^2+t^2} &= -\frac{s^2+t^2-2t^2}{(s^2+t^2)^2} = \frac{t^2-s^2}{(s^2+t^2)^2};\\
\frac{\partial}{\partial s}\frac{s}{s^2+t^2} &= \frac{s^2+t^2-2s^2}{(s^2+t^2)^2} = \frac{t^2-s^2}{(s^2+t^2)^2};\\
\frac{\partial}{\partial t}\frac{s}{s^2+t^2} &= -\frac{2st}{(s^2+t^2)^2}.
\end{aligned}$$

Hence the Jacobian matrix of $\boldsymbol{g}$ is

$$J_{\boldsymbol{g}}(s,t) = \frac{1}{(s^2+t^2)^2}\begin{bmatrix} 2st & t^2-s^2 \\ t^2-s^2 & -2st \end{bmatrix}.$$

☞ **Example 1.4.21** Consider the function $f(x,t) := xe^{-t}$. Then the partial derivatives are

$$\begin{aligned}
\frac{\partial}{\partial x}f(x,t) &= e^{-t}\\
\frac{\partial}{\partial t}f(x,t) &= -xe^{-t}
\end{aligned}$$

If we assume that x is a function of t as well, then by the chain rule

$$\begin{aligned}
\frac{d}{dt}f(x(t),t) &= \frac{\partial}{\partial x}f(x(t),t)x'(t) + \frac{\partial}{\partial t}f(x(t),t)\frac{dt}{dt}\\
&= e^{-t}x'(t) - x(t)e^{-t} = \big(x'(t) - x(t)\big)e^{-t}.
\end{aligned}$$

Hence,

$$\frac{\partial}{\partial t}f(x,t) = -xe^{-t} \qquad \text{and} \qquad \frac{d}{dt}f(x,t) = (x'-x)e^{-t}$$

are NOT the same. The first is the partial derivative, and the second is called the *total derivative* of f with respect to t. (See also Remark 1.4.2 on page 34.)

1.4.4 Directional Derivatives

As explained in Section 1.4.1 the partial derivatives of a function of several variables represent the change of the function along a cross-section through its graph parallel to one of the coordinate axes. These directions are quite arbitrary as, at one stage, we made

a choice of coordinates. It is therefore natural to look at cross-sections through the graph of a function in other directions, not just the coordinate directions. We want to derive a formula to compute such derivatives. To do so let $\boldsymbol{v} \in \mathbb{R}^N$ be a unit vector, that is, a vector of length one. Given a real valued function f defined on a set $D \subset \mathbb{R}^N$ and $\boldsymbol{a} \in D$ the graph of the function

$$g_{\boldsymbol{v}}(t) := f(\boldsymbol{a} + t\boldsymbol{v})$$

represents the cross-section through the graph of f at $\boldsymbol{a}$ in the direction of $\boldsymbol{v}$. The situation is depicted in Figure 1.4.3.

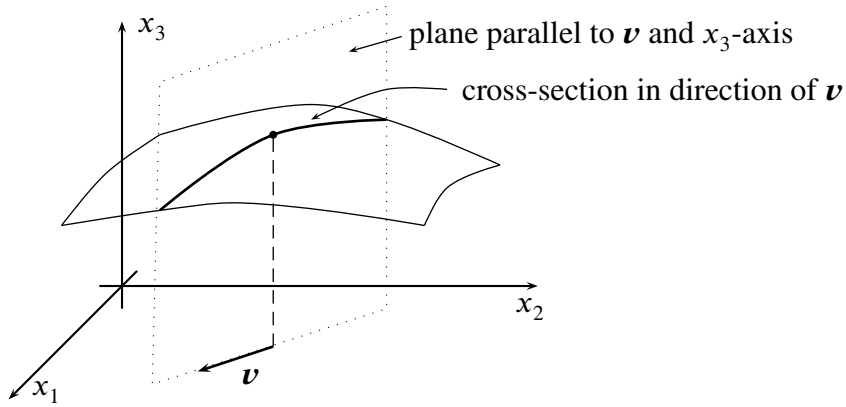


Figure 1.4.3: Cross section in the direction of $\boldsymbol{v}$ through the graph of $\boldsymbol{f}$.

Definition 1.4.22 (Directional derivative) Suppose $\boldsymbol{v}$ is a unit vector. If $\boldsymbol{a}$ is an interior point of the domain $D \subset \mathbb{R}^N$ of $\boldsymbol{f}$ then, if the limit exists,

$$\frac{\partial}{\partial \boldsymbol{v}} f(\boldsymbol{a}) := g_{\boldsymbol{v}}'(0) = \lim_{t \to 0} \frac{1}{t}\Big(f(\boldsymbol{a} + t\boldsymbol{v}) - f(\boldsymbol{a}) \Big)$$

is called the *directional derivative of f at $\boldsymbol{a}$ in the direction of $\boldsymbol{v}$.*

☞ **Remark 1.4.23** 1. Sometimes the notation

$$\frac{\partial}{\partial \boldsymbol{v}} f(\boldsymbol{a}) = \frac{\partial f}{\partial \boldsymbol{v}}(\boldsymbol{a}) = f_{\boldsymbol{v}}(\boldsymbol{a}) = \partial_{\boldsymbol{v}} f(\boldsymbol{a})$$

is used for the directional derivative at $\boldsymbol{a}$ in the direction of $\boldsymbol{v}$.
2. For $i = 1, \dots, N$ denote by

$$\boldsymbol{e}_i := (0, \dots, 0, \underset{\substack{\uparrow \\ i\text{-th coordinate}}}{1}, 0, \dots, 0)$$

the i-th vector of the standard basis of $\mathbb{R}^N$. Clearly $\|\boldsymbol{e}_i\| = 1$, so $\boldsymbol{e}_i$ is a unit vector. By comparing the definitions we see that

$$\frac{\partial}{\partial \boldsymbol{e}_i} f(\boldsymbol{a}) = \frac{\partial}{\partial x_i} f(\boldsymbol{a}),$$

so that the notion of a directional derivative is a direct generalisation of the partial derivatives as motivated at the start of this section.

We next want to derive a formula to compute the directional derivatives of a function.

Proposition 1.4.24 *Let $\boldsymbol{a}$ be an interior point of the domain $D \subset \mathbb{R}^N$ of f. If $\boldsymbol{v}$ is a unit vector, and* grad f *is continuous at $\boldsymbol{a}$, then*

$$\boxed{\frac{\partial}{\partial \boldsymbol{v}} f(\boldsymbol{a}) = \big(\operatorname{grad} f(\boldsymbol{a})\big) \cdot \boldsymbol{v}.} \tag{1.4.3}$$

Proof. We use the chain rule to prove the formula. To do so set $\boldsymbol{g}(t) := \boldsymbol{a} + t\boldsymbol{v}$. Then we have $\boldsymbol{g}'(t) = \boldsymbol{v}$ for all t. Hence by the definition of directional derivatives and the chain rule (see Theorem 1.4.15) we have

$$\frac{\partial}{\partial \boldsymbol{v}} f(\boldsymbol{a}) = \frac{d}{dt}(f \circ \boldsymbol{g})(0) = \big(\operatorname{grad} f(\boldsymbol{g}(0))\big) \cdot \boldsymbol{g}'(0) = \big(\operatorname{grad} f(\boldsymbol{a})\big) \cdot \boldsymbol{v}$$

as required. ■

Remark 1.4.25 We saw in Remark 1.4.10 that the existence of the partial derivatives at a point does not imply that the function is continuous at that point. The same function as considered there shows that the existence of *all* directional derivatives is not in general enough to guarantee the continuity of a function! As an exercise show that the function (1.3.4) has directional derivatives in every direction at $(0, 0)$. From Example 1.3.22 we know that f is not continuous at $(0, 0)$.

1.4.5 Tangents and Normals

In this section we want to derive two properties of the gradient. The first will show that it always points into the direction of the steepest ascent (the most rapid increase) of the given function. The second shows that it is always perpendicular to the level set.

The Direction of Most Rapid Increase

If $N = 2$ then the graph of a function looks like the topography of a hilly area. At a point $(\boldsymbol{x}, f(\boldsymbol{x}))$ on the graph of f the directional derivative of f at $\boldsymbol{x}$ in the direction of a unit vector $\boldsymbol{v}$ tells us how steep the hill is when moving in the direction of $\boldsymbol{v}$ from that point. We next want to determine the directions of steepest ascent and descent. Mathematically we have to find those unit vectors, $\boldsymbol{v}$, for which the directional derivatives are maximal or minimal. Using formula (1.4.3) for the directional derivative we need to determine $\boldsymbol{v}$ such that

$$\frac{\partial}{\partial \boldsymbol{v}} f(\boldsymbol{x}) = \big(\nabla f(\boldsymbol{x})\big) \cdot \boldsymbol{v}.$$

is maximal or minimal. Using the Cauchy-Schwarz inequality (Theorem 1.1.4 on page 7) we see that

$$\left|\frac{\partial}{\partial \boldsymbol{v}} f(\boldsymbol{x})\right| = \left|\big(\nabla f(\boldsymbol{x})\big) \cdot \boldsymbol{v}\right| \leq \|\nabla f(\boldsymbol{x})\| \|\boldsymbol{v}\|$$

with equality if and only if $\boldsymbol{v}$ is parallel to $\nabla f(\boldsymbol{x})$. Hence the directional derivative is maximal or minimal if and only if

$$\boldsymbol{v} = \pm \frac{\nabla f(\boldsymbol{x})}{\|\nabla f(\boldsymbol{x})\|}$$

if $\nabla f(\boldsymbol{x}) \neq 0$. It follows that the maximal directional derivative is

$$\frac{\partial}{\partial \boldsymbol{v}} f(\boldsymbol{x}) = \big(\nabla f(\boldsymbol{x})\big) \cdot \frac{\nabla f(\boldsymbol{x})}{\|\nabla f(\boldsymbol{x})\|} = \frac{\|\nabla f(\boldsymbol{x})\|^2}{\|\nabla f(\boldsymbol{x})\|} = \|\nabla f(\boldsymbol{x})\|,$$

and the minimal directional derivative

$$\frac{\partial}{\partial \boldsymbol{v}} f(\boldsymbol{x}) = -\|\nabla f(\boldsymbol{x})\|.$$

Hence the direction of the steepest ascent is in the direction of $\nabla f(\boldsymbol{x})$. From the above considerations we get the following interpretation of the gradient. Note that the above arguments do not depend on the particular dimension $\boldsymbol{N} = 2$, but hold for arbitrary N.

Theorem 1.4.26 *At every point the gradient of a function of several variables points in the direction of the most rapid increase of that function.*

Normals to Level Sets

We next want to make a connection to the level sets considered in Section 1.2.3. We just saw that the gradient of a given function points in the direction of the most rapid increase of that function. Hence we would expect that, in every perpendicular direction, the

function does not increase, that is, the gradient should be perpendicular to the level set under reasonable assumptions. To show this we start by looking at a function, f, of two variables defined on $D \subset \mathbb{R}^2$. Fix a point $\boldsymbol{a}$ in the interior of D, and assume that grad f is continuous at $\boldsymbol{a}$. Furthermore, assume that the level set (or contour line)$f^{-1}[c]$ forms a differentiable curve near $\boldsymbol{a}$. This means that there exists a vector valued, differentiable function $\boldsymbol{g}(t) = (g_1(t), g_2(t))$, $t \in (-\varepsilon, \varepsilon)$, such that

$$f(\boldsymbol{g}(t)) = c$$

for all $t \in (-\varepsilon, \varepsilon)$, and $\boldsymbol{g}(0) = \boldsymbol{a}$. As c is a constant, and by the chain rule (see Theorem 1.4.15 on page 40) we get

$$\frac{d}{dt}c = 0 = \frac{d}{dt}(f \circ \boldsymbol{g})(0) = \big(\operatorname{grad} f(\boldsymbol{g}(0))\big) \cdot \boldsymbol{g}'(0) = \big(\operatorname{grad} f(\boldsymbol{a})\big) \cdot \boldsymbol{g}'(0)$$

This shows that grad $f(\boldsymbol{a})$ is perpendicular (normal) to $\boldsymbol{g}'(0)$. Recall that $\boldsymbol{g}'(0)$ is parallel to the tangent to the contour line at $\boldsymbol{g}(0) = \boldsymbol{a}$ (see Section 1.4.2). Hence we proved the following:

grad $f(\boldsymbol{a})$ is perpendicular (to the tangent line) to the contour line $f^{-1}[c]$ at $\boldsymbol{a}$.

The situation is depicted in Figure 1.4.4.

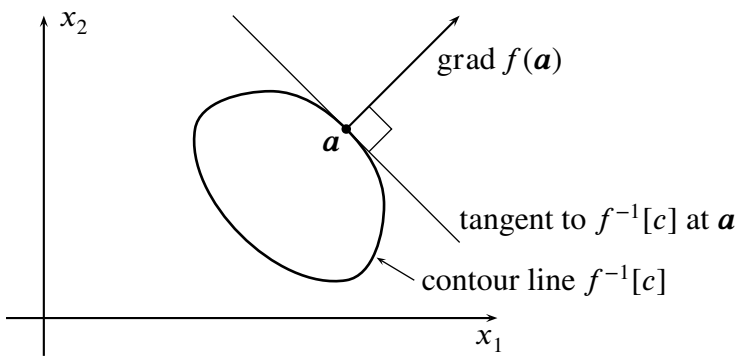


Figure 1.4.4: Contour line with tangent and normal.

We next generalise the above ideas to functions of three variables. Assume that f is such a function defined on $D \subset \mathbb{R}^3$. We assume that $\boldsymbol{a}$ lies on the level set $f^{-1}[c]$. Suppose that $\boldsymbol{g}(t) = (g_1(t), g_2(t), g_3(t))$, $t \in (-\varepsilon, \varepsilon)$, is a curve on that level set with $\boldsymbol{g}(0) = \boldsymbol{a}$. Then $\boldsymbol{g}'(t)$ is parallel to the tangent to that curve at $\boldsymbol{a}$, and thus tangent to the level set $f^{-1}[c]$ at $\boldsymbol{a}$. Applying the chain rule as before we see that

$$\frac{d}{dt}c = 0 = \frac{d}{dt}(f \circ \boldsymbol{g})(0) = \big(\operatorname{grad} f(\boldsymbol{g}(0))\big) \cdot \boldsymbol{g}'(0) = \big(\operatorname{grad} f(\boldsymbol{a})\big) \cdot \boldsymbol{g}'(0)$$

as before. Hence, similarly as before we have the following fact:

grad $f(\boldsymbol{a})$ is perpendicular to (all tangent lines to) the level set $f^{-1}[c]$ at $\boldsymbol{a}$.

The relevant part of a level surface with two curves, their tangents (dashed) and the gradient at a point are shown in Figure 1.4.5. The situation remains essentially the same

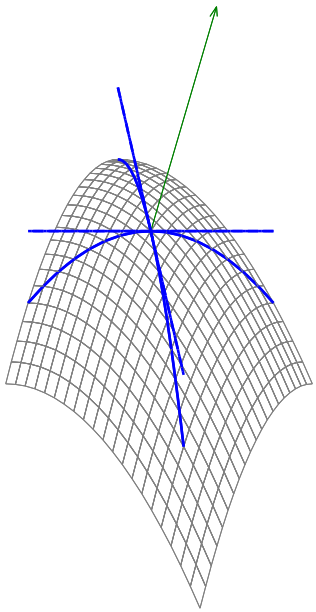


Figure 1.4.5: A level surface with two curves, their tangents and the gradient.

for general N, so we get the following theorem.

Theorem 1.4.27 *Suppose that f is a real valued function defined on $D \subset \mathbb{R}^N$. Suppose that $\boldsymbol{a}$ is an interior point of D lying on the level set $S := f^{-1}[c]$ for some $c \in \mathbb{R}$. Moreover, assume that* grad f *is continuous at $\boldsymbol{a}$. Then* grad $f(\boldsymbol{a})$ *is (normal) perpendicular to the level set S. More precisely,* grad $f(\boldsymbol{a})$ *is perpendicular to every tangent vector to S at $\boldsymbol{a}$.*

One of the essential assumptions above was that the level set $f^{-1}[c]$ is a differentiable curve or surface, respectively. In general we do not know whether this is the case. There is a very convenient criterion which guarantees that $f^{-1}[c]$ is a curve or surface, respectively. The following theorem, usually referred to as the *implicit function theorem* provides such a criterion. We only give a precise formulation for functions of two variables.

Theorem 1.4.28 (Implicit function theorem) *Suppose that f has continuous partial derivatives at a point $\boldsymbol{a}$ of its domain. Moreover, suppose that $f(\boldsymbol{a}) = c$, and that*

grad $f(\boldsymbol{a}) \neq \mathbf{0}$. *Then, there exist an open interval* $(-\varepsilon, \varepsilon)$, *a continuously differentiable function* $\boldsymbol{g}\colon I \to \mathbb{R}^2$ *and an open disc* $B := B_r(\boldsymbol{a})$ *such that*

$$f^{-1}[c] \cap B = \{\boldsymbol{g}(t) : t \in I\} \cap B.$$

The proof of the implicit function theorem is rather lengthy and is omitted here.

☞ **Remark 1.4.29** One could be more specific about the function $\boldsymbol{g}$ whose existence is claimed in the implicit function theorem. The condition grad $f(\boldsymbol{a}) \neq \mathbf{0}$ means that at least one of the partial derivatives of f at $\boldsymbol{a}$ are nonzero. Assume that

$$\frac{\partial}{\partial x_2} f(\boldsymbol{a}) \neq 0.$$

Then, in a neighbourhood of $\boldsymbol{a}$ we can write x_2 as a differentiable function of x_1, so that $\boldsymbol{g} = (x_1, x_2(x_1))$. The situation is depicted in Figure 1.4.6.

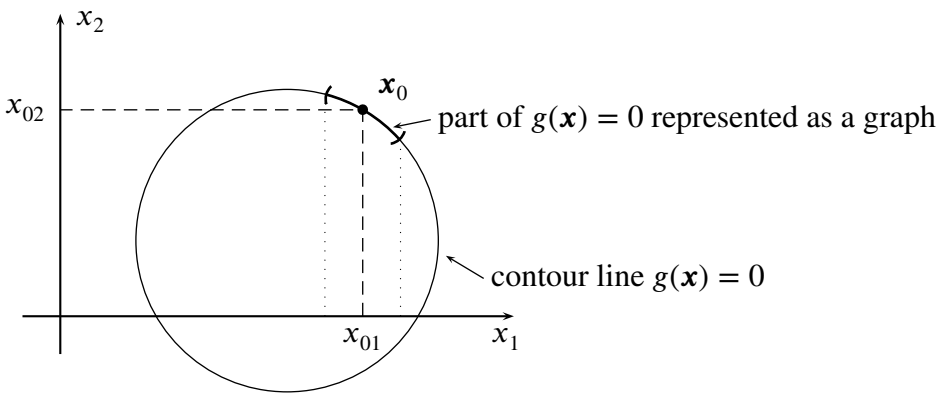


Figure 1.4.6: The fat part of $g(\boldsymbol{x}) = 0$ near $\boldsymbol{x}_0$ is the graph of a function.

Remark 1.4.30 Theorem 1.4.28 can be generalised to functions defined on subsets of $\mathbb{R}^3$. If $f(\boldsymbol{a}) = c$ the condition grad $f(\boldsymbol{a}) \neq 0$ makes sure that $f^{-1}[c]$ forms a 'nice' surface in a neighbourhood of $\boldsymbol{a}$.

1.4.6 Higher Derivatives and Taylor polynomials

The aim of this section is to discuss the main properties of higher partial derivatives, and then to apply these results to introduce Taylor polynomials for functions of several variables.

Higher Derivatives

The partial derivative of a function f defined on $D \subset \mathbb{R}^N$ is again a function on D. Hence we can again take partial derivatives, if they exist. We call such derivatives higher order partial derivatives. We write

$$\frac{\partial^2}{\partial x_j \partial x_i} f(\boldsymbol{a}) := \frac{\partial^2 f}{\partial x_j \partial x_i}(\boldsymbol{a}) := \frac{\partial}{\partial x_j}\Big(\frac{\partial f}{\partial x_i}\Big)(\boldsymbol{a}),$$

and

$$\frac{\partial^2}{\partial x_i^2} f(\boldsymbol{a}) := \frac{\partial^2 f}{\partial x_i^2}(\boldsymbol{a}) := \frac{\partial}{\partial x_i}\Big(\frac{\partial f}{\partial x_i}\Big)(\boldsymbol{a}).$$

for the second order partial derivatives. The exponent indicates how many derivatives we take. A fourth order derivative is for instance given by

$$\frac{\partial^4 f}{\partial x_1 \partial x_3 \partial x_2 \partial x_1}(\boldsymbol{a}) := \frac{\partial}{\partial x_1}\Big(\frac{\partial}{\partial x_3}\Big(\frac{\partial}{\partial x_2}\Big(\frac{\partial f}{\partial x_1}\Big)\Big)\Big)(\boldsymbol{a}).$$

Example 1.4.31 Let $f(x, y) := x^3 - 3x^2 y$ for all $(x, y) \in \mathbb{R}^2$. Then the first order partial derivatives are

$$\frac{\partial f}{\partial x}(x, y) = 3x^2 - 6xy,$$
$$\frac{\partial f}{\partial y}(x, y) = -3x^2.$$

Hence the second order partial derivatives are

$$\frac{\partial^2 f}{\partial x^2}(x, y), = 6x - 6y \qquad \frac{\partial^2 f}{\partial y \partial x}(x, y), = -6x$$
$$\frac{\partial^2 f}{\partial x \partial y}(x, y), = -6x \qquad \frac{\partial^2 f}{\partial y^2}(x, y). = 0$$

In the above example we see that

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y},$$

that is, interchanging the order of the partial derivative leads to the same answer. This is not accidental as the following proposition shows, the proof of which will be omitted.

Proposition 1.4.32 *Suppose that f, defined on $D \subset \mathbb{R}^N$ has continuous second order partial derivatives. Then*

$$\frac{\partial^2 f}{\partial x_j \partial x_i} = \frac{\partial^2 f}{\partial x_i \partial x_j}$$

for all $i, j = 1, \dots, N$.

Remark 1.4.33 Examples show that the assumption that the partial derivatives be continuous is essential for the above result to be true!

As a natural generalisation of partial derivatives we studied directional derivatives. We now want to look at higher order directional derivatives. Given a function f defined on $D \subset \mathbb{R}^N$ and a unit vector $\boldsymbol{v} = (v_1, \dots, v_N)$ we set

$$\frac{\partial^2}{\partial \boldsymbol{v}^2} f(\boldsymbol{x}) := \frac{\partial}{\partial \boldsymbol{v}}\Big(\frac{\partial f}{\partial \boldsymbol{v}}\Big)(\boldsymbol{x}), \qquad \frac{\partial^3}{\partial \boldsymbol{v}^3} f(\boldsymbol{x}) := \frac{\partial}{\partial \boldsymbol{v}}\Big(\frac{\partial}{\partial \boldsymbol{v}}\Big(\frac{\partial f}{\partial \boldsymbol{v}}\Big)\Big)(\boldsymbol{x}), \qquad \text{etc.}$$

In Proposition 1.4.24 we derived a formula for the directional derivative. We found that

$$\frac{\partial f}{\partial \boldsymbol{v}}(\boldsymbol{x}) = \big(\operatorname{grad} f(\boldsymbol{x})\big) \cdot \boldsymbol{v}$$

if $\operatorname{grad} f$ is continuous at $\boldsymbol{x}$. To compute the second directional derivative we can apply the same formula to the function $\big(\operatorname{grad} f(\boldsymbol{x})\big) \cdot \boldsymbol{v}$. Doing that we get

$$\frac{\partial^2}{\partial \boldsymbol{v}^2} f(\boldsymbol{x}) = \operatorname{grad}\Big(\big(\operatorname{grad} f(\boldsymbol{x})\big) \cdot \boldsymbol{v}\Big) \cdot \boldsymbol{v}.$$

To derive a more explicit formula for the above expression we compute the partial derivatives of $\big(\operatorname{grad} f(\boldsymbol{x})\big) \cdot \boldsymbol{v}$:

$$\frac{\partial}{\partial x_i}\big(\operatorname{grad} f(\boldsymbol{x})\big) \cdot \boldsymbol{v} = \frac{\partial}{\partial x_i} \sum_{j=1}^{N} \frac{\partial f}{\partial x_j}(\boldsymbol{x}) v_j = \sum_{j=1}^{N} \frac{\partial^2 f}{\partial x_i \partial x_j}(\boldsymbol{x}) v_j.$$

Therefore,

$$\frac{\partial^2}{\partial \boldsymbol{v}^2} f(\boldsymbol{x}) = \operatorname{grad}\Big(\big(\operatorname{grad} f(\boldsymbol{x})\big) \cdot \boldsymbol{v}\Big) \cdot \boldsymbol{v} = \sum_{i=1}^{N} \sum_{j=1}^{N} \frac{\partial^2 f}{\partial x_i \partial x_j}(\boldsymbol{x}) v_i v_j.$$

If we set

$$H_f(\boldsymbol{x}) := \begin{bmatrix} \frac{\partial^2}{\partial x_1^2} f(\boldsymbol{x}) & \dots & \frac{\partial^2}{\partial x_1 \partial x_n} f(\boldsymbol{x}) \\ \vdots & \ddots & \vdots \\ \frac{\partial^2}{\partial x_n \partial x_1} f(\boldsymbol{x}) & \dots & \frac{\partial^2}{\partial x_n^2} f(\boldsymbol{x}) \end{bmatrix}, \tag{1.4.4}$$

and

$$\boldsymbol{v} = \begin{bmatrix} v_1 \\ \vdots \\ v_N \end{bmatrix} \qquad \text{and} \qquad \boldsymbol{v}^T = \begin{bmatrix} v_1 & \dots & v_N \end{bmatrix},$$

then, using matrix multiplications, we can rewrite the second directional derivative by

$$\frac{\partial^2}{\partial \boldsymbol{v}^2} f(\boldsymbol{x}) = \boldsymbol{v}^T H_f(\boldsymbol{x}) \boldsymbol{v}.$$

This motivates the following definition.

Definition 1.4.34 The matrix $H_f(\boldsymbol{x})$ given by (1.4.4) is called the *Hessian matrix* of f at $\boldsymbol{x}$.

☞ **Remark 1.4.35** It follows from Proposition 1.4.32 that the Hessian matrix $H_f(\boldsymbol{x})$ is symmetric if grad f is continuous at $\boldsymbol{x}$.

Example 1.4.36 The Hessian matrix of the function in Example 1.4.31 is

$$H_f(x, y) = \begin{bmatrix} 6x - 6y & -6x \\ -6x & 0 \end{bmatrix}.$$

Note that the matrix is symmetric.

Let us summarise what we just found.

Proposition 1.4.37 *Suppose that $\boldsymbol{v}$ is a unit vector, and that f has continuous first and second order partial derivatives at $\boldsymbol{x}$. Then the second directional derivative in the direction of $\boldsymbol{v}$ is given by*

$$\boxed{\frac{\partial^2}{\partial \boldsymbol{v}^2} f(\boldsymbol{x}) = \boldsymbol{v}^T H_f(\boldsymbol{x}) \boldsymbol{v},}$$

where $H_f(\boldsymbol{x})$ is the Hessian matrix of f at $\boldsymbol{x}$ defined by (1.4.4).

In principle we could continue to apply (1.4.3) to compute the third, fourth and higher directional derivatives. However, for later purposes we only need the second derivative. We next want to use what we learnt to find 'Taylor polynomials' for functions of several variables.

Taylor Polynomials

If g is a function of one variable then the Taylor polynomial of order n about a is defined by

$$T_n(a + h) = g(a) + g'(a)h + \frac{g''(a)}{2!}h^2 + \cdots + \frac{g^{(n)}(a)}{n!}h^n.$$

As you know from first year calculus T_n is the best approximation of f near a by an n-th order polynomial. From the above formula we now want to derive the Taylor polynomials of order one and two for a function, f, defined on $D \subset \mathbb{R}^N$. Given an interior point of $\boldsymbol{a} \in D$ and $h \in \mathbb{R}^N$ we consider the function

$$g_{\boldsymbol{h}}(t) := f(\boldsymbol{a} + t\boldsymbol{h}),$$

and write down its Taylor polynomial of order 2 about $t = 0$:

$$g_{\boldsymbol{h}}(0) + g'_{\boldsymbol{h}}(0)t + \frac{g''_{\boldsymbol{h}}(0)}{2!}t^2.$$

Applying the chain rule (see Theorem 1.4.15 on page40) we get

$$g'_{\boldsymbol{h}}(0) = \big(\operatorname{grad} f(\boldsymbol{a})\big) \cdot \boldsymbol{h}.$$

If $\boldsymbol{h} \neq 0$ then $\boldsymbol{v} := \boldsymbol{h}/\|\boldsymbol{h}\|$ is a unit vector, and by the formula (1.4.3) the above becomes

$$g'_{\boldsymbol{h}}(0) = \|h\|\big(\operatorname{grad} f(\boldsymbol{a})\big) \cdot \boldsymbol{v} = \|h\|\frac{\partial f}{\partial \boldsymbol{v}}(\boldsymbol{a}).$$

Applying the same arguments to $g'_{\boldsymbol{h}}(t)$ we get

$$g''_{\boldsymbol{h}}(0) = \|h\|^2\frac{\partial^2 f}{\partial \boldsymbol{v}^2}(\boldsymbol{a}).$$

Using the formula for the second directional derivative from Proposition 1.4.37 and the definition of $\boldsymbol{v}$ we get

$$g''_{\boldsymbol{h}}(0) = \|h\|^2\boldsymbol{v}^T H_f(\boldsymbol{a})\boldsymbol{v} = \|h\|^2\frac{\boldsymbol{h}^T}{\|h\|}H_f(\boldsymbol{a})\frac{\boldsymbol{h}}{\|h\|} = \boldsymbol{h}^T H_f(\boldsymbol{a})\boldsymbol{h},$$

where $H_f(\boldsymbol{a})$ is the Hessian matrix as defined in Definition 1.4.34. The above motivates the following definition.

Definition 1.4.38 The functions

$$\begin{aligned} T_1(\boldsymbol{a}+\boldsymbol{h}) &:= f(\boldsymbol{a}) + \big(\operatorname{grad} f(\boldsymbol{a})\big) \cdot \boldsymbol{h} \\ T_2(\boldsymbol{a}+\boldsymbol{h}) &:= f(\boldsymbol{a}) + \big(\operatorname{grad} f(\boldsymbol{a})\big) \cdot \boldsymbol{h} + \frac{1}{2}\boldsymbol{h}^T H_f(\boldsymbol{a})\boldsymbol{h} \end{aligned}$$

are called the *Taylor polynomial of f of order one and two*, respectively, centred at $\boldsymbol{a}$.

By construction T_1 and T_2 are the Taylor polynomials of $g_{\boldsymbol{h}}$ at $t = 1$, so we can apply Taylor's theorem to $g_{\boldsymbol{h}}$ at $t = 1$. Doing this, with some effort involved in dealing with the remainder term, we get the following version of Taylor's Theorem for a function of several variables. For a rigorous proof we refer to Apostol [2, pages 308-310].

Theorem 1.4.39 (Taylor's Theorem) *Suppose that all first and second order partial derivatives of f are continuous at $\boldsymbol{a}$. Then (for $n = 1, 2$)*

$$f(\boldsymbol{a}+\boldsymbol{h}) = T_n(\boldsymbol{h}) + R_n(\boldsymbol{h}),$$

where the remainder term $R_n(\boldsymbol{h})$ has the property that

$$\lim_{\boldsymbol{h}\to\boldsymbol{0}} \frac{R_n(\boldsymbol{h})}{\|\boldsymbol{h}\|^n} = 0,$$

where $n = 1, 2$ as before.

We conclude this section by giving one example.

Example 1.4.40 Suppose that $f(x, y) := x^3 - 3x^2y$ for all $(x, y) \in \mathbb{R}^2$ is as in Example 1.4.31. Then we saw that

$$\operatorname{grad} f(x, y) = (3x^2 - 6xy, -3x^2) \qquad \text{and} \qquad H_f(x, y) = \begin{bmatrix} 6x - 6y & -6x \\ -6x & 0 \end{bmatrix}.$$

Hence the Taylor polynomial of order one centred at $\boldsymbol{a} = (1, 1)$ is

$$T_1(1 + h_1, 1 + h_2) = -2 + (-3, -3) \cdot (h_1, h_2) = -2 - 3(h_1 + h_2)$$

for all $(h_1, h_2) \in \mathbb{R}^2$. The Taylor polynomial of order two centred at $\boldsymbol{a} = (1, 1)$ is

$$\begin{aligned} T_2(1 + h_1, 1 + h_2) = T_1(1 + h_1, 1 + h_2) + \frac{1}{2} \begin{bmatrix} h_1 & h_2 \end{bmatrix} \begin{bmatrix} 0 & -6 \\ -6 & 0 \end{bmatrix} \begin{bmatrix} h_1 \\ h_2 \end{bmatrix} \\ = -2 - 3(h_1 + h_2) - 6h_1h_2. \end{aligned}$$

1.5 Maxima and Minima

If g is a differentiable function of one variable defined on an open interval, then we know that $g'(t) = 0$ at every point t where it attains a (local) maximum or minimum. To find the local maxima and minima we therefore solve the equation $g'(t) = 0$ to find the possible candidates for maxima and minima. We then need criteria which guarantee that g attains in fact a maximum or minimum at such a point. One possible such criterion is that $g''(t) < 0$ for a maximum, and $g''(t) > 0$ for a minimum.

1.5.1 Critical Points: The First Derivative Test

Using the above facts we want to derive conditions for maxima and minima of functions of several variables. To do so assume that f is defined on a domain D in $\mathbb{R}^N$. Suppose that f attains a (local) maximum (or minimum) at some point $\boldsymbol{a} = (a_1, \dots, a_N)$. For every $i = 1, \dots, N$ we can then define a function of one variable, t, by setting

$$g_i(t) = f(a_1, \dots, a_{i-1}, a_i + t, a_{i+1}, \dots, a_N).$$

The graph of this function is a cross-section of the graph of f in the i-th coordinate direction as seen in Figure 1.4.1 for a function of two variables. If f attains a maximum (or minimum) at $\boldsymbol{a}$ then g_i attains a (local) maximum (or minimum) at $t = 0$ for all

$i = 1, \dots, N$. If f has partial derivatives in all directions then by the criterion for single variable functions and the definition of the partial derivatives we have

$$0 = g_i'(0) = \frac{\partial}{\partial x_i} f(\boldsymbol{a})$$

for all $i = 1, \dots, N$. We therefore have the following theorem.

Theorem 1.5.1 *Suppose that f is a function defined on a domain D in $\mathbb{R}^N$ attaining a maximum or minimum at the interior point $\boldsymbol{a}$ of D. Then*

$$\boxed{\operatorname{grad} f(\boldsymbol{a}) = \mathbf{0}} \tag{1.5.1}$$

if all partial derivatives exist.

Definition 1.5.2 (Critical point) A point $\boldsymbol{a}$ in the domain of f is called a *critical point of f* if $\operatorname{grad} f(\boldsymbol{a}) = \mathbf{0}$.

Hence, all local maxima are among the critical point of a function, but as in case of functions of one variable, not all critical points are maxima or minima. As mentioned at the start of the section a function g of one variable attains a local maximum in an open interval if $g'(t) = 0$ and $g''(t) < 0$. Next we look at in more detail what happens in several dimensions.

1.5.2 The Hessian Matrix: The Second Derivative Test

As with the first derivative test for maxima and minima we can look at cross sections of the graph of the function f in the coordinate direction. If $\operatorname{grad} f(\boldsymbol{a}) = 0$, then

$$\frac{\partial^2}{\partial x_i^2} f(\boldsymbol{a}) < 0.$$

implies that the cross sections in the coordinate directions have a maximum at $\boldsymbol{a}$. The question is whether this *guarantees* a maximum at $f(\boldsymbol{a})$. Unfortunately, the answer is NO! The reason is that we only look at cross-sections of the graph in the coordinate directions. The condition only tells us that the function attains a maximum along these cross-section. In other directions it may be decreasing. As an example look at the function $f(x, y) = 4x^2y^2 - x^4 - y^4$, whose graph is displayed in Figure 1.5.1. Along the cross-sections parallel to the axes through $(0, 0)$ the function attains a maximum, but along the cross-sections in the diagonal directions it attains a minimum. You can easily verify that formally. Hence we have to make sure that we have a maximum along *every*

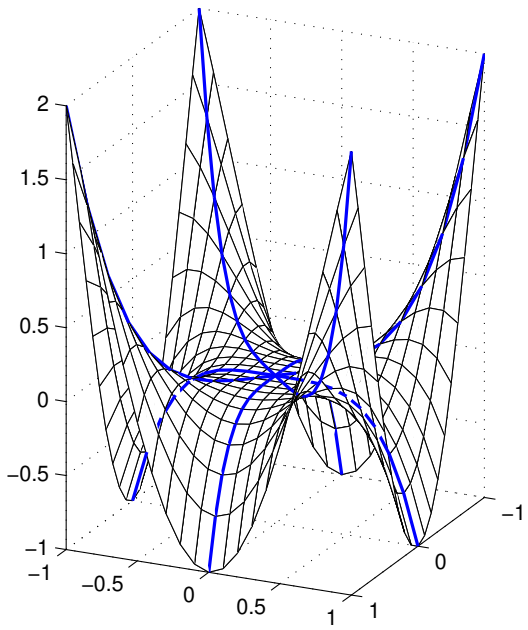


Figure 1.5.1: Not a maximum or a minimum at $(0, 0)$.

cross-section. This means that *all* first directional derivatives must be zero, and *all* second directional derivatives of f at $\boldsymbol{a}$ must be negative in order to ensure a maximum. Hence let $\boldsymbol{v} = (v_1, \dots, v_n)$ be an arbitrary unit vector, and define

$$g_{\boldsymbol{v}}(t) := f(\boldsymbol{a} + t\boldsymbol{v}).$$

Assume that all first and second order partial derivatives of f are continuous at $\boldsymbol{a}$. According to the previous reasoning f attains a maximum at $\boldsymbol{a}$ if $g'_{\boldsymbol{v}}(0) = 0$ and $g''_{\boldsymbol{v}}(0) < 0$ for *every* unit vector $\boldsymbol{v}$. By Proposition 1.4.24 we have

$$g'_{\boldsymbol{v}}(0) = \big(\operatorname{grad} f(\boldsymbol{a})\big) \cdot \boldsymbol{v}.$$

Hence $g'_{\boldsymbol{v}}(0) = 0$ if and only if $\operatorname{grad} f(\boldsymbol{a}) = \boldsymbol{0}$. Now let us look at the second order derivative. By Proposition 1.4.37 we know that

$$g''_{\boldsymbol{v}}(0) = \frac{\partial^2 f}{\partial \boldsymbol{v}^2}(\boldsymbol{a}) = \boldsymbol{v}^T H_f(\boldsymbol{a})\boldsymbol{v},$$

where $H_f(\boldsymbol{a})$ is the Hessian matrix of f at $\boldsymbol{a}$ defined by (1.4.4). Hence, if

$$\boldsymbol{v}^T H_f(\boldsymbol{a})\boldsymbol{v} < 0$$

for all unit vectors $\boldsymbol{v}$, then f attains a maximum at $\boldsymbol{a}$. The last conclusion is not obvious, an additional argument is required. Before we formulate a theorem we introduce some definitions from linear algebra.

Definition 1.5.3 (Positive definite) A symmetric $N \times N$-matrix A is called *positive definite* if there exists a constant $c > 0$ such that $\boldsymbol{v}^T A\boldsymbol{v} \geq c\|\boldsymbol{v}\|^2$ for all $\boldsymbol{v} \in \mathbb{R}^N$. The matrix A is called *negative definite* if $-A$ is positive definite. If $\boldsymbol{v}^T A\boldsymbol{v} \geq 0$ (or ≤ 0) for all $\boldsymbol{v} \in \mathbb{R}^N$ then A is called *positive* (or *negative*) *semi-definite*. If A is none of the above then A is called *indefinite*.

With this terminology we can formulate our main result about maxima and minima.

Theorem 1.5.4 *Suppose that all first and second order partial derivatives of f are continuous at the interior point $\boldsymbol{a}$ of the domain of f. If* $\operatorname{grad} f(\boldsymbol{a}) = 0$ *then*

(i) *f attains a maximum at $\boldsymbol{a}$ if $H_f(\boldsymbol{a})$ is negative definite;*

(ii) *f attains a minimum at $\boldsymbol{a}$ if $H_f(\boldsymbol{a})$ is positive definite;*

(iii) *f has a saddle point at $\boldsymbol{a}$ if $H_f(\boldsymbol{a})$ is indefinite.*

Proof. We can give a proof of the above using Taylor's Theorem 1.4.39. If $H_f(\boldsymbol{a})$ is positive definite, then there exists $c > 0$ such that

$$\begin{aligned} f(\boldsymbol{a}+\boldsymbol{h}) - f(\boldsymbol{a}) &= \operatorname{grad} f(\boldsymbol{a}) + \boldsymbol{h}^T H_f(\boldsymbol{a})\boldsymbol{h} + R_2(\boldsymbol{h}) \\ &= \boldsymbol{h}^T H_f(\boldsymbol{a})\boldsymbol{h} + R_2(\boldsymbol{h}) \geq c\|\boldsymbol{h}\|^2 + R_2(\boldsymbol{h}) \end{aligned}$$

for $\|\boldsymbol{h}\|$ small enough. Dividing by $\|\boldsymbol{h}\|^2$ we get

$$\frac{1}{\|\boldsymbol{h}\|^2}\Big(f(\boldsymbol{a}+\boldsymbol{h}) - f(\boldsymbol{a})\Big) \geq c + \frac{R_2(\boldsymbol{h})}{\|\boldsymbol{h}\|^2}.$$

By Taylor's Theorem 1.4.39 $R_2(\boldsymbol{h})/\|\boldsymbol{h}\|^2 \to 0$ as $\|h\| \to 0$. Since $c > 0$ we get

$$\frac{1}{\|\boldsymbol{h}\|^2}\Big(f(\boldsymbol{a}+\boldsymbol{h}) - f(\boldsymbol{a})\Big) \geq c/2 > 0$$

for $\|h\|$ small enough. Hence, if $\operatorname{grad} f(\boldsymbol{a}) = 0$ and $H_f(\boldsymbol{a})$ is positive definite, then f has a local minimum at $\boldsymbol{a}$. This proves (ii). Now (i) follows from (ii) by applying it to $-f$.

If $H_f(\boldsymbol{a})$ is indefinite there exist unit vectors $\boldsymbol{v}, \boldsymbol{w}$ such that $g''_{\boldsymbol{v}} = \boldsymbol{v}^T H_f(\boldsymbol{a})\boldsymbol{v} < 0$, and $g''_{\boldsymbol{w}} = \boldsymbol{w}^T H_f(\boldsymbol{a})\boldsymbol{w} > 0$. This means that along some cross-section through the graph of f we have a maximum, and in some we have a minimum. ■

Remark 1.5.5 A point such that (iii) in the above theorem is satisfied is usually called a *saddle point*. Examples are shown in Figure 1.2.5 or Figure 1.5.1

To decide whether a matrix is positive definite we need to use some linear algebra. Recall that every symmetric $N \times N$ matrix A is diagonalisable by means of an orthogonal matrix. An orthogonal matrix is an invertible matrix P such that $P^{-1} = P^T$, that is, $P^T P = PP^T = I$. You learn in linear algebra that for every symmetric matrix A there exists an orthogonal matrix P such that

$$P^T AP = D := \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & \dots & 0 & \lambda_N \end{bmatrix}$$

is diagonal and $\lambda_1, \dots, \lambda_N$ are the eigenvalues of A. By definition of the Euclidean norm we have

$$\|P\boldsymbol{x}\|^2 = (P\boldsymbol{x})^T P\boldsymbol{x} = \boldsymbol{x}^T P^T P\boldsymbol{x} = \boldsymbol{x}^T \boldsymbol{x} = \|\boldsymbol{x}\|^2.$$

Similarly we get $\|P^T\boldsymbol{x}\|^2 = \|\boldsymbol{x}^2\|$. If $\boldsymbol{y} = (y_1, \dots, y_N)$, then a simple calculation shows that

$$\boldsymbol{y}^T D\boldsymbol{y} = \sum_{k=1}^{N} \lambda_k y_k^2$$

and hence

$$\boldsymbol{y}^T D\boldsymbol{y} \geq \lambda_{\min}\|\boldsymbol{y}\|^2, \qquad \boldsymbol{y}^T D\boldsymbol{y} \leq \lambda_{\max}\|\boldsymbol{y}\|^2,$$

where

$$\lambda_{\min} := \min\{\lambda_1, \dots, \lambda_N\} \quad \text{and} \quad \lambda_{\max} := \max\{\lambda_1, \dots, \lambda_N\}.$$

Combining all the above the above we get

$$\begin{aligned} \boldsymbol{x}^T A\boldsymbol{x} &= \boldsymbol{x}^T PDP^T\boldsymbol{x} \geq \lambda_{\min}\|P^T\boldsymbol{x}\|^2 = \lambda_{\min}\|\boldsymbol{x}\|^2 \\ \boldsymbol{x}^T A\boldsymbol{x}, &= \boldsymbol{x}^T PDP^T\boldsymbol{x} \geq \lambda_{\max}\|P^T\boldsymbol{x}\|^2 = \lambda_{\max}\|\boldsymbol{x}\|^2 \end{aligned}$$

for all $\boldsymbol{x} \in \mathbb{R}^N$. Note that there is equality if $\boldsymbol{x}$ happens to be an eigenvector corresponding to $\lambda_{\max}$ and $\lambda_{\min}$, respectively. This leads to the following criterion for positive and negative matrices.

Proposition 1.5.6 *Suppose that A is a symmetric $N \times N$ matrix with smallest and largest eigenvalues $\lambda_{\min}$ and $\lambda_{\max}$. Then*

(i) *A is positive definite if and only if $\lambda_{\min} > 0$;*

(ii) *A is negative definite if and only if $\lambda_{\max} < 0$;*

(iii) *A is indefinite if and only if $\lambda_{\min} < 0$ and $\lambda_{\max} > 0$.*

From the above we can get a simple criterion in case of $N = 2$.

Corollary 1.5.7 *Suppose that* $A = \begin{bmatrix} a & c \\ c & b \end{bmatrix}$ *is a symmetric matrix. Then*

- *A is positive definite if and only if $a > 0$ and* $\det A > 0$*;*
- *A is negative definite if and only if $a < 0$ and* $\det A > 0$*;*
- *A is indefinite if* $\det A < 0$.

Proof. To compute the eigenvalues we determine the characteristic polynomial of A given by

$$\det(A - \lambda I) = \det \begin{bmatrix} a - \lambda & c \\ c & b - \lambda \end{bmatrix} = (a - \lambda)(b - \lambda) - c^2$$
$$= \lambda^2 - (a + b)\lambda + ab - c^2 = \lambda^2 - (a + b)\lambda + \det A.$$

Hence the eigenvalues of A are

$$\lambda_1 = \frac{1}{2}\Big(a + b + \sqrt{(a + b)^2 - 4 \det A}\Big)$$
$$\lambda_2 = \frac{1}{2}\Big(a + b - \sqrt{(a + b)^2 - 4 \det A}\Big),$$

and therefore,

$$\lambda_1 + \lambda_2 = a + b, \tag{1.5.2}$$
$$\lambda_1 \lambda_2 = \det A. \tag{1.5.3}$$

If $\det A > 0$, then (1.5.3) shows that either both eigenvalues are positive or both are negative. By (1.5.2) they are both positive if $a > 0$ (or $b > 0$) and both negative if $a < 0$. If $\det A < 0$, then (1.5.3) shows that the eigenvalues must have different sign. To finish the proof we use Proposition 1.5.6. ■

Example 1.5.8

1. Let $A = \begin{bmatrix} 9 & -1 \\ -1 & 2 \end{bmatrix}$. Then $a = 9 > 0$ and $\det A = 18 - 1 = 17 > 0$, so A is positive definite.
2. Let $A = \begin{bmatrix} -2 & 3 \\ 3 & 1 \end{bmatrix}$. Then $\det A = -2 - 9 = -11 < 0$, so A is indefinite.

3. Let $A = \begin{bmatrix} -5 & 2 \\ 2 & -1 \end{bmatrix}$. Then $a = -5 < 0$ and $\det A = 5 - 4 = 1 > 0$, so A is negative definite.

With the above we can give a convenient criterion for maxima and minima of functions of two variables.

Theorem 1.5.9 *Suppose that all first and second partial derivatives of f defined on $D \subset \mathbb{R}^2$ are continuous at the interior point $\boldsymbol{a} \in D$, and that* $\operatorname{grad} f(\boldsymbol{a}) = 0$. *Then*

- *f attains a (local) maximum at $\boldsymbol{a}$ if* $\dfrac{\partial^2 f}{\partial x_1^2}(\boldsymbol{a}) < 0$ *and* $\det H_f(\boldsymbol{a}) > 0$*;*
- *f attains a (local) minimum at $\boldsymbol{a}$ if* $\dfrac{\partial^2 f}{\partial x_1^2}(\boldsymbol{a}) > 0$ *and* $\det H_f(\boldsymbol{a}) > 0$*;*
- *f has a saddle point at $\boldsymbol{a}$ if* $\det H_f(\boldsymbol{a}) < 0$*;*
- *the test is inconclusive if* $\det H_f(\boldsymbol{a}) = 0$.

Proof. The first three assertions are a direct consequence of Theorem 1.5.4 and Corollary 1.5.7. The last assertion can be shown by constructing suitable examples: If we set $f(x, y) = \alpha x^4 + \beta y^4$ then $\operatorname{grad} f(x, y) = 4(\alpha x^3, \beta y^3)$, and thus $(0, 0)$ is a critical point. The Hessian matrix is

$$H_f(x, y) = \begin{bmatrix} 12\alpha x^2 & 0 \\ 0 & 12\beta y^2 \end{bmatrix},$$

and so

$$\det H_f(0, 0) = \det \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0.$$

However, it is rather obvious that $(0, 0)$ is a minimum if $\alpha = \beta = 1$, a maximum if $\alpha = \beta = -1$, and a saddle point if $\alpha = 1$ and $\beta = -1$. Sketch the graphs to see this! ■

Example 1.5.10 Determine the critical points of $f(x, y) = x^3 + 3xy^2 - 3x^2 - 3y^2 + 4$, and decide whether f attains a maximum, minimum or a saddle point at each of them.
Solution: To determine the critical points we have to solve the system of equations

$$\frac{\partial f}{\partial x}(x, y) = 3x^2 + 3y^2 - 6x = 0,$$
$$\frac{\partial f}{\partial y}(x, y) = 6xy - 6y = 0.$$

From the second equation we get that $y = 0$ or $x = 1$. If $y = 0$ then from the first equation $x(3x-6) = 0$ and so $x = 0$ or $x = 2$. Hence $(0,0)$ and $(2,0)$ are critical points. If $x = 1$ then from the first equation $3 + 3y^2 - 6 = 0$, and so $y = \pm 1$. Hence the critical points of f are

$$(0,0) \qquad (2,0), \qquad (1,1) \qquad \text{and} \qquad (1,-1).$$

To decide the behaviour of f at these points we determine its Hessian matrix:

$$\begin{aligned}
\frac{\partial^2 f}{\partial x^2}(x,y) &= \frac{\partial}{\partial x}(3x^2 + 3y^2 - 6x) = 6x - 6, \\
\frac{\partial^2 f}{\partial y^2}(x,y) &= \frac{\partial}{\partial y}(6xy - 6y) = 6x - 6, \\
\frac{\partial^2 f}{\partial y \partial x}(x,y) = \frac{\partial^2 f}{\partial x \partial y}(x,y) &= \frac{\partial}{\partial x}(6xy - 6y) = 6y.
\end{aligned}$$

Hence the Hessian matrix is

$$H_f(x,y) = 6 \begin{bmatrix} x-1 & y \\ y & x-1 \end{bmatrix}.$$

We next look at $H_f(x,y)$ at every critical point:

- We have $H_f(0,0) = 6 \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$, and so $-1 < 0$ and $\det H_f(0,0) = 1$, showing that f attains a maximum at $(0,0)$.
- Next we have $H_f(2,0) = 6 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, and so $1 > 0$ and $\det H_f(0,0) = 1$, showing that f attains a minimum at $(2,0)$.
- Next we have $H_f(1,1) = 6 \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, and so $\det H_f(0,0) = -1 < 0$, showing that f has a saddle point at $(1,1)$.
- Finally we have $H_f(1,-1) = 6 \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$, and so $\det H_f(0,0) = -1 < 0$, showing that f has another saddle point at $(1,-1)$.

Chapter 2

Multiple Integrals

The purpose of this chapter is to outline the theory of integrals of functions of more than one variable, called multiple integrals. We discuss the theory for functions in two variables in some detail. The only difference to more variables is that it is possible to visualise what we are doing. We do not give rigorous proofs but explain in quite some detail the ideas behind the theory. Rigorous proofs can for instance be found in Apostol [2].

2.1 Double Integrals

We first define double integrals, and then show how to compute them over certain domains. The ideas for defining double integrals are very similar to the ordinary integrals you know. To prepare for this section look at the chapter on Riemann sums from first year calculus.

2.1.1 Definition of Double Integrals

Before we introduce double integrals let us recall how the integral of a function of one variable, say t, was defined. Geometrically, the definite integral of a function f over an interval $I = [a, b]$ represents the area of the region between the graph and the t-axis. To find an approximation of this area the idea is to use a partition of I given by $a = t_0 < t_1 < t_2 < \cdots < t_n = b$. Then we choose a point t_i^* in every interval $[t_{i-1}, t_i]$. Now we look at the rectangle of width $\Delta t_i := t_i - t_{i-1}$ and height $f(t_i^*)$. Such a rectangle is highlighted in Figure 2.1.1. The area of this rectangle is $f(t_i^*)\Delta t_i$. Hence the total area

of all rectangles as shown in Figure 2.1.1 is

$$\sum_{i=1}^{n} f(t_i^*)\Delta t_i. \tag{2.1.1}$$

This total area is an approximation for the area under the graph lightly shaded in Figure 2.1.1. The above sum is called a *Riemann sum* of f over I. Taking a finer partition, that is, max Δt_i smaller, we expect that the corresponding Riemann sum provides a better approximation. Indeed, for continuous functions the limit

$$\int_a^b f(t)dt := \lim_{\max \Delta t_i \to 0} \sum_{i=1}^{n} f(t_i^*)\Delta t_i$$

exists, and is independent of the intermediate points t_i^* chosen. The limit is called the *Riemann integral* or simply the *integral* of f over the interval $[a, b]$. If $f(t_i^*)$ is negative then the area is considered negative.

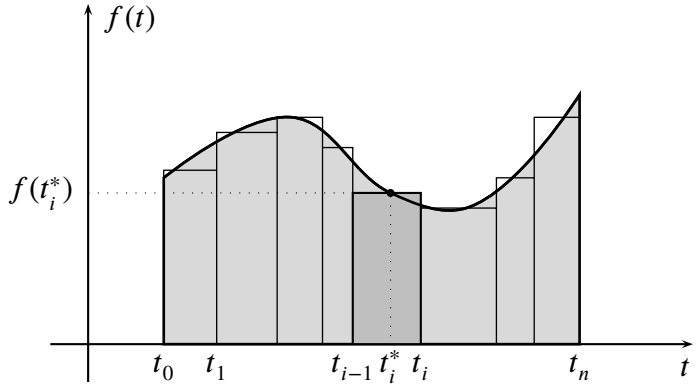
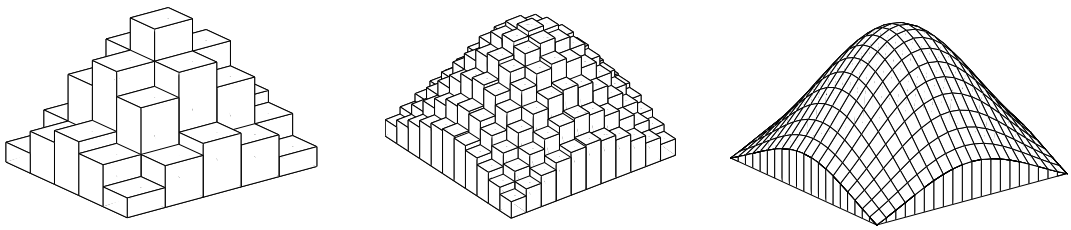


Figure 2.1.1: Approximation of the area under the graph of a function.

The idea to define integrals of functions of two variables (double integrals) is very similar to the above. The area under the graph of a function of one variable corresponds to the volume of the region between the graph of a function of two variables and the xy-plane. To get an approximation of that volume we replace the rectangles by rectangular boxes, and then pass to the limit by taking smaller and smaller rectangular boxes. Two such approximations are shown in Figure 2.1.2, together with the limiting case. To be more precise let us look at a function, f, defined on a rectangle $[a, b] \times [c, d]$ of the plane. By choosing partitions $a = x_0 < x_1 < \cdots < x_m = b$ and $c = y_0 < y_2 < \cdots < y_n = d$ of the intervals $[a, b]$ and $[c, d]$ we get a subdivision of the rectangle $[a, b] \times [c, d]$ into small rectangles $[x_{i-1}, x_i] \times [y_{j-1}, y_j]$, one of which is highlighted in Figure 2.1.3. The lengths of the edges of such a rectangle is Δx_i and Δy_j, respectively. Now we choose an arbitrary point $\boldsymbol{x}_{ij}^* := (x_i^*, y_j^*)$ in that rectangle similarly as done in the one dimensional



Rough approximation Finer approximation Limit

Figure 2.1.2: Approximation of the volume under the graph by rectangular boxes

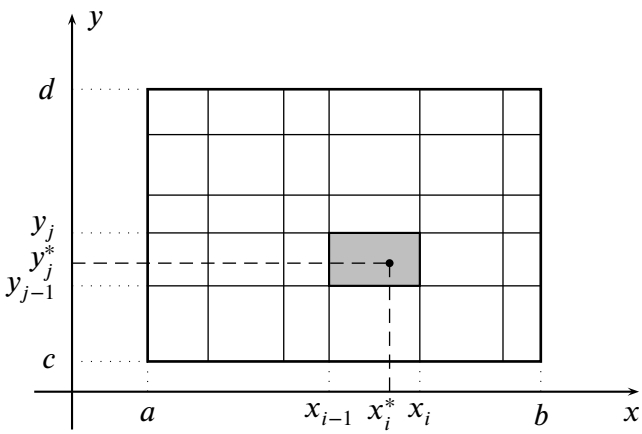


Figure 2.1.3: Partition of a rectangle

case with t_i^*. Then we look at the rectangular box with basis $[x_{i-1}, x_i] \times [y_{j-1}, y_j]$ and height $f(x_i^*, y_j^*)$. Such a box is shown in Figure 2.1.4. Its volume is the product of the lengths of its edges, namely $f(x_i^*, y_j^*)\Delta x_i \Delta y_j$. If $f(x_i^*, y_j^*)$ is negative then the volume is considered negative, as in the case of a function of one variable. Hence the total volume of all boxes is

$$\sum_{i=1}^{m} \sum_{j=1}^{n} f(x_i^*, y_j^*)\Delta x_i \Delta y_j. \tag{2.1.2}$$

This sum corresponds to the Riemann sums we looked at before, and is also called a Riemann sum. It turns out that, if f is continuous on the closed rectangle, then one can pass to the limit as $\max \Delta x_i \to 0$ and $\max \Delta y_j \to 0$. This leads to the following definition.

Definition 2.1.1 Suppose that f is continuous on the rectangle $R = [a, b] \times [c, d]$. Then

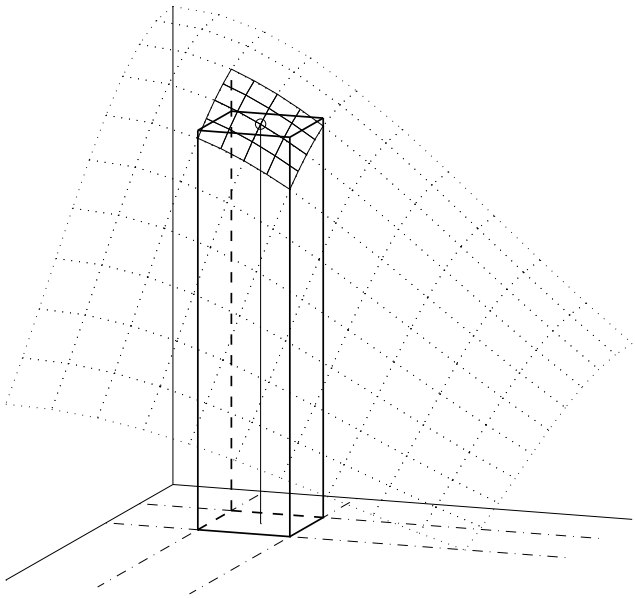


Figure 2.1.4: The graph of a function and the box over a small rectangle.

we call

$$\int_R f(\boldsymbol{x})\,d\boldsymbol{x} := \iint_R f(x,y)\,dxdy := \lim_{\substack{\max \Delta x_i \to 0 \\ \max \Delta y_j \to 0}} \sum_{i=1}^{m} \sum_{j=1}^{n} f(x_i^*, y_j^*)\Delta x_i \Delta y_j$$

the *double integral* or simply the *integral of f over R*.

Remark 2.1.2 The notation for integrals used in the literature varies. In older or more applied texts the notation

$$\iint_R f(x,y)\,dxdy$$

is used for double integrals. In the more modern literature one simply writes

$$\int_R f(\boldsymbol{x})\,d\boldsymbol{x}.$$

In this course we will mainly use the second, more concise form.

We would like to integrate over more general regions than just rectangles. We can certainly integrate over a union of finitely many rectangles. If we want to integrate over more general sets then we need to approximate the sets by a union of small rectangles as well. To get upper and lower bounds we approximate from the inside and the outside as shown in Figure 2.1.5. We then form a sum similar to (2.1.2) over the relevant rectangles. If the set is reasonably 'nice,' that is, its boundary is not too wild, then the limits of

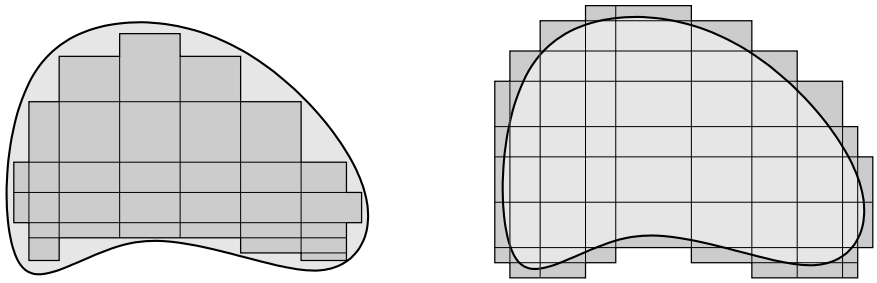


Figure 2.1.5: Approximation of a domain from the in- and outside by rectangles.

the Riemann sums from the inside and the outside are the same, and we can define the double integral over such regions. It is quite difficult to characterise domains for which this is possible, and we omit such a discussion. All domains we encounter in this course have the required properties.
We next collect some properties of the integral. They follow directly from the definition as limits of Riemann sums.

Theorem 2.1.3 *Suppose that $D, D_1, D_2 \subset \mathbb{R}^2$ are domains on which the integral of every continuous function is well defined. If f, g are continuous functions on D and $\alpha \in \mathbb{R}$ then*

- $\displaystyle\int_D (f(\boldsymbol{x}) + g(\boldsymbol{x}))\, d\boldsymbol{x} = \int_D f(\boldsymbol{x})\, d\boldsymbol{x} + \int_D g(\boldsymbol{x})\, d\boldsymbol{x};$
- $\displaystyle\int_D \alpha f(\boldsymbol{x})\, d\boldsymbol{x} = \alpha \int_D f(\boldsymbol{x})\, d\boldsymbol{x};$
- *If $D = D_1 \cup D_2$ and $D_1 \cap D_2 = \emptyset$ then*

$$\int_D f(\boldsymbol{x})\, d\boldsymbol{x} = \int_{D_1} f(\boldsymbol{x})\, d\boldsymbol{x} + \int_{D_2} f(\boldsymbol{x})\, d\boldsymbol{x}.$$

☞ **Remark 2.1.4** We introduced the double integral as the volume of the region between the graph of a function and the xy-plane. Note, however, that this is not the only possible interpretation. If we choose f to be the constant function with value one, that is, $f(x, y) = 1$, then the integral represents the area of the domain:

$$\text{Area of the domain } D = \int_D 1\, d\boldsymbol{x}. \tag{2.1.3}$$

Another physical interpretation is the following. Let D represent the area occupied by a thin plate made of some material. At every point (x, y) on that plate let $\varrho(x, y)$ denote the mass density of the material. Then the

$$\boxed{\text{Total mass of the plate occupying } D = \int_D \varrho(\boldsymbol{x})\, d\boldsymbol{x}.}$$

The theory developed in this section does not allow us to compute the double integral of a given function. We will see how to do this in the next section.

2.1.2 Evaluation of Double Integrals

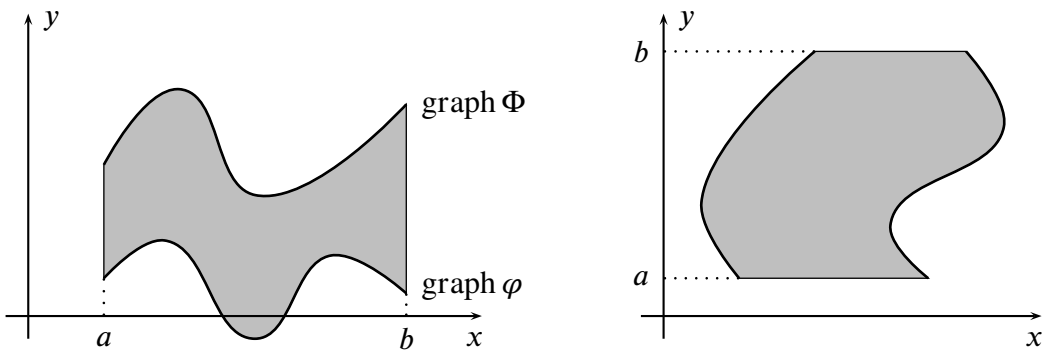
In this section we will learn how to compute a double integral over some domains. The idea is to reduce the computation to the evaluation of two integrals over functions of one variable only. We consider special domains of the form

$$D_1 = \left\{(x, y) \in \mathbb{R}^2 : \ a \le x \le b, \varphi(x) \le y \le \Phi(x)\right\}, \tag{2.1.4}$$

or the same situation with x and y switched

$$D_2 = \left\{(x, y) \in \mathbb{R}^2 : \ a \le y \le b, \varphi(y) \le x \le \Phi(y)\right\}, \tag{2.1.5}$$

where φ and Φ are continuous functions on the interval $[a, b]$, and $\varphi(x) \le \Phi(x)$ for all $x \in [a, b]$. The domain D is the region between the graphs of φ and Φ. Both possibilities are shown in Figure 2.1.6 Assume now that we have a domain of the type D_1. Recall



A domain of the form D_1. A domain of the form D_2.

Figure 2.1.6: The two types of special domains.

that, the double integral of a function f can be interpreted as the volume of the region under its graph. To get an approximation of that volume we can 'cut' the region parallel

to the y-axis into thin slices, and then add the volumes of the slices up. More precisely, we choose a partition $a = x_0 < x_1 < \cdots < x_n = b$. We then look at the slice between x_{i-1} and x_i as shown in Figure 2.1.7. Its thickness is $\Delta x_i := x_i - x_{i-1}$. We next choose

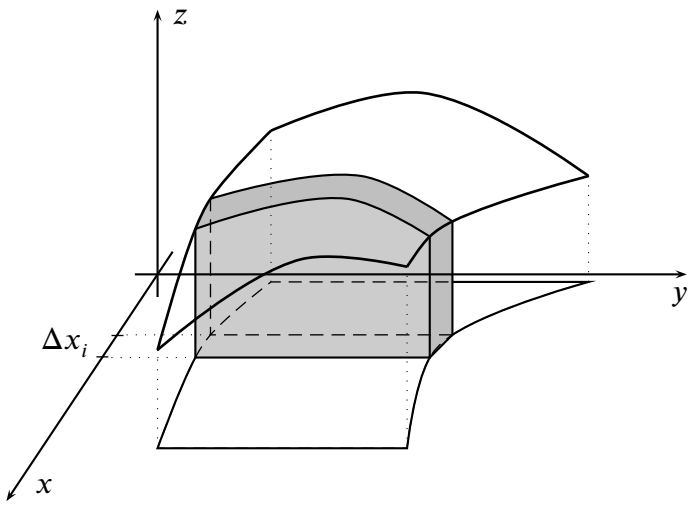


Figure 2.1.7: Cutting a domain into thin slices

$x_i^* \in [x_{i-1}, x_i]$. The approximate surface area of one side of the slice is then

$$F(x_i^*) := \int_{\varphi(x_i^*)}^{\Phi(x_i^*)} f(x_i^*, y)\, dy.$$

Hence the approximate volume of one of the slices is $F(x_i^*)\Delta x_i$. Adding up we conclude that

$$\sum_{i=1}^{n} F(x_i^*)\Delta x_i$$

is an approximation of the volume under the graph of f. The last expression is a Riemann sum for F. Passing to the limit as $\max \Delta x_i \to 0$ the above suggests that

$$\iint_{D_1} f(x, y)\, dx\, dy = \int_a^b \Big(\int_{\varphi(x)}^{\Phi(x)} f(x, y)\, dy \Big)\, dx.$$

The integral

$$\int_{\varphi(x)}^{\Phi(x)} f(x, y)\, dy$$

is usually called the *inner integral*, the integral over $[a, b]$ the *outer integral*, and the full expression an *iterated integral*. The above is not a rigorous proof, but strong evidence that a formula like the above holds. One can show that the formula is true, at least if f, φ, Φ are all continuous. Similar arguments can be applied with x and y interchanged, leading to a formula for domains of the form D_2. We state the result as a theorem.

Theorem 2.1.5 (Fubini's Theorem) *Suppose that φ, Φ are continuous functions on the interval $[a, b]$, and that f is continuous on a domain of the form D_1 or D_2 as defined in* (2.1.4) *and* (2.1.5)*, respectively. Then*

$$\boxed{\iint_{D_1} f(x, y)\, dx\, dy = \int_a^b \Big(\int_{\varphi(x)}^{\Phi(x)} f(x, y)\, dy \Big)\, dx}$$

and

$$\boxed{\iint_{D_2} f(x, y)\, dx\, dy = \int_a^b \Big(\int_{\varphi(y)}^{\Phi(y)} f(x, y)\, dx \Big)\, dy.}$$

Example 2.1.6 Let D be the domain below $y = 2x$ and above $y = x^2$. Write $\int_D f(\boldsymbol{x})\, d\boldsymbol{x}$ as an iterated integral.

Solution: We first determine where $2x \geq x^2$. To do so we solve the equation $2x = x^2$. One solution certainly is $x = 0$. If $x \neq 0$ we can divide by x and get $2 = x$, providing another solution. Hence we have $2x \geq x^2$ for $0 \leq x \leq 2$. Make a sketch of the domain to visualise the situation. We therefore have $a = 0$, $b = 2$, $\varphi(x) = x^2$ and $\Phi(x) = 2x$. Hence, by Fubini's Theorem

$$\int_D f(\boldsymbol{x})\, d\boldsymbol{x} = \int_0^2 \Big(\int_{x^2}^{2x} f(x, y)\, dy \Big)\, dx,$$

which is an iterated integral as required.

Example 2.1.7 Compute $\iint_D x^2 y\, dx\, dy$, where D is the domain from the previous example.

Solution: By the result from the previous example we have

$$\iint_D f(x, y)\, dx\, dy = \int_0^2 \Big(\int_{x^2}^{2x} x^2 y\, dy \Big)\, dx.$$

We first evaluate the inner integral:

$$\int_{x^2}^{2x} x^2 y\, dy = x^2 \int_{x^2}^{2x} y\, dy = x^2 \frac{y^2}{2}\Big|_{x^2}^{2x} = \frac{x^2}{2}\big((2x)^2 - (x^2)^2\big) = 2x^4 - \frac{x^6}{2}.$$

Then we evaluate the outer integral:

$$\iint_D f(x,y)\,dx\,dy = \int_0^2 \Big(\int_{x^2}^{2x} x^2 y\,dy\Big)\,dx = \int_0^2 2x^4 - \frac{x^6}{2}\,dx$$
$$= \frac{2x^5}{5}\Big|_0^2 - \frac{x^7}{2\cdot 7}\Big|_0^2 = \frac{2^6}{5} - 0 - \frac{2^7}{2\cdot 7} + 0 = \frac{7\cdot 2^6 - 5\cdot 2^6}{35} = \frac{2\cdot 2^6}{35} = \frac{2^7}{35},$$

which is the required result.

2.1.3 The Transformation Formula

In this section we want to derive a formula generalising the substitution formula

$$\int_a^b f(g(s))g'(s)\,ds = \int_{g(a)}^{g(b)} f(t)\,dt \tag{2.1.6}$$

you know from first year calculus. In the context of multiple integral such a formula is usually called the *transformation formula* or the *area formula* for multiple integrals. Make sure you understand the ideas presented in this section. A similar procedure can be applied to triple integrals. More importantly the ideas are essential when defining and discussing surface integrals!

The function g in the above formula transforms the interval $[a, b]$ into another interval, hence the different limits. When looking at double integrals this corresponds to a deformation of the domain D, over which we integrate our function, as shown in Figure 2.1.8. The function deforming one domain into the other is a vector valued function, taking a

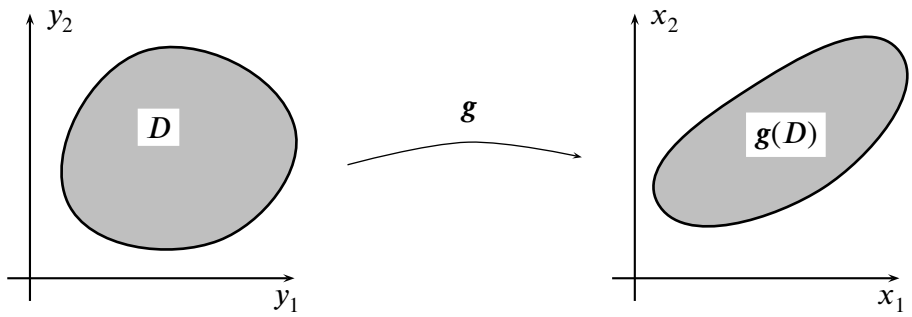


Figure 2.1.8: Deformation of a domain into another domain

point (y_1, y_2) to the point $x_1 = g_1(y_1, y_2)$ and $x_2 = g_2(y_1, y_2)$. We assume that

$$\boldsymbol{g}(y_1, y_2) = \big(g_1(y_1, y_2), g_2(y_1, y_2)\big)$$

defines a function on the closure of D. The deformed domain is then the set

$$\boldsymbol{g}(D) := \big\{\boldsymbol{g}(\boldsymbol{y}) : \ \boldsymbol{y} \in D\big\}.$$

To derive a substitution formula we look at a small rectangle

$$R := [y_1, y_1 + \Delta y_1] \times [y_2, y_2 + \Delta y_2]$$

in D as done in the construction of the double integral in Section 2.1.1. We want to estimate the area of its image $\boldsymbol{g}(R)$. The rectangle and its image are shown in Figure 2.1.9. We assume that $\boldsymbol{g}$ has continuous first order derivatives on D. For fixed y_2 the image

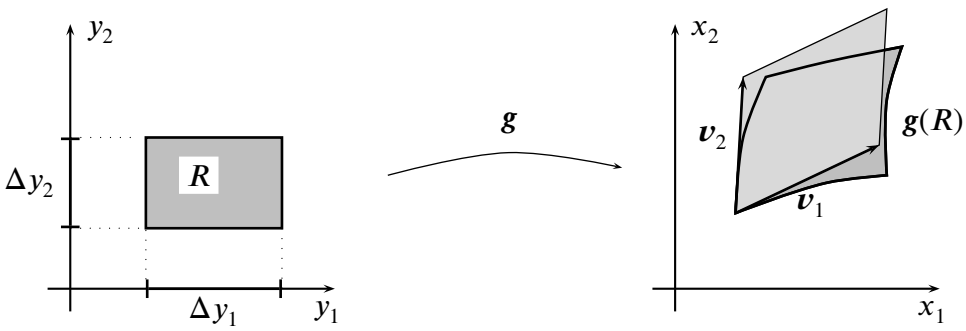


Figure 2.1.9: Deformation of a small rectangle by $\boldsymbol{g}$

of the map $y_1 \to \boldsymbol{g}(y_1, y_2)$ is a curve containing the lower edge of $\boldsymbol{g}(R)$. According to Remark 1.4.12 the length of that lower edge is approximately the length of the vector

$$\boldsymbol{v}_1 := \frac{\partial}{\partial y_1}\boldsymbol{g}(\boldsymbol{y})\Delta y_1.$$

Likewise the left edge of $\boldsymbol{g}(R)$ has approximately the length of

$$\boldsymbol{v}_2 := \frac{\partial}{\partial y_2}\boldsymbol{g}(\boldsymbol{y})\Delta y_2.$$

The surface area of $\boldsymbol{g}(R)$ is therefore approximately the surface area of the parallelogram spanned by $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$. In Proposition 1.1.7 we saw that the

$$\text{area of the parallelogram spanned by } \boldsymbol{v}_1 \text{ and } \boldsymbol{v}_2 = \left|\det\begin{bmatrix}\boldsymbol{v}_1 & \boldsymbol{v}_2\end{bmatrix}\right|.$$

Using the definition of $\boldsymbol{v}_1$, $\boldsymbol{v}_2$ and the properties of the determinant we have

$$\begin{aligned}\det\begin{bmatrix}\boldsymbol{v}_1 & \boldsymbol{v}_2\end{bmatrix} &= \det\begin{bmatrix}\frac{\partial g_1}{\partial y_1}(\boldsymbol{y})\Delta y_1 & \frac{\partial g_1}{\partial y_2}(\boldsymbol{y})\Delta y_2 \\ \frac{\partial g_2}{\partial y_1}(\boldsymbol{y})\Delta y_1 & \frac{\partial g_2}{\partial y_2}(\boldsymbol{y})\Delta y_2\end{bmatrix} \\ &= \det\begin{bmatrix}\frac{\partial g_1}{\partial y_1}(\boldsymbol{y}) & \frac{\partial g_1}{\partial y_2}(\boldsymbol{y}) \\ \frac{\partial g_2}{\partial y_1}(\boldsymbol{y}) & \frac{\partial g_2}{\partial y_2}(\boldsymbol{y})\end{bmatrix}\Delta y_1 \Delta y_2 = \det\big(J_{\boldsymbol{g}}(\boldsymbol{y})\big)\Delta y_1 \Delta y_2,\end{aligned}$$

where $J_g(\boldsymbol{y})$ is the Jacobian matrix of $\boldsymbol{g}$ at $\boldsymbol{y}$ as introduced in Definition 1.4.16. Hence the

$$\text{area of the parallelogram spanned by } \boldsymbol{v}_1 \text{ and } \boldsymbol{v}_2 = \Big|\det\big(J_g(\boldsymbol{y})\big)\Big|\Delta y_1 \Delta y_2. \tag{2.1.7}$$

Intuitively, $\Big|\det\big(J_g(\boldsymbol{y})\big)\Big|$ is factor by which the area of a small rectangle R is distorted by the map $\boldsymbol{g}$.

Definition 2.1.8 (Jacobian determinant) The determinant of the Jacobian matrix, $\det\big(J_g(\boldsymbol{y})\big)$, is called the *Jacobian determinant* or simply the *Jacobian* of $\boldsymbol{g}$ at $\boldsymbol{y}$.

☞ **Remark 2.1.9** In the more traditional literature the Jacobian determinant is often denoted by

$$\frac{\partial(g_1, g_2)}{\partial(y_1, y_2)} := \det\big(J_g(\boldsymbol{y})\big).$$

We will occasionally use this notation.

Now we partition D into small rectangles. Let us denote the collection of rectangles covering D by $\mathcal{R}$. If $\boldsymbol{y}$ denotes the left lower corner of each $R \in \mathcal{R}$ the sum

$$\sum_{R\in\mathcal{R}} f(\boldsymbol{g}(\boldsymbol{y}))\,\mathrm{area}(\boldsymbol{g}(R))$$

is an approximation for the volume of the region between $\boldsymbol{g}(D)$ and the graph of f. If we replace $\mathrm{area}(\boldsymbol{g}(R))$ by the approximation (2.1.7) this volume is approximately

$$\sum_{R\in\mathcal{R}} f(\boldsymbol{g}(\boldsymbol{y}))\Big|\det\big(J_g(\boldsymbol{y})\big)\Big|\Delta y_1 \Delta y_2.$$

The last expression is a Riemann sum for the function $\boldsymbol{y} \mapsto f(\boldsymbol{g}(\boldsymbol{y}))\Big|\det\big(J_g(\boldsymbol{y})\big)\Big|$. Passing to the limit this suggests that the volume of the region between $\boldsymbol{g}(D)$ and the graph of f is given by

$$\int_D f(\boldsymbol{g}(\boldsymbol{y}))\Big|\det\big(J_g(\boldsymbol{y})\big)\Big|\,d\boldsymbol{y}.$$

By the way we constructed double integrals we know that the volume of the region between $\boldsymbol{g}(D)$ and the graph of f is also given by

$$\int_{\boldsymbol{g}(D)} f(\boldsymbol{x})\,d\boldsymbol{x},$$

and so must be equal to the previous integral. The above procedure is not a proof, but with some effort all arguments can be made rigorous. In contrast to (2.1.6) we need that $\boldsymbol{g}$ is one-to-one. The reason is that double integrals are not 'oriented.' Hence we have the following generalisation of the substitution formula, called *transformation formula* or *area formula* for double integrals.

Theorem 2.1.10 (Transformation formula) *Suppose that $D \subset \mathbb{R}^2$ is a closed set on which the double integral for every continuous function is well defined. Assume that $\boldsymbol{g}\colon D \to \mathbb{R}^2$ is a one-to-one function with continuous first order derivatives. Then for every continuous function $f : \boldsymbol{g}(D) \to \mathbb{R}$ we have*

$$\boxed{\int_D f(\boldsymbol{g}(\boldsymbol{y}))\left|\det\left(J_{\boldsymbol{g}}(\boldsymbol{y})\right)\right| d\boldsymbol{y} = \int_{\boldsymbol{g}(D)} f(\boldsymbol{x})\, d\boldsymbol{x},}$$

where $J_{\boldsymbol{g}}(\boldsymbol{y})$ is the Jacobian matrix of $\boldsymbol{g}$ at $\boldsymbol{y}$.

Remark 2.1.11 In more traditional notation the transformation formula reads

$$\boxed{\iint_D f(x_1, x_2)\left|\frac{\partial(x_1, x_2)}{\partial(y_1, y_2)}\right| dy_1\, dy_2 = \iint_{\boldsymbol{g}(D)} f(x_1, x_2)\, dx_1\, dx_2,}$$

where x_1, x_2 are considered to be functions of y_1 and y_2.

Example 2.1.12 Let us write down the transformation formula if $\boldsymbol{g}$ is an *affine map*, that is, a linear map up to a translation. If $\boldsymbol{a}$ is a vector and A an invertible 2×2-matrix, then the map

$$\boldsymbol{g}(\boldsymbol{y}) := \boldsymbol{a} + A\boldsymbol{y}$$

is an affine map. Then (check this as an exercise), $J_{\boldsymbol{g}}(\boldsymbol{y}) = A$ for all $\boldsymbol{y}$. Hence the Jacobian determinant is given by $\det J_{\boldsymbol{g}}(\boldsymbol{y}) = \det A$. As $\det A$ is a constant the transformation formula reduces to

$$\boxed{\int_{\boldsymbol{a}+A(D)} f(\boldsymbol{x})\, d\boldsymbol{x} = |\det A| \int_D f(\boldsymbol{a} + A\boldsymbol{y}) d\boldsymbol{y}.} \tag{2.1.8}$$

2.1.4 Application: Polar Coordinates

Cartesian coordinates such as discussed in Section 1.1.1 are not the only way to represent points in the plane. We can also represent them by indicating the distance, r, from the origin and the angle, φ, from the x-axis as shown in Figure 2.1.10. If we assume that $r > 0$ and $\varphi \in [0, 2\pi)$ then every point except the origin has the unique representation

$$x = r\cos\varphi$$
$$y = r\sin\varphi.$$

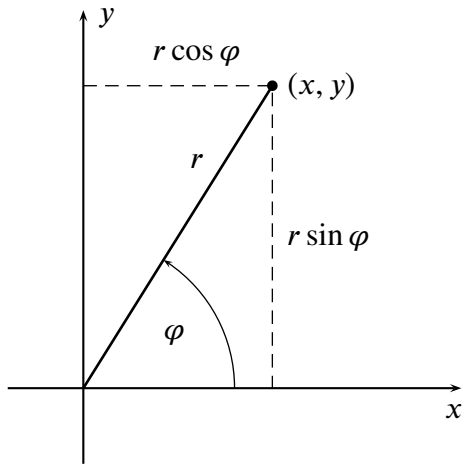


Figure 2.1.10: Polar coordinates of a point in the plane

Hence $\mathbb{R}^2$ is the image of the strip $[0, \infty) \times [0, 2\pi)$ under the transformation

$$\boldsymbol{g}(r, \varphi) := (r\cos\varphi, r\sin\varphi).$$

We want to apply the transformation formula from the previous section to find out how to integrate a function given in polar coordinates. To do so we need to compute the Jacobian determinant of $\boldsymbol{g}$. The Jacobian matrix is given by

$$J_g(r, \varphi) = \begin{bmatrix} \frac{\partial}{\partial r} r\cos\varphi & \frac{\partial}{\partial \varphi} r\cos\varphi \\ \frac{\partial}{\partial r} r\sin\varphi & \frac{\partial}{\partial \varphi} r\sin\varphi \end{bmatrix} = \begin{bmatrix} \cos\varphi & -r\sin\varphi \\ \sin\varphi & r\cos\varphi \end{bmatrix}.$$

Hence the Jacobian determinant is

$$\begin{aligned} \det J_g(r, \varphi) &= \det \begin{bmatrix} \cos\varphi & -r\sin\varphi \\ \sin\varphi & r\cos\varphi \end{bmatrix} \\ &= r \det \begin{bmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{bmatrix} = r(\cos^2\varphi + \sin^2\varphi) = r \geq 0. \end{aligned}$$

Applying the transformation formula we see that

$$\boxed{\iint_{g(D)} f(x, y)\, dx\, dy = \iint_D f(r\cos\varphi, r\sin\varphi) r\, dr\, d\varphi.} \tag{2.1.9}$$

Example 2.1.13 Compute the area of a disc of radius R.

Solution: A disc, D, with radius R is given in polar coordinates by $\{(r,\varphi):\ 0 \le r \le R, 0 \le \varphi < 2\pi\}$. Hence by (2.1.3), (2.1.9) and Fubini's theorem

$$\text{area}(D) = \iint_D 1\,dx\,dy = \int_0^R \int_0^{2\pi} r\,dr\,d\varphi = \int_0^R \frac{r^2}{2}\Big|_0^R d\varphi = \int_0^{2\pi} \frac{R^2}{2}\,d\varphi = \pi R^2.$$

Example 2.1.14 Compute the area of the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} \le 1.$$

Solution: We aim to find a function mapping a circle onto the given ellipse. If we set $x = au$ and $y = bv$ then

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{a^2u^2}{a^2} + \frac{b^2v^2}{b^2} = u^2 + v^2 = 1,$$

showing that the ellipse, E, is the image of the disc, D, given by $u^2 + v^2 \le 1$, under the (linear) transformation

$$\boldsymbol{g}(u,v) := (au, bv) = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix}.$$

According to the transformation formula (2.1.8) we get

$$\text{area}(E) = \iint_E 1\,dx\,dy = \det \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \iint_D 1\,du\,dv = ab \iint_D 1\,du\,dv.$$

Using the result from the previous example to compute the last integral we see that the area of the ellipse is $ab\pi$.

2.2 More Variables

In this section we aim to extend the theory of integrals of functions of two variables to functions of more variables. As we cannot draw the graph of a function of three or more variables it may not seem reasonable to do so. Who wants to know about 'volumes' of sets in four, five higher 'dimensions,' sets which we cannot even draw? To answer this question recall that the interpretation of the double integral as the volume of a body is *only one of many other* possible interpretation! We saw in Remark 2.1.4 that if $\varrho(\boldsymbol{x})$ is the mass density of a plate occupying D then $\int_D \varrho(\boldsymbol{x})\,d\boldsymbol{x}$ is the total mass of the plate. This is a problem with a natural extension to a body of mass in space. If $\varrho(x,y,z)$ is the mass density of a body in space at (x,y,z) then computing the total mass leads to

the integral of a function, ϱ, of three variables, often called a *triple integral*. Hence we have one model problem for integrals of functions of three variables. The problem however does not make sense to motivate integrals of functions of four or more variables. We will give one (very simplified) example from physics. A radio transmitter produces electro-magnetic waves which can be received by a radio, television or a mobile phone. The antenna of one of these devices absorbs the energy of the electro-magnetic field, amplifies the signal and turns it into something we can understand. We want to compute how much energy an antenna absorbs from the electro-magnetic field in a given time interval. If $\boldsymbol{E}(x, y, z, t)$ is the electric and $\boldsymbol{B}(x, y, z, t)$ the magnetic field at the point (x, y, z) at time t, then it is shown in physics that its 'energy density' (energy per volume) at (x, y, z) at time t is given by

$$\frac{1}{2}\Big(\|\boldsymbol{E}(x, y, z, t)\|^2 + \|\boldsymbol{B}(x, y, z, t)\|^2\Big).$$

Assume that the antenna of the receiving device occupies the region $R \subset \mathbb{R}^3$. To compute how much energy the antenna absorbs in the time interval $[a, b]$ we first compute the energy of the electro-magnetic field in R at every instance $t \in [a, b]$. It is the 'sum' of the density over the region D, represented by the triple integral

$$\iiint_D \frac{1}{2}\Big(\|\boldsymbol{E}(x, y, z, t)\|^2 + \|\boldsymbol{B}(x, y, z, t)\|^2\Big)\, dx\, dy\, dz.$$

To get the energy absorbed in the time interval $[a, b]$ we have to take the 'sum' over $[a, b]$, that is, integrate over $[a, b]$. Hence the total energy absorbed is

$$\iiiint_{[a,b]\times D} \frac{1}{2}\Big(\|\boldsymbol{E}(x, y, z, t)\|^2 + \|\boldsymbol{B}(x, y, z, t)\|^2\Big)\, dx\, dy\, dz\, dt,$$

which is the integral of a function of four variables. Other problems require taking integrals over functions of even more variables, but we hope the above provides enough motivation for the theory developed below.

2.2.1 Triple Integrals

As with double integrals we first define triple integrals, and then show how to evaluate them.

Definition of Triple Integrals

Suppose that $D \subset \mathbb{R}^3$ is a closed bounded set, and $f : D \to \mathbb{R}$ a continuous function. We now subdivide D into small rectangular boxes, either approaching D from the in- or the outside, similarly as shown in Figure 2.1.5 for plane domains. Suppose the partition has n boxes. Number them in some order from 1 to n, and denote the i-th box by R_i. These boxes then have all the form

$$R_i = [x_i, x_i + \Delta x_i] \times [y_i, y_i + \Delta y_i] \times [z_i, z_i + \Delta z_i],$$

and their volume is $\Delta x_i \Delta y_i \Delta z_i$. We then choose an arbitrary point $(x_i^*, y_i^*, z_i^*) \in R_i$ and consider the sum

$$\sum_{i=1}^{n} f(x_i^*, y_i^*, z_i^*) \Delta x_i \Delta y_i \Delta z_i,$$

which is the counterpart of the Riemann sum (2.1.2). If D is 'reasonably' smooth and f is continuous then one can show that the above sum, also called a Riemann sum, has a limit as $\max_i\{\Delta x_i, \Delta y_i, \Delta z_i\} \to 0$. Moreover, the limit is independent of the choice of $(x_i^*, y_i^*, z_i^*) \in R_i$. The limit is denoted by

$$\iiint_D f(x, y, z)\, dx\, dy\, dz,$$

or in more modern notation simply by

$$\int_D f(\boldsymbol{x})\, d\boldsymbol{x},$$

and called the (*triple*) *integral of f over D*.

☞ **Remark 2.2.1** If we choose f to be the constant function with value one, that is, $f(x, y, z) = 1$, then the integral represents the volume of the domain:

$$\boxed{\text{Volume of the domain } D = \int_D 1\, d\boldsymbol{x}.} \tag{2.2.1}$$

Another physical interpretation is the following. Let D represent the region occupied by a solid made of some material. At every point (x, y, z) on that plate let $\varrho(x, y, z)$ denote

the mass density of the material. Then the

$$\boxed{\text{Total mass of the solid occupying } D = \int_D \varrho(\boldsymbol{x})\, d\boldsymbol{x}.}$$

In the next section we see how to evaluate triple integrals over certain classes of domains.

Evaluation of Triple Integrals

As with double integrals, triple integrals can be evaluated by computing iterated integrals if the domain is given in a suitable form. The form is

$$D = \big\{(x, y, z) : \ a \le x \le b, \varphi(x) \le y \le \Phi(x), \psi(x, y) \le z \le \Psi(x, y)\big\}. \tag{2.2.2}$$

Note that the projection of D along the z-axis onto the xy-axis is given by

$$D_0 := \big\{(x, y, z) : \ a \le x \le b, \varphi(x) \le y \le \Phi(x)\},$$

and that the condition $\psi(x, y) \le z \le \Psi(x, y)$ indicates that D is between the graph of ψ and Ψ over D_0. By slicing the domain into thin domains similarly as done in Section 2.1.2 one can show the following generalised version of Fubini's theorem.

Theorem 2.2.2 (Fubini's Theorem) *Suppose that D has the form* (2.2.2) *with φ, Φ continuous on $[a, b]$, and ψ, Ψ continuous on D_0 as defined above. If f is continuous on D then*

$$\boxed{\iiint_D f(x, y, z)\, dx\, dy\, dz = \int_a^b \Big(\int_{\varphi(x)}^{\Phi(x)} \Big(\int_{\psi(x,y)}^{\Psi(x,y)} f(x, y, z)\, dz \Big)\, dy \Big)\, dx.}$$

Note that we can interchange the roles of x, y, z arbitrarily.

Example 2.2.3 Let D be the domain bounded below by $z = \sqrt{x^2 + y^2}$, and bounded from above by $9x^2 + 4y^2 + z^2 = 5$. Find the limits for the triple integral of a function $f(x, y, z)$ over the domain D.

Solution: We first find the projection, D_0, of the domain onto the xy-plane. The intersection of $z = \sqrt{x^2 + y^2}$ and $9x^2 + 4y^2 + z^2 = 5$ forms a curve in space. Its projection onto the xy-plane is the boundary of D_0. To find that projection we eliminate z from the two equations. If we substitute the first into the second equation we get

$$9x^2 + 4y^2 + z^2 = 9x^2 + 4y^2 + x^2 + y^2 = 10x^2 + 5y^2 = 5,$$

which we can write as $2x^2 + y^2 = 1$. The last equation describes an ellipse, so we have that D_0 is given by

- $-\dfrac{1}{\sqrt{2}} \le x \le \dfrac{1}{\sqrt{2}}$,
- $-\sqrt{1-2x^2} \le y \le \sqrt{1-2x^2}$.

From the conditions on z we finally get

- $\sqrt{x^2+y^2} \le z \le \sqrt{5-9x^2-4y^2}$.

Hence by Fubini's Theorem we can write

$$\int_D f(\boldsymbol{x})\,d\boldsymbol{x} = \int_{-1/\sqrt{2}}^{1/\sqrt{2}} \left(\int_{-\sqrt{1-2x^2}}^{\sqrt{1-2x^2}} \left(\int_{\sqrt{x^2+y^2}}^{\sqrt{5-9x^2-4y^2}} f(x,y,z)\,dz \right) dy \right) dx.$$

If f is given we need to compute three integrals, starting with the innermost.

2.2.2 The Transformation Formula

For triple integrals we can derive a transformation formula similar to the one given in Theorem 2.1.10. To obtain such a formula we simply generalise the method presented in Section 2.1.3 to three dimensions. The situation is the following. We want to integrate a function, f, defined on the image of a domain $D \subset \mathbb{R}^3$ under a map $\boldsymbol{g}$. The domain D and the deformed domain $\boldsymbol{g}(D)$ are as shown in Figure 2.1.8, but in space. We denote the coordinates in D by (y_1, y_2, y_3), and those in its image, $\boldsymbol{g}(D)$, by (x_1, x_2, x_3). The map $\boldsymbol{g}$ has now three components:

$$\boldsymbol{g}(y_1, y_2, y_3) = \big(g_1(y_1, y_2, y_3), g_2(y_1, y_2, y_3), g_3(y_1, y_2, y_3)\big)$$

Now we look at the image of a small rectangular box

$$R = [y_1, y_1 + \Delta y_1] \times [y_2, y_2 + \Delta y_2] \times [y_3, y_3 + \Delta y_3].$$

As in two dimensions we find an approximation of the volume of the deformed rectangular box $\boldsymbol{g}(R)$. By similar arguments as used in Section 2.1.3 we see that the volume of $\boldsymbol{g}(R)$ is close to the volume of the parallelepiped spanned by

$$\boldsymbol{v}_1 := \frac{\partial}{\partial y_1}\boldsymbol{g}(\boldsymbol{y})\Delta y_1, \qquad \boldsymbol{v}_2 := \frac{\partial}{\partial y_2}\boldsymbol{g}(\boldsymbol{y})\Delta y_2 \qquad \text{and} \qquad \boldsymbol{v}_3 := \frac{\partial}{\partial y_3}\boldsymbol{g}(\boldsymbol{y})\Delta y_3.$$

The two dimensional analogue of the present situation is shown in Figure 2.1.9. Now we know from Application 1.1.14 that the

$$\text{volume of the paralellepiped spanned by } \boldsymbol{v}_1, \boldsymbol{v}_2 \text{ and } \boldsymbol{v}_3 = \left|\det \begin{bmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \boldsymbol{v}_3 \end{bmatrix}\right|.$$

Using the definition of $\boldsymbol{v}_1$, $\boldsymbol{v}_2$, $\boldsymbol{v}_3$ and the properties of the determinant we have

$$\begin{aligned}
\det\begin{bmatrix}\boldsymbol{v}_1 & \boldsymbol{v}_2 & \boldsymbol{v}_3\end{bmatrix} &= \det\begin{bmatrix}\frac{\partial g_1}{\partial y_1}(\boldsymbol{y})\Delta y_1 & \frac{\partial g_1}{\partial y_2}(\boldsymbol{y})\Delta y_2 & \frac{\partial g_1}{\partial y_3}(\boldsymbol{y})\Delta y_3\\ \frac{\partial g_2}{\partial y_1}(\boldsymbol{y})\Delta y_1 & \frac{\partial g_2}{\partial y_2}(\boldsymbol{y})\Delta y_2 & \frac{\partial g_2}{\partial y_3}(\boldsymbol{y})\Delta y_3\\ \frac{\partial g_3}{\partial y_1}(\boldsymbol{y})\Delta y_1 & \frac{\partial g_3}{\partial y_2}(\boldsymbol{y})\Delta y_2 & \frac{\partial g_3}{\partial y_3}(\boldsymbol{y})\Delta y_3\end{bmatrix}\\
&= \det\begin{bmatrix}\frac{\partial g_1}{\partial y_1}(\boldsymbol{y}) & \frac{\partial g_1}{\partial y_2}(\boldsymbol{y}) & \frac{\partial g_1}{\partial y_3}(\boldsymbol{y})\\ \frac{\partial g_2}{\partial y_1}(\boldsymbol{y}) & \frac{\partial g_2}{\partial y_2}(\boldsymbol{y}) & \frac{\partial g_2}{\partial y_3}(\boldsymbol{y})\\ \frac{\partial g_3}{\partial y_1}(\boldsymbol{y}) & \frac{\partial g_3}{\partial y_2}(\boldsymbol{y}) & \frac{\partial g_3}{\partial y_3}(\boldsymbol{y})\end{bmatrix}\Delta y_1\Delta y_2\Delta y_3\\
&= \det\big(J_{\boldsymbol{g}}(\boldsymbol{y})\big)\Delta y_1\Delta y_2\Delta y_3,
\end{aligned}$$

where $J_{\boldsymbol{g}}(\boldsymbol{y})$ is the Jacobian matrix of $\boldsymbol{g}$ at $\boldsymbol{y}$ as introduced in Definition 1.4.16. Hence the

$$\begin{aligned}
&\text{volume of the parallelepiped spanned by } \boldsymbol{v}_1, \boldsymbol{v}_2 \text{ and } \boldsymbol{v}_3\\
&\qquad = \Big|\det\big(J_{\boldsymbol{g}}(\boldsymbol{y})\big)\Big|\Delta y_1\Delta y_2\Delta y_3, \qquad (2.2.3)
\end{aligned}$$

so as with double integrals the Jacobian determinant appears. Again, the Jacobian determinant is the factor by which the volume of a small rectangle is distorted by the map $\boldsymbol{g}$.

☞ **Remark 2.2.4** In the more traditional literature the Jacobian determinant is often denoted by

$$\frac{\partial(g_1, g_2, g_3)}{\partial(y_1, y_2, y_3)} := \det\big(J_{\boldsymbol{g}}(\boldsymbol{y})\big).$$

We will occasionally use this notation.

Now we partition D into small rectangular boxes. Let us denote the collection of rectangles covering D by $\mathcal{R}$. If $\boldsymbol{y}$ denotes one corner of each $R \in \mathcal{R}$ the sum

$$\sum_{R\in\mathcal{R}} f(\boldsymbol{g}(\boldsymbol{y}))\,\mathrm{vol}(\boldsymbol{g}(R))$$

is an approximation for the integral of f over $\boldsymbol{g}(D)$. If we substitute $\mathrm{vol}(\boldsymbol{g}(R))$ by (2.2.3) then the above sum is very close to

$$\sum_{R\in\mathcal{R}} f(\boldsymbol{g}(\boldsymbol{y}))\Big|\det\big(J_{\boldsymbol{g}}(\boldsymbol{y})\big)\Big|\Delta y_1\Delta y_2\Delta y_3.$$

The last expression is a Riemann sum for the function $\boldsymbol{y} \mapsto f(\boldsymbol{g}(\boldsymbol{y}))\left|\det\left(J_{\boldsymbol{g}}(\boldsymbol{y})\right)\right|$. Passing to the limit this suggests that the integral of f over $\boldsymbol{g}(D)$ is given by

$$\int_D f(\boldsymbol{g}(\boldsymbol{y}))\left|\det\left(J_{\boldsymbol{g}}(\boldsymbol{y})\right)\right| d\boldsymbol{y}.$$

The above procedure is not a proof, but with some effort all arguments can be made rigorous. Hence we have the following result, completely analogous to Theorem 2.1.10.

Theorem 2.2.5 (Transformation formula) *Suppose that $D \subset \mathbb{R}^3$ is a closed set on which the integral for every continuous function is well defined. Assume that $\boldsymbol{g} : D \to \mathbb{R}^3$ is a one-to-one function with continuous first order derivatives. Then for every continuous function $f : \boldsymbol{g}(D) \to \mathbb{R}$ we have*

$$\boxed{\int_D f(\boldsymbol{g}(\boldsymbol{y}))\left|\det\left(J_{\boldsymbol{g}}(\boldsymbol{y})\right)\right| d\boldsymbol{y} = \int_{\boldsymbol{g}(D)} f(\boldsymbol{x})\, d\boldsymbol{x},}$$

where $J_{\boldsymbol{g}}(\boldsymbol{y})$ is the Jacobian matrix of $\boldsymbol{g}$ at $\boldsymbol{y}$.

Remark 2.2.6 In more traditional notation the transformation formula reads

$$\boxed{\begin{aligned}\iiint_D f(x_1, x_2, x_3)\left|\frac{\partial(x_1, x_2, x_3)}{\partial(y_1, y_2, y_3)}\right| dy_1\, dy_2\, dy_3 \\ = \iiint_{\boldsymbol{g}(D)} f(x_1, x_2, x_3)\, dx_1\, dx_2\, dx_3,\end{aligned}}$$

where x_1, x_2, x_3 are considered to be functions of y_1, y_2 and y_3.

Note that the special case considered in Example 2.1.12 carries over to the present situation.

2.2.3 Application: Spherical and Cylindrical Coordinates

Sometimes it is useful to work with other 'coordinates' than Cartesian coordinates. The most commonly used such coordinates are spherical and cylindrical coordinates. We want to see how to integrate functions given in these coordinates. As for the polar coordinates in Section 2.1.4 we need to compute the corresponding Jacobian determinant.

Spherical Coordinates

A point in space can be determined by giving the distance, r, from the origin and two angles, θ and φ, as shown in Figure 2.2.1. For every point off the z-axis the triple (r, θ, φ) is uniquely determined if we choose $r > 0$, $\theta \in [0, \pi]$ and $\varphi \in [0, 2\pi)$. We call r, θ, φ the *spherical coordinates* of the given point. Our notation follows the naming convention used in applied mathematics and mathematical physics. Looking at Figure 2.2.1 we can

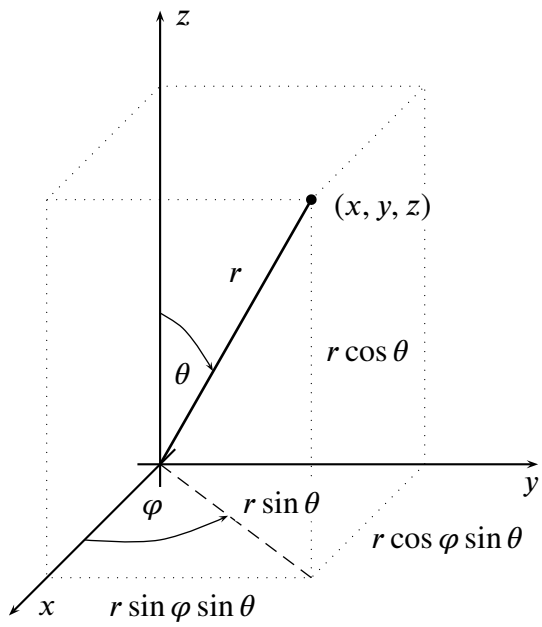


Figure 2.2.1: Spherical coordinates of a point

express x, y, z by

$$\begin{aligned} x &= r\cos\varphi\sin\theta, \\ y &= r\sin\varphi\sin\theta, \\ z &= r\cos\theta. \end{aligned}$$

Hence the transformation

$$\boldsymbol{g}(r, \theta, \varphi) := \big(r\cos\varphi\sin\theta, r\sin\varphi\sin\theta, r\cos\theta\big) \tag{2.2.4}$$

maps $[0,\infty)\times[0,\pi]\times[0,2\pi)$ onto $\mathbb{R}^3$. We now compute the Jacobian matrix of that transformation:

$$\begin{aligned}
J_g(r,\theta,\varphi) &= \begin{bmatrix} \frac{\partial}{\partial r} r\cos\varphi\sin\theta & \frac{\partial}{\partial \theta} r\cos\varphi\sin\theta & \frac{\partial}{\partial \varphi} r\cos\varphi\sin\theta \\ \frac{\partial}{\partial r} r\sin\varphi\sin\theta & \frac{\partial}{\partial \theta} r\sin\varphi\sin\theta & \frac{\partial}{\partial \varphi} r\sin\varphi\sin\theta \\ \frac{\partial}{\partial r} r\cos\theta & \frac{\partial}{\partial \theta} r\cos\theta & \frac{\partial}{\partial \varphi} r\cos\theta \end{bmatrix} \\
&= \begin{bmatrix} \cos\varphi\sin\theta & r\cos\varphi\cos\theta & -r\sin\varphi\sin\theta \\ \sin\varphi\sin\theta & r\sin\varphi\cos\theta & r\cos\varphi\sin\theta \\ \cos\theta & -r\sin\theta & 0 \end{bmatrix}.
\end{aligned}$$

Expanding along the last row we get

$$\begin{aligned}
\det J_g(r,\theta,\varphi) &= r^2 \det \begin{bmatrix} \cos\varphi\sin\theta & \cos\varphi\cos\theta & -\sin\varphi\sin\theta \\ \sin\varphi\sin\theta & \sin\varphi\cos\theta & \cos\varphi\sin\theta \\ \cos\theta & -\sin\theta & 0 \end{bmatrix} \\
&= r^2\big(\cos\theta(\cos^2\varphi+\sin^2\varphi)\sin\theta\cos\theta \\
&\qquad \sin\theta(\cos^2\varphi\sin^2\theta+\sin^2\varphi\sin^2\theta) \\
&= r^2\sin\theta(\cos^2\theta+\sin^2\theta) \\
&= r^2\sin\theta.
\end{aligned}$$

The fact that $J_g(r,\theta,\varphi)$ is positive for all $\theta\in(0,\pi)$ tells us that the change of variables preserves the orientation of the coordinate systems, and that is why we take the variables in the order (r,θ,φ). Hence we have the following lemma, providing the Jacobian determinant for spherical coordinates.

Lemma 2.2.7 (Jacobian of spherical coordinates) *Suppose that* $\boldsymbol{g}$ *is given by* (2.2.4). *Then*

$$\boxed{\det J_g(r,\theta,\varphi) = \frac{\partial(x,y,z)}{\partial(r,\theta,\varphi)} = r^2\sin\theta.}$$

Applying the transformation formula from Theorem 2.2.5 we see that

$$\boxed{\begin{aligned}
&\iiint_D f(x,y,z)\,dx\,dy\,dz \\
&\qquad = \iiint_{g(D)} f(r\cos\varphi\sin\theta, r\sin\varphi\sin\theta, r\cos\theta) r^2\sin\theta\,dr\,d\varphi\,d\theta.
\end{aligned}} \tag{2.2.5}$$

Remark 2.2.8 Sometimes, spherical coordinates are defined measuring the angle θ not from the z-axis but from the xy-plane. If done so then $\theta \in [-\pi/2, \pi/2]$, and

$$\begin{aligned} x &= r\cos\varphi\cos\theta, \\ y &= r\sin\varphi\cos\theta, \\ z &= r\sin\theta. \end{aligned}$$

Moreover, the corresponding Jacobian determinant is $J_g(r, \varphi, \theta) = r^2\cos\theta$.

Example 2.2.9 Compute the volume of a ball of radius R in $\mathbb{R}^3$.
Solution: In polar coordinates we can describe the ball, B, of radius R as the set of all (r, θ, φ) for which $r \in [0, R]$, $\varphi \in [0, 2\pi]$ and $\theta \in [0, \pi]$. Hence, applying Fubini's theorem and the formula for spherical coordinates, the volume of the ball is

$$\begin{aligned} \int_B 1 d\boldsymbol{x} &= \int_0^{2\pi}\left(\int_0^{\pi}\left(\int_0^{R} r^2\sin\theta\, dr\right) d\theta\right) d\varphi = \int_0^{2\pi}\left(\int_0^{\pi}\frac{R^3}{3}\sin\theta\, d\theta\right) d\varphi \\ &= \frac{R^3}{3}\int_0^{2\pi}\left(-\cos\theta\Big|_0^{\pi}\right) d\varphi = \frac{R^3}{3}\int_0^{2\pi} 2\, d\varphi = \frac{4\pi}{3}R^3. \end{aligned}$$

Cylindrical Coordinates

Given a point in space we can express the first two coordinates in polar coordinates, (r, φ), and leave the third one, z, as shown in Figure 2.2.2. Choosing $r > 0$ and $\varphi \in [0, 2\pi)$ the triple (r, φ, z) is uniquely determined for every point not on the z-axis. We call r, φ, z the *cylindrical coordinates* of a point. Looking at Figure 2.2.2 we can express x, y, z by

$$\begin{aligned} x &= r\cos\varphi, \\ y &= r\sin\varphi, \\ z &= z. \end{aligned}$$

Hence the transformation

$$\boldsymbol{g}(r, \varphi, z) := \big(r\cos\varphi, r\sin\varphi, z\big) \tag{2.2.6}$$

maps $[0, \infty) \times [0, 2\pi) \times \mathbb{R}$ onto $\mathbb{R}^3$. We now compute the Jacobian matrix of that transformation:

$$J_g(r, \varphi, z) = \begin{bmatrix} \frac{\partial}{\partial r} r\cos\varphi & \frac{\partial}{\partial \varphi} r\cos\varphi & \frac{\partial}{\partial z} r\cos\varphi \\ \frac{\partial}{\partial r} r\sin\varphi & \frac{\partial}{\partial \varphi} r\sin\varphi & \frac{\partial}{\partial z} r\sin\varphi \\ \frac{\partial}{\partial r} z & \frac{\partial}{\partial \varphi} z & \frac{\partial}{\partial z} z \end{bmatrix} = \begin{bmatrix} \cos\varphi & -r\sin\varphi & 0 \\ \sin\varphi & r\cos\varphi & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

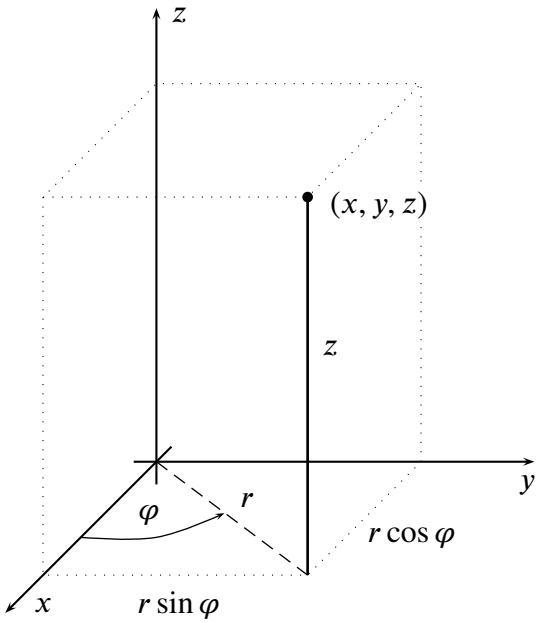


Figure 2.2.2: Cylindrical coordinates of a point

Expanding along the last row we get

$$\det J_g(r,\varphi,z) = \det \begin{bmatrix} \cos\varphi & -r\sin\varphi & 0 \\ \sin\varphi & r\cos\varphi & 0 \\ 0 & 0 & 1 \end{bmatrix} = r(\cos^2\varphi + \sin^2\varphi) = r$$

Hence we have the following lemma, providing the Jacobian determinant for cylindrical coordinates.

Lemma 2.2.10 (Jacobian of cylindrical coordinates) *Suppose* $\boldsymbol{g}$ *is given by* (2.2.6). *Then*

$$\boxed{\det J_g(r,\varphi,z) = \frac{\partial(x,y,z)}{\partial(r,\varphi,z)} = r.}$$

Applying the transformation formula from Theorem 2.2.5 we see that

$$\boxed{\iiint_D f(x,y,z)\,dx\,dy\,dz = \iiint_{g(D)} f(r\cos\varphi, r\sin\varphi, z)r\,dr\,d\varphi\,dz.} \tag{2.2.7}$$

2.2.4 Comments on More Variables

There is no real difficulty about generalising the theory in this section to more than three variables. The integral is defined subdividing a domain in $\mathbb{R}^N$ into small 'boxes,' that is, Cartesian products of N rather than three small intervals, and then doing the same kind of limiting process. Everything, including the transformation formula in Theorem 2.2.5 remains valid. Also the proof of the transformation formula does not require new ideas, and relies on the facts on N-dimensional volumes in Remark 1.1.15. Also spherical coordinates as well as various cylindrical coordinates have a generalisation to N variables, $N > 3$.

Chapter 3

Line Integrals

This chapter deals with a generalisation of integrals of a function of one variable. We want to learn how to integrate certain quantities along a line. Applications of the theory include computing the work done moving along a path in a force field or the circulation of a vector field.

3.1 Basic Definitions

In this section we aim to give a definition of 'curves,' and then to discuss some possible properties. Suppose that $\boldsymbol{\gamma}$ is a continuous vector valued function defined on an interval $I \subset \mathbb{R}$ with values in $\mathbb{R}^N$. By definition, the image,

$$C := \{\boldsymbol{\gamma}(t) : t \in I\} \subset \mathbb{R}^N,$$

is called a *curve* in $\mathbb{R}^N$. As discussed already in Section 1.2.4, the function $\boldsymbol{\gamma}$ has N component functions, namely

$$\boldsymbol{\gamma}(t) = \big(\gamma_1(t), \gamma_2(t), \dots, \gamma_N(t)\big)$$

for $t \in I$. If $N = 2$ or $N = 3$ we can visualise the curve C as shown in Figure 1.2.8. We can think of C as the path followed by a particle moving in space. If C has no points of intersection with itself, then we call C a *simple curve*, if C has the same start and endpoints it is called a *closed curve*. See Figure 3.1.1 to visualise this. Obviously there are many ways to travel along a given path. We can move slower or faster, backward and forward, etc. Hence there are many different continuous functions describing the same curve. We call every one-to-one function describing C a *parametrisation of* C.

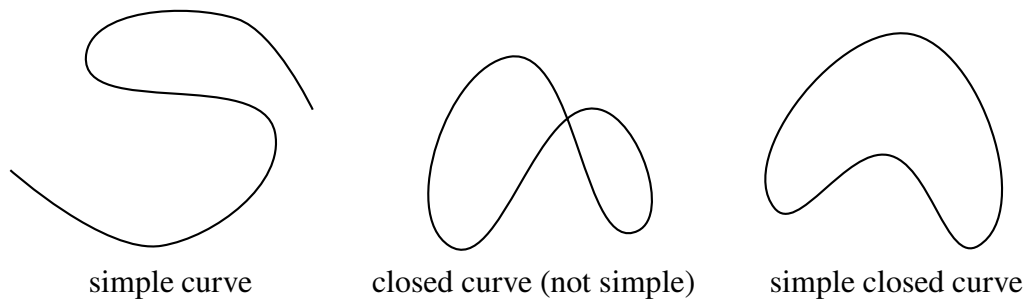


Figure 3.1.1: Different types of curves.

Definition 3.1.1 (Smooth curve) We call C a *smooth curve* if it admits a continuously differentiable parametrisation $\boldsymbol{\gamma}(t)$, $t \in I$, such that $\boldsymbol{\gamma}'(t) \neq \mathbf{0}$ for all $t \in I$. A parametrisation as the above is called a *regular parametrisation* of C. We call C a *piecewise smooth curve* if it can be written as a union of finitely many smooth curves.

Figure 3.1.2 shows a piecewise smooth curve, the curves shown in Figure 3.1.1 are all smooth. In this course we will always work with smooth or piecewise smooth curves. We saw in Section 1.4.2 that $\boldsymbol{\gamma}'(t)$ points in the direction of the tangent of C at $\boldsymbol{\gamma}(t)$. If

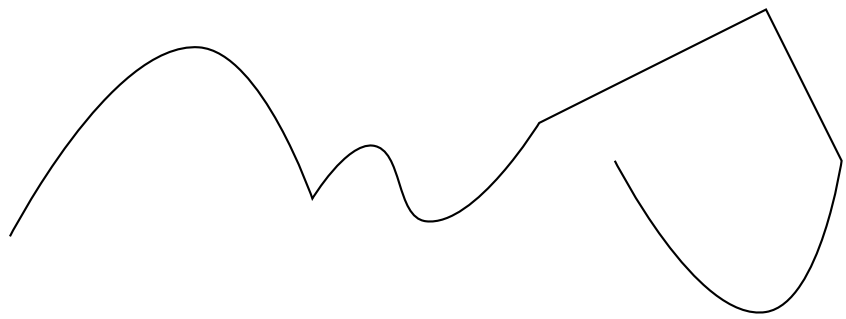


Figure 3.1.2: A piecewise smooth curve.

C is a smooth curve and $\boldsymbol{\gamma}(t)$, $t \in I$, is a regular parametrisation of C we can compute a unit tangent vector at every point of C. As $\boldsymbol{\gamma}'(t) \neq \mathbf{0}$ for all t we can set

$$\boxed{\boldsymbol{\tau}(t) := \frac{\boldsymbol{\gamma}'(t)}{\|\boldsymbol{\gamma}'(t)\|}, \qquad t \in I.} \tag{3.1.1}$$

At each point of the curve there are exactly two unit tangent vectors pointing in opposite directions. If $\boldsymbol{\tau}(t)$ is given as above then $\boldsymbol{\tau}(t)$ and $-\boldsymbol{\tau}(t)$ are the only continuous unit tangent vectors. One of them we define to be the *positive field of unit tangents*. The opposite field is called the *negative field of unit tangents*. The fields are depicted in Figure 3.1.3. The two possibilities reflect the fact that we can run along the curve in two different directions, one called positive, the opposite one called negative.

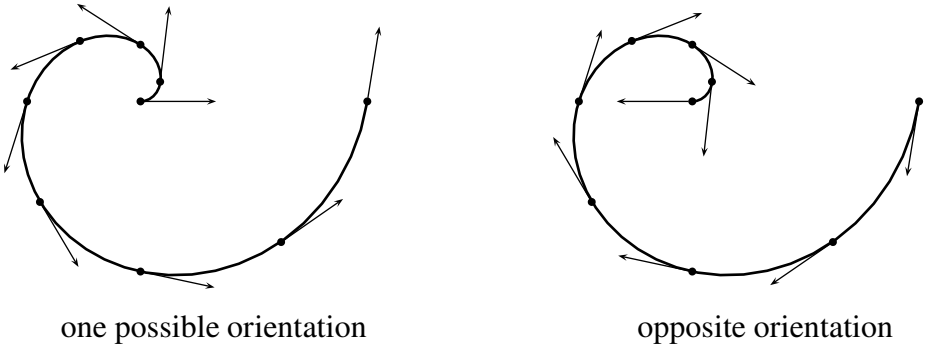


Figure 3.1.3: The two possible orientations of a curve.

Definition 3.1.2 A smooth curve, C, with the positive unit tangent vector field is called *positively oriented*. Equipped with the opposite unit tangent vector field C is called negatively oriented. We often write

- C for the positively oriented curve;
- $-C$ for the negatively oriented curve;
- $|C|$ for the curve without a particular orientation fixed.

We call a parametrisation, $\gamma(t)$, $t \in I$ *consistent with the orientation* of the curve if (3.1.1) defines the positive field of tangents.

3.2 Arc Length

In this section we want to define *arc length*, that is, the length of a curve. We suppose that $\gamma(t)$, $t \in [a, b]$, is a regular parametrisation of a smooth curve C. To find its length we choose a partition $a =: t_0 < t_1 < t_2 < \cdots < t_n := b$. We then approximate C by the polygon with vertices $\gamma(t_i)$ as shown in Figure 3.2.1. The length of the line segment

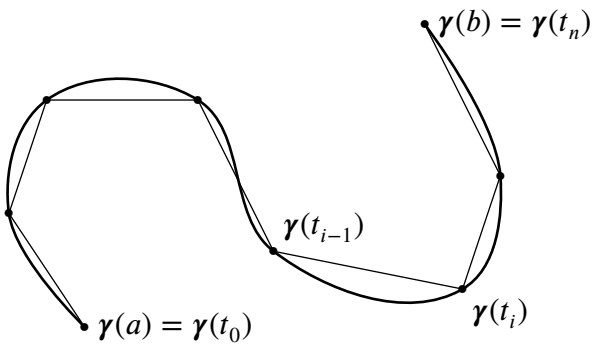


Figure 3.2.1: Approximating a curve by a polygon.

connecting $\gamma(t_{i-1})$ to $\gamma(t_i)$ is $\|\gamma(t_i) - \gamma(t_{i-1})\|$, so the total length of the polygon is

$$\sum_{i=1}^{n} \|\gamma(t_i) - \gamma(t_{i-1})\|. \tag{3.2.1}$$

The length of the polygon will always be smaller than the length of the curve as we always take a 'shortcut' from one point to the next on the polygon. If we take a finer partition, that is, make $\max(t_i - t_{i-1})$ smaller, we expect to get a closer approximation of the length of C. This motivates the following definition.

Definition 3.2.1 (arc length) We define the *arc length* of the curve C to be the least upper bound of (3.2.1) over all partitions of the interval $[a, b]$.

The above definition is not very handy to compute the length of C. Using that the curve is smooth we want to derive a formula to compute its length. Setting $\Delta t_i := t_i - t_{i-1}$ we can rewrite (3.2.1) as

$$\sum_{i=1}^{n} \left\| \frac{\gamma(t_{i-1} + \Delta t_i) - \gamma(t_{i-1})}{\Delta t_i} \right\| \Delta t_i.$$

As $\Delta t_i \to 0$ the i-th fraction in the above expression tends to $\gamma'(t_{i-1})$, so the above sum is very close to

$$\sum_{i=1}^{n} \|\gamma'(t_{i-1})\| \Delta t_i$$

if $\max \Delta t_i$ is small. The last sum is a Riemann sum for the function $t \mapsto \|\gamma'(t)\|$, and converges to

$$\int_a^b \|\gamma'(t)\| \, dt$$

as $\max \Delta t_i$ tends to zero. With some effort involved to estimate the error terms in the above procedure one can show that in fact the following is true.

Proposition 3.2.2 *Suppose that $\boldsymbol{\gamma}(t)$, $t \in [a,b]$ is a regular parametrisation of the smooth curve C. Then the length, $L(C)$, of C is given by*

$$\boxed{L(C) = \int_a^b \|\boldsymbol{\gamma}'(t)\|\,dt = \int_a^b \sqrt{\left(\gamma_1'(t)\right)^2 + \left(\gamma_2'(t)\right)^2 + \cdots + \left(\gamma_N'(t)\right)^2}\,dt.}$$

Note that the length of a curve is independent of the orientation of the curve. To compute the length of a piecewise smooth curve we add up the lengths of all its smooth parts.

Example 3.2.3 We can use the above formula to compute the circumference of a circle of radius R. A possible parametrisation of such a circle is given by $\boldsymbol{\gamma}(t) = (R\cos t, R\sin t)$, $t \in [0, 2\pi]$. Hence $\boldsymbol{\gamma}'(t) = (-R\sin t, R\cos t)$, and thus the circumference is

$$\int_0^{2\pi} \sqrt{R^2\sin^2 t + R^2\cos^2 t}\,dt = R\int_0^{2\pi} 1\,dt = 2\pi R,$$

which is the well known result.

Let us finally look at two special cases of the above formula. First assume that f is a continuously differentiable function on the interval $[a,b]$. Then the function

$$\boldsymbol{\gamma}(t) := (t, f(t))$$

defines a smooth curve in $\mathbb{R}^2$, namely the graph of f. It follows that $\boldsymbol{\gamma}'(t) := (1, f'(t))$, and applying Proposition 3.2.2 we see that the

$$\boxed{\text{Length of the graph of } f = \int_a^b \sqrt{1 + (f'(t))^2}\,dt.}$$

Next we suppose that the curve is given in polar coordinates, that is, the radius as a function of the angle: $r = f(\varphi)$, $\varphi \in [\alpha, \beta]$. Then $x = r\cos\varphi = f(\varphi)\cos\varphi$ and $y = r\sin\varphi = f(\varphi)\sin\varphi$, so

$$\boldsymbol{\gamma}(\varphi) = (f(\varphi)\cos\varphi, f(\varphi)\sin\varphi), \qquad \varphi \in [\alpha, \beta]$$

is a parametrisation of the curve. If f is continuously differentiable then by the product rule

$$\boldsymbol{\gamma}'(\varphi) = (f'(\varphi)\cos\varphi - f(\varphi)\sin\varphi, f'(\varphi)\sin\varphi + f(\varphi)\cos\varphi),$$

and thus

$$\begin{aligned}
\|\boldsymbol{\gamma}'(\varphi)\|^2 &= \left(f'(\varphi)\cos\varphi - f(\varphi)\sin\varphi\right)^2 + \left(f'(\varphi)\sin\varphi + f(\varphi)\cos\varphi\right)^2 \\
&= \left(f'(\varphi)\right)^2(\cos^2\varphi + \sin^2\varphi) - 2f'(\varphi)f(\varphi)\cos\varphi\sin\varphi \\
&\qquad + 2f'(\varphi)f(\varphi)\cos\varphi\sin\varphi + \left(f(\varphi)\right)^2(\sin^2\varphi + \cos^2\varphi) \\
&= \left(f(\varphi)\right)^2 + \left(f'(\varphi)\right)^2.
\end{aligned}$$

Hence, applying Proposition 3.2.2 we see that the

$$\boxed{\text{Length of the curve } r = f(\varphi) \text{ is } \int_\alpha^\beta \sqrt{\big(f(\varphi)\big)^2 + \big(f'(\varphi)\big)^2}\, d\varphi.}$$

3.3 Integrals of a Scalar Function

We may consider a curve as a deformed interval. We know how to integrate a function over an interval. In this section we want to learn how to integrate functions defined on a deformed interval, that is, on a curve. Denote the curve by C, and assume that a function, f, with values in $\mathbb{R}$ is defined on C. If P is a point on the curve, denote by $s = s(P)$ the distance measured along the curve from one end of the curve to P. Suppose that the total length of the curve is L. The function f is now only a function of s. Hence we may look at f as a function on the interval $[0, L]$ and define

$$\int_C f\, ds := \int_0^L f(s)\, ds.$$

Now suppose that C is a smooth curve, and that $\boldsymbol{\gamma}(t)$, $t \in [a, b]$, is a regular parametrisation of C. As $\boldsymbol{\gamma}$ is a regular parametrisation, every point, $P \in C$, has a representation $\gamma(t)$ for a unique $t \in [a, b]$ (recall that a parametrisation is one-to-one by definition). According to Proposition 3.2.2 the length of the curve between $\boldsymbol{\gamma}(a)$ and $\boldsymbol{\gamma}(t) = P$ is given by

$$s(t) = \int_a^t \|\boldsymbol{\gamma}'(\tau)\|\, d\tau.$$

By the fundamental theorem of calculus it follows that

$$s'(t) = \|\boldsymbol{\gamma}'(t)\|,$$

and thus by the substitution formula (2.1.6) we see that

$$\int_0^L f(s)\, ds = \int_a^b f(s(t))s'(t)\, dt = \int_a^b f(\boldsymbol{\gamma}(t))\|\boldsymbol{\gamma}'(t)\|\, dt.$$

Hence we make the following definition.

Definition 3.3.1 (Line integral of a scalar function) Suppose that C is a smooth positively oriented curve, and that $\boldsymbol{\gamma}(t)$, $t \in [a, b]$, a regular parametrisation of C. Then for

every continuous scalar function on C we define

$$\boxed{\int_C f\,ds := \int_a^b f(\boldsymbol{\gamma}(t))\|\boldsymbol{\gamma}'(t)\|\,dt.}$$

The integral over a piecewise smooth curve is defined to be the sum over the integrals over the smooth parts of the curve.

One can show that the above definition of line integral is independent of the particular parametrisation $\boldsymbol{\gamma}$ choosen.

Example 3.3.2 Compute $\int_C (x^2 - y)\,ds$ if C is given by $\gamma_1(t) = t$ and $\gamma_2(t) = \cosh t$ with $t \in [0, 1]$.
Solution: We first compute $\|\boldsymbol{\gamma}(t)\| = \sqrt{\left(\gamma_1'(t)\right)^2 + \left(\gamma_2'(t)\right)^2}$. As $\gamma_1'(t) = 1$ and $\gamma_2'(t) = \sinh t$ we get

$$\|\boldsymbol{\gamma}'(t)\| = \sqrt{1 + \sinh^2 t} = \sqrt{\cosh^2 t} = \cosh t.$$

Hence,

$$\begin{aligned}\int_C (x^2 - y)\,ds &= \int_0^1 ((\gamma_1(t))^2 - \gamma_2(t))\sqrt{(\gamma_1'(t))^2 + (\gamma_2'(t))^2)}\,dt \\ &= \int_0^1 (t^2 - \cosh t)\cosh t\,dt = \int_0^1 t^2 \cosh t\,dt - \int_0^1 \cosh^2 t\,dt\end{aligned}$$

We now compute the last two integrals separately. For the first we integrate by parts twice:

$$\begin{aligned}\int_0^1 t^2 \cosh t\,dt &= \left[t^2 \sinh t\right]_0^1 - 2\int_0^1 t \sinh t\,dt \\ &= \sinh 1 - 2\left(\left[t\cosh t\right]_0^1 - \int_0^1 \cosh t\,dt\right) \\ &= \sinh 1 - 2\cosh 1 + 2\left[\sinh t\right]_0^1 \\ &= \sinh 1 - 2\cosh 1 + 2\sinh 1 = 3\sinh 1 - 2\cosh 1\end{aligned}$$

Using a well known identity and integration of parts we have

$$\begin{aligned}\int \cosh^2 t\,dt = \sinh t\cosh t - \int \sinh^2 t\,dt &= \sinh t\cosh t - \int (\cosh^2 t - 1)\,dt \\ &= \sinh t\cosh t + t - \int \cosh^2 t\,dt.\end{aligned}$$

Hence,

$$\int_0^1 \cosh^2 t\, dt = \frac{1}{2}\Big[t + \sinh t \cosh t\Big]_0^1 = \frac{1 + \sinh 1 \cosh 1}{2},$$

and therefore

$$\int_C (x^2 - y)\, ds = 3\sinh 1 - 2\cosh 1 - \frac{1 + \sinh 1 \cosh 1}{2}.$$

This is the required result.

Physical Interpretation of Line Integrals

We finish this section by giving some physical interpretations of the line integral. First consider the case $N = 2$. We think of the curve, C, in the plane to be the boundary of a block of land. We want to put up a fence along that boundary. Its height at a point $(x, y) \in \mathbb{R}^2$ is supposed to be $h(x, y)$, and may vary along different parts. (The fence to the bad neighbours is a bit higher than the fence to the good neighbours.) Before we go to the hardware store we want to know how much fencing we need to buy, so we want to compute the surface area of the fence. This is exactly what the line integral gives us, so the area is

$$\int_C h\, ds.$$

Another possible physical interpretation is the following. Suppose the curve C has the shape of a bent wire. The wire is made of different materials, or its thickness slightly varies. Fix a point, P, on C and let S be a piece of length Δs of C containing P. If Δm is the mass of S then $\Delta m/\Delta s$ is the average mass density of S. Shrinking Δs to zero we get the mass density of the wire at P, that is, the mass per length. Denote that mass density at $(x, y, z) \in C$ by $\varrho(x, y, z)$. Then the total mass of the wire is given by

$$\int_C \varrho\, ds.$$

In the next section we motivate and define line integrals of vector fields. The definition will be reduced to the one given here.

3.4 Integrals of a Vector Field

We now study integrals along curves of vector valued functions. If $\boldsymbol{f}$ is a function defined on $D \subset \mathbb{R}^N$ to $\mathbb{R}^N$ we can "attach" the image $\boldsymbol{f}(\boldsymbol{x})$ of every point $\boldsymbol{x} \in D$ to $\boldsymbol{x}$. As a result we get what we call a *vector field*. There are many physical interpretations of vector fields. For instance $\boldsymbol{f}(\boldsymbol{x})$ is the velocity of a fluid at the point $\boldsymbol{x}$, or the force

acting on a particle. In the latter case we talk about a 'force field.' Examples of force fields are the electric, magnetic or gravitational fields. As an example of a vector field consider $\boldsymbol{f}(x, y) := (xy, x^2 - y^2)$ shown in Figure 3.4.1.

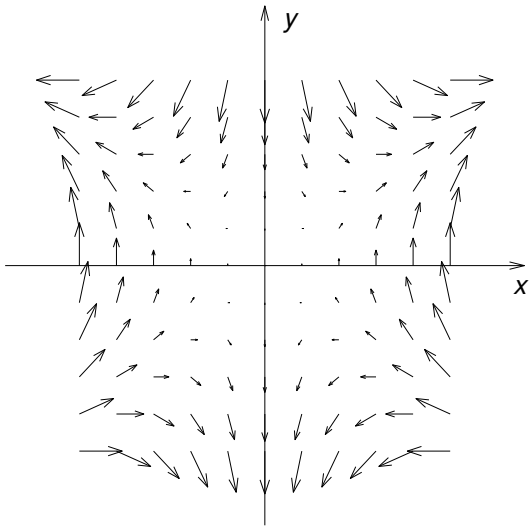


Figure 3.4.1: The vector field $\boldsymbol{f}(x, y) := (xy, x^2 - y^2)$ in $\mathbb{R}^2$.

To motivate the definition of line integrals we consider $\boldsymbol{f}$ to be a force field. We want to estimate the work done moving along the curve from one end to the other. Quite obviously it matters in which direction we move. Hence we look at a positively oriented curve as defined at the end of Section 3.1, and assume that a particle travels in the positive direction from one end of the curve to the other. Recall that in physics work is defined by the force times the distance travelled. Only the component in the direction travelled counts. To estimate the work done moving along C we partition C by n points P_0 to P_n. Denote the distance along C from P_{i-1} to P_i by Δs_i. If $\boldsymbol{f}$ is a continuous vector field, then $\boldsymbol{f}(P)$ is almost constant between P_{i-1} and P_i provided Δs_i is sufficiently small. Moreover, the segment of the curve between P_{i-1} and P_i is almost on the tangent to C at P_{i-1}. The situation is depicted in Figure 3.4.2. The component of $\boldsymbol{f}(P_{i-1})$ into the direction of the tangent is given by $\|\boldsymbol{f}(P_{i-1})\| \cos\varphi$. If $\boldsymbol{\tau}(P_{i-1})$ denotes the unit tangent vector at P_{i-1} to C as shown in Figure 3.4.2 we can rewrite this component as a scalar product $\boldsymbol{f}(P_{i-1}) \cdot \boldsymbol{\tau}(P_{i-1})$ (see Remark 1.1.3). Hence the work done travelling from P_{i-1} to P_i is approximately $\boldsymbol{f}(P_{i-1}) \cdot \boldsymbol{\tau}(P_{i-1})\Delta s_i$. The total work moving along the curve is therefore close to

$$\sum_{i=1}^{n} f(P_{i-1}) \cdot \boldsymbol{\tau}(P_{i-1})\Delta s_i.$$

The above sum looks like a Riemann sum for the scalar valued function $\boldsymbol{f} \cdot \boldsymbol{\tau}$ along the

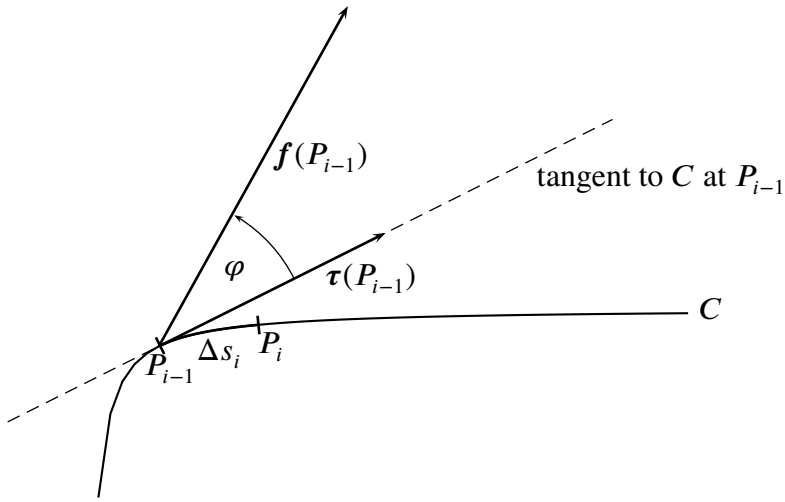


Figure 3.4.2: Work done moving along a small segment of the curve C.

curve C. Using the theory from the previous section the limit of the sum as $\max \Delta s_i$ tends to zero is $\int_C \boldsymbol{f} \cdot \boldsymbol{\tau}\, ds$. If $\boldsymbol{\gamma}(t)$, $t \in [a, b]$, is a regular parametrisation consistent with the orientation of C then the positive unit tangent vector $\boldsymbol{\tau}(t)$ at $\boldsymbol{\gamma}(t)$ is given by (3.1.1). Hence applying Definition 3.3.1 to $\boldsymbol{f} \cdot \boldsymbol{\tau}$ we get

$$\int_C \boldsymbol{f} \cdot \boldsymbol{\tau}\, ds = \int_a^b \boldsymbol{f}(\boldsymbol{\gamma}(t)) \cdot \frac{\boldsymbol{\gamma}'(t)}{\|\boldsymbol{\gamma}'(t)\|} \|\boldsymbol{\gamma}'(t)\|\, dt = \int_a^b \boldsymbol{f}(\boldsymbol{\gamma}(t)) \cdot \boldsymbol{\gamma}'(t)\, dt.$$

Setting $x_i = \gamma_i(t)$ we can formally write $dx_i = \gamma_i'(t)dt$. Hence, if $\boldsymbol{x} = (x_1, \dots, x_N)$ and $\boldsymbol{\gamma} = (\gamma_1, \dots, \gamma_N)$ as usual, we see that formally

$$\int_C \boldsymbol{f} \cdot \boldsymbol{\tau}\, ds = \int_a^b \sum_{i=1}^N f_i(\boldsymbol{\gamma}(t))\gamma_i'(t)dt = \int_C \sum_{i=1}^N f_i(\boldsymbol{x})dx_i = \int_C \boldsymbol{f}(\boldsymbol{x}) \cdot\, d\boldsymbol{x}.$$

The last expression is obtained by formally taking the scalar product of $\boldsymbol{f}(\boldsymbol{x})$ with $d\boldsymbol{x} := (dx_1, \dots, dx_N)$. The above motivates the following definition.

Definition 3.4.1 (Line integral of vector fields) Let C be a smooth positively oriented curve, and $\boldsymbol{\gamma}(t)$, $t \in [a, b]$, a regular parametrisation of C consistent with the orientation of C. Then for every continuous vector field $\boldsymbol{f}$ on C we define

$$\boxed{\int_C \boldsymbol{f} \cdot\, d\boldsymbol{x} := \int_C \boldsymbol{f} \cdot \boldsymbol{\tau}\, ds = \int_a^b \boldsymbol{f}(\boldsymbol{\gamma}(t)) \cdot \boldsymbol{\gamma}'(t)\, dt.}$$

The integral over a piecewise smooth curve is defined to be the sum over the integrals over the smooth parts of the curve.

As we integrate the tangential component along the curve we also talk about the *flux of* $\boldsymbol{f}$ *along the curve* C.

Remark 3.4.2 (Physical interpretation) If $\boldsymbol{f}$ is a force field, then, as shown before, $\int_C \boldsymbol{f} \cdot d\boldsymbol{x}$ is the work done by a particle moving along C. If $\boldsymbol{f}$ describes the motion of a medium (fluid or gas) then the same arguments as above show that $\int_C \boldsymbol{f} \cdot d\boldsymbol{x}$ is the net flow along the curve C. If C is a closed curve we talk about the *circulation* of the field along C.

In the special case of a function of three variables, x, y, z and a vector field $\boldsymbol{f} = (P, Q, R)$ the above reads

$$\int_C P\,dx + Q\,dy + R\,dz = \int_a^b Px'(t) + Qy'(t) + Rz'(t)\,dt$$

if $(x(t), y(t), z(t))$, $t \in [a, b]$, is a parametrisation of C consistent with the orientation of C.

☞ **Remark 3.4.3** We saw that it matters how the curve is oriented. As the opposite orientation of a curve C is given by reversing the sign of the unit tangent vectors we have that

$$\boxed{\int_{-C} \boldsymbol{f} \cdot d\boldsymbol{x} = -\int_C \boldsymbol{f} \cdot d\boldsymbol{x}.}$$

Remark 3.4.4 If C is a closed curve then we often write

$$\oint_C \boldsymbol{f} \cdot d\boldsymbol{x} := \int_C \boldsymbol{f} \cdot d\boldsymbol{x}.$$

The circle is really redundant but often useful to remind us that C is a closed curve.

Example 3.4.5 Consider the curve C shown in Figure 3.4.3. The arc is a half circle. Compute the line integral $\displaystyle\int_C 2xy\,dx + x^2\,dy$.
Solution: We compute the integral over the smooth parts, C_1 and C_2, of C separately. We first look at C_1. The function $\boldsymbol{\gamma}(t) := (t, t)$, $t \in [-1, 1]$ is a parametrisation consistent with the orientation of C_1. Hence $\boldsymbol{\gamma}'(t) = (1, 1)$ and therefore

$$\begin{aligned}\int_{C_1} 2xy\,dx + x^2\,dy &= \int_{-1}^1 2\gamma_1(t)\gamma_2(t)\gamma_1'(t) + (\gamma_1(t))^2\gamma_2'(t)\,dt \\ &= \int_{-1}^1 2t^2 + t^2\,dt = \int_{-1}^1 3t^2\,dt = [t^3]_{-1}^1 = 1 - (-1) = 2.\end{aligned}$$

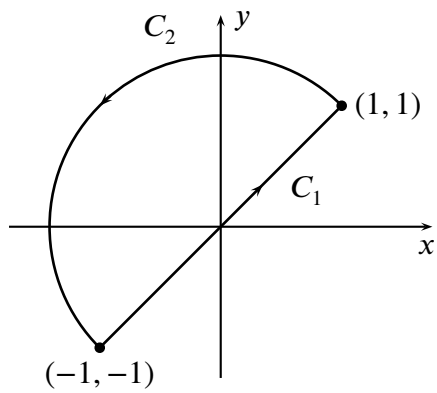


Figure 3.4.3: A piecewise smooth positively oriented closed curve.

Next we compute the integral over C_2. A regular parametrisation, consistent with the orientation, is given by $\boldsymbol{\gamma}(t) := \sqrt{2}(\cos t, \sin t)$, $t \in [\pi/4, 5\pi/4]$. We have $\boldsymbol{\gamma}'(t) = \sqrt{2}(-\sin t, \cos t)$, and so

$$\begin{aligned}
\int_{C_2} 2xy\,dx + x^2\,dy &= \int_{\pi/4}^{5\pi/4} 2\gamma_1(t)\gamma_2(t)\gamma_1'(t) + (\gamma_1(t))^2\gamma_2'(t)\,dt \\
&= \int_{\pi/4}^{5\pi/4} 2(\sqrt{2}\cos t)(\sqrt{2}\sin t)(-\sqrt{2}\sin t) + 2\cos^2 t\sqrt{2}\cos t\,dt \\
&= \int_{\pi/4}^{5\pi/4} -4\sqrt{2}\sin^2 t\cos t + 2\sqrt{2}\cos t(1-\sin^2 t)\,dt \\
&= \int_{\pi/4}^{5\pi/4} -6\sqrt{2}\sin^2 t\cos t + 2\sqrt{2}\cos t\,dt \\
&= \left[-2\sqrt{2}\sin^3 t + 2\sqrt{2}\sin t\right]_{\pi/4}^{5\pi/4} = -2.
\end{aligned}$$

Hence we have

$$\int_C 2xy\,dx + x^2\,dy = \int_{C_1} 2xy\,dx + x^2\,dy + \int_{C_2} 2xy\,dx + x^2\,dy = 2 - 2 = 0.$$

3.5 Conservative Vector Fields

Many vector fields arising in physics have the property that the integral along a curve only depends on the two endpoints of the curve. We say that path independence holds. Such vector fields are called *conservative vector fields*. The reason is that in such fields *conservation laws* hold. Examples include the gravitational field, where we have conservation of energy, that is, the energy of a particle moving along some curve is constant in time. The force field of a ball shaped mass sitting at the origin of a Cartesian coordinate

system in $\mathbb{R}^3$ for instance is given by

$$\boldsymbol{f}(\boldsymbol{x}) = -c\frac{\boldsymbol{x}}{\|\boldsymbol{x}\|^3},$$

where c is a constant depending on the mass of that body and the system of units used. You can check that

$$\boldsymbol{f}(\boldsymbol{x}) = \operatorname{grad}\frac{c}{\|\boldsymbol{x}\|},$$

so $\boldsymbol{f}$ is the gradient of $V(\boldsymbol{x}) := c/\|\boldsymbol{x}\|$. The function V is called the *potential* of the gravitational field. Fields with similar properties include the electrostatic field (but not the magnetic field). A field which is the gradient of a function, V, is generally called a *gradient field*, and the function V a potential. In this section we want to investigate the relationship between path independence, gradient fields, potentials and related properties of vector fields.

3.5.1 Gradient Vector Fields and Potentials

In this section we want to characterise vector fields in which path independence holds. A precise definition of path independence is given below.

Definition 3.5.1 (Path independence) Suppose that $\boldsymbol{f}$ is a continuous vector field on an open set $D \subset \mathbb{R}^N$. We say that line integrals are *path independent* in $\boldsymbol{f}$ if

$$\int_C \boldsymbol{f} \cdot d\boldsymbol{x}$$

has the same value for every piecewise smooth curve C in D with the same start and end points. A vector field for which line integrals are path independent is called a *conservative vector field*, or simply a *conservative field*.

Using Remark 3.4.3 we easily deduce the following proposition.

Proposition 3.5.2 *A vector field, $\boldsymbol{f}$, on $D \subset \mathbb{R}^N$ is conservative if and only if*

$$\oint_C \boldsymbol{f} \cdot d\boldsymbol{x} = 0$$

for every piecewise smooth closed curve C in D.

We will show in this section that vector fields in which line integrals are path independent are what we call gradient vector fields.

Definition 3.5.3 (Gradient field/Potential) Suppose that $D \subset \mathbb{R}^N$ is open, and that $\boldsymbol{f}$ is a continuous vector field on D. We say that $\boldsymbol{f}$ is a *gradient vector field*, or simply a *gradient field*, if there exists a function $V : D \to \mathbb{R}$ such that

$$\boldsymbol{f}(\boldsymbol{x}) = \operatorname{grad} V(\boldsymbol{x})$$

for all $\boldsymbol{x} \in D$. The function V is called a *potential for* $\boldsymbol{f}$.

Note that a potential of a vector field is *never unique*! It is determined only up to a constant. We next show that path independence holds in gradient vector fields.

Theorem 3.5.4 (Gradient fields are conservative) *Suppose that $\boldsymbol{f}$ is a continuous gradient vector field on the open set $D \subset \mathbb{R}^N$. If V is a potential for $\boldsymbol{f}$ then for every piecewise smooth curve, C, connecting $\boldsymbol{a}$ to $\boldsymbol{b}$ in D we have*

$$\int_C \boldsymbol{f} \cdot d\boldsymbol{x} = V(\boldsymbol{b}) - V(\boldsymbol{a}).$$

In particular, $\boldsymbol{f}$ is a conservative vector field.

Proof. The statement of the theorem looks very similar to the fundamental theorem of calculus. In fact, its proof is based on the fundamental theorem of calculus. Suppose that $\boldsymbol{\gamma}(t)$, $t \in [\alpha, \beta]$ is a regular parametrisation of a smooth curve, C, connecting $\boldsymbol{a}$ and $\boldsymbol{b}$ in D. Hence $\boldsymbol{\gamma}(\alpha) = \boldsymbol{a}$ and $\boldsymbol{\gamma}(\beta) = \boldsymbol{b}$. We then define the function φ by

$$\varphi(t) := V(\boldsymbol{\gamma}(t))$$

for all $t \in [\alpha, \beta]$. By the chain rule (see Theorem 1.4.15) we deduce that

$$\frac{d}{dt}\varphi(t) = \operatorname{grad} V(\boldsymbol{\gamma}(t)) \cdot \boldsymbol{\gamma}'(t).$$

As V is a potential for $\boldsymbol{f}$ we have $\boldsymbol{f} = \operatorname{grad} V$ and thus

$$\frac{d}{dt}\varphi(t) = \operatorname{grad} V(\boldsymbol{\gamma}(t)) \cdot \boldsymbol{\gamma}'(t) = \boldsymbol{f}(\boldsymbol{\gamma}(t)) \cdot \boldsymbol{\gamma}'(t).$$

Now by definition of the line integral and the fundamental theorem of calculus we see that

$$\int_C \boldsymbol{f} \cdot d\boldsymbol{x} = \int_\alpha^\beta \boldsymbol{f}(\boldsymbol{\gamma}(t)) \cdot \boldsymbol{\gamma}'(t)\, dt = \int_\alpha^\beta \frac{d}{dt}\varphi(t)\, dt = \varphi(\beta) - \varphi(\alpha).$$

By definition of φ and the fact that $\boldsymbol{\gamma}(\alpha) = \boldsymbol{a}$ and $\boldsymbol{\gamma}(\beta) = \boldsymbol{b}$ we finally have

$$\int_C \boldsymbol{f} \cdot d\boldsymbol{x} = \varphi(\beta) - \varphi(\alpha) = V(\boldsymbol{\gamma}(\beta)) - V(\boldsymbol{\gamma}(\alpha)) = V(\boldsymbol{b}) - V(\boldsymbol{a})$$

as required. If C is only piecewise smooth then we do the above calculation for each smooth part. Adding up the results the intermediate terms cancel, and only the first and the last remain, whence the assertion of the theorem. ■

We just saw that gradient vector fields are conservative fields. We now ask whether a conservative field, $\boldsymbol{f}$, is a gradient field. This in fact turns out to be the case. What we need to do is to construct a potential for $\boldsymbol{f}$. The idea to do so is as follows. To start with suppose that $\boldsymbol{f}$ is a gradient vector field. Fix a point, $\boldsymbol{a}$, in the domain, $D \subset \mathbb{R}^N$, of $\boldsymbol{f}$, and suppose that $C_{\boldsymbol{x}}$ is a smooth curve connecting $\boldsymbol{a}$ to $\boldsymbol{x}$. According to the above theorem

$$V(\boldsymbol{x}) = V(\boldsymbol{a}) + \int_{C_{\boldsymbol{x}}} \boldsymbol{f} \cdot d\boldsymbol{x},$$

so up to a constant V is a line integral along a curve connecting $\boldsymbol{a}$ to $\boldsymbol{x}$. Replacing $V(\boldsymbol{x})$ by $V(\boldsymbol{x}) - V(\boldsymbol{a})$ we have

$$V(\boldsymbol{x}) = \int_{C_{\boldsymbol{x}}} \boldsymbol{f} \cdot d\boldsymbol{x} \tag{3.5.1}$$

The point $\boldsymbol{a}$ is called a *reference point* for the potential as $V(\boldsymbol{a}) = 0$. If $\boldsymbol{f}$ is a conservative vector field, that is, a field for which we have path independence, we can define a function, V, by (3.5.1) by fixing $\boldsymbol{a}$ and choosing for every $\boldsymbol{x} \in D$ a smooth curve, $C_{\boldsymbol{x}}$, connecting $\boldsymbol{a}$ to $\boldsymbol{x}$. By assumption the function V is well defined as it does not depend on the particular choice of $C_{\boldsymbol{x}}$. Of course, we need to assume that such a curve $C_{\boldsymbol{x}}$ exists for each $\boldsymbol{x} \in D$. Open sets having this property are called *connected domains*. We hope that V defined as above is a potential for $\boldsymbol{f}$. The following theorem, a converse to Theorem 3.5.4, shows that this is in fact the case.

Theorem 3.5.5 (Conservative fields are gradient fields) *Every conservative field, $\boldsymbol{f}$, defined on an open connected domain, $D \subset \mathbb{R}^N$, is a gradient vector field. If we fix a reference point $\boldsymbol{a} \in D$, and for every $\boldsymbol{x} \in D$ a smooth curve $C_{\boldsymbol{x}}$, connecting $\boldsymbol{a}$ to $\boldsymbol{x}$, then a potential is defined by* (3.5.1).

Proof. Denote by $\boldsymbol{e}_i$ the i-th standard basis vector of $\mathbb{R}^N$. Given V as above we want to show that for all $x \in D$

$$\frac{\partial}{\partial x_i} V(\boldsymbol{x}) = \lim_{s \to 0} \frac{V(\boldsymbol{x} + s\boldsymbol{e}_i) - V(\boldsymbol{x})}{s} = f_i(\boldsymbol{x}), \tag{3.5.2}$$

exists for all $i = 1, \dots, N$. Here, as usual, f_i is the i-th component function of $\boldsymbol{f}$. If we are able to prove the above then clearly $\boldsymbol{f} = \operatorname{grad} V$, and we are done. Now fix $\boldsymbol{x} \in D$ and $i \in \{1, \dots, N\}$. Suppose that C_0 and C_s are smooth curves connecting the reference point $\boldsymbol{a}$ to $\boldsymbol{x}$ and $\boldsymbol{x} + s\boldsymbol{e}_i$, respectively. Moreover, let Σ_s denote the line segment joining

$\boldsymbol{x}$ and $\boldsymbol{x} + s\boldsymbol{e}_i$. As D is open, Σ_s is completely contained in D if s is sufficiently small. The situation is depicted in Figure 3.5.1.

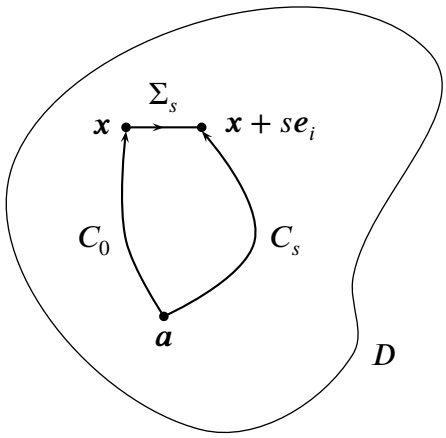


Figure 3.5.1: Paths connecting to the reference point.

As line integrals are path independent in $\boldsymbol{f}$ by assumption, we have

$$\int_{C_s} \boldsymbol{f} \cdot d\boldsymbol{x} = \int_{C_0} \boldsymbol{f} \cdot d\boldsymbol{x} + \int_{\Sigma_s} \boldsymbol{f} \cdot d\boldsymbol{x}.$$

Using the definition (3.5.1) of V this implies that

$$V(\boldsymbol{x} + s\boldsymbol{e}_i) = V(\boldsymbol{x}) + \int_{\Sigma_s} \boldsymbol{f} \cdot d\boldsymbol{x},$$

and therefore, rearranging the terms and dividing by s we get

$$\frac{V(\boldsymbol{x} + s\boldsymbol{e}_i) - V(\boldsymbol{x})}{s} = \frac{1}{s} \int_{\Sigma_s} \boldsymbol{f} \cdot d\boldsymbol{x}. \tag{3.5.3}$$

We next compute the limit of the right hand side as s tends to zero. To do so we parametrise Σ_s by $\boldsymbol{\gamma}(t) := \boldsymbol{x} + t\boldsymbol{e}_i$, $t \in [0, s]$. Then $\boldsymbol{\gamma}'(t) = \boldsymbol{e}_i$ for all t. As $\boldsymbol{f} \cdot \boldsymbol{e}_i = f_i$ we get

$$\frac{1}{s} \int_{\Sigma_s} \boldsymbol{f} \cdot d\boldsymbol{x} = \frac{1}{s} \int_0^s \boldsymbol{f}(\boldsymbol{\gamma}(t)) \cdot \boldsymbol{\gamma}'(t)\, dt = \frac{1}{s} \int_0^s f_i(\boldsymbol{x} + t\boldsymbol{e}_i)\, dt.$$

The limit of the last expression as $s \to 0$ is the derivative of the function $g(s) := \int_0^s f_i(\boldsymbol{x} + t\boldsymbol{e}_i)\, dt$ at $s = 0$. By the fundamental theorem of calculus we therefore have that

$$\lim_{s \to 0} \frac{1}{s} \int_0^s f_i(\boldsymbol{x} + t\boldsymbol{e}_i)\, dt = g'(0) = f_i(\boldsymbol{x} + 0\boldsymbol{e}_i) = f_i(\boldsymbol{x}).$$

Together with (3.5.3) it follows that (3.5.2) is true, completing the proof of the theorem.

■

The above is not very useful to determine whether a given vector field is conservative or not. In the next section we want to study conditions which guarantee that a vector field is a gradient field.

3.5.2 Closed Vector Fields and the Curl

Checking path independence for a given vector field by evaluating the integral over *every* closed curve is an impossible task! Hence, in practice, we need other criteria to check whether a given vector field is conservative. In this section we want to introduce such a criterion. Assume for the moment that $\boldsymbol{f} = (f_1, \dots, f_N)$ is a continuously differentiable gradient vector field with potential V on some open subset D of $\mathbb{R}^N$. By the symmetry of the second partial derivatives (see Proposition 1.4.32) and since $\boldsymbol{f} = \operatorname{grad} V$ we have

$$\frac{\partial}{\partial x_i} f_j(\boldsymbol{x}) = \frac{\partial}{\partial x_i}\frac{\partial}{\partial x_j} V(\boldsymbol{x}) = \frac{\partial}{\partial x_j}\frac{\partial}{\partial x_i} V(\boldsymbol{x}) = \frac{\partial}{\partial x_j} f_i(\boldsymbol{x})$$

for all $\boldsymbol{x} \in D$ and $1 \le i, j \le N$. We have seen in the previous section that gradient and conservative fields are the same. Hence we have the following proposition.

Proposition 3.5.6 *Suppose that $\boldsymbol{f} = (f_1, \dots, f_N)$ is a continuously differentiable conservative vector field on $D \subset \mathbb{R}^N$. Then*

$$\boxed{\frac{\partial}{\partial x_i} f_j(\boldsymbol{x}) = \frac{\partial}{\partial x_j} f_i(\boldsymbol{x})} \tag{3.5.4}$$

for all $\boldsymbol{x} \in D$ and $1 \le i, j \le N$.

Hence for a vector field $\boldsymbol{f}$ to be conservative, (3.5.4) must be satisfied. The question arising is whether (3.5.4) *guarantees* that $\boldsymbol{f}$ is conservative. Unfortunately, the answer is NO in general (see Example 3.5.9 below)! However, under additional assumptions on D we will be able to show that (3.5.4) is sufficient to ensure that $\boldsymbol{f}$ is a gradient field. Because of their significance in any 'field theory,' vector fields satisfying condition (3.5.4) have a special name.

Definition 3.5.7 (Closed vector field) A vector field, $\boldsymbol{f} = (f_1, \dots, f_N)$ is said to be *closed* on $D \subset \mathbb{R}^N$ if (3.5.4) holds for all $\boldsymbol{x} \in D$ and $1 \le i, j \le N$.

We mentioned above that it depends on the domain D whether every closed vector field on D is conservative or not. The condition on D is a topological one. We need to make sure that every closed curve in D can be continuously shrunk to a point in D without

ever leaving D. Figure 3.5.2 shows a simply connected domain in the plane. If a plane domain has a hole like the one in Figure 3.5.3, a curve around the hole cannot be shrunk to a point in D. One can then prove the following theorem.

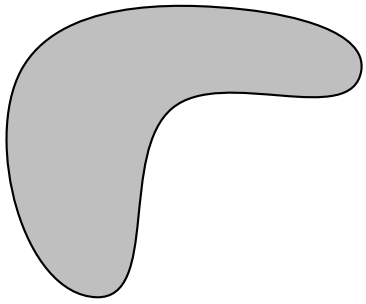


Figure 3.5.2: A simply connected domain.

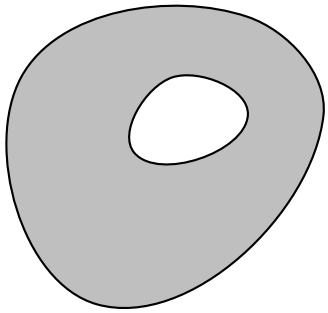


Figure 3.5.3: A domain not simply connected.

Theorem 3.5.8 *Every closed vector field on a simply connected domain in $\mathbb{R}^N$ is conservative.*

A proof of the above theorem in full generality is out of the scope of these notes. Some special cases can easily be obtained as applications of the theorems of Green and Stokes in Chapter 5. For $N = 2$ and $N = 3$ we will give a complete proof in Section 5.1.4 and 5.4.3. We now look at the special cases of $N = 2$ and $N = 3$.

Special Case: Vector Fields in the Plane

If $\boldsymbol{f} = (f_1, f_2)$ is a vector field on a subset of the plane condition then (3.5.4) reduces to

$$\boxed{\frac{\partial}{\partial x_2} f_1(\boldsymbol{x}) = \frac{\partial}{\partial x_1} f_2(\boldsymbol{x})} \tag{3.5.5}$$

for all $\boldsymbol{x}$ in the domain of $\boldsymbol{f}$. We now give an example of a closed but not conservative vector field in the plane. The example will occur many times in the sequel, so make sure you understand it well.

☞ **Example 3.5.9** Let $\boldsymbol{f}$ be the vector field given by

$$\boldsymbol{f}(x, y) = \left(\frac{-y}{x^2 + y^2}, \frac{x}{x^2 + y^2}\right)$$

for $(x, y) \neq (0, 0)$. It is shown in Figure 3.5.4. Note that $\|\boldsymbol{f}\| = \|(x, y)\|^{-1}$, so the length

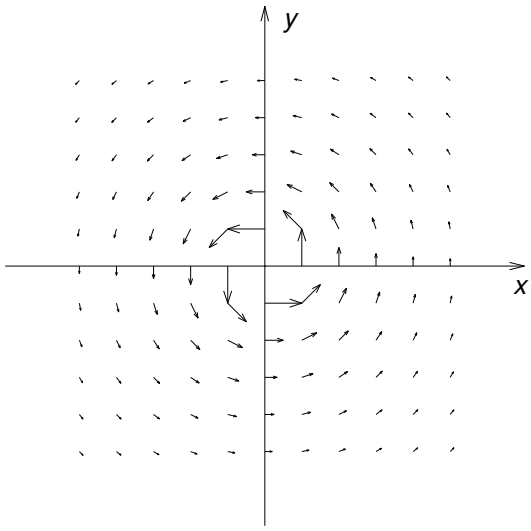


Figure 3.5.4: The vector field $\boldsymbol{f}(x, y) := \left(\dfrac{-y}{x^2+y^2}, \dfrac{x}{x^2+y^2}\right)$ in $\mathbb{R}^2$.

of the vectors go to infinity as (x, y) approaches $(0, 0)$. We say that $\boldsymbol{f}$ has a singularity at $(0, 0)$. We check whether $\boldsymbol{f}$ is closed:

$$\frac{\partial}{\partial y}\frac{-y}{x^2+y^2} = \frac{-(x^2+y^2)+2y^2}{(x^2+y^2)^2} = \frac{y^2-x^2}{(x^2+y^2)^2};$$
$$\frac{\partial}{\partial x}\frac{x}{x^2+y^2} = \frac{x^2+y^2-2y^2}{(x^2+y^2)^2} = \frac{y^2-x^2}{(x^2+y^2)^2}.$$

Hence $\boldsymbol{f}$ is indeed a closed vector field on $\mathbb{R}^2 \setminus \{\mathbf{0}\}$. We now want to compute the integral along the circle of radius R centred at $(0, 0)$. Assume that the circle is oriented counterclockwise. We parametrise it by

$$\boldsymbol{\gamma}(t) = (R\cos t, R\sin t), \qquad t \in [0, 2\pi].$$

Then $\boldsymbol{\gamma}'(t) = (-R\sin t, R\cos t)$, and using the definition of the line integral we see that

$$\int_C \boldsymbol{f}\cdot d\boldsymbol{x} = \int_0^{2\pi} \frac{R\sin t}{R^2\sin^2 + R^2\cos^t} R\sin t + \int_0^{2\pi} \frac{R\cos t}{R^2\sin^2 + R^2\cos^t} R\cos t\, dt$$
$$= \int_0^{2\pi} \frac{R^2\cos t + R^2\sin^2}{R^2\sin^2 + R^2\cos^t}\, dt = \int_0^{2\pi} 1\, dt = 2\pi.$$

The circle is a closed curve. Hence if $\boldsymbol{f}$ were conservative the integral would be zero. As it is not zero, $\boldsymbol{f}$ cannot be conservative, even though it is closed. Note that $\mathbb{R}^2 \setminus \{\mathbf{0}\}$ is not simply connected, so Theorem 3.5.8 does not apply!

Special Case: Vector Fields in Space

If $N = 3$ and and $\boldsymbol{f} = (f_1, f_2, f_3)$ is a vector field then $\boldsymbol{f}$ is closed if

$$\frac{\partial}{\partial x_2} f_3(\boldsymbol{x}) = \frac{\partial}{\partial x_3} f_2(\boldsymbol{x}), \quad \frac{\partial}{\partial x_3} f_1(\boldsymbol{x}) = \frac{\partial}{\partial x_1} f_3(\boldsymbol{x}), \quad \frac{\partial}{\partial x_2} f_1(\boldsymbol{x}) = \frac{\partial}{\partial x_1} f_2(\boldsymbol{x}).$$

If we introduce a new differential expression on vector fields in $\mathbb{R}^3$ we can state the above conditions in a very concise form.

Definition 3.5.10 (Curl of a vector field) If $\boldsymbol{f} = (f_1, f_2, f_3)$ is a vector field defined on a subset of $\mathbb{R}^3$ we set

$$\boxed{\operatorname{curl} \boldsymbol{f}(\boldsymbol{x}) := \nabla \times \boldsymbol{f}(\boldsymbol{x}) := \begin{bmatrix} \frac{\partial}{\partial x_2} f_3(\boldsymbol{x}) - \frac{\partial}{\partial x_3} f_2(\boldsymbol{x}) \\ \frac{\partial}{\partial x_3} f_1(\boldsymbol{x}) - \frac{\partial}{\partial x_1} f_3(\boldsymbol{x}) \\ \frac{\partial}{\partial x_1} f_2(\boldsymbol{x}) - \frac{\partial}{\partial x_2} f_1(\boldsymbol{x}) \end{bmatrix}.} \tag{3.5.6}$$

☞ **Remark 3.5.11** The symbol ∇ is called the *nabla operator*, and can be thought of as the vector

$$\nabla = \left(\frac{\partial}{\partial x_1}, \frac{\partial}{\partial x_2}, \frac{\partial}{\partial x_3} \right).$$

Then the curl of $\boldsymbol{f}$ is formally the vector product of ∇ with $\boldsymbol{f}$ as defined in (1.1.8). Hence the notation $\nabla \times \boldsymbol{f}$. Moreover, the gradient of a scalar function, f, is formally the multiplication by scalars of the 'vector' ∇ and the scalar f, so we write ∇f for the gradient. In the more applied literature most of the time the 'nabla operator' is used.

With these definitions we have the following.

> A vector field, $\boldsymbol{f}$, defined on a subset of $\mathbb{R}^3$ is closed if and only if $\operatorname{curl} \boldsymbol{f} = 0$ on its domain.

The curl will appear again later when discussing the Theorem of Stokes.

☞ **Remark 3.5.12** Let us make a remark about the relationship of the curl with the condition (3.5.5) for a plane vector field to be closed. If (f_1, f_2) is a plane vector field we can always define a vector field $\boldsymbol{f}(x_1, x_2, x_3) := (f_1(x_1, x_2), f_2(x_1, x_2), 0)$ in $\mathbb{R}^3$. (The vectors $\boldsymbol{f}(\boldsymbol{x})$ do not depend on the third variable.) By definition of the curl we get

$$\operatorname{curl} \boldsymbol{f}(\boldsymbol{x}) = \begin{bmatrix} 0 \\ 0 \\ \frac{\partial}{\partial x_1} f_2(\boldsymbol{x}) - \frac{\partial}{\partial x_2} f_1(\boldsymbol{x}) \end{bmatrix}.$$

Hence curl $\boldsymbol{f}(\boldsymbol{x}) = 0$ if and only if (3.5.5) holds. Hence,

$$\frac{\partial}{\partial x_1} f_2(\boldsymbol{x}) - \frac{\partial}{\partial x_2} f_1(\boldsymbol{x})$$

is often called the *curl of a plane vector field*. Using the Theorem of Stokes we will give a physical interpretation of the curl. For details see Section 5.2.2 on page 141.

Chapter 4

Surface Integrals

Intuitively, a surface is a deformed plane domain lying in space. As a model you can think of a piece of paper (or a flat stretchable material) which you pick up from the table and deform it, possibly gluing together some edges. In this section we want to learn how to integrate functions defined on surfaces. This yields a generalisation of double integrals over plane domains to double integrals over curved domains, that is, surfaces. Similarly, in Chapter 3, line integrals were a generalisation of integrals over an interval to integrals along a "bent" interval, that is, a curve.

4.1 Representation of Surfaces

In this section we introduce possible ways of describing a given surface, the parametric, the implicit and the explicit representations. For 'smooth' surfaces it turns out that it is possible, at least in principle, to describe the surface by any of the three. It depends on the purpose which form is more convenient.

4.1.1 Parametric Representation of Surfaces

As noted in the introduction to the present chapter, a surface, S, is a deformed plane domain, $D \subset \mathbb{R}^2$, lying in space. Hence to describe S mathematically we need to say which point in D corresponds to what point on S, just as a map of the world (represented by D) shows the geographic locations in the real world (the surface S). This correspondence defines a (vector valued) one-to-one function $\boldsymbol{g}$ from D to $\mathbb{R}^3$ whose image is S. The function $\boldsymbol{g}: D \to \mathbb{R}^3$ is called a *parametrisation* of S. Some surfaces cannot be represented by one single parametrisation, or it is not convenient to do so. (To represent the world we often use a collection of maps, an atlas, and not a single map of

the whole world.) As with parametrisations of a curve, a parametrisation of a surface is never unique! We call a representation by means of a parametrisation a *parametric representation* of a surface.

The above class of surfaces is too big. We only want to work with smooth (or at least piecewise smooth) surfaces. To define exactly what we mean by a smooth surface let us assume that $\boldsymbol{g}: D \to \mathbb{R}^3$ is the parametrisation of (part of) a surface. Suppose that $\boldsymbol{a} = (a_1, a_2) \in D$, so that $\boldsymbol{g}(\boldsymbol{a})$ is the corresponding point on S. Then the map

$$x_1 \to \boldsymbol{g}(x_1, a_2)$$

is the parametrisation of a curve on S through $\boldsymbol{g}(\boldsymbol{a})$. Similarly, the map

$$x_2 \to \boldsymbol{g}(a_1, x_2)$$

is the parametrisation of another curve on S through $\boldsymbol{g}(\boldsymbol{a})$. In order for S to be smooth we assume that these curves are smooth. More precisely this means that the above are regular parametrisations, so in particular we require that

$$\boldsymbol{v}_i := \frac{\partial}{\partial x_i}\boldsymbol{g}(\boldsymbol{a}) \neq \boldsymbol{0}$$

for $i = 1, 2$. For S to be smooth we require more. If we are very unlucky, the image of D is only a curve, and not a surface. In that case the vectors $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$ are parallel. To prevent this we require in addition to the above that $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$ not be parallel. Both of the above requirements are satisfied if and only if the area of the parallelogram spanned by $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$ is nonzero. In Proposition 1.1.7 on page 11 we derived a formula to compute the area of a parallelogram. We have to form the matrix with columns $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$, which in our present case is the Jacobian matrix $J_{\boldsymbol{g}}(\boldsymbol{a})$ of $\boldsymbol{g}$ at $\boldsymbol{a}$. According to Proposition 1.1.7 the surface area spanned by $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$ is $\sqrt{\det\left(\left(J_{\boldsymbol{g}}(\boldsymbol{y})\right)^T J_{\boldsymbol{g}}(\boldsymbol{y})\right)}$. Because of its significance in the theory of surfaces, or more generally in differential geometry, the the square root has a name.

Definition 4.1.1 (Jacobian of a parametrisation) Suppose that $\boldsymbol{g}: \mathbb{R}^k \to \mathbb{R}^N$ is a differentiable map. Then the expression

$$\sqrt{\det\left(\left(J_{\boldsymbol{g}}(\boldsymbol{y})\right)^T J_{\boldsymbol{g}}(\boldsymbol{y})\right)}$$

is called the *Jacobian* of the parametrisation $\boldsymbol{g}$ at $\boldsymbol{y}$.

Definition 4.1.2 (Smooth surface) Suppose that $\boldsymbol{g}: D \to \mathbb{R}^3$ is the parametrisation of (part of) a surface. We say that $\boldsymbol{g}$ is a *regular parametrisation* of (part of) S, if the

Jacobian of $\boldsymbol{g}$ is nowhere zero. Finally, S is called a *smooth surface*, if it can be fully described by one (or several) regular parametrisation. If several parametrisations are required then their images on S must not overlap by more than just a line.

We also sometimes look at *piecewise smooth* surfaces which are finite unions of smooth surfaces.

☞ **Example 4.1.3** Suppose that S is the sphere of radius R centred at the origin in $\mathbb{R}^3$. We can describe every point on that sphere by indicating two angles, one from the vertical, the other from the horizontal. (Called latitude and longitude on a map of the world). According to the considerations in Section 2.2.3 a parametrisation is given by

$$\boldsymbol{g}(\theta, \varphi) := \big(R\cos\varphi\sin\theta, R\sin\varphi\sin\theta, R\cos\theta\big),$$

where $(\theta, \varphi) \in D := [0, \pi]\times[0, 2\pi)$. This is the same as (2.2.4), except that we keep the radius constant. We next show that $\boldsymbol{g}$ is a regular parametrisation. To do so we compute the Jacobian matrix

$$J_g(\theta, \varphi) = R\begin{bmatrix} \cos\varphi\cos\theta & -\sin\varphi\sin\theta \\ \sin\varphi\cos\theta & \cos\varphi\sin\theta \\ -\sin\theta & 0 \end{bmatrix}.$$

Now we compute the product

$$\begin{aligned}\big(J_g\big)^T J_g &= R^2\begin{bmatrix} \cos\varphi\cos\theta & \sin\varphi\cos\theta & -\sin\theta \\ -\sin\varphi\sin\theta & \cos\varphi\sin\theta & 0 \end{bmatrix}\begin{bmatrix} \cos\varphi\cos\theta & -\sin\varphi\sin\theta \\ \sin\varphi\cos\theta & \cos\varphi\sin\theta \\ -\sin\theta & 0 \end{bmatrix} \\ &= R^2\begin{bmatrix} 1 & 0 \\ 0 & \sin^2\theta \end{bmatrix}\end{aligned}$$

and thus

$$\det\big(\big(J_g\big)^T J_g\big) = R^4\det\begin{bmatrix} 1 & 0 \\ 0 & \sin^2\theta \end{bmatrix} = R^4\sin^2\theta.$$

As $\sin\theta \neq 0$ for $(\theta, \varphi) \in (0, \pi) \times [0, 2\pi)$, the Jacobian is nonzero except for the points at the point on top and the bottom of the sphere ($\theta = 0, \pi$). For most purposes these exceptional points are irrelevant.

Remark 4.1.4 It is the subject of differential geometry to study k-dimensional surfaces in $\mathbb{R}^N$ ($k = 1, \dots, N$). Our definition of regular parametrisation can be generalised to

this context. In particular, if $k = 1$, it is consistent with the notion of regular parametrisation of curves given in Definition 3.1.1. Indeed, if $\boldsymbol{\gamma}(t)$, $t \in I$, is the parametrisation of a curve, C, then the Jacobian matrix of $\boldsymbol{\gamma}(t)$ is

$$J_{\boldsymbol{\gamma}}(t) = \boldsymbol{\gamma}'(t) = \begin{bmatrix} \gamma_1'(t) \\ \vdots \\ \gamma_N'(t), \end{bmatrix}$$

and we get

$$\left(J_{\boldsymbol{\gamma}}(t)\right)^T J_{\boldsymbol{\gamma}}(t) = \boldsymbol{\gamma}'(t)^T \boldsymbol{\gamma}'(t) = \|\boldsymbol{\gamma}'(t)\|^2.$$

We said that $\boldsymbol{\gamma}(t)$, $t \in I$, is a regular parametrisation if $\gamma'(t) \neq 0$ for all $t \in I$, showing that the two definitions are consistent.

4.1.2 Implicit and Explicit Representations

Quite often, surfaces are given as the set of points $\boldsymbol{x} = (x_1, x_2, x_3) \in \mathbb{R}^3$ such that $f(\boldsymbol{x}) = 0$ for some given function $\boldsymbol{f} : \mathbb{R}^3 \to \mathbb{R}$. We call this an *implicit representation* of a surface. In some cases it is possible to solve $f(\boldsymbol{x}) = 0$ for one of the variables. For instance we can solve for x_3. If we can solve with one single function then S is the graph of a function. Graphs of a function h defined on a subset $D \subset \mathbb{R}^3$ are the simplest possible surfaces. We call $x_3 = h(x_1, x_2)$, $(x_1, x_2) \in D$ an *explicit representation* of the surface. In many cases, however, solving for one variable will only be possible locally, even for very simple surfaces like the sphere.

Example 4.1.5 Suppose that S is a sphere of radius R. By definition S is the set of points in $\mathbb{R}^3$ with distance R from the origin. Hence S is the set of all points $\boldsymbol{x} = (x_1, x_2, x_3) \in \mathbb{R}^3$ such that

$$f(\boldsymbol{x}) := x_1^2 + x_2^2 + x_3^2 - R^2 = 0,$$

providing an implicit representation of the sphere. We can solve the above equation for x_3 to get

$$x_3 = \pm\sqrt{R^2 - x_1^2 - x_2^2}.$$

Note that we need two different functions to describe the upper and the lower half of the sphere. The above provides an *explicit representation* of S.

Not for every function f does the equation $f(\boldsymbol{x}) = 0$ describe a smooth surface. For instance the only solution to $x_1^2 + x_2^2 + x_3^2 = 0$ is $(0, 0, 0)$, that is, a single point. Hence

we need to find criteria which guarantee that the solutions of $f(\boldsymbol{x}) = 0$ form a surface. We have the following theorem, which is a generalised form of the implicit function Theorem 1.4.28, where also the ideas for a more general result are indicated.

Theorem 4.1.6 (Implicit function theorem) *Suppose that $f : \mathbb{R}^3 \to \mathbb{R}$ is a function with continuous gradient. Moreover, suppose that* $\operatorname{grad} f(\boldsymbol{x}) \neq \mathbf{0}$ *for all $\boldsymbol{x} \in \mathbb{R}^3$ satisfying $f(\boldsymbol{x}) = 0$. Then the set of points $\boldsymbol{x} \in \mathbb{R}^3$ for which $f(\boldsymbol{x}) = 0$ forms a smooth surface.*

In the above example of the sphere we have $\operatorname{grad} f(\boldsymbol{x}) = (2x_1, 2x_2, 2x_3)$, which is clearly nonzero if $R > 0$ and $x_1^2 + x_2^2 + x_3^2 = R$.
As mentioned already smooth surfaces, in principle, can be represented by all three means. We consider some examples.

Example 4.1.7 (Sphere) We already saw three different representations of the sphere with radius R centred at $\mathbf{0}$:

- parametric representation by spherical coordinates:

$$\boldsymbol{g}(\theta, \varphi) := \big(R\cos\varphi\sin\theta, R\sin\varphi\sin\theta, R\cos\theta\big),$$

  where $(\theta, \varphi) \in D := [0, \pi] \times [0, 2\pi)$;

- implicit representation by $x_1^2 + x_2^2 + x_3^2 - R^2 = 0$;

- explicit representation by $x_3 = \sqrt{R^2 - x_1^2 - x_2^2}$ and $x_3 = -\sqrt{R^2 - x_1^2 - x_2^2}$.

Example 4.1.8 (Graph of a function) Suppose that $h : D \to \mathbb{R}$ is continuously differentiable on the open set $D \subset \mathbb{R}^2$. Then the graph

$$S := \{(x_1, x_2, h(x_1, x_2)) : (x_1, x_2) \in D\}$$

forms a smooth surface in $\mathbb{R}^3$. Then the three possible representations are as follows:

- explicit representation is $x_3 = h(x_1, x_2)$, $(x_1, x_2) \in D$;

- implicit representation is $f(\boldsymbol{x}) := x_3 - h(x_1, x_2) = 0$;

- parametric representation is $\boldsymbol{g}(x_1, x_2) = (x_1, x_2, h(x_1, x_2))$ for $(x_1, x_2) \in D$.

To check that S is smooth just note that $\operatorname{grad} f(\boldsymbol{x}) = (-\operatorname{grad} h(x_1, x_2), 1)$, which is always nonzero. Hence by the above theorem the surface is smooth.

Example 4.1.9 (Plane) Most commonly, a plane in $\mathbb{R}^3$ is given in implicit form by the equation

$$ax + by + cz + d = 0.$$

At least one of a, b, c is nonzero. If for instance $a \neq 0$ we get the explicit representation

$$x = -\frac{1}{a}\Big(by + cz + d\Big).$$

A parametric representation can be obtained as follows. Suppose that $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$ are linearly independent vectors, lying on the plane $ax + by + cz = 0$ (the original plane translated to the origin). If $\boldsymbol{a}$ is a point on the original plane, then

$$\boldsymbol{g}(s,t) := \boldsymbol{a} + s\boldsymbol{v}_1 + t\boldsymbol{v}_2, \qquad (s,t) \in \mathbb{R}^2$$

is a parametric representation of the plane.

4.1.3 Unit Normals and Orientation

We were able to endow a curve with an orientation by using the fact that we can move along it in two opposite directions. This does not work for surfaces. Hence we have to orient them in a different way. If we pick up a sheet of paper then it has two sides, the front and the back. If the sheet is blank then it is not clear which is the front and which is the back. We need to make a choice, and lay it down one way, and say, the side facing up is the front. (Maybe you mark the front by writing a headline on it.) If we have that piece of paper, S, one side 'faces up' and the other side 'faces down.' Mathematically we can express this by attaching to every point of the sheet a vector perpendicular to S pointing 'upwards'. Having done this, we can move the sheet and, looking at the vectors, it is still clear what the 'front' and the 'back' of S are. Hence the vector field chosen 'marks' the front of S, as the headline you put on a sheet of paper, and thus gives it an 'orientation.' The two possible normal fields are shown in Figure 4.1.1.

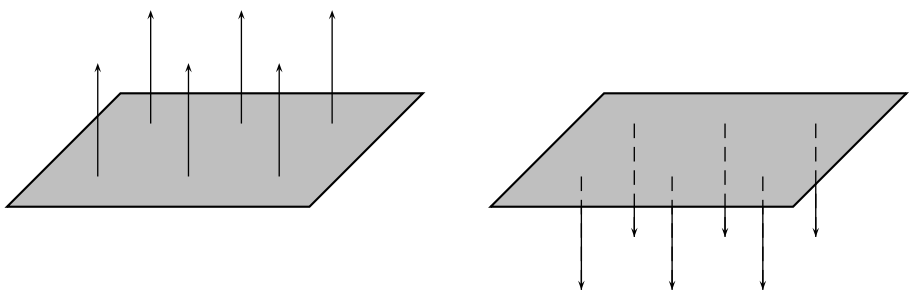


Figure 4.1.1: The two possible orientations of a "sheet of paper."

To define an orientation for a 'curved' smooth surface, S, we proceed the same way as with the sheet of paper. We attach to every point a vector perpendicular to S and normalise it to length one. Such a vector is called a *unit normal vector to* S. As usual normal to S means normal to the tangent plane to S. At every point there are exactly two such vectors. Sometimes it is possible to choose these normal vectors in such a way that they vary continuously as we move around on the surface. As with the sheet of paper these fields can be used to give the S an orientation. Fix one of the vector field and call it the *positive unit normal vector field*. A surface with both possible continuous unit normal vector fields is shown in Figure 4.1.2.

Definition 4.1.10 (Orientation of a surface) Suppose that S is a smooth surface in $\mathbb{R}^3$. The surface is called *orientable* if there exists a continuous unit normal vector field on S. We can fix one of them and call it the *positive unit normal vector field*. Equipped with this vector field S is called *positively oriented*. Equipped with the opposite orientation S is called *negatively oriented*.

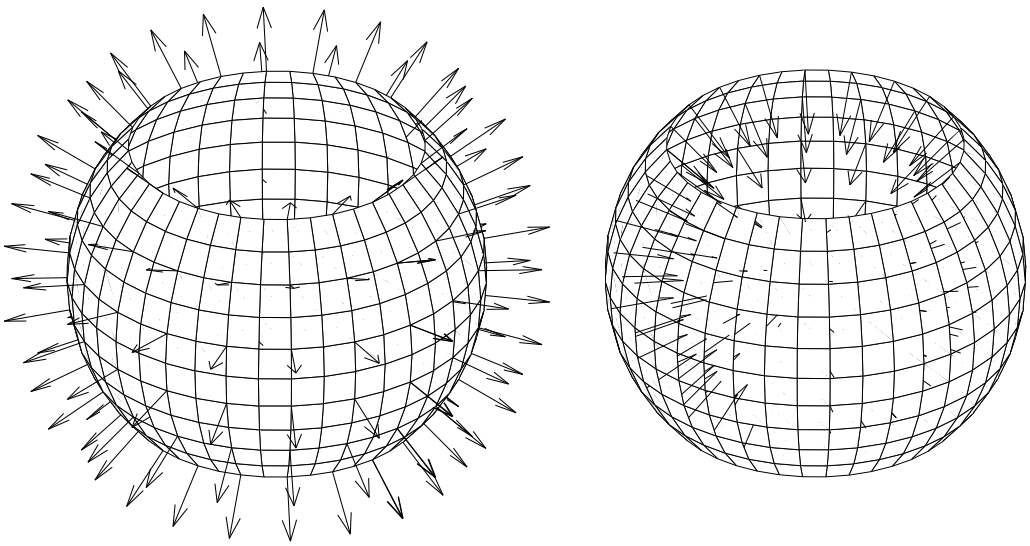


Figure 4.1.2: The two possible unit normal vector fields on a surface giving rise to opposite orientations.

Remark 4.1.11 As opposed to curves, not every surface is orientable! The Möbius strip is the standard example of non-orientable surface. It is impossible to get a *continuous* unit normal vector field. We can start putting unit normals at one point and move continuously along the band back to the same point. Arriving there the unit normal will point in the opposite direction to that in which we started. A Möbius strip and a (discontinu-

ous) unit normal vector field on it is shown in Figure 4.1.3. The reason a Möbius strip is not orientable is that it has only "one side."

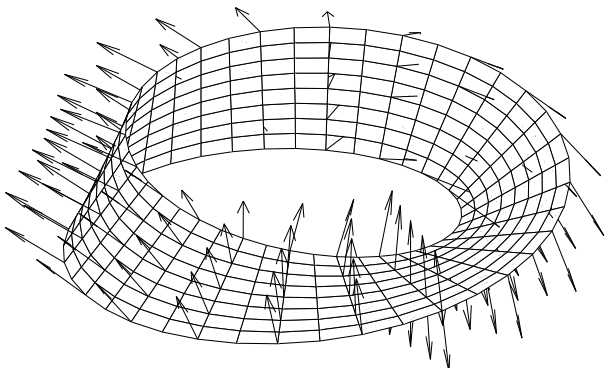


Figure 4.1.3: A non-orientable surface: The Möbius band

Up to now we did not worry about how to compute unit normals to a surface, but want to do so now. There are two possibilities: using an implicit representation, or using a parametric representation of the surface. The first is much easier than the second, but we will be required to use both of them.

Computation of unit normals using implicit representations

Suppose that $\boldsymbol{g}\colon \mathbb{R}^3 \to \mathbb{R}$ is a smooth function, and that the level set $g(\boldsymbol{x}) = 0$ defines a smooth surface. We saw in Theorem 1.4.27 that $\operatorname{grad} f(\boldsymbol{x})$ is normal to S for all $\boldsymbol{x} \in S$. Hence we have the following theorem.

Theorem 4.1.12 (Normal to implicitly given surface) *Suppose that S is a surface given implicitly by $g(\boldsymbol{x}) = 0$, and that $\operatorname{grad} g(\boldsymbol{x}) \neq \mathbf{0}$ for all $\boldsymbol{x} \in S$. Then*

$$\boxed{\boldsymbol{n}(\boldsymbol{x}) = \frac{\operatorname{grad} g(\boldsymbol{x})}{\|\operatorname{grad} g(\boldsymbol{x})\|}} \tag{4.1.1}$$

defines a continuous field of unit normal vectors on S.

As an important special case we might look at a surface which is given explicitly as the graph of a function $h\colon D \to \mathbb{R}$. Then $x_3 - h(x_1, x_2) = 0$ is an implicit representation and therefore

$$\boxed{\boldsymbol{n} = \frac{\big(-\operatorname{grad} h(\boldsymbol{x}), 1\big)}{\sqrt{1 + \|\operatorname{grad} h(\boldsymbol{x})\|^2}}.} \tag{4.1.2}$$

defines a unit normal vector field to the graph of h. The above normal field is pointing upwards, that is, has a positive x_3-component.

Computation of unit normals using parametric representations

Suppose that S is an orientable curve, and that $\boldsymbol{g}: D \to \mathbb{R}^3$ is a regular parametrisation of S. If we fix $\boldsymbol{a} = (a_1, a_2) \in D$, then the maps $x_1 \to \boldsymbol{g}(x_1, a_2)$ and $x_2 \to \boldsymbol{g}(a_1, x_2)$ define two curves on S crossing at $\boldsymbol{g}(\boldsymbol{a})$. The vectors

$$\boldsymbol{v}_i := \frac{\partial \boldsymbol{g}}{\partial x_i}(\boldsymbol{a}),$$

$i = 1, 2$, are tangent vectors to S at $\boldsymbol{g}(\boldsymbol{a})$. As $\boldsymbol{g}$ is a regular parametrisation (see Definition 4.1.2 on page 112) we have that $\boldsymbol{v}_1 \times \boldsymbol{v}_2 \neq \boldsymbol{0}$. Recall that the vector product of two vectors is perpendicular to the original vectors (see Proposition 1.1.11 on page 13). Hence we have the following theorem.

Theorem 4.1.13 (Normal to a parametrised surface) *Suppose that S is a smooth orientable surface with regular parametrisation $\boldsymbol{g}$. Then*

$$\boxed{\boldsymbol{n}(\boldsymbol{x}) = \frac{\dfrac{\partial \boldsymbol{g}}{\partial x_1}(\boldsymbol{x}) \times \dfrac{\partial \boldsymbol{g}}{\partial x_2}(\boldsymbol{x})}{\left\| \dfrac{\partial \boldsymbol{g}}{\partial x_1}(\boldsymbol{x}) \times \dfrac{\partial \boldsymbol{g}}{\partial x_2}(\boldsymbol{x}) \right\|}} \tag{4.1.3}$$

defines a continuous field of unit normal vectors on S.

Let us write down the components of the above vector product product in components. Using (1.1.9) and the definition of the Jacobian determinant in Remark 2.1.9 we can write

$$\frac{\partial}{\partial x_1}\boldsymbol{g}(\boldsymbol{x}) \times \frac{\partial}{\partial x_2}\boldsymbol{g}(\boldsymbol{x}) = \begin{bmatrix} \dfrac{\partial(g_2, g_3)}{\partial(x_1, x_2)} \\ \dfrac{\partial(g_3, g_1)}{\partial(x_1, x_2)} \\ \dfrac{\partial(g_1, g_2)}{\partial(x_1, x_2)} \end{bmatrix}$$

and

$$\left\| \frac{\partial \boldsymbol{g}}{\partial x_1}(\boldsymbol{x}) \times \frac{\partial \boldsymbol{g}}{\partial x_2}(\boldsymbol{x}) \right\| = \sqrt{\left| \frac{\partial(g_2, g_3)}{\partial(x_1, x_2)} \right|^2 + \left| \frac{\partial(g_3, g_1)}{\partial(x_1, x_2)} \right|^2 + \left| \frac{\partial(g_1, g_2)}{\partial(x_1, x_2)} \right|^2}.$$

There is another way to express the norm of the above vector product. We know from Proposition 1.1.11 on page 13 that the norm of the vector product is the area of the

parallelogram spanned by the two vectors. In Proposition 1.1.7 on page 11 we derived another formula for the area of a parallelogram spanned by two vectors. We have to form the matrix with the two vectors as columns. In our case this turns out to be the Jacobian matrix of $\boldsymbol{g}$:

$$J_{\boldsymbol{g}}(\boldsymbol{x}) = \begin{bmatrix} \dfrac{\partial \boldsymbol{g}}{\partial x_1}(\boldsymbol{x}) & \dfrac{\partial \boldsymbol{g}}{\partial x_2}(\boldsymbol{x}) \end{bmatrix}.$$

As a consequence of Proposition 1.1.7 we can express the norm of the above vector product in terms of the Jacobian of $\boldsymbol{g}$.

Proposition 4.1.14 *If $D \subset \mathbb{R}^2$ and $\boldsymbol{g}\colon D \to \mathbb{R}^3$ is a differentiable function, then*

$$\boxed{\left\| \frac{\partial \boldsymbol{g}}{\partial x_1}(\boldsymbol{x}) \times \frac{\partial \boldsymbol{g}}{\partial x_2}(\boldsymbol{x}) \right\| = \sqrt{\det\left(\left(J_{\boldsymbol{g}}(\boldsymbol{x})\right)^T J_{\boldsymbol{g}}(\boldsymbol{x})\right)}.} \tag{4.1.4}$$

4.2 The Definition of Surface Integrals

In Section 2.1.3 we derived a formula telling us how to integrate a function over a deformed plane domain. We called it the transformation formula. The original domain, and its image, the deformed domain, were both lying in the plane. A surface, at least locally, is the image of a plane domain lying not in the plane but in space. Hence to construct surface integrals we proceed in *exactly the same way* as we derived the transformation formula. Assume now that S is a smooth surface, and for simplicity we assume that it is parametrised by one function $\boldsymbol{g}\colon D \to S$, where $D \subset \mathbb{R}^2$. We again look at the image, $\boldsymbol{g}(R)$, of a small rectangle $R \subset D$ as shown in Figure 4.2.1 (compare with Figure 2.1.9

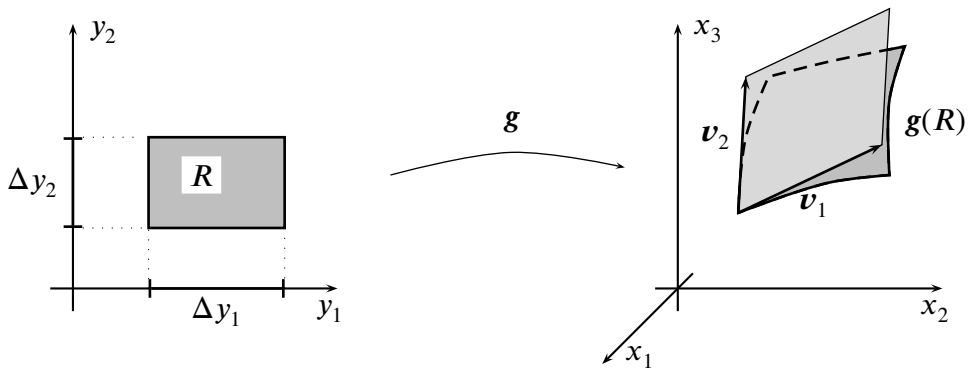


Figure 4.2.1: Deformation of a small rectangle by $\boldsymbol{g}$

on page 72), with the only difference that $\boldsymbol{g}(R) \subset \mathbb{R}^3$. The vectors $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$ are given

exactly the same way:

$$\boldsymbol{v}_1 = \frac{\partial \boldsymbol{g}}{\partial y_1}(\boldsymbol{y})\Delta y_1 \qquad \text{and} \qquad \boldsymbol{v}_2 = \frac{\partial \boldsymbol{g}}{\partial y_2}(\boldsymbol{y})\Delta y_2. \tag{4.2.1}$$

The difference is that they have three and not two components, which means that we need a different formula to compute the area of the parallelogram they span. To compute that area we use Proposition 1.1.7. Note that the matrix with columns

$$\frac{\partial \boldsymbol{g}}{\partial y_1}(\boldsymbol{y}) \qquad \text{and} \qquad \frac{\partial \boldsymbol{g}}{\partial y_2}(\boldsymbol{y})$$

is the Jacobian matrix, $J_{\boldsymbol{g}}(\boldsymbol{y})$, of $\boldsymbol{g}$ at $\boldsymbol{y}$. Hence the approximate area of $\boldsymbol{g}(R)$ in our present situation is

$$\text{area}(\boldsymbol{g}(R)) \approxeq \sqrt{\det\left(\left(J_{\boldsymbol{g}}(\boldsymbol{y})\right)^T J_{\boldsymbol{g}}(\boldsymbol{y})\right)}\,\Delta y_1 \Delta y_2.$$

We encountered the above determinant already when defining smooth surfaces, and called it the Jacobian of the parametrisation $\boldsymbol{g}$. The Jacobian is the factor by which the area of a small rectangle is distorted when mapped by $\boldsymbol{g}$ onto the surface. To define surface integrals we replace $|\det(J_{\boldsymbol{g}}(\boldsymbol{y}))|$ by the Jacobian of the parametrisation $\boldsymbol{g}$. This motivates the following definition.

Definition 4.2.1 (Surface integral of a scalar function) Suppose that S is a smooth surface parametrised by $\boldsymbol{g}\colon D \to S$. If f is a continuous function on S then we define

$$\boxed{\int_S f\,dS := \int_D f(\boldsymbol{g}(\boldsymbol{y}))\sqrt{\det\left(\left(J_{\boldsymbol{g}}(\boldsymbol{y})\right)^T J_{\boldsymbol{g}}(\boldsymbol{y})\right)}\,d\boldsymbol{y}.} \tag{4.2.2}$$

If S is piecewise smooth, or has to be parametrised by more than one function, we just add up the corresponding integrals.

☞ **Remark 4.2.2** Note that we can compute the Jacobian of $\boldsymbol{g}$ by using Proposition 4.1.14.

☞ **Remark 4.2.3** If we set $f = 1$ on S then the surface integral yields the area of the surface. Hence, if S is parametrised by $\boldsymbol{g}$ then

$$\boxed{\text{area}(S) = \int_S 1\,dS = \int_D \sqrt{\det\left(\left(J_{\boldsymbol{g}}(\boldsymbol{y})\right)^T J_{\boldsymbol{g}}(\boldsymbol{y})\right)}\,d\boldsymbol{y}.}$$

Remark 4.2.4 Let us note that the above definition of surface integrals can be generalised to more general surfaces, namely to k-dimensional "surfaces" in $\mathbb{R}^N$. In particular it is consistent with the definition of integrals over a one dimensional "surface," that is, a curve, where $k = 1$. If we parametrise a curve C by $\boldsymbol{\gamma}(t)$, $t \in I$, then it was shown in Remark 4.1.4 that

$$\big(J_\gamma(t)\big)^T J_\gamma(t) = \boldsymbol{\gamma}'(t)^T \boldsymbol{\gamma}'(t) = \|\boldsymbol{\gamma}'(t)\|^2.$$

If we apply (4.2.2) to the present situation we get

$$\int_C f\,ds := \int_I f(\boldsymbol{\gamma}(t))\sqrt{\det\Big(\big(J_\gamma(t)\big)^T J_\gamma(t)\Big)}\,dt = \int_I f(\boldsymbol{\gamma}(t))\|\boldsymbol{\gamma}'(t)\|\,dt,$$

which coincides with the formula given in Definition 3.3.1.

Remark 4.2.5 Let us emphasise that Definition 4.2.1 reduces to the transformation formula in Theorem 2.1.10 if S is flat, that is, lies in $\mathbb{R}^2$! One could ask why the transformation formula was a theorem, and the above is a definition. The reason is that, if $\boldsymbol{g}(D)$ is in $\mathbb{R}^2$, then the integral over $\boldsymbol{g}(D)$ is already defined. The theorem tells us that we can compute it using the transformation formula. For surfaces, however, the integral is not already defined, so we use (4.2.2) to do so.
It was not a trivial task to *define* the surface area for general surfaces such that it has the properties we expect, and some attempts failed. To define the length of a curve we approximated the given curve by polygons. Mathematicians tried to generalise this idea to surfaces, approximating them by polyhedra. We choose a grid of points and form triangles as shown in Figure 4.2.2, add up their areas and try to pass to the limit! However, as an example due to Hermann Amandus Schwarz (1843–1921) shows, the limit could be infinite for a surface as simple as a finite cylinder! (see [4, pages 75–77] for details). Hence it is not possible to define surface area by approximation by polyhedra. In our approach we approximated the surface by small scales like on the skin of a fish. Doing so we get a "surface area" having the expected properties. In particular it behaves well under transformation of coordinates.

We now want to compute the Jacobians for two important cases, namely for spherical coordinates, and for surfaces given as the graph of a function.

4.2.1 The Jacobian for Spherical Coordinates

We can parametrise a sphere by spherical coordinates

$$\boldsymbol{g}(\theta, \varphi) := \big(R\cos\varphi\sin\theta, R\sin\varphi\sin\theta, R\cos\theta\big),$$

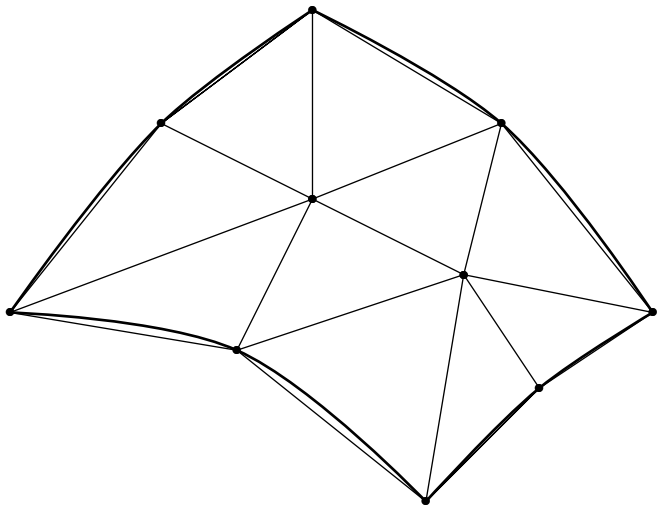


Figure 4.2.2: Approximating a surface by a polyhedron

where $(\theta, \varphi) \in D := [0, \pi] \times [0, 2\pi)$. Applying Example 4.1.3 we have the following proposition.

Proposition 4.2.6 *For spherical coordinates as defined above we have*

$$\boxed{\sqrt{\det\left(\left(J_g(\boldsymbol{y})\right)^T J_g(\boldsymbol{y})\right)} = R^2 \sin\theta.}$$

Note that if we measure the angle from the xy plane then we need to replace $\sin\theta$ by $\cos\theta$ in the above formula. We can use this to compute the surface area of the sphere.

Example 4.2.7 (Surface area of sphere) We can use the above to compute the surface area of a sphere of radius R. It is most convenient to use spherical coordinates as introduced in Example 4.1.3. We deduce from Example 4.1.3 and Proposition 4.1.14 that the surface area is

$$\begin{aligned}\int_S 1\, dS = \int_0^{2\pi} \int_0^{\pi} R^2 \sin\theta \, d\varphi \, d\theta = 2\pi R^2 [-\cos\theta]_0^{\pi} \\ = 2\pi R^2(-(-1) + 1) = 4\pi R^2,\end{aligned}$$

which is the well known result.

4.2.2 The Jacobian for Graphs

Suppose that S is the graph of a smooth function $h\colon D \to \mathbb{R}$. We want to compute Jacobian of the corresponding parametrisation

$$\boldsymbol{g}(x_1, x_2) := (x_1, x_2, h(x_1, x_2)), \qquad x \in D$$

of S. We first compute the Jacobian matrix of $\boldsymbol{g}$:

$$J_{\boldsymbol{g}}(\boldsymbol{x}) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ \dfrac{\partial h}{\partial x_1}(\boldsymbol{x}) & \dfrac{\partial h}{\partial x_2}(\boldsymbol{x}) \end{bmatrix}.$$

Then

$$\begin{aligned}
\big(J_{\boldsymbol{g}}(\boldsymbol{x})\big)^T J_{\boldsymbol{g}}(\boldsymbol{x}) &= \begin{bmatrix} 1 & 0 & \dfrac{\partial h}{\partial x_1}(\boldsymbol{x}) \\ 0 & 1 & \dfrac{\partial h}{\partial x_2}(\boldsymbol{x}) \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ \dfrac{\partial h}{\partial x_1}(\boldsymbol{x}) & \dfrac{\partial h}{\partial x_2}(\boldsymbol{x}) \end{bmatrix} \\
&= \begin{bmatrix} 1 + \Big(\dfrac{\partial h}{\partial x_1}(\boldsymbol{x})\Big)^2 & \Big(\dfrac{\partial h}{\partial x_1}(\boldsymbol{x})\Big)\Big(\dfrac{\partial h}{\partial x_2}(\boldsymbol{x})\Big) \\ \Big(\dfrac{\partial h}{\partial x_1}(\boldsymbol{x})\Big)\Big(\dfrac{\partial h}{\partial x_2}(\boldsymbol{x})\Big) & 1 + \Big(\dfrac{\partial h}{\partial x_2}(\boldsymbol{x})\Big)^2 \end{bmatrix}
\end{aligned}$$

Now we compute the determinant of the above matrix:

$$\begin{aligned}
&\det \begin{bmatrix} 1 + \Big(\dfrac{\partial h}{\partial x_1}(\boldsymbol{x})\Big)^2 & \Big(\dfrac{\partial h}{\partial x_1}(\boldsymbol{x})\Big)\Big(\dfrac{\partial h}{\partial x_2}(\boldsymbol{x})\Big) \\ \Big(\dfrac{\partial h}{\partial x_1}(\boldsymbol{x})\Big)\Big(\dfrac{\partial h}{\partial x_2}(\boldsymbol{x})\Big) & 1 + \Big(\dfrac{\partial h}{\partial x_2}(\boldsymbol{x})\Big)^2 \end{bmatrix} \\
&\qquad = \Big(1 + \Big(\frac{\partial h}{\partial x_1}(\boldsymbol{x})\Big)^2\Big)\Big(1 + \Big(\frac{\partial h}{\partial x_2}(\boldsymbol{x})\Big)^2\Big) - \Big(\frac{\partial h}{\partial x_1}(\boldsymbol{x})\Big)^2\Big(\frac{\partial h}{\partial x_2}(\boldsymbol{x})\Big)^2 \\
&\qquad\qquad = 1 + \Big(\frac{\partial h}{\partial x_1}(\boldsymbol{x})\Big)^2 + \Big(\frac{\partial h}{\partial x_2}(\boldsymbol{x})\Big)^2 = 1 + \|\operatorname{grad} h(\boldsymbol{x})\|^2.
\end{aligned}$$

Hence we have the following proposition.

Proposition 4.2.8 *For the graph of a function h we have*

$$\boxed{\sqrt{\det\Big(\big(J_{\boldsymbol{g}}(\boldsymbol{y})\big)^T J_{\boldsymbol{g}}(\boldsymbol{y})\Big)} = \sqrt{1 + \|\operatorname{grad} h(\boldsymbol{x})\|^2}.}$$

Example 4.2.9 Calculate the surface area of the graph of $f(x, y) = xy$ over the disc D given by $x^2 + y^2 \leq 1$.
Solution: We have grad $f(x, y) = (y, x)$, and so by the above proposition the Jacobian is $1 + x^2 + y^2$. Hence the surface area is

$$\iint_D \sqrt{1 + x^2 + y^2}\, dx\, dy.$$

Changing to polar coordinates, using (2.1.9), we get

$$\iint_D \sqrt{1 + x^2 + y^2}\, dx\, dy = \int_0^{2\pi} \int_0^1 \sqrt{1 + r^2}\, r\, dr\, d\varphi$$
$$= 2\pi \left[\frac{1}{2}\frac{2}{3}(1 + r^2)^{3/2}\right]_0^1 = \frac{2\pi}{3}\left(2\sqrt{2} - 1\right).$$

Hence the surface area is $\frac{2\pi}{3}(2\sqrt{2} - 1)$ square units.

4.3 The Flux Across a Surface

Imagine that $\boldsymbol{f}$ models the velocity of a fluid at any given point in space. We want to know how much fluid passes through a surface in a unit time interval. The first thing we must do is to decide in which direction we want to measure the flux. Mathematically this means that we need to give S an orientation. Hence suppose that S is an orientable surface, and that $\boldsymbol{n}(\boldsymbol{x})$ is the positive unit normal at every point $\boldsymbol{x} \in S$. Assume that $\boldsymbol{g}\colon D \to S$ is a parametrisation of S. Given a small rectangle $R \subset D$ with edges of length Δy_1 and Δy_2 we want to estimate the volume of fluid going across $\boldsymbol{g}(R)$ in a unit time interval. The volume flowing across is approximately the volume of the parallelepiped spanned by $\boldsymbol{f}(\boldsymbol{g}(\boldsymbol{y}))$ and the vectors $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$ given by (4.2.1). The volume of that parallelepiped equals the product of the area of its base and its height. The situation is depicted in Figure 4.3.1. We saw that the area of the base involves the Jacobian, and is given in this situation by

$$\text{area}(\boldsymbol{g}(R)) \approxeq \sqrt{\det\left(\left(J_{\boldsymbol{g}}(\boldsymbol{y})\right)^T J_{\boldsymbol{g}}(\boldsymbol{y})\right)}\, \Delta y_1 \Delta y_2.$$

We get the height by taking the projection onto the direction of the unit normal $\boldsymbol{n}$. Hence the height is $\boldsymbol{f} \cdot \boldsymbol{n}$, and the approximate volume of fluid crossing $\boldsymbol{g}(R)$ in a unit time interval is

$$\boldsymbol{f}(\boldsymbol{g}(\boldsymbol{y})) \cdot \boldsymbol{n}(\boldsymbol{g}(\boldsymbol{y}))\sqrt{\det\left(\left(J_{\boldsymbol{g}}(\boldsymbol{y})\right)^T J_{\boldsymbol{g}}(\boldsymbol{y})\right)}\, \Delta y_1 \Delta y_2.$$

Adding up the above terms over a partition of D into small rectangles we get a Riemann sum for the function $\boldsymbol{f} \cdot \boldsymbol{n}\sqrt{\det\left(\left(J_{\boldsymbol{g}}\right)^T J_{\boldsymbol{g}}\right)}$. Hence we make the following definition.

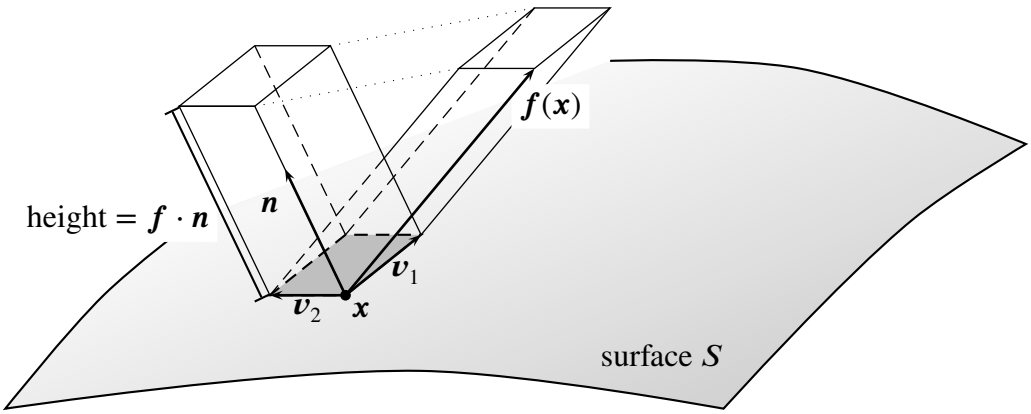


Figure 4.3.1: Flux through a small "surface element" in a unit time interval. The volume of the two parallelepipeds is the same.

Definition 4.3.1 (Surface integral of a vector field) Suppose that $\boldsymbol{f}$ is a continuous vector field over the smooth orientable surface S with positive unit normal $\boldsymbol{n}$. Then we say that

$$\boxed{\int_S \boldsymbol{f} \cdot \boldsymbol{n}\, dS = \int_D \boldsymbol{f}(\boldsymbol{g}(\boldsymbol{y})) \cdot \boldsymbol{n}(\boldsymbol{g}(\boldsymbol{y}))\sqrt{\det\left(\left(J_{\boldsymbol{g}}(\boldsymbol{y})\right)^T J_{\boldsymbol{g}}(\boldsymbol{y})\right)}\, d\boldsymbol{y}.} \tag{4.3.1}$$

is the integral of $\boldsymbol{f}$ over S, or the flux of $\boldsymbol{f}$ across S.

We next derive a formula allowing us to evaluate the flux of a vector field across a surface. We assume that S is a smooth orientable surface with a regular parametrisation $\boldsymbol{g}\colon D \to S$ consistent with the orientation. By this we mean that the positive field of normals is given by (4.1.3). By taking into account (4.1.3) and Proposition 4.1.14 we see that

$$\begin{aligned}
&f(\boldsymbol{g}(\boldsymbol{y})) \cdot \boldsymbol{n}(\boldsymbol{g}(\boldsymbol{y}))\sqrt{\det\left(\left(J_{\boldsymbol{g}}(\boldsymbol{y})\right)^T J_{\boldsymbol{g}}(\boldsymbol{y})\right)} \\
&\qquad = f(\boldsymbol{g}(\boldsymbol{y})) \cdot \frac{\dfrac{\partial \boldsymbol{g}}{\partial x_1}(\boldsymbol{x}) \times \dfrac{\partial \boldsymbol{g}}{\partial y_2}(\boldsymbol{y})}{\left\|\dfrac{\partial \boldsymbol{g}}{\partial y_1}(\boldsymbol{y}) \times \dfrac{\partial \boldsymbol{g}}{\partial y_2}(\boldsymbol{y})\right\|}\sqrt{\det\left(\left(J_{\boldsymbol{g}}(\boldsymbol{y})\right)^T J_{\boldsymbol{g}}(\boldsymbol{y})\right)} \\
&\qquad = f(\boldsymbol{g}(\boldsymbol{y})) \cdot \frac{\dfrac{\partial \boldsymbol{g}}{\partial y_1}(\boldsymbol{x}) \times \dfrac{\partial \boldsymbol{g}}{\partial y_2}(\boldsymbol{y})}{\sqrt{\det\left(\left(J_{\boldsymbol{g}}(\boldsymbol{y})\right)^T J_{\boldsymbol{g}}(\boldsymbol{y})\right)}}\sqrt{\det\left(\left(J_{\boldsymbol{g}}(\boldsymbol{y})\right)^T J_{\boldsymbol{g}}(\boldsymbol{y})\right)} \\
&\qquad\qquad = f(\boldsymbol{g}(\boldsymbol{y})) \cdot \left(\frac{\partial \boldsymbol{g}}{\partial y_1}(\boldsymbol{x}) \times \frac{\partial \boldsymbol{g}}{\partial y_2}(\boldsymbol{y})\right)
\end{aligned}$$

Using (4.1.3) we finally get

$$f(\boldsymbol{g}(\boldsymbol{y}))\cdot\boldsymbol{n}(\boldsymbol{g}(\boldsymbol{y}))\sqrt{\det\big(\big(J_g(\boldsymbol{y})\big)^T J_g(\boldsymbol{y})\big)}$$
$$= f_1(\boldsymbol{g}(\boldsymbol{y}))\frac{\partial(g_2,g_3)}{\partial(y_1,y_2)} + f_2(\boldsymbol{g}(\boldsymbol{y}))\frac{\partial(g_3,g_1)}{\partial(y_1,y_2)} + f_3(\boldsymbol{g}(\boldsymbol{y}))\frac{\partial(g_1,g_2)}{\partial(y_1,y_2)}.$$

Hence we get the following proposition.

Proposition 4.3.2 *Suppose that S is an orientable smooth surface, and that $\boldsymbol{g}\colon D\to S$ is a regular parametrisation consistent with the orientation of S. Then*

$$\boxed{\begin{aligned}&\int_S \boldsymbol{f}\cdot\boldsymbol{n}\,dS\\ &\quad= \iint_D f_1(\boldsymbol{g}(\boldsymbol{y}))\frac{\partial(g_2,g_3)}{\partial(y_1,y_2)} + f_2(\boldsymbol{g}(\boldsymbol{y}))\frac{\partial(g_3,g_1)}{\partial(y_1,y_2)} + f_3(\boldsymbol{g}(\boldsymbol{y}))\frac{\partial(g_1,g_2)}{\partial(y_1,y_2)}\,dy_1\,dy_2.\end{aligned}}$$

Let us now look at the special case of an explicitly given surface, that is, S is the graph of a function over $h\colon D\to\mathbb{R}$. By convention we orient a graph upwards. From (4.1.2) and Proposition 4.2.8 we see that

$$f(\boldsymbol{g}(\boldsymbol{x}))\cdot\boldsymbol{n}(\boldsymbol{g}(\boldsymbol{x}))\sqrt{\det\big(\big(J_g(\boldsymbol{x})\big)^T J_g(\boldsymbol{x})\big)} = \boldsymbol{f}(\boldsymbol{x},h(\boldsymbol{x}))\cdot(-\operatorname{grad}h(\boldsymbol{x}),1).$$

Hence we have the following result.

Proposition 4.3.3 *Suppose that S is the graph of the smooth function $h\colon D\to\mathbb{R}$. Then the flux of the continuous vector field $\boldsymbol{f}$ across S in the upward direction is*

$$\boxed{\begin{aligned}\int_S \boldsymbol{f}\cdot\boldsymbol{n}\,dS = \iint_D\Big(&-f_1\big(\boldsymbol{x},h(\boldsymbol{x})\big)\frac{\partial h}{\partial x_1}(\boldsymbol{x})\\ &- f_2\big(\boldsymbol{x},h(\boldsymbol{x})\big)\frac{\partial h}{\partial x_2}(\boldsymbol{x}) + f_3\big(\boldsymbol{x},h(\boldsymbol{x})\big)\Big)\,dx_1\,dx_2.\end{aligned}}$$

Example 4.3.4 Calculate the flux of $\boldsymbol{f}(x,y,z)=(2x,2y,3)$ across the part of the paraboloid $z=4-x^2-y^2$ lying above the xy-plane. By convention we orient the surface upwards.
Solution: By assumption, $z\geq 0$ so the surface is the graph of $4-x^2-y^2$ over the disc, D, given by $x^2+y^2\leq 4$. According to the above proposition, the flux across the surface

is given by

$$\begin{aligned}\int_S \boldsymbol{f}\cdot\boldsymbol{n}\,dS &= \int_D -2x(-2x)-2y(-2y)+3\,dx\,dy\\ &= \int_D 4x^2+4y^2+3\,dx\,dy = \int_0^{2\pi}\int_0^2 (4r^2+3)r\,dr\,d\varphi\\ &= 2\pi\left[r^2+\frac{3}{2}r^2\right]_0^2 = 2\pi(16+3\cdot 2-0) = 44\pi.\end{aligned}$$

☞ **Example 4.3.5** Calculate the flux of

$$\boldsymbol{f}(x,y,z) := \left(\frac{x}{(x^2+y^2+z^2)^{3/2}}, \frac{y}{(x^2+y^2+z^2)^{3/2}}, \frac{z}{(x^2+y^2+z^2)^{3/2}}\right)$$

out of the sphere with radius R centred at the origin.

Solution: The outward pointing unit normal to a sphere of radius R centred at the origin is given by

$$\boldsymbol{n}(x,y,z) = \frac{1}{R}(x,y,z).$$

Moreover, for points (x,y,z) on the sphere we have $x^2+y^2+z^2 = R^2$ and so

$$\boldsymbol{f}(x,y,z) = \frac{1}{R^3}(x,y,z).$$

Using the two facts we get

$$\boldsymbol{f}\cdot\boldsymbol{n} = \frac{1}{R}\frac{1}{R^3}(x,y,z)\cdot(x,y,z) = \frac{1}{R^4}(x^2+y^2+z^2) = \frac{1}{R^2}.$$

Hence using spherical coordinates we have

$$\int_S \boldsymbol{f}\cdot\boldsymbol{n}\,dS = \frac{1}{R^2}\int_0^{2\pi}\int_0^{\pi} R^2\sin\theta\,d\theta\,d\varphi = 2\pi[-\cos\theta]_0^{\pi} = 4\pi.$$

The flux across the sphere is therefore 4π. Note that it is independent of R.

Chapter 5

The Integral Theorems of Vector Calculus

In this chapter we introduce the main integral theorems in vector calculus, namely the theorems of Green, Gauss and Stokes. They have many applications within and outside mathematics. In particular they appear, possibly in a generalised form, in every "field theory" such as for instance gravitation or electro-magnetism (Maxwell equations). They can also be used to derive many other equations including the transport equation, the heat equation and the wave equation.

5.1 Green's Theorem

If f is a continuously differentiable function of one variable, defined on a closed bounded interval $[a, b]$, then the *fundamental theorem of calculus* asserts that

$$\int_a^b f'(x)\,dx = f(b) - f(a).$$

The above formula expresses the integral of the derivative f' over an interval by means of the values of f at the boundary points a and b of the interval $[a, b]$.
Green's formula provides a generalisation of this fact to functions of two variables. For a function of two variables, the domain is a region, D, in the plane $\mathbb{R}^2$, and the boundary, ∂D, of D is a curve. The formula will apply to vector fields $\boldsymbol{f}$ and have the following form.

$$\iint_D \text{“suitable derivative of } \boldsymbol{f}\text{”} = \int_{\partial D} \text{“vector field”}$$

The interval is replaced by the domain D, and the boundary points a and b by the rim (or boundary) ∂D of D. In two dimensions the degree of freedom is much higher than

in one dimension, and therefore we need more "structure" for a formula like the above to be true. Not surprisingly, the key ingredient in the proof of Green's Theorem will be the fundamental theorem of calculus.

5.1.1 Domains and their Boundaries

In order to be able to compute an integral over the boundary of a domain we need to assume that it is not too wild. We will work with the following class of domains.

Definition 5.1.1 (Piecewise smooth domain) We call $D \subset \mathbb{R}^2$ a *piecewise smooth domain* if its boundary can be written as a finite union of piecewise smooth curves. More precisely, $\partial D = C_1 \cup C_2 \cup \cdots \cup C_n$, where C_i admits a regular parametrisation $\boldsymbol{\gamma}_i(t)$, $t \in [a_i, b_i]$, for all $i = 1, \ldots, n$.

Given a continuous function f defined on the boundary of a piecewise smooth domain ∂D we set

$$\int_{\partial D} f\, ds := \int_{C_1} f\, ds + \cdots + \int_{C_n} f\, ds.$$

We orient the boundary of D in such a way that if we walked along ∂D in the positive direction then the right arm points out of D as shown in Figure 5.1.1. Equipped with

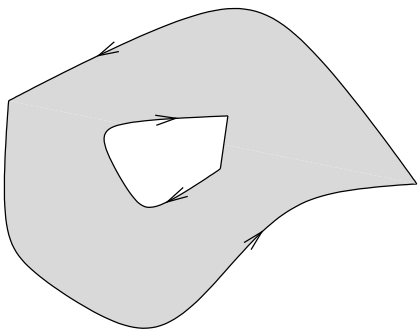


Figure 5.1.1: A piecewise smooth domain with positively oriented boundary.

the above orientation we say that ∂D (or simply D) is *positively oriented*. If we talk about ∂D we always mean the *positively oriented boundary of* D! Note that we can also have domains with holes like the one in Figure 5.1.1. Then the orientation of the inner boundary is opposite to that of the outer boundary.

5.1.2 Green's Theorem

Recall that by a smooth vector field we mean a vector field $\boldsymbol{f} = (f_1, f_2)$, where f_1 and f_2 have continuous partial derivatives. We will always assume that $\boldsymbol{f}$ is defined on an open set U, and that $\overline{D} \subset U$.

Theorem 5.1.2 (Green's Theorem) *Suppose that D is a piecewise smooth bounded domain, and $\boldsymbol{f} = (f_1, f_2)$ a smooth vector field on D. Then*

$$\boxed{\int_D \left(\frac{\partial}{\partial x_1} f_2(\boldsymbol{x}) - \frac{\partial}{\partial x_2} f_1(\boldsymbol{x}) \right) d\boldsymbol{x} = \int_{\partial D} f_1 \, dx_1 + f_2 \, dx_2.} \tag{5.1.1}$$

☞ **Remark 5.1.3** Before giving the proof let us emphasise the importance of the assumption that $\boldsymbol{f}$ be smooth in D! If $\boldsymbol{f}$ has a singularity in D then (5.1.1) does not need to be true. As an example consider the vector field from Example 3.5.9 on page 106. From the calculations made there it is clear that the integral in (5.1.1) over D is zero, whereas the line integral is 2π. The reason is that $\boldsymbol{f}$ has a singularity at $(0, 0)$.

Proof. To prove Green's Theorem we first consider very special cases and put them together to obtain the general formula.

Case 1: Suppose that D is a domain which can be described as

$$D = \{(x_1, x_2) \in \mathbb{R}^2 \mid a \le x_1 \le b, \varphi(x_1) \le x_2 \le \psi(x_1)\},$$

where $a < b$ are constants, and $\varphi < \psi$ are continuously differentiable functions. Further assume that $f_2 = 0$, that is,

$$\boldsymbol{f} = (f_1, 0).$$

To compute

$$\int_{\partial D} f_1 \, dx_1 := \int_{\partial D} f_1 \, dx_1 + 0 \, dx_2$$

we determine the path integral over the curves C_1–C_4 indicated in Figure 5.1.2, and then add them up to get the total integral. We first look at C_1, which we can parametrise by (b, t), $t \in [\varphi(b), \psi(b)]$. According to the definition of the line integral (see Definition 3.4.1 on page 98) we get

$$\int_{C_1} f_1 \, dx_1 = \int_{\varphi(b)}^{\psi(b)} (f_1(b, t) \cdot 0 + 0 \cdot 1) \, dt = 0.$$

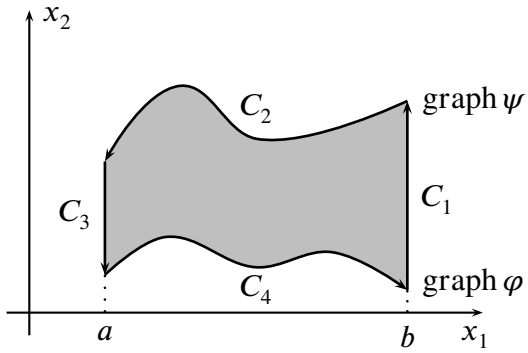


Figure 5.1.2: Special Domain

In much the same way we compute the integral over C_3, the reverse of which we can parametrise by (a, t), $t \in [\varphi(a), \psi(a)]$. As before we get

$$\int_{C_3} f_1 \, dx_1 = -\int_{\varphi(a)}^{\psi(a)} (f_1(a,t) \cdot 0 + 0 \cdot 1) \, dt = 0.$$

We next compute the integral over C_2. We can parametrise the reverse of C_2 by $(t, \psi(t))$, $t \in [a, b]$. Note that this parametrisation is against the orientation of C_2, so we get

$$\int_{C_2} f_1 \, dx_1 = -\int_a^b \big(f_1(t, \psi(t)) \cdot 1 + 0 \cdot \psi'(t)\big) \, dt = -\int_a^b f_1(t, \psi(t)) \, dt.$$

In a similar manner we get the integral over C_4. Note however, that the orientation of the parametrisation $(t, \varphi(t))$, $t \in [a, b]$ is consistent with the orientation. Hence we get

$$\int_{C_4} f_1 \, dx_1 = \int_a^b \big(f_1(t, \varphi(t)) \cdot 1 + 0 \cdot \varphi'(t)\big) \, dt = \int_a^b f_1(t, \varphi(t)) \, dt.$$

Adding the four integrals together we finally get

$$\int_{\partial D} f_1 \, dx_1 = -\int_a^b \big(f_1(t, \psi(t)) - f_1(t, \varphi(t))\big) \, dt. \tag{5.1.2}$$

We next want to rewrite the right hand side of the above equation. To do so we set for every fixed $t \in [a, b]$

$$g(x_2) := f_1(t, x_2)$$

where g is defined for all $x_2 \in [\varphi(t), \psi(t)]$. By the fundamental theorem of calculus we can write

$$g(\psi(t)) - g(\varphi(t)) = \int_{\varphi(t)}^{\psi(t)} \frac{d}{dx_2} g(x_2) \, dx_2.$$

By the definition of partial derivatives we have

$$\frac{d}{dx_2} g(x_2) = \frac{\partial}{\partial x_2} f_1(t, x_2).$$

Substituting all this into (5.1.2) we get

$$\int_{\partial D} f_1\, dx_1 = -\int_a^b \Big(\int_{\varphi(t)}^{\psi(t)} \frac{\partial}{\partial x_2} f_1(t, x_2)\, dx_2 \Big)\, dt.$$

Note that we can rename t by x_1 (this was our choice for the parameter anyway). Then the formula becomes

$$\int_{\partial D} f_1\, dx_1 = -\int_a^b \Big(\int_{\varphi(x_1)}^{\psi(x_1)} \frac{\partial}{\partial x_2} f_1(x_1, x_2)\, dx_2 \Big)\, dx_1$$

Applying Fubini's Theorem 2.1.5 we finally get

$$\int_{\partial D} f_1\, dx_1 = -\int_D \frac{\partial}{\partial x_2} f_1(\boldsymbol{x})\, d\boldsymbol{x}$$

Case 2: We next consider the special case with D of the form

$$D = \{(x_1, x_2) \in \mathbb{R}^2 \mid a < x_2 < b, \varphi(x_2) < x_1 < \psi(x_2)\},$$

where $a < b$ are constants and $\varphi < \psi$ two continuously differentiable functions on $[a, b]$. Moreover, let us assume that $f_1 = 0$, so that $\boldsymbol{f} = (0, f_2)$. We can reduce this case to the previous one by interchanging the role of x_1 and x_2. Note however, that this changes the

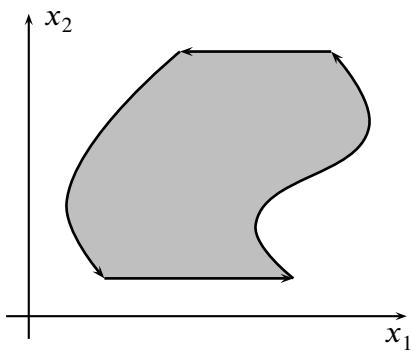


Figure 5.1.3: Another special domain

orientation of the boundary of D, changing the sign in the line integral. Hence

$$\int_{\partial D} f_2\, dx_2 = \int_D \frac{\partial}{\partial x_1} f_2(\boldsymbol{x})\, d\boldsymbol{x}$$

Case 3: Let us now assume that D admits a representation of the form considered in Case 1 and Case 2 simultaneously. This is the case if every horizontal and vertical line intersect the domain in exactly two points or exactly one line segment. First we use the

fact that D admits a representation considered in Case 1. Applying this to the vector field $(f_1, 0)$ we get

$$-\int_D \frac{\partial}{\partial x_2} f_1(\boldsymbol{x})\, d\boldsymbol{x} = \int_{\partial D} f_1\, dx_1 + 0\, dx_1.$$

Next we use the fact that D admits a representation considered in Case 2. Applying this to the vector field $(0, f_2)$ we get

$$\int_D \frac{\partial}{\partial x_1} f_2(\boldsymbol{x})\, d\boldsymbol{x} = \int_{\partial D} 0\, dx_1 - f_2\, dx_2.$$

Adding these formluas we get equation (5.1.1). This proves Green's theorem for the special domains considered.

Case 4: We now consider a general domain. It can be shown that every piecewise smooth domain can be decomposed into finitely many domains satisfying the conditions of Case 3, at least in suitably chosen coordinates. For a rigorous proof we refer to [1, Section 10–14]. We only illustrate the idea by some examples shown in Figure 5.1.4 and 5.1.5. Note that the line integrals along the interior boundaries appear twice in opposite

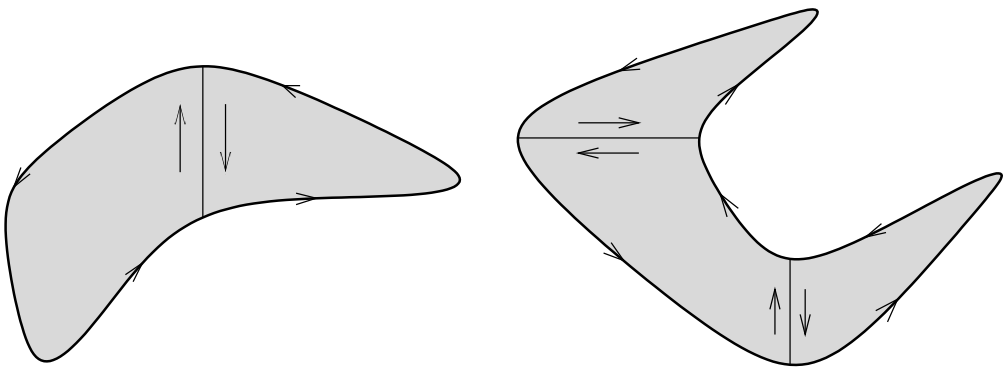


Figure 5.1.4: Decomposition of some non-convex domains.

directions. Hence if we add up the line integrals over the boundaries of the sub-domains they cancel. This completes the proof of Green's theorem. ■

5.1.3 Application: The Area of a Domain.

Choosing an appropriate vector field we can derive a formula to compute the area of domain D with a piecewise boundary. The idea is to choose a vector field (f_1, f_2) in

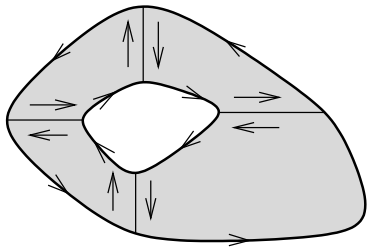


Figure 5.1.5: Decomposition of a domain with a hole.

such a way that

$$\frac{\partial f_2}{\partial x_1} - \frac{\partial f_1}{\partial x_2} = 1.$$

One possibility is to choose $(f_1, f_2) := (0, x_1)$. Hence by Green's theorem

$$\text{area}(D) = \iint_D 1\, dx_1\, dx_2 = \iint_D \frac{\partial}{\partial x_1} x_1 - \frac{\partial}{\partial x_2} 0\, dx_1\, dx_2 = \int_{\partial D} x_1 dx_2$$

Another possibility is to choose $(f_1, f_2) := (-x_2, 0)$. Again Green's theorem yields

$$\text{area}(D) = \iint_D 1\, dx_2\, dx_1 = \iint_D \frac{\partial}{\partial x_1} 0 + \frac{\partial}{\partial x_2} x_2\, dx_1\, dx_2 = -\int_{\partial D} x_2 dx_1$$

Adding these two formulas and dividing by two we obtain the following formula for the area of a domain.

Proposition 5.1.4 *Suppose that D is a domain with a piecewise smooth boundary. Then its surface area is given by*

$$\boxed{\text{area}(D) = \frac{1}{2} \int_{\partial D} x_1\, dx_2 - x_2\, dx_1.}$$

We use the above formula because it is invariant under rotations of the coordinate system, whereas the other two are not.

Example 5.1.5 Determine the area of the ellipse given by

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1$$

Solution: A possible parametrisation of the boundary of the given ellipse is

$$(a \cos t, b \sin t), \qquad t \in [0, 2\pi].$$

According to the above formula the area is therefore given by

$$\begin{aligned}\frac{1}{2}\int_{\frac{x^2}{a^2}+\frac{y^2}{b^2}=1} x\,dy - y\,dx &= \frac{1}{2}\int_0^{2\pi} a\cos t(b\cos t) - b\sin t(-a\sin t)\,dt \\ &= \frac{ab}{2}\int_0^{2\pi} \cos^2 t + \sin^2 t\,dt \\ &= \frac{ab}{2}\int_0^{2\pi} 1\,dt = \frac{ab}{2}2\pi = ab\pi\end{aligned}$$

Hence the area of the ellipse is πab, the same we got in Example 2.1.14 using polar coordinates and the transformation formula.

5.1.4 Application: Conservative Vector Fields

We saw in Section 3.5.2 that every conservative vector field is closed. We also mentioned that the converse is true if D is simply connected. In this section we want to give a proof of this fact for plane vector fields. Recall from (3.5.5) that closed for a plane vector field, $\boldsymbol{f} = (f_1, f_2)$, means that

$$\frac{\partial}{\partial x_1} f_2 - \frac{\partial}{\partial x_2} f_1 = 0. \tag{5.1.3}$$

This is exactly the term appearing in (5.1.1). Simply connected means that every closed curve, C, in D can be continuously shrunk to a point in D without ever leaving D (see Figure 3.5.2). In particular this means that the domain, D_C, enclosed by C is a subset of D. Hence it follows from (5.1.3) and Green's Theorem that

$$0 = \iint_{D_C} \frac{\partial}{\partial x_1} f_2 - \frac{\partial}{\partial x_2} f_1\,dx_1\,dx_2 = \oint_C \boldsymbol{f}\cdot d\boldsymbol{x}.$$

This shows that if D is simply connected and $\boldsymbol{f}$ is closed then the integral over every closed piecewise continuous curve is zero. By Proposition 3.5.2 the field $\boldsymbol{f}$ is conservative.

Example 5.1.6 Show that $\boldsymbol{f}(x, y) = (2xy, x^2 - y^3)$ is a conservative vector field on $\mathbb{R}^2$.
Solution: $\mathbb{R}^2$ is clearly simply connected, so we have to check whether $\boldsymbol{f}$ is closed. We compute the relevant derivatives:

$$\begin{aligned}\frac{\partial}{\partial x}(x^2 - y^3) &= 2x, \\ \frac{\partial}{\partial y} 2xy &= 2x.\end{aligned}$$

As the two derivatives are the same $\boldsymbol{f}$ is closed and thus conservative.

5.2 Two Interpretations of Green's Theorem

There are several possibilities to interpret Green's theorem. We can interpret the line integral as the flux of a field across the boundary of a domain. This leads to what is usually called the Divergence Theorem (or Gauss' Theorem). The other possibility is to look at the flow of the vector field along the "rim" of the surface and to see how it relates to the integral over the interior of the domain. This will lead to a special case of Stokes' Theorem.

The fact that both the Divergence Theorem and Stokes' Theorem can be obtained from the same formula shows that both are closely related. In fact, using differential forms, one can indeed unify the two and write them in the form

$$\int_S d\omega = \int_{\partial S} \omega.$$

(note the similarity to the fundamental theorem of calculus.) However this is beyond the scope of this course. A good reference for this theory is the book by Spivak [5].

5.2.1 The Divergence Theorem in the Plane

Suppose that $\boldsymbol{f} = (f_1, f_2)$ is a smooth vector field. Applying Green's theorem to the field $(-f_2, f_1)$ we obtain

$$\iint_D \left(\frac{\partial}{\partial x_1} f_1 + \frac{\partial}{\partial x_2} f_2 \right) dx_1\, dx_2 = \int_{\partial D} f_1\, dx_2 - f_2\, dx_1.$$

We want to get an interpretation of the integral on the right hand side. To do so suppose that

$$\boldsymbol{\gamma}(t) = (\gamma_1(t), \gamma_2(t)), \qquad t \in [a, b]$$

is a regular parametrisation of some part $C \subset \partial D$ consistent with the orientation of ∂D. By definition of the line integral we have

$$\int_C f_1\, dx_2 - f_2\, dx_1 = \int_a^b f_1(\gamma_1(t), \gamma_2(t))\gamma_2'(t) - f_2((\gamma_1(t), \gamma_2(t))\gamma_1'(t)\, dt.$$

We can view the integrand on the right hand side as a dot product of $\boldsymbol{f}$ with the vector $(\gamma_2'(t), -\gamma_1'(t))$. We rescale the latter to unit length and define

$$\boldsymbol{n}(t) := \frac{1}{\|\boldsymbol{\gamma}'(t)\|} \begin{bmatrix} \gamma_2'(t) \\ -\gamma_1'(t) \end{bmatrix}.$$

Observe that $\boldsymbol{n}(t)$ is perpendicular to the positive unit tangent vector $\boldsymbol{\tau}(t) = \boldsymbol{\gamma}'(t)/\|\gamma'(t)\|$. As it is outward pointing, it is called the *outward pointing unit normal* to D. Both vectors

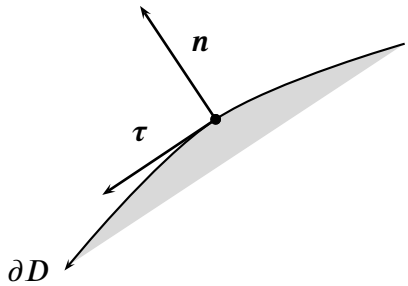


Figure 5.2.1: Positive tangent and outward pointing unit normal vectors.

are shown in Figure 5.2.1.
Hence we can rewrite our integral as

$$\int_C f_1\,dx_2 - f_2\,dx_1 = \int_a^b \boldsymbol{f}(\boldsymbol{\gamma}(t)) \cdot \boldsymbol{n}(\boldsymbol{\gamma}(t))\|\boldsymbol{\gamma}'(t)\|\,dt$$

By our Definition 3.4.1 of line integrals we see that

$$\int_C f_1\,dx_2 - f_2\,dx_1 = \int_C \boldsymbol{f} \cdot \boldsymbol{n}\,ds. \tag{5.2.1}$$

Let Δs denote a very small line element. If $\boldsymbol{f}$ models the motion of a fluid (direction and velocity of fluid at a given point) then $\boldsymbol{f} \cdot \boldsymbol{n}\Delta s$ is the flux across C through the line element Δs, and $\boldsymbol{f} \cdot \boldsymbol{n}\Delta s$ is the approximate amount of fluid flowing across the boundary ∂D through ds. The situation is depicted in Figure 5.2.2. Note that the area of the parallelogram and the rectangle are the same. Hence the integral on the right hand side

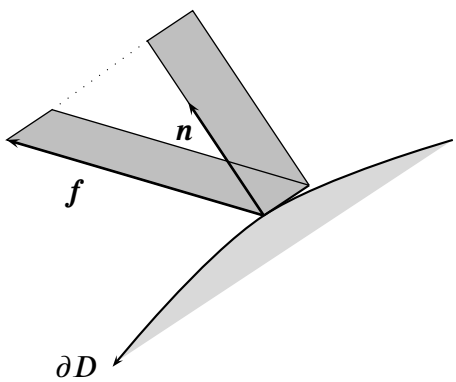


Figure 5.2.2: Flux across a boundary element of ∂D. The parallelogram and the rectangle have the same area.

of (5.2.1) represents the total flux of $\boldsymbol{f}$ across $C \subset \partial D$. Note that the above arguments are completely analogous to those used in Section 4.3 to find the flux of a vector field in

space across a surface. Going back to the original formula we see that

$$\iint_D \left(\frac{\partial}{\partial x_1} f_1 + \frac{\partial}{\partial x_2} f_2 \right) dx_1\, dx_2 = \int_{\partial D} \boldsymbol{f} \cdot \boldsymbol{n}\, ds$$

is the total loss (or increase) of the fluid from D through ∂D. We make the following definition.

Definition 5.2.1 If $\boldsymbol{f} = (f_1, \dots, f_N)$ is a vector field in $\mathbb{R}^N$ then

$$\boxed{\operatorname{div} \boldsymbol{f} := \frac{\partial}{\partial x_1} f_1 + \frac{\partial}{\partial x_2} f_2 + \cdots + \frac{\partial}{\partial x_N} f_N}$$

is called the *divergence* of the vector field.

☞ **Remark 5.2.2** Often the divergence of a vector field is written using the nabla operator in Remark 3.5.11. Formally, div $\boldsymbol{f}$ is the scalar product of ∇ with $\boldsymbol{f}$, and so we write

$$\nabla \cdot \boldsymbol{f} := \operatorname{div} \boldsymbol{f}.$$

We can now rewrite Green's theorem. In that form it has a different name.

Theorem 5.2.3 (Divergence Theorem) *If $\boldsymbol{f}$ is a smooth vector field defined on a piecewise smooth domain D then*

$$\boxed{\int_D \operatorname{div} \boldsymbol{f}(\boldsymbol{x})\, d\boldsymbol{x} = \int_{\partial D} \boldsymbol{f} \cdot \boldsymbol{n}\, ds,}$$

where $\boldsymbol{n}$ is the outward pointing unit normal vector to ∂D.

There is a direct generalisation of the Divergence Theorem to three or more dimensions. We will discuss it in Section 5.3.

Physical Interpretation of the Divergence

We next want to derive a physical interpretation of div $\boldsymbol{f}$. Let $\boldsymbol{f}$ be the velocity of a fluid or a gas in a region D. Fix a point $\boldsymbol{a} = (a_1, a_2)$ in D, and denote by B_δ a disc centred at $\boldsymbol{a}$ with radius δ. Then

$$\int_{\partial B_\delta} \boldsymbol{f} \cdot \boldsymbol{n}\, ds$$

is the net flow out of B_δ (if it is negative then the net flow is into B_δ). By the Divergence Theorem

$$\frac{1}{\text{area}(B_\delta)}\int_{B_\delta} \operatorname{div} \boldsymbol{f}\, d\boldsymbol{x} = \frac{1}{\text{area}(B_\delta)}\int_{\partial B_\delta} \boldsymbol{f}\cdot\boldsymbol{n}\, ds$$

models the net loss or increase of fluid (or gas) from B_δ per area. If we shrink B_δ to $\boldsymbol{a}$ we get a density for the loss or increase of fluid (or gas) at the point $\boldsymbol{a}$. It indicates how much fluid (or gas) disappears or appears at that point. It turns out that (see Lemma 5.2.4 below)

$$\lim_{\delta\to 0}\frac{1}{\text{area}(B_\delta)}\int_{B_\delta} \operatorname{div} \boldsymbol{f}\, d\boldsymbol{x} = \operatorname{div} \boldsymbol{f}(\boldsymbol{a})$$

Hence div $\boldsymbol{f}$ is often called the "*source strength*" of the vector field. If we have a gas, then by compressing it is possible that less flows out than in, or if we expand it more flows out than in. Hence if the sign of div $\boldsymbol{f}$ is positive there is a net flow out, if it is negative there is a net flow in. For that reason div $\boldsymbol{f}$ is also sometimes called the "*compressibility*" of a vector field. In any case we talk about a *source* if div $\boldsymbol{f} > 0$, and about a *sink* if div $\boldsymbol{f} < 0$.

An Auxiliary Lemma

The proof of the following lemma is a generalisation of the proof of the fundamental theorem of calculus to higher dimensions. It also works for $N \geq 2$, we only need to replace area by volume.

Lemma 5.2.4 *Suppose that $D \subset \mathbb{R}^2$ is an open set, and that $g : D \to \mathbb{R}$ is continuous at $\boldsymbol{a} \in D$. Moreover, let $B_\delta := B_\delta(\boldsymbol{a})$. Then*

$$\lim_{\delta\to 0}\frac{1}{\text{area}(B_\delta)}\int_{B_\delta} g(\boldsymbol{x})\, d\boldsymbol{x} = g(\boldsymbol{a}) \tag{5.2.2}$$

Proof. As $\displaystyle\int_{B_\delta} g(\boldsymbol{a})\, d\boldsymbol{x} = \text{area}(B_\delta)g(\boldsymbol{a})$ we have that

$$\frac{1}{\text{area}(B_\delta)}\int_{B_\delta} g(\boldsymbol{x})\, d\boldsymbol{x} - g(\boldsymbol{a}) = \frac{1}{\text{area}(B_\delta)}\int_{B_\delta} g(\boldsymbol{x}) - g(\boldsymbol{a})\, d\boldsymbol{x}.$$

Now fix $\varepsilon > 0$ arbitrary. As g is continuous at $\boldsymbol{a}$ there exists $\delta_0 > 0$ such that $|g(\boldsymbol{x}) - g(\boldsymbol{a})| < \varepsilon$ whenever $\|\boldsymbol{x} - \boldsymbol{a}\| < \delta_0$ (see Proposition 1.3.17). Hence for all $\delta < \delta_0$ we have

$$\begin{aligned}\Big|\frac{1}{\text{area}(B_\delta)}\int_{B_\delta} g(\boldsymbol{x})\, d\boldsymbol{x} - g(\boldsymbol{a})\Big| &\leq \frac{1}{\text{area}(B_\delta)}\int_{B_\delta} |g(\boldsymbol{x}) - g(\boldsymbol{a})|\, d\boldsymbol{x}\\ &\leq \frac{1}{\text{area}(B_\delta)}\int_{B_\delta} \varepsilon\, d\boldsymbol{x} = \frac{\text{area}(B_\delta)}{\text{area}(B_\delta)}\varepsilon = \varepsilon.\end{aligned}$$

As $\varepsilon > 0$ was arbitrary (5.2.2) follows. ■

5.2.2 The Theorem of Stokes in the Plane

Given a vector field $\boldsymbol{f} = (f_1, f_2)$, Green's theorem asserts that

$$\iint_D \left(\frac{\partial}{\partial x_1} f_2 - \frac{\partial}{\partial x_2} f_1 \right) dx_1\, dx_2 = \int_{\partial D} f_1\, dx_1 + f_2\, dx_2.$$

As we saw in Section 3.4 the line integral on the right hand side represents the circulation of the vector field $\boldsymbol{f}$ along the curve ∂D in the positive direction. In Remark 3.5.12 we called the integrand on the left hand side the curl of a plane vector field, so we set

$$\operatorname{curl} \boldsymbol{f} := \frac{\partial}{\partial x_1} f_2 - \frac{\partial}{\partial x_2} f_1$$

in this section. With this we can reformulate Green's theorem as follows.

Theorem 5.2.5 (Theorem of Stokes) *If $\boldsymbol{f}$ is a smooth vector field defined on a piecewise smooth domain D then*

$$\boxed{\int_D \operatorname{curl} \boldsymbol{f}(\boldsymbol{x})\, d\boldsymbol{x} = \int_{\partial D} \boldsymbol{f} \cdot \boldsymbol{\tau}\, ds,} \tag{5.2.3}$$

where $\boldsymbol{\tau}$ is the positive unit tangent to ∂D.

We next want to obtain a physical interpretation of $\operatorname{curl} \boldsymbol{f}$.

Physical Interpretation of the Curl

We next want to derive a physical interpretation of $\operatorname{curl} \boldsymbol{f}$. Let $\boldsymbol{f}$ be the velocity of a fluid or a gas in a region D. Fix a point $\boldsymbol{a} = (a_1, a_2)$ in D, and denote by B_δ a disc centred at $\boldsymbol{a}$ with radius δ. Then

$$\int_{\partial B_\delta} \boldsymbol{f} \cdot \boldsymbol{\tau}\, ds$$

is the net flow of $\boldsymbol{f}$ along ∂B_δ. In other words it is the circulation of the field along ∂B_δ. By the theorem of Stokes

$$\frac{1}{\operatorname{area}(B_\delta)} \int_{B_\delta} \operatorname{curl} \boldsymbol{f}\, d\boldsymbol{x} = \frac{1}{\operatorname{area}(B_\delta)} \int_{\partial B_\delta} \boldsymbol{f} \cdot \boldsymbol{\tau}\, ds.$$

The above is a density of the circulation of $\boldsymbol{f}$ in B_δ. If we shrink B_δ to $\boldsymbol{a}$ we get (see Lemma 5.2.4)

$$\lim_{\delta \to 0} \frac{1}{\operatorname{area}(B_\delta)} \int_{B_\delta} \operatorname{curl} \boldsymbol{f}\, d\boldsymbol{x} = \operatorname{curl} \boldsymbol{f}(\boldsymbol{a})$$

Hence curl $\boldsymbol{f}$ is a measure for the circulation of the vector field $\boldsymbol{f}$ at $\boldsymbol{a}$. If we fix a little paddle wheel at $\boldsymbol{a}$ then curl $\boldsymbol{f}(\boldsymbol{a})$ tells us how it turns. If curl $\boldsymbol{f}(\boldsymbol{a}) > 0$ it turns in the counterclockwise direction, if curl $\boldsymbol{f}(\boldsymbol{a}) < 0$ it turns in the clockwise direction, and if curl $\boldsymbol{f}(\boldsymbol{a}) = 0$ it does not turn at all.

Example 5.2.6 Let $\boldsymbol{f} = (y, 0)$ for $(x, y) \in \mathbb{R}^2$. Then the vector field looks as shown in Figure 5.2.3. A simple computation shows that curl $\boldsymbol{f} = -1$, so a little paddle wheel would turn in the clockwise direction. This is clear as the current is different on the top and the bottom.

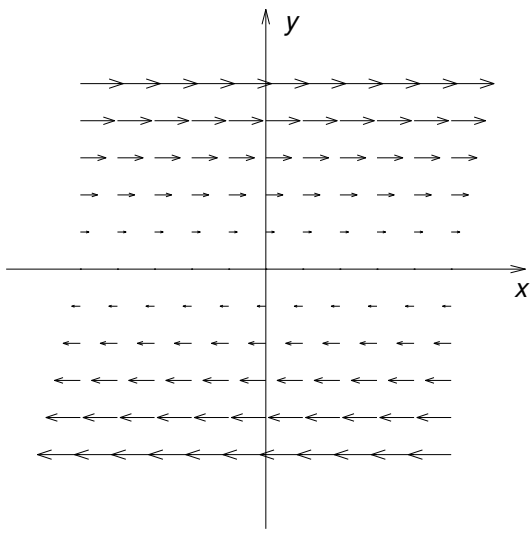


Figure 5.2.3: The vector field $\boldsymbol{f}(x, y) := (y, 0)$ in $\mathbb{R}^2$.

Example 5.2.7 Let $\boldsymbol{f} = (-y, x)$ for $(x, y) \in \mathbb{R}^2$. Then the vector field looks as shown in Figure 5.2.4. A simple computation shows that curl $\boldsymbol{f} = 2$, so a little paddle wheel would turn in the counterclockwise direction.

As the vector field circles around the origin it seems 'obvious' that the curl is positive. Note however, that the curl is a *local property* of the vector field, and that a global picture can be quite deceptive as the following example shows!

Example 5.2.8 Let $\boldsymbol{f}(x, y) := \left(\dfrac{-y}{x^2 + y^2}, \dfrac{x}{x^2 + y^2}\right)$ be the vector field studied in Example 3.5.9. It looks as shown in Figure 3.5.4 on page 107. It follows from the computations there that curl $\boldsymbol{f} = 0$ except at the origin, where $\boldsymbol{f}$ is not defined. On the other hand the field looks at first quite similar to the one from the previous example, where the curl was positive.

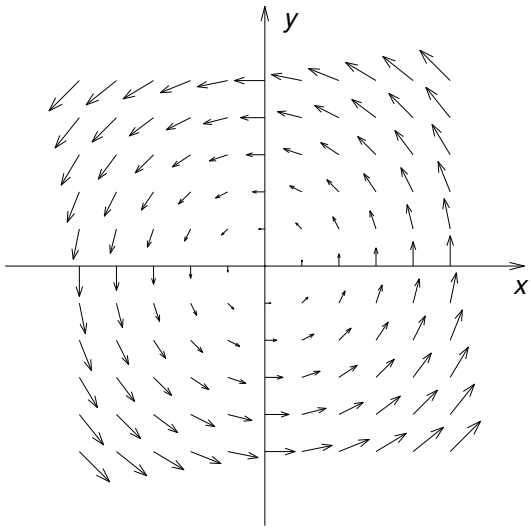


Figure 5.2.4: The vector field $\boldsymbol{f}(x, y) := (-y, x)$ in $\mathbb{R}^2$.

5.3 The Divergence Theorem and Green's Identities

We already mentioned in Section 5.2.1 that the divergence theorem extends to higher dimensions. In this section we prove an extension to three dimensions and use it to derive some useful identities, called *Green's first and second identities*.

5.3.1 The Divergence Theorem

We stated the divergence theorem in Section 5.2.1 for piecewise smooth plane domains. We omitted the quite difficult proof that every piecewise smooth domain can be decomposed into domains of the form as those shown in Figure 5.1.2 Figure 5.1.3. Subsets D of $\mathbb{R}^3$ are more complicated, so it is not clear what definition of piecewise smooth we should use. By smooth we mean that ∂D is a smooth surface, possibly having several components. This is not general enough, as for instance a cube does not have a smooth boundary. Instead of introducing piecewise smooth domains we just introduce *simple domains*.

Definition 5.3.1 (Simple domains) We call a $D \subset \mathbb{R}^3$ an xy-simple domain if there exist continuously differentiable functions $\varphi(x, y) \leq \psi(x, y)$ such that

$$D = \big\{(x, y, z) \in D_0 : \ \varphi(x, y) \leq z \leq \psi(x, y)\big\}, \tag{5.3.1}$$

where D_0 is the projection of D onto the xy-plane. Similarly we define yz-simple and

xz-simple. We call a domain *simple* if it is xy-simple, yz-simple and xz-simple at the same time.

An xy-simple surface is depicted in Figure 5.3.1. We will prove the divergence theorem for disjoint unions of simple domains. Let us note that every bounded domain whose

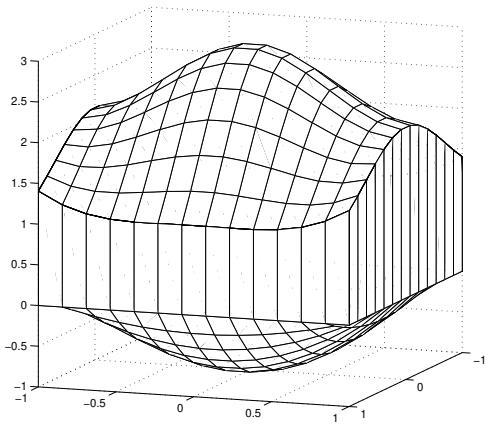


Figure 5.3.1: An xy-simple domain.

boundary is a smooth surface can be written as a finite union of simple domains, but this is not so clear for 'piecewise smooth' domains in general. As an example consider a decomposition of a torus (or doughnut) into a union of four simple domains as shown in Figure 5.3.2. We gave plane domains an orientation, and we want to do that for piecewise

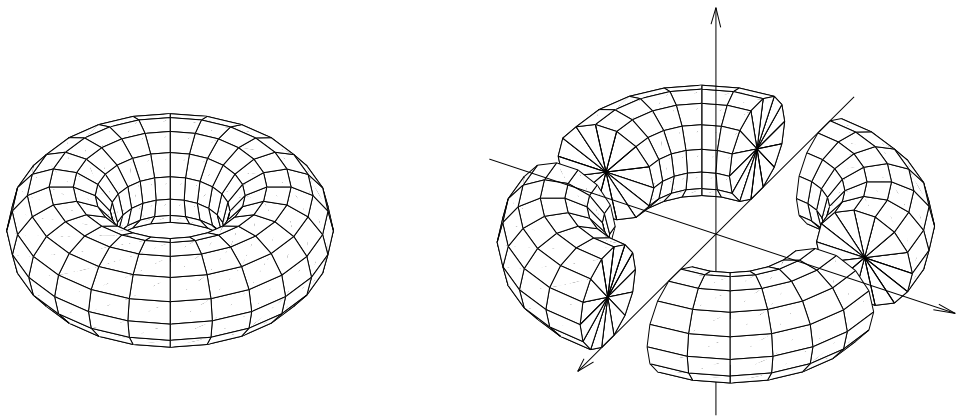


Figure 5.3.2: A torus and a possible decomposition into simple domains.

smooth domains $D \subset \mathbb{R}^3$ as well. The boundary ∂D consists of finitely many smooth surfaces. We take the orientation given by the unit normal field pointing out of D. We call this the *outward pointing unit normal field*.

Theorem 5.3.2 (Divergence Theorem) *Suppose $\boldsymbol{f}$ is a smooth vector field defined on a bounded domain $D \subset \mathbb{R}^3$. Moreover, suppose we can write D as a finite disjoint union of simple domains. Then*

$$\boxed{\int_D \operatorname{div} \boldsymbol{f}(\boldsymbol{x})\, d\boldsymbol{x} = \int_{\partial D} \boldsymbol{f} \cdot \boldsymbol{n}\, dS,}$$

where $\boldsymbol{n}$ is the outward pointing unit normal vector to ∂D.

Proof. The proof of the divergence theorem is very similar to the proof of Green's theorem. We start with very simple vector fields and domains, and then combine everything to get the general formula. Hence suppose that D is an xy-simple domain $D \subset \mathbb{R}^3$ with projection D_0 onto the xy-plane. Moreover let φ, ψ be smooth functions such that D has the form (5.3.1). Finally assume that

$$\boldsymbol{f} = (0, 0, f_3).$$

We then compute the surface integral

$$\int_{\partial D} \boldsymbol{f} \cdot \boldsymbol{n}\, dS$$

The boundary of D can be written as the union of three surfaces, namely $S_1 := \operatorname{graph}(\psi)$, $S_2 := \operatorname{graph}(\varphi)$ and the vertical pieces, S_3. Denote the unit normal vector to ∂D by $\boldsymbol{n} = (n_1, n_2, n_3)$. As $n_3 = 0$ on S_3 it follows that $\boldsymbol{f} \cdot \boldsymbol{n} = f_3 n_3 = 0$. Hence

$$\int_{S_3} \boldsymbol{f} \cdot \boldsymbol{n}\, dS = \int_{S_3} f_3 n_3\, dS = 0$$

We next compute the integral over S_1. According to Proposition 4.3.3 we have

$$\int_{S_1} f_3 n_3\, dS = \int_{S_1} \boldsymbol{f} \cdot \boldsymbol{n}\, dS = \iint_{D_0} f_3\big(x_1, x_2, \psi(x_1, x_2)\big)\, dx_1\, dx_2$$

Using the same formula we get

$$\int_{S_2} f_3 n_3\, dS = \int_{S_2} \boldsymbol{f} \cdot \boldsymbol{n}\, dS = -\iint_{D_0} f_3\big(x_1, x_2, \varphi(x_1, x_2)\big)\, dx_1\, dx_2,$$

where the negative sign appears because S_2 is oriented downwards. Adding up the three integrals we get

$$\int_{\partial D} f_3 n_3\, dS = \iint_{D_0} f_3\big(x_1, x_2, \psi(x_1, x_2)\big) - f_3\big(x_1, x_2, \varphi(x_1, x_2)\big)\, dx_1\, dx_2.$$

Setting $g(x_3) := f_3(x_1, x_2, x_3)$ it follows from the fundamental theorem of calculus and the definition of partial derivatives that

$$\begin{aligned} f_3\big(x_1, x_2, \psi(x_1, x_2)\big) - f_3\big(x_1, x_2, \varphi(x_1, x_2)\big) &= g(\psi(x_1, x_2)\big) - g\big(\varphi(x_1, x_2)\big) \\ &= \int_{\varphi(x_1,x_2)}^{\psi(x_1,x_2)} \frac{d}{dx_3} g(x_3)\, dx_3 = \int_{\varphi(x_1,x_2)}^{\psi(x_1,x_2)} \frac{\partial}{\partial x_3} f_3(x_1, x_2, x_3)\, dx_3. \end{aligned}$$

If we substitute the above into the previous formula and apply Fubini's Theorem 2.2.2 we get

$$\begin{aligned} \int_{\partial D} f_3 n_3\, dS &= \iint_{D_0} \Big(\int_{\varphi(x_1,x_2)}^{\psi(x_1,x_2)} \frac{\partial}{\partial x_3} f_3(x_1, x_2, x_3)\, dx_3 \Big)\, dx_1\, dx_2 \\ &= \int_D \frac{\partial}{\partial x_3} f_3(\boldsymbol{x})\, d\boldsymbol{x}. \quad (5.3.2) \end{aligned}$$

If D is xz-simple we can apply the same procedure with x_1, x_2 replaced by x_1, x_3 and the vector field $\boldsymbol{f} = (0, f_2, 0)$. Hence

$$\int_{\partial D} f_2 n_2\, dS = \int_D \frac{\partial}{\partial x_2} f_2(\boldsymbol{x}) n_2\, d\boldsymbol{x}. \quad (5.3.3)$$

Likewise we get for a yz-simple domain and $\boldsymbol{f} = (f_1, 0, 0)$ that

$$\int_{\partial D} f_1 n_1\, dS = \int_D \frac{\partial}{\partial x_1} f_1(\boldsymbol{x}) n_1\, d\boldsymbol{x}. \quad (5.3.4)$$

Now assume that D is simple. By definition D is xy-simple, xz-simple and yz-simple at the same time. Hence if $\boldsymbol{f} = (f_1, f_2, f_3)$ we can apply (5.3.2) to $(0, 0, f_3)$, then (5.3.3) to $(0, f_2, 0)$ and finally (5.3.4) to $(f_1, 0, 0)$. Adding up the three identities and using the definition of the divergence (see Definition 5.2.1 on page 139) we get

$$\begin{aligned} \int_{\partial D} \boldsymbol{f} \cdot \boldsymbol{n}\, dS &= \int_{\partial D} f_1 n_1 + f_2 n_2 + f_3 n_3\, dS \\ &= \int_D \frac{\partial}{\partial x_1} f_1(\boldsymbol{x}) + \frac{\partial}{\partial x_2} f_2(\boldsymbol{x}) + \frac{\partial}{\partial x_3} f_3(\boldsymbol{x})\, d\boldsymbol{x} = \int_D \operatorname{div} \boldsymbol{f}\, d\boldsymbol{x}. \end{aligned}$$

This proves the divergence theorem for simple domains. If D is a domain which can be decomposed into simple domains then the integrals over all adjoining interior boundaries appear twice in opposite orientations. Hence the corresponding surface integrals have opposite signs and therefore cancel when we add them up over all sub-domains. Hence the Divergence Theorem follows. ■

☞ **Remark 5.3.3** As with Green's theorem the assumption that $\boldsymbol{f}$ be smooth in D is essential. Consider the vector field

$$\boldsymbol{f}(\boldsymbol{x}) := \frac{\boldsymbol{x}}{\|\boldsymbol{x}\|^3}$$

which has a singularity at $\mathbf{0}$. Computing the partial derivative of the i-th component we get

$$\frac{\partial}{\partial x_i}\frac{x_i}{\|\boldsymbol{x}\|^3} = \frac{\|\boldsymbol{x}\|^2 - 3x_i^2}{\|\boldsymbol{x}\|^5},$$

so the divergence of $\boldsymbol{f}$ is

$$\begin{aligned}\operatorname{div}\boldsymbol{f}(\boldsymbol{x}) = \frac{1}{\|\boldsymbol{x}\|^5}\sum_{i=1}^{3}\left(\|\boldsymbol{x}\|^2 - 3x_i^2\right) &= \frac{1}{\|\boldsymbol{x}\|^5}\left(3\|\boldsymbol{x}\|^2 - 3\sum_{i=1}^{3}x_i^2\right)\\ &= \frac{3}{\|\boldsymbol{x}\|^5}\left(\|\boldsymbol{x}\|^2 - \|\boldsymbol{x}\|^2\right) = 0.\end{aligned}$$

Hence if B is a ball of radius R centred at the origin then

$$\int_B \operatorname{div}\boldsymbol{f}(\boldsymbol{x})\,d\boldsymbol{x} = \int_B 0\,d\boldsymbol{x} = 0.$$

On the other hand we saw in Example 4.3.5 that

$$\int_{\partial B}\boldsymbol{f}\cdot\boldsymbol{n}\,dS = 4\pi.$$

The two results are obviously not equal, and the reason is, that $\boldsymbol{f}$ has a singularity in B. The example given here is not a pathological one. The field is, up to a constant, that of the gravitational field of a 'point mass' concentrated at the origin. From the present example one can derive *Gauss' law* for the gravitational field. Physicists say that the mass is the source of the gravitational field. A similar phenomenon appears for the electric field in electrostatics.

5.3.2 Green's First and Second Identity

We want to derive two useful identities from the divergence theorem. In some sense they are extensions of the formula of 'integration by parts' to higher dimensions. We first introduce the Laplace operator.

Definition 5.3.4 (Laplace operator) Suppose that $D \subset \mathbb{R}^N$, and that $u: D \to \mathbb{R}$ is a function. Then we set

$$\boxed{\Delta u := \operatorname{div}(\operatorname{grad} u) = \sum_{i=1}^{N} \frac{\partial^2 u}{\partial x_i^2} = \frac{\partial^2 u}{\partial x_1^2} + \cdots + \frac{\partial^2 u}{\partial x_N^2},}$$

and call Δ the *Laplace operator* or the *Laplacian*.

Remark 5.3.5 Using the nabla operator from Remark 3.5.11 and the notation introduced in Remark 5.2.2 we can formally write Δ as the scalar product $\nabla \cdot \nabla$. For this reason one often writes

$$\Delta u =: \nabla \cdot \nabla u =: \nabla^2 u.$$

We need the following "product rule."

Lemma 5.3.6 *Suppose that $\boldsymbol{f}$ is a vector field, and u a scalar valued function. Then*

$$\boxed{\operatorname{div}(u\boldsymbol{f}) = \boldsymbol{f} \cdot \operatorname{grad} u + u \operatorname{div} \boldsymbol{f}}$$

whenever all derivatives exist.

Proof. By the product rule we have

$$\operatorname{div} u = \sum_{i=1}^{N} \frac{\partial}{\partial x_i}(u f_i) = \sum_{i=1}^{N} f_i \frac{\partial u}{\partial x_i} + u \sum_{i=1}^{N} \frac{\partial f_i}{\partial x_i} = \boldsymbol{f} \cdot \operatorname{grad} u + u \operatorname{div} \boldsymbol{f}$$

as required. ■

Theorem 5.3.7 (Green's first identity) *Suppose that D is a bounded set on which the divergence theorem can be applied. If $u, v : D \to \mathbb{R}$ have continuous second order partial derivatives on $\overline{D}$ then*

$$\boxed{\int_D u\Delta v + \nabla u \cdot \nabla v \, d\boldsymbol{x} = \int_{\partial D} u \nabla v \cdot \boldsymbol{n} \, dS,} \tag{5.3.5}$$

where $\boldsymbol{n}$ is the outward pointing unit normal to ∂D.

Proof. Applying Lemma 5.3.6 with $\boldsymbol{f} := \nabla v$ we get

$$\begin{aligned} \operatorname{div}(u\nabla v) = \operatorname{div}(u\boldsymbol{f}) &= \boldsymbol{f} \cdot \operatorname{grad} u + u \operatorname{div} \boldsymbol{f} \\ &= \nabla u \cdot \nabla v + u \operatorname{div}(\nabla u) = \nabla u \cdot \nabla v + u\Delta u. \end{aligned}$$

Hence by the Divergence Theorem

$$\int_D u\Delta u + \nabla u \cdot \nabla v \, d\boldsymbol{x} = \int_D \operatorname{div}(u\nabla v) \, d\boldsymbol{x} = \int_{\partial D} u\nabla v \cdot \boldsymbol{n} \, dS,$$

completing the proof of the theorem. ■

Theorem 5.3.8 (Green's second identity) *Suppose that D is a bounded set on which the divergence theorem can be applied. If $u, v : D \to \mathbb{R}$ have continuous second order partial derivatives on $\overline{D}$ then*

$$\boxed{\int_D u\Delta v - v\Delta u \, d\boldsymbol{x} = \int_{\partial D} u\nabla v \cdot \boldsymbol{n} - v\nabla u \cdot \boldsymbol{n} \, dS,} \tag{5.3.6}$$

where $\boldsymbol{n}$ is the outward pointing unit normal to ∂D.

Proof. Apply Green's first identity to $u\Delta v$ and then to $v\Delta u$ and subtract the two. The term

$$\int_D \nabla u \cdot \nabla v \, d\boldsymbol{x}$$

then cancels as it appears twice with opposite signs. ■

5.4 The Theorem of Stokes

Green's Theorem and the Divergence Theorem both relate an integral over the interior of a set to an integral over the boundary of a set. The theorem of Stokes does the same, but with surfaces. It relates a surface integral to a line integral along the "rim" or "boundary" of the surface.

5.4.1 Surfaces with a Boundary

Some surfaces have a rim or boundary as for instance a half sphere. We assume that the rim is the finite union of piecewise smooth curves. Given a surface S we denote its boundary, or rim by ∂S. Assume now that S is an orientable surface and fix an orientation. Denote by $\boldsymbol{n}$ the positive field of unit normal vectors to S. We now want to orient the boundary ∂S consistent with the orientation of S. We do this as follows. Suppose you stand at the boundary of S with the upright position given by the positive unit normal vector to S. Then walk along ∂S in such a way that your right hand points away from the surface as shown in Figure 5.4.1.

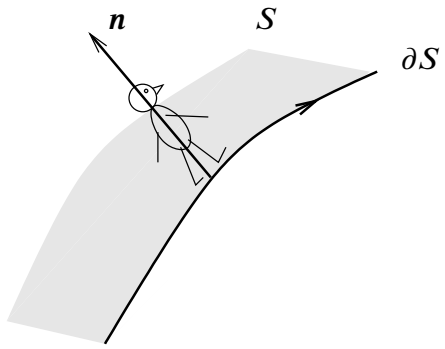


Figure 5.4.1: An oriented surface with a positively oriented boundary.

5.4.2 The Theorem of Stokes

We have already discussed a version of Stokes' theorem in the plane in Section 5.2.2. We can view a plane domain as a (flat) surface in space. Here we want to extend the theorem given there to "curved" surfaces lying in space. In Section 5.2.2 we have used the "curl" for plane vector fields instead of the "real" curl. If we view a plane domain as a surface in $\mathbb{R}^3$ then we can consider the vector field $\boldsymbol{f} = (f_1, f_2, 0)$. We saw in Remark 3.5.12 that in this case

$$\operatorname{curl} \boldsymbol{f}(\boldsymbol{x}) = \begin{bmatrix} 0 \\ 0 \\ \dfrac{\partial}{\partial x_1} f_2(\boldsymbol{x}) - \dfrac{\partial}{\partial x_2} f_1(\boldsymbol{x}) \end{bmatrix}.$$

If we assume that $\boldsymbol{n} = (0, 0, 1)$ then the boundary of a plane domain is oriented counterclockwise by convention. Then the integrand on the left hand side of (5.2.3) becomes $(\operatorname{curl} \boldsymbol{f}) \cdot \boldsymbol{n}$. For curved surfaces this turns out to be the right expression.

Theorem 5.4.1 (Stokes' Theorem) *Suppose that $S \subset \mathbb{R}^3$ is a smooth orientable surface with piecewise smooth boundary ∂S. Furthermore, let $\boldsymbol{f}$ be a smooth vector field. Finally assume that ∂S is oriented as explained in the previous section, and denote the positive unit normal to S by $\boldsymbol{n}$, and the positive unit tangent to ∂S by $\boldsymbol{\tau}$. Then*

$$\boxed{\iint_S (\operatorname{curl} \boldsymbol{f}) \cdot \boldsymbol{n} \, dS = \int_{\partial S} \boldsymbol{f} \cdot \boldsymbol{\tau} \, ds.} \tag{5.4.1}$$

Recall from Definition 3.4.1 on page 98 that

$$\int_C \boldsymbol{f} \cdot \boldsymbol{\tau} \, ds = \int_C \boldsymbol{f} \cdot d\boldsymbol{x}.$$

Proof. For simplicity we only give a proof for surfaces that can be completely described by one parametrisation. We therefore assume that

$$\boldsymbol{g}(\boldsymbol{y}) = \big(g_1(y_1, y_2), g_2(y_1, y_2), g_3(y_1, y_2)\big), \qquad (y_1, y_2) \in D$$

is a regular parametrisation of S. Suppose that $(y_1(t), y_2(t))$, $t \in [a, b]$, is a parametrisation of ∂D which is regular on every smooth part of ∂D. Then set

$$\boldsymbol{\gamma}(t) := \boldsymbol{g}(\boldsymbol{y}(t)), \qquad t \in [a, b].$$

Clearly the image of $\boldsymbol{\gamma}$ contains ∂S, but it could be more. When deforming D into S we have to paste part of the edges together. Note however that $\boldsymbol{\gamma}(t)$ runs twice along these 'cuts' but in opposite directions. Hence

$$\int_{\partial S} \boldsymbol{f} \cdot \boldsymbol{\tau}\, ds = \int_a^b \boldsymbol{f}(\boldsymbol{\gamma}(t)) \cdot \boldsymbol{\gamma}'(t)\, dt. \tag{5.4.2}$$

Examples are shown in Figure 5.4.2 and (5.4.3). In the first example we cut a cylinder. The dotted parts of the boundary of D and S correspond to each other. Figure 5.4.3 shows a half sphere parametrised by spherical coordinates. We cut the half sphere along the dotted line and then deform it into a rectangle. As with the proof of Green's theorem

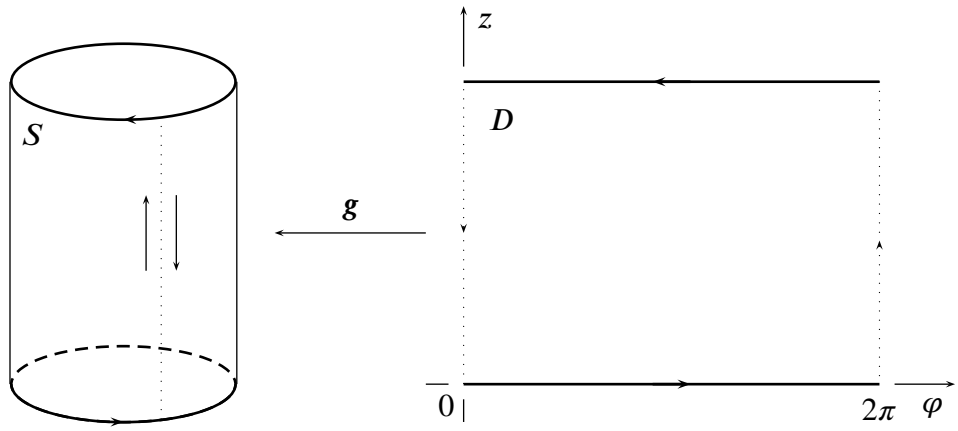


Figure 5.4.2: Cylindrical coordinates: the dotted lines correspond to each other.

and the divergence theorem we start with very simple vector fields, and then add up the results to get the formula for a general vector field. Hence assume that $\boldsymbol{f} = (f_1, 0, 0)$. Then by the previous formula

$$\int_{\partial S} \boldsymbol{f} \cdot \boldsymbol{\tau}\, ds = \int_a^b \boldsymbol{f}_1(\boldsymbol{\gamma}(t)) \cdot \boldsymbol{\gamma}_1'(t)\, dt.$$

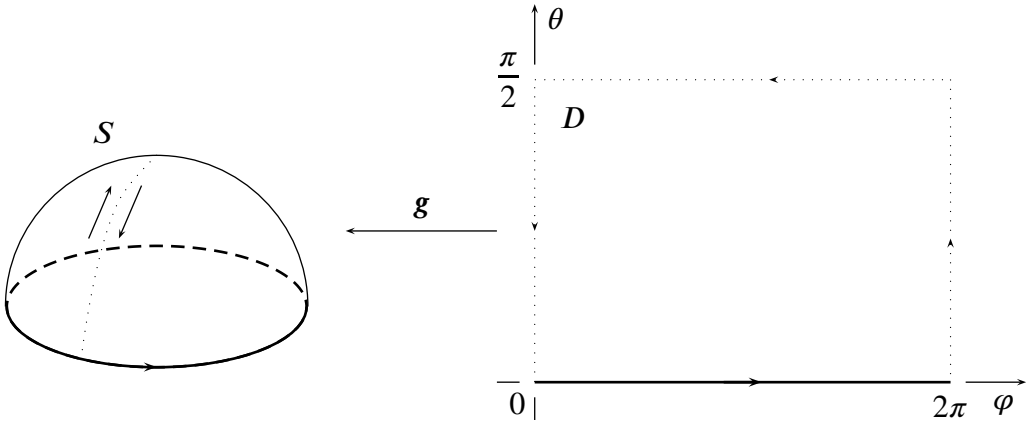


Figure 5.4.3: Spherical coordinates: the dotted lines correspond to each other.

Using the definition of $\boldsymbol{\gamma}(t)$ and the chain rule (see Theorem 1.4.15 on page 40) we get

$$\gamma_1'(t) = \operatorname{grad} f(\boldsymbol{y}(t)) \cdot \boldsymbol{y}'(t).$$

Hence we get

$$\int_{\partial S} \boldsymbol{f} \cdot \boldsymbol{\tau}\, ds = \int_a^b f_1(\boldsymbol{g}(\boldsymbol{y}(t))) \operatorname{grad} g_1(\boldsymbol{y}(t)) \cdot \boldsymbol{y}'(t)\, dt = \int_{\partial D} (f_1 \circ \boldsymbol{g}) \operatorname{grad} g_1 \cdot d\boldsymbol{y}.$$

Applying Green's Theorem 5.1.2 to the vector field

$$(f_1 \circ \boldsymbol{g}) \operatorname{grad} g_1 = \left(f_1 \circ \boldsymbol{g} \frac{\partial g_1}{\partial y_1}, f_1 \circ \boldsymbol{g} \frac{\partial g_1}{\partial y_1} \right)$$

we see that

$$\begin{aligned} \int_{\partial S} \boldsymbol{f} \cdot \boldsymbol{\tau}\, ds &= \int_{\partial D} (f_1 \circ \boldsymbol{g}) \operatorname{grad} g_1 \cdot d\boldsymbol{y} \\ &= \int_D \frac{\partial}{\partial y_1} \left(f_1(\boldsymbol{g}(\boldsymbol{y})) \frac{\partial g_1}{\partial y_2}(\boldsymbol{y}) \right) - \frac{\partial}{\partial y_2} \left(f_1(\boldsymbol{g}(\boldsymbol{y}) \frac{\partial g_1}{\partial y_1}(\boldsymbol{y}) \right) d\boldsymbol{y}. \end{aligned}$$

Now we compute the derivatives in the last integrand. Using the chain and product rules we get

$$\frac{\partial}{\partial y_1} \left(f_1(\boldsymbol{g}(\boldsymbol{y})) \frac{\partial g_1}{\partial y_2}(\boldsymbol{y}) \right) = \operatorname{grad} f_1(\boldsymbol{g}(\boldsymbol{y})) \cdot \frac{\partial \boldsymbol{g}}{\partial y_1}(\boldsymbol{y}) \frac{\partial g_1}{\partial y_2}(\boldsymbol{y}) + f_1(\boldsymbol{g}(\boldsymbol{y})) \frac{\partial^2 \boldsymbol{g}_1}{\partial y_1 \partial y_2}(\boldsymbol{y})$$

and similarly

$$\frac{\partial}{\partial y_2} \left(f_1(\boldsymbol{g}(\boldsymbol{y})) \frac{\partial g_1}{\partial y_1}(\boldsymbol{y}) \right) = \operatorname{grad} f_1(\boldsymbol{g}(\boldsymbol{y})) \cdot \frac{\partial \boldsymbol{g}}{\partial y_2}(\boldsymbol{y}) \frac{\partial g_1}{\partial y_1}(\boldsymbol{y}) + f_1(\boldsymbol{g}(\boldsymbol{y})) \frac{\partial^2 \boldsymbol{g}_1}{\partial y_2 \partial y_1}(\boldsymbol{y}).$$

Taking into account the symmetry of the second order partial derivatives (Proposition 1.4.32 on page 50) we conclude that

$$\begin{aligned}
&\frac{\partial}{\partial y_1}\Big(f_1(\boldsymbol{g}(\boldsymbol{y}))\frac{\partial g_1}{\partial y_2}(\boldsymbol{y})\Big) - \frac{\partial}{\partial y_2}\Big(f_1(\boldsymbol{g}(\boldsymbol{y})\frac{\partial g_1}{\partial y_1}(\boldsymbol{y})\Big)\\
&\qquad = \operatorname{grad} f_1(\boldsymbol{g}(\boldsymbol{y}))\cdot\Big(\frac{\partial \boldsymbol{g}}{\partial y_1}(\boldsymbol{y})\frac{\partial g_1}{\partial y_2}(\boldsymbol{y}) - \frac{\partial \boldsymbol{g}}{\partial y_2}(\boldsymbol{y})\frac{\partial g_1}{\partial y_1}(\boldsymbol{y})\Big)\\
&\qquad = \frac{\partial f_1}{\partial x_1}(g(\boldsymbol{y}))\Big(\frac{\partial g_1}{\partial y_1}(\boldsymbol{y})\frac{\partial g_1}{\partial y_2}(\boldsymbol{y}) - \frac{\partial g_1}{\partial y_2}(\boldsymbol{y})\frac{\partial g_1}{\partial y_1}(\boldsymbol{y})\Big)\\
&\qquad + \frac{\partial f_1}{\partial x_2}(g(\boldsymbol{y}))\Big(\frac{\partial g_2}{\partial y_1}(\boldsymbol{y})\frac{\partial g_1}{\partial y_2}(\boldsymbol{y}) - \frac{\partial g_2}{\partial y_2}(\boldsymbol{y})\frac{\partial g_1}{\partial y_1}(\boldsymbol{y})\Big)\\
&\qquad + \frac{\partial f_1}{\partial x_3}(g(\boldsymbol{y}))\Big(\frac{\partial g_3}{\partial y_1}(\boldsymbol{y})\frac{\partial g_1}{\partial y_2}(\boldsymbol{y}) - \frac{\partial g_3}{\partial y_2}(\boldsymbol{y})\frac{\partial g_1}{\partial y_1}(\boldsymbol{y})\Big)\\
&\qquad\qquad = -\frac{\partial f_1}{\partial x_2}(g(\boldsymbol{y}))\frac{\partial(g_1,g_2)}{\partial(y_1,y_2)}(\boldsymbol{y}) + \frac{\partial f_1}{\partial x_3}(g(\boldsymbol{y}))\frac{\partial(g_3,g_1)}{\partial(y_1,y_2)}(\boldsymbol{y}).
\end{aligned}$$

Now observe that

$$\operatorname{curl}\begin{bmatrix} f_1\\ 0\\ 0\end{bmatrix} = \begin{bmatrix} 0\\ \dfrac{\partial f_1}{\partial x_3}\\ -\dfrac{\partial f_1}{\partial x_2}\end{bmatrix}.$$

Hence it follows from Proposition 4.3.2 that

$$\begin{aligned}
\int_{\partial S}(f_1,0,0)\cdot\boldsymbol{\tau}\,ds &= \int_D \frac{\partial f_1}{\partial x_3}(g(\boldsymbol{y}))\frac{\partial(g_3,g_1)}{\partial(y_1,y_2)}(\boldsymbol{y}) - \frac{\partial f_1}{\partial x_2}(g(\boldsymbol{y}))\frac{\partial(g_1,g_2)}{\partial(y_1,y_2)}(\boldsymbol{y})\,d\boldsymbol{y}\\
&= \int_S \operatorname{curl}(f_1,0,0)\cdot\boldsymbol{n}\,dS.
\end{aligned}$$

Doing a cyclic permutation of the indices $1\to 2\to 3\to 1$ we get

$$\int_{\partial S}(0,f_2,0)\cdot\boldsymbol{\tau}\,ds = \int_S \operatorname{curl}(0,f_2,0)\cdot\boldsymbol{n}\,dS$$

and

$$\int_{\partial S}(0,0,f_3)\cdot\boldsymbol{\tau}\,ds = \int_S \operatorname{curl}(0,0,f_3)\cdot\boldsymbol{n}\,dS$$

Adding the three identities we get (5.4.1), completing the proof of the theorem. ■

5.4.3 Application: Conservative Vector Fields in Space

In Section 3.5.2 we stated without a proof that a closed vector field, $\boldsymbol{f}$, on a simply connected domain $D\subset\mathbb{R}^3$ is conservative. We now want to apply the theorem of

Stokes to sketch a proof. Simply connected means that every closed curve in D can be continuously deformed into a point in D without ever leaving D as shown in Figure 3.5.2. It turns out that this is equivalent to the following. Suppose that C_0 and C_1 are smooth curves in D connecting $\boldsymbol{x}_0 \in D$ to $\boldsymbol{x}_1 \in D$. Then it is possible to deform C_0 smoothly into C_1 without leaving D. When doing this the "trace" left by the moving curve defines a surface, S, with boundary $\partial S = C_0 \cup C_1$ as shown in Figure 5.4.4. Choosing consistent

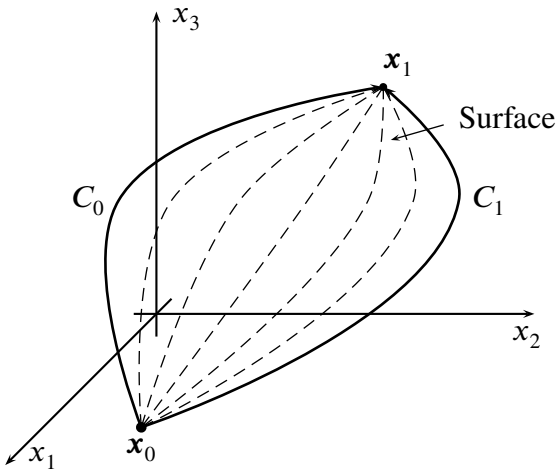


Figure 5.4.4: A surface traced out by deforming one curve smoothly into another.

orientations for S and ∂S we get from Stokes' Theorem

$$\int_{C_0} \boldsymbol{f} \cdot d\boldsymbol{x} - \int_{C_1} \boldsymbol{f} \cdot d\boldsymbol{x} = \int_{\partial S} \boldsymbol{f} \cdot d\boldsymbol{x} = \int_S (\operatorname{curl} \boldsymbol{f}) \cdot \boldsymbol{n}\, dS.$$

We saw in Section 3.5.2 that a vector field defined on a subset D of $\mathbb{R}^3$ is closed if and only if $\operatorname{curl} \boldsymbol{f} = 0$ on D. Hence, if $\boldsymbol{f}$ is closed, then the above surface integral is zero, and therefore

$$\int_{C_0} \boldsymbol{f} \cdot d\boldsymbol{x} = \int_{C_1} \boldsymbol{f} \cdot d\boldsymbol{x}.$$

This shows that line integrals are path independent and thus $\boldsymbol{f}$ is conservative.

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